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September 10, 2026

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VIA EDIS

The Honorable Lisa R. Barton
Secretary to the Commission
U.S. International Trade Commission
500 E Street, SW, Room 112
Washington, DC 20436

Re: *In the Matter of Certain Wearable Devices, ITC Inv. No. 337-TA-1478*

Dear Secretary Barton,

Complainants Samsung Electronics Co., Ltd. and Samsung Electronics America, Inc. ("Samsung") submit this supplement to Samsung's Verified First Amended Complaint (Doc. ID No. 867832), dated January 5, 2026. In compliance with Commission Rules 210.12(a)(9)(i) and 210.12(a)(9)(ii), Samsung respectfully encloses the following replacement exhibit and appendix to be substituted for the non-certified versions originally submitted with Samsung's Verified First Amended Complaint:

Exhibit	Description
1	Certified Copy of U.S. Patent No. 10,978,789
Appendix	Description
A	Certified Copy of the Prosecution History for U.S. Patent No. 10,978,789

Please contact me with any questions regarding this submission. Thank you for your attention to this matter.

Respectfully submitted,

/s/ D. Sean Trainor
D. Sean Trainor

O'MELVENY & MYERS LLP

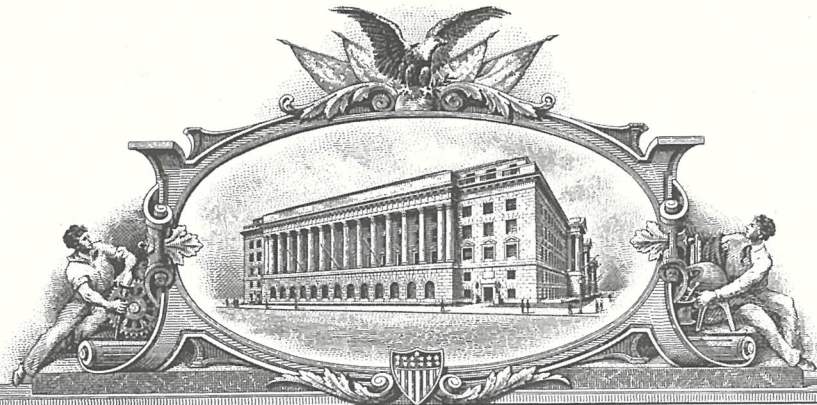
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*Counsel for Complainants Samsung Electronics
Co., Ltd. and Samsung Electronics America, Inc..*

EXHIBIT 1

8660338



THE UNITED STATES OF AMERICA

TO ALL TO WHOM THESE PRESENTS SHALL COME:

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office

February 21, 2026

THIS IS TO CERTIFY THAT ANNEXED HERETO IS A TRUE COPY FROM
THE RECORDS OF THIS OFFICE OF:

PATENT NUMBER: 10,978,789

ISSUE DATE: April 13, 2021

By Authority of the
Under Secretary of Commerce for Intellectual Property
and Director of the United States Patent and Trademark Office

Paula Smith

Paula Smith
Certifying Officer





US010978789B2

(12) **United States Patent**
Cho et al.

(10) **Patent No.:** **US 10,978,789 B2**
(45) **Date of Patent:** **Apr. 13, 2021**

(54) **ELECTRONIC DEVICE INCLUDING
HIGH-FREQUENCY TRANSMISSION
CIRCUIT**

(58) **Field of Classification Search**
USPC 361/790
See application file for complete search history.

(71) Applicant: **Samsung Electronics Co., Ltd.,**
Gyeonggi-do (KR)

(56) **References Cited**

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Byoung-Il Son, Gyeonggi-do (KR);
Sung-Chul Park, Seoul (KR)

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(Continued)

(73) Assignee: **Samsung Electronics Co., Ltd**

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

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part application No. PCT/KR2017/006070, 7 pages.

(21) Appl. No.: **16/939,708**

(Continued)

(22) Filed: **Jul. 27, 2020**

(65) **Prior Publication Data**

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Primary Examiner — Andargie M Aychillhum

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Related U.S. Application Data

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Jun. 13, 2017, now Pat. No. 10,727,568.

(51) **Int. Cl.**
H05K 7/00 (2006.01)
H01Q 1/24 (2006.01)

(Continued)

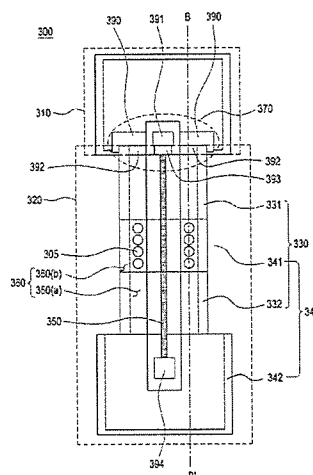
(52) **U.S. Cl.**
CPC **H01Q 1/243** (2013.01); **H01Q 1/38**
(2013.01); **H01Q 5/35** (2015.01); **H01Q 21/28**
(2013.01);

(Continued)

(57) **ABSTRACT**

Disclosed is an electronic device including a housing, a
printed circuit board (PCB) in the housing, wherein the PCB
includes a plurality of layers with one or more conductive
and insulation layers, a first electrical component formed as
at least a part of or in the housing, a second electrical
component above or near the PCB in the housing, wherein
the first and second electrical components are separated, and
at least one electrical path extending from the first electrical
component to the second electrical component, wherein at
least a portion of the electrical path runs on or inside the
PCB, wherein the PCB includes a region including a pattern
of conductive vias, wherein each of the vias extends through
at least part of the plurality of layers to contact at least one

(Continued)



of the one or more conductive layers, and wherein the electrical path runs through the region without contacting the vias.

20 Claims, 14 Drawing Sheets

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H01Q 1/38 (2006.01)
H01R 12/79 (2011.01)
H01R 13/66 (2006.01)
H05K 1/11 (2006.01)
H05K 1/14 (2006.01)
H05K 1/18 (2006.01)
H05K 7/14 (2006.01)

(52) U.S. Cl.

CPC *H01R 12/79* (2013.01); *H01R 13/6691* (2013.01); *H05K 1/028* (2013.01); *H05K 1/0218* (2013.01); *H05K 1/0243* (2013.01); *H05K 1/115* (2013.01); *H05K 1/142* (2013.01); *H05K 1/148* (2013.01); *H05K 1/18* (2013.01); *H05K 7/1427* (2013.01); *H05K 2201/0999* (2013.01); *H05K 2201/09609* (2013.01); *H05K 2201/10098* (2013.01)

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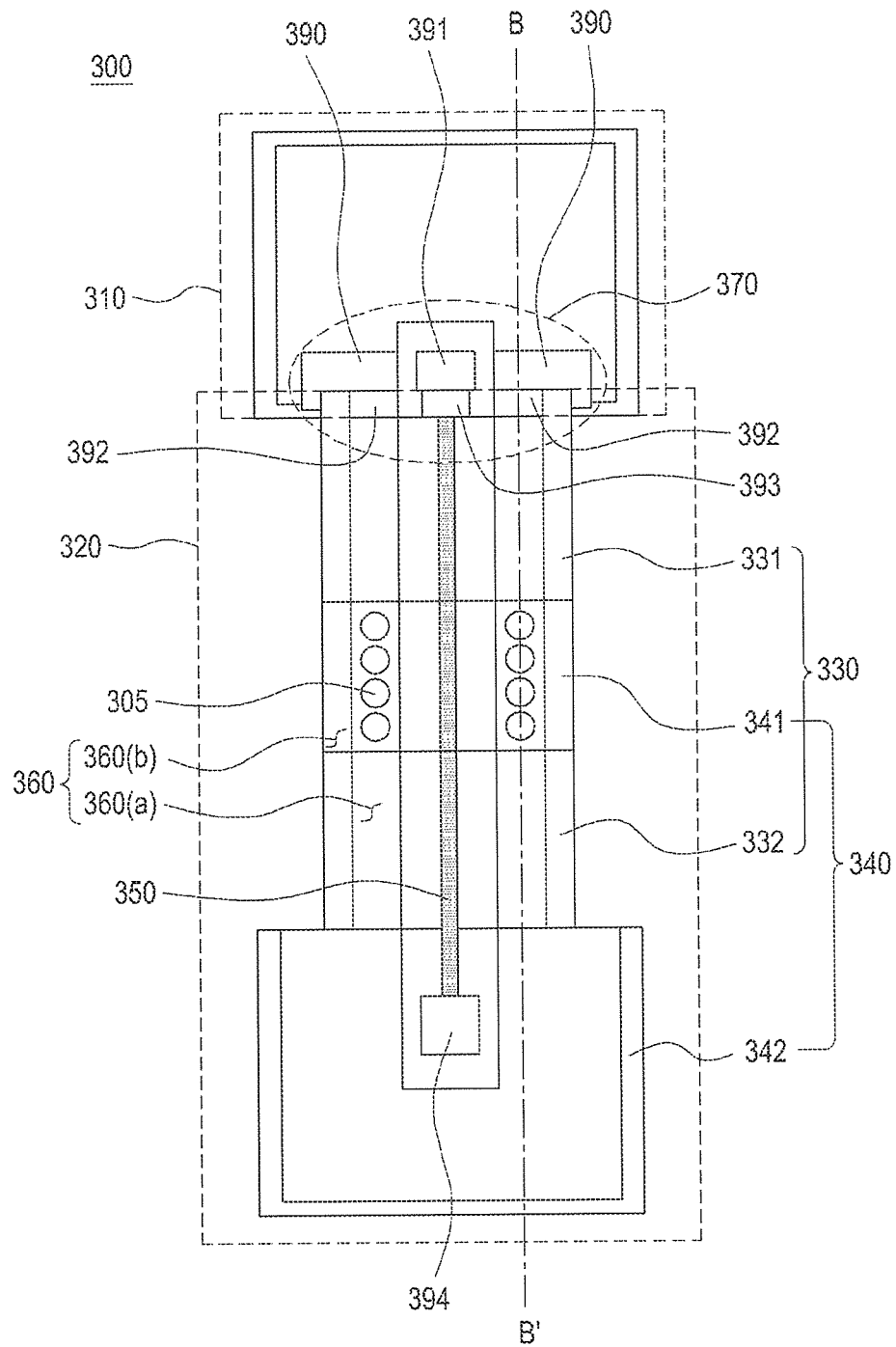


FIG. 1

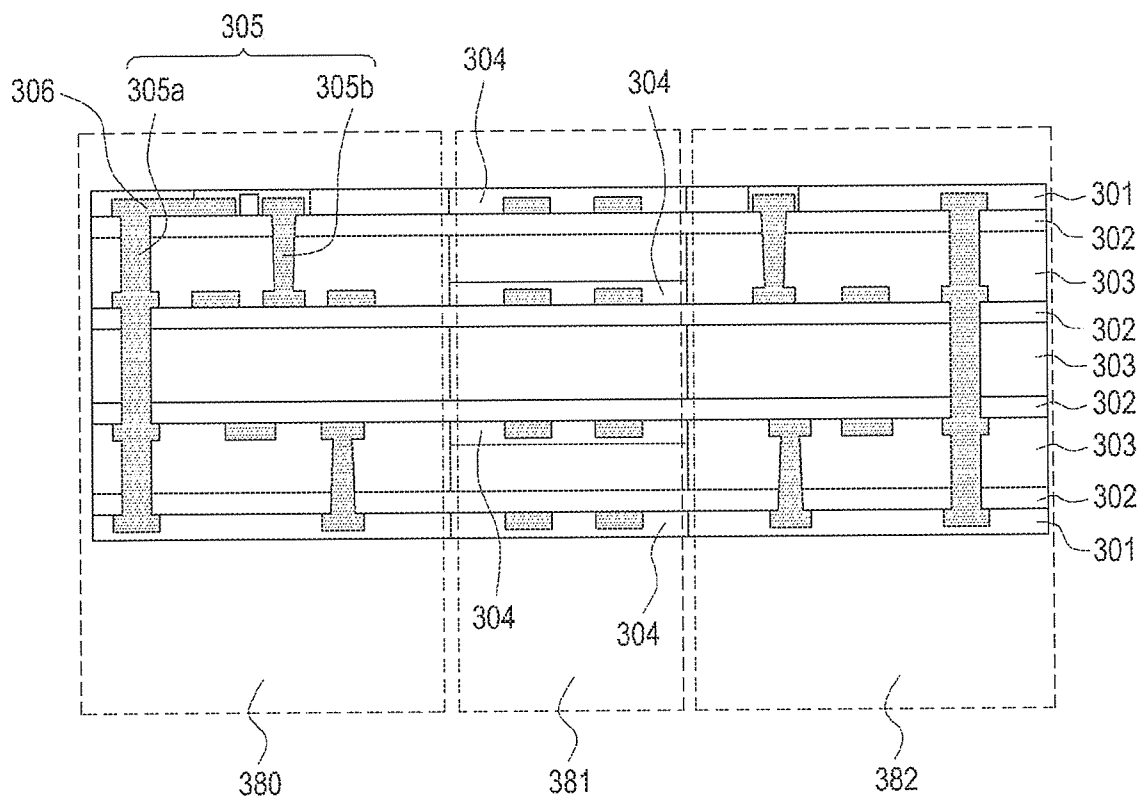


FIG. 2

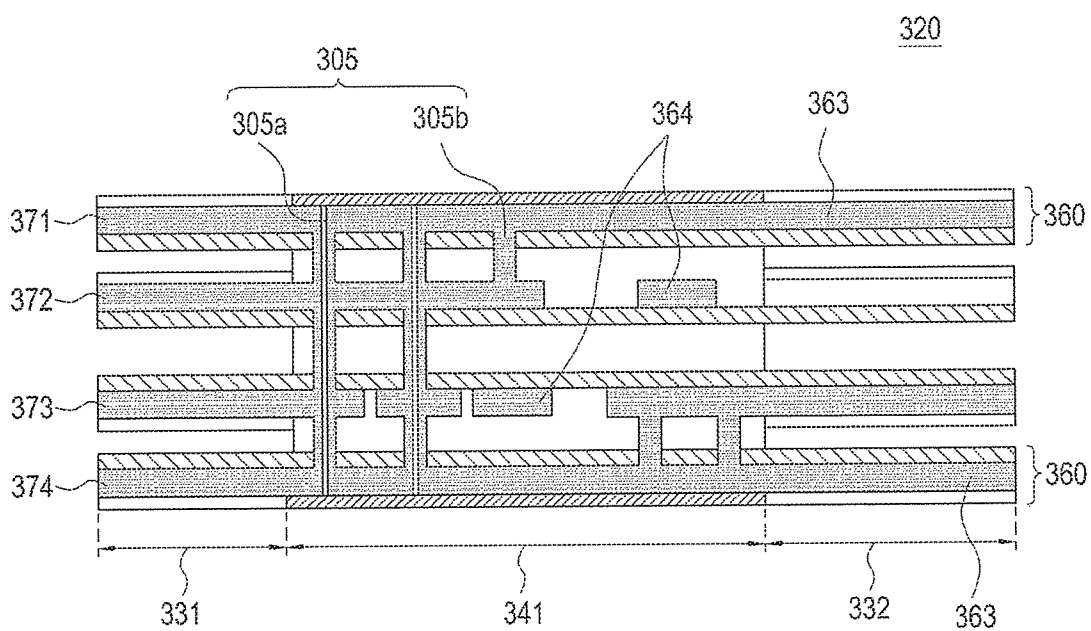


FIG.3

FIG. 4

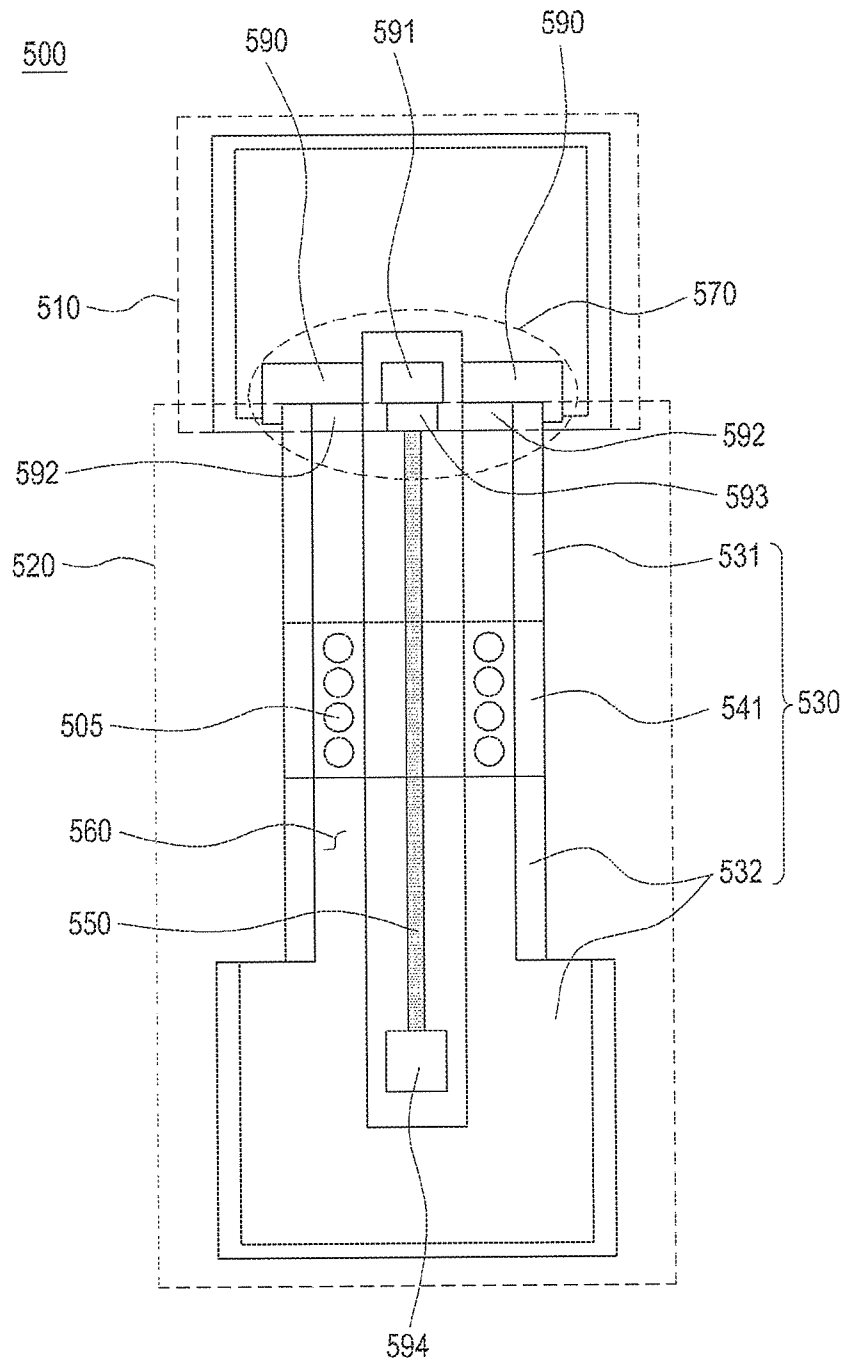


FIG. 5

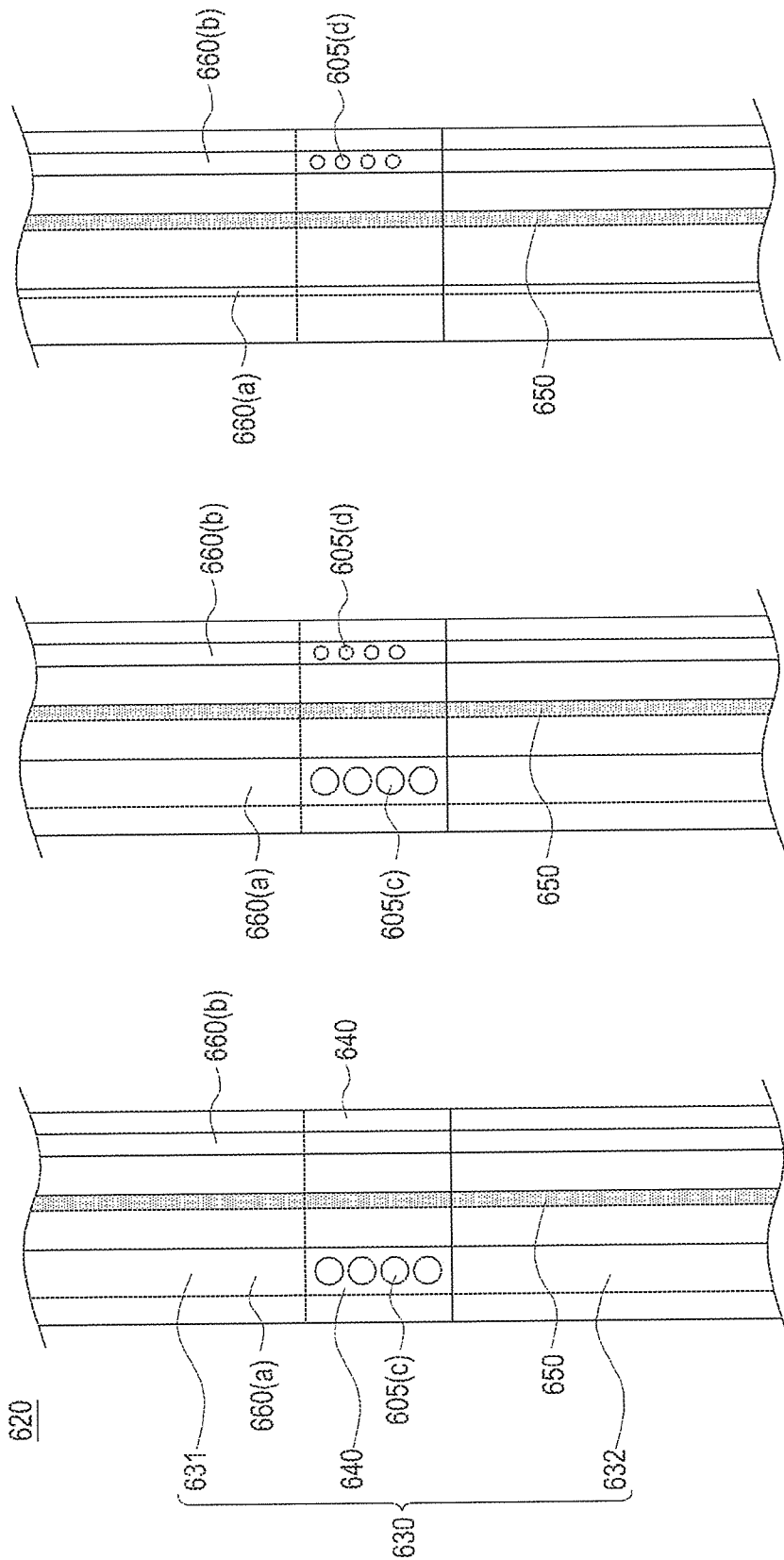


FIG.6C

FIG.6B

FIG.6A

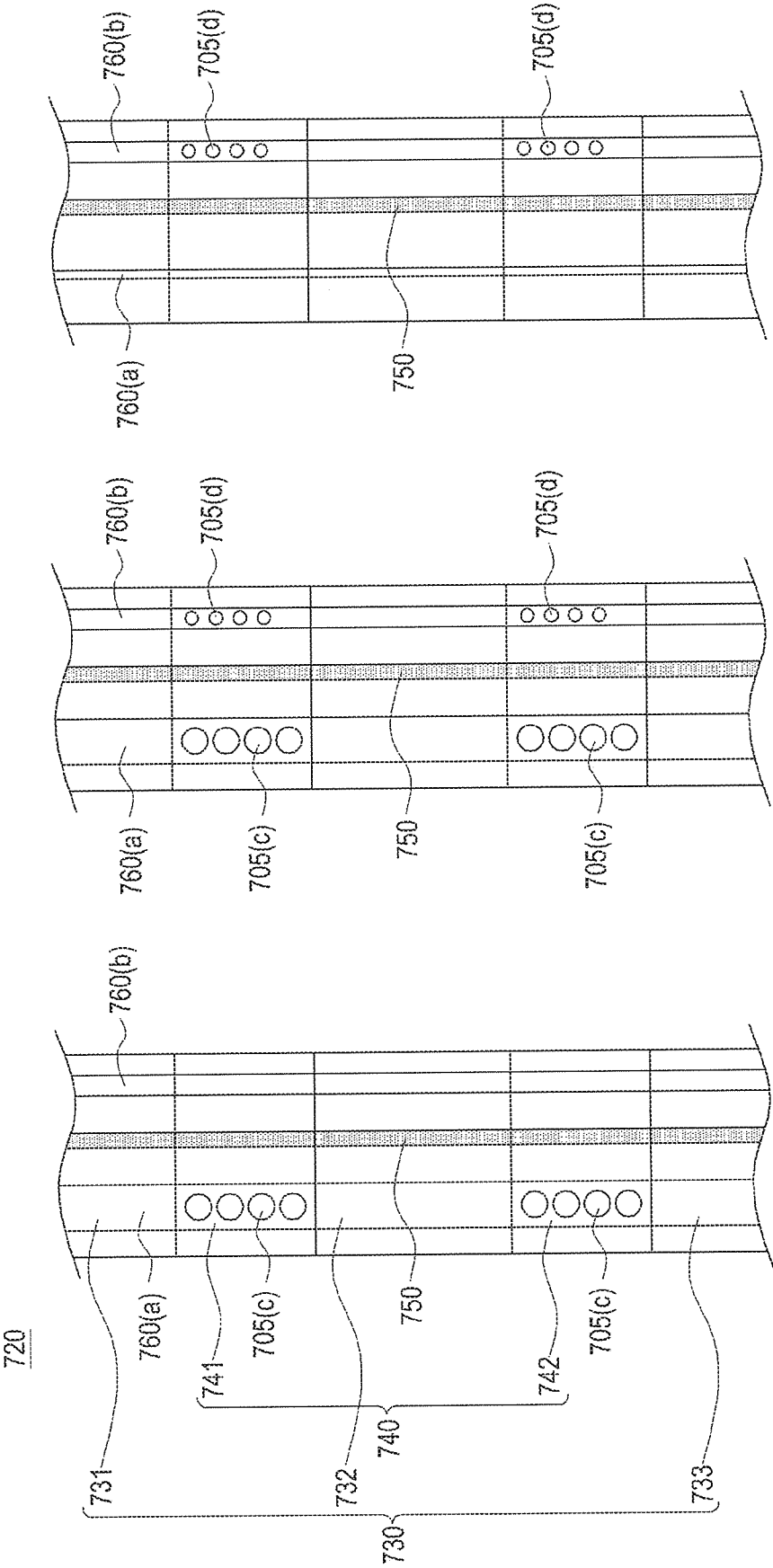


FIG. 7C

FIG. 7B

FIG. 7A

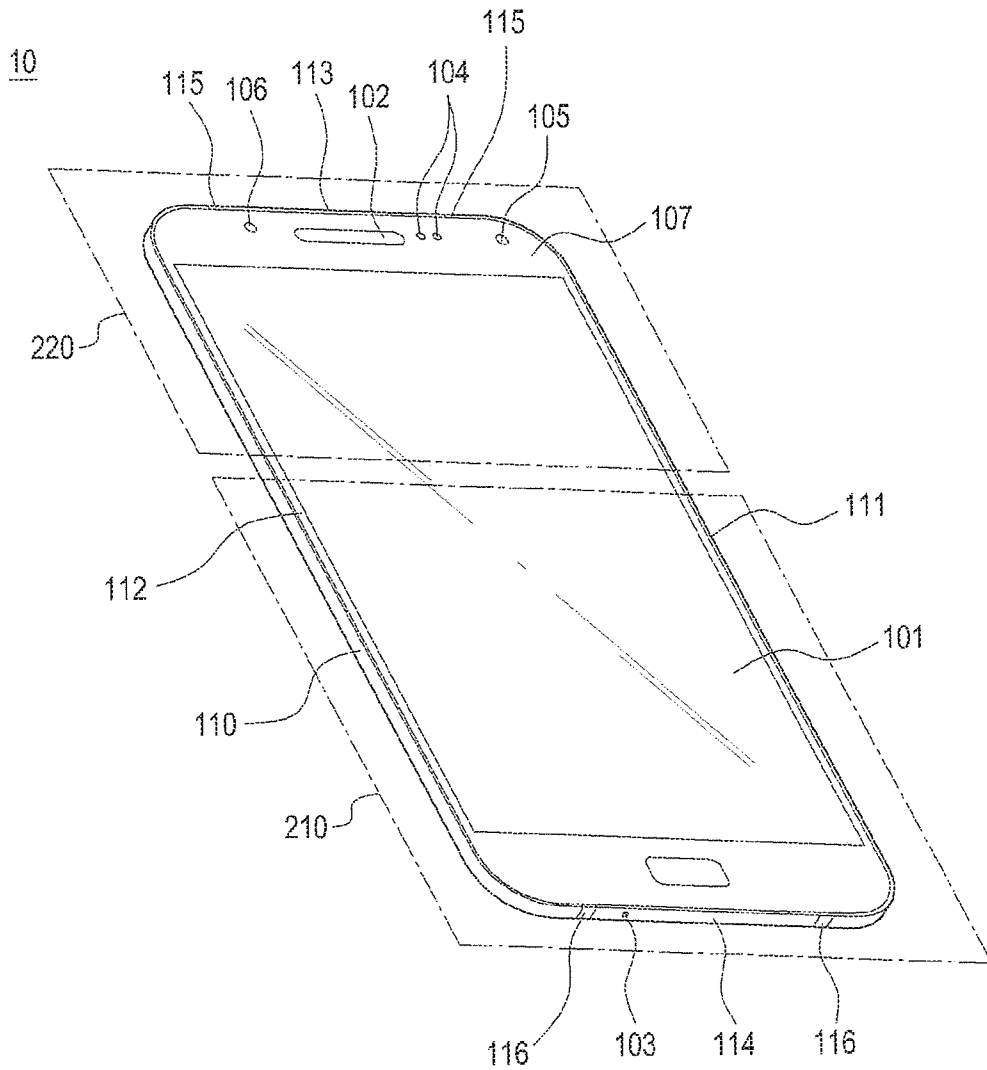


FIG. 8

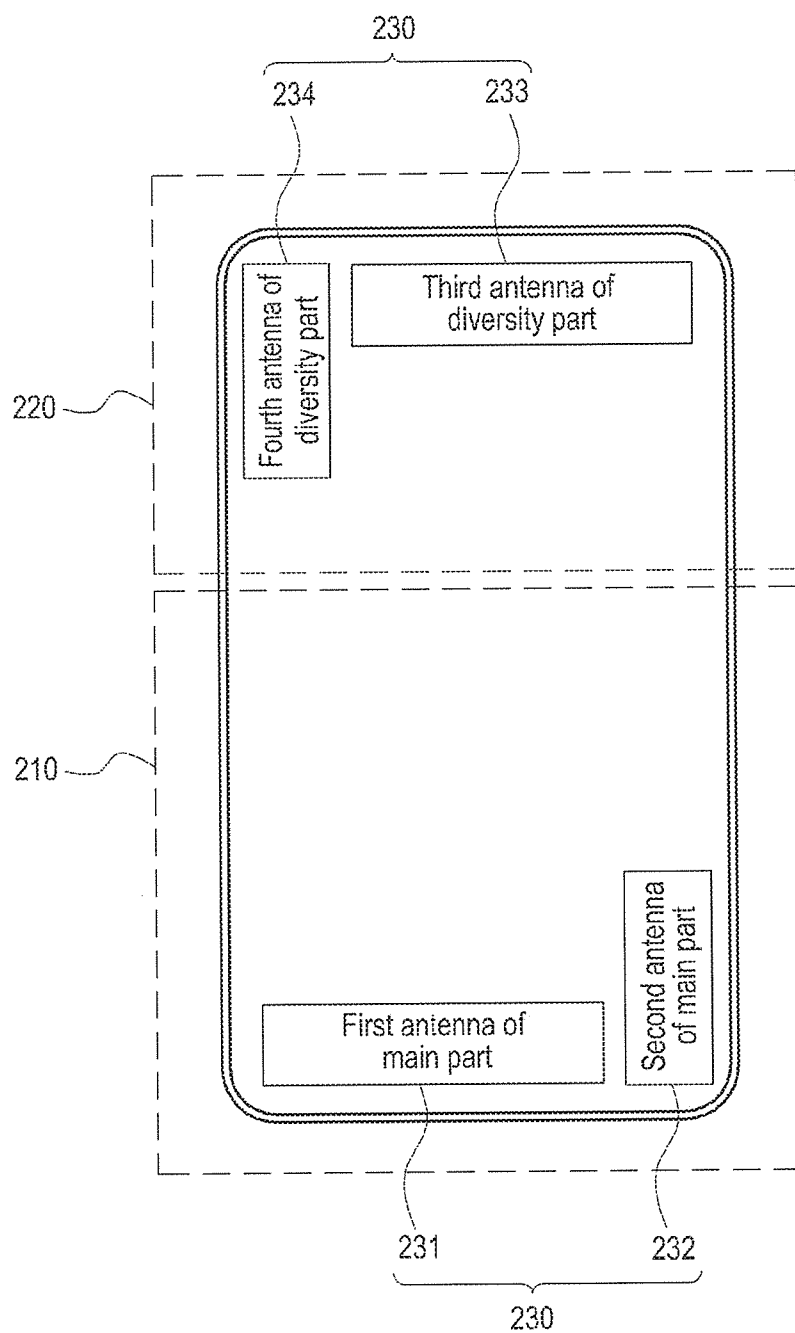


FIG. 9

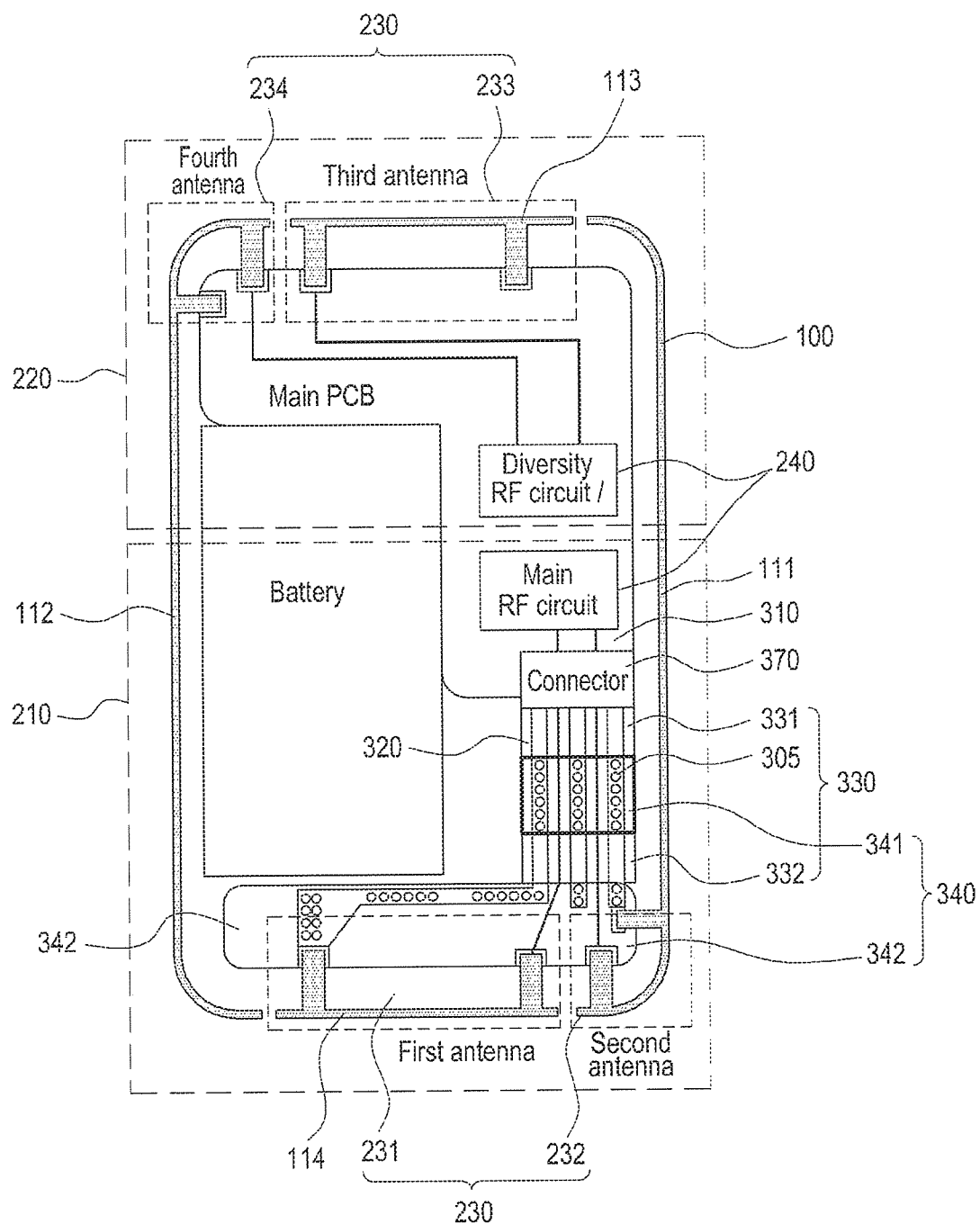


FIG. 10

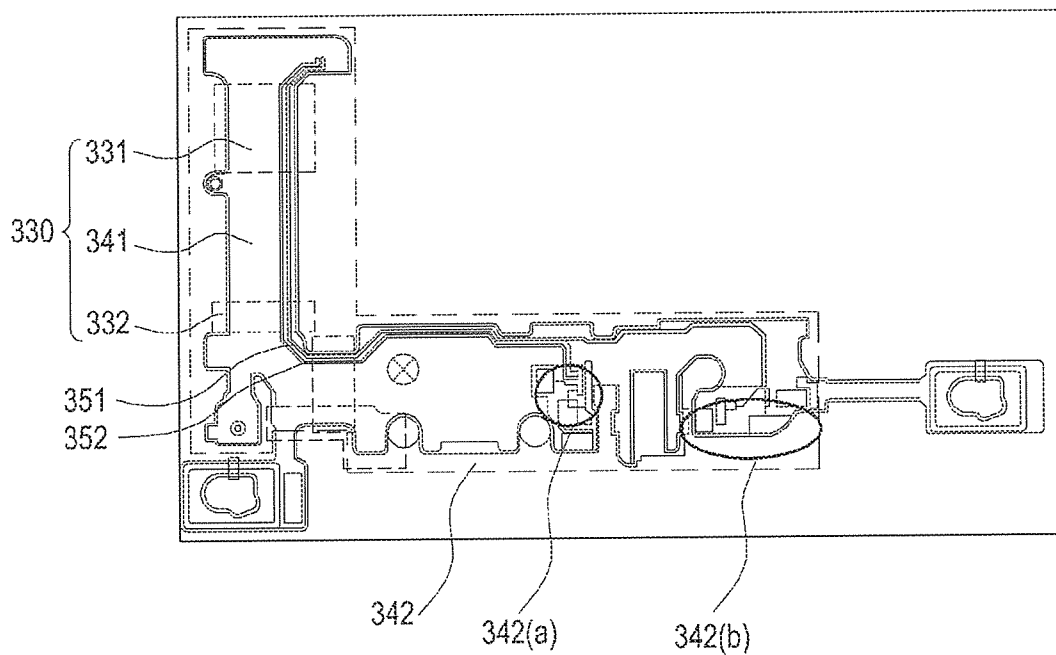


FIG. 11

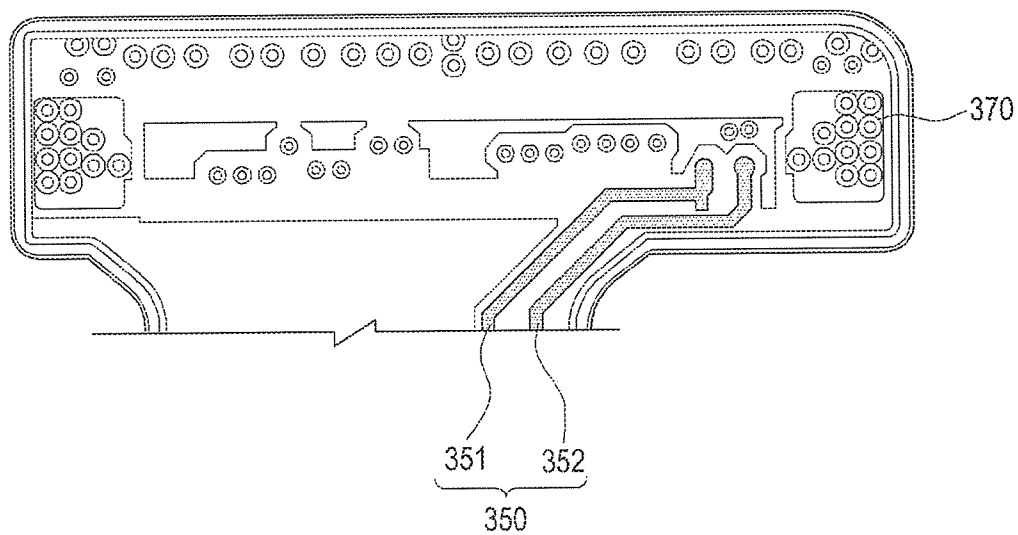


FIG. 12

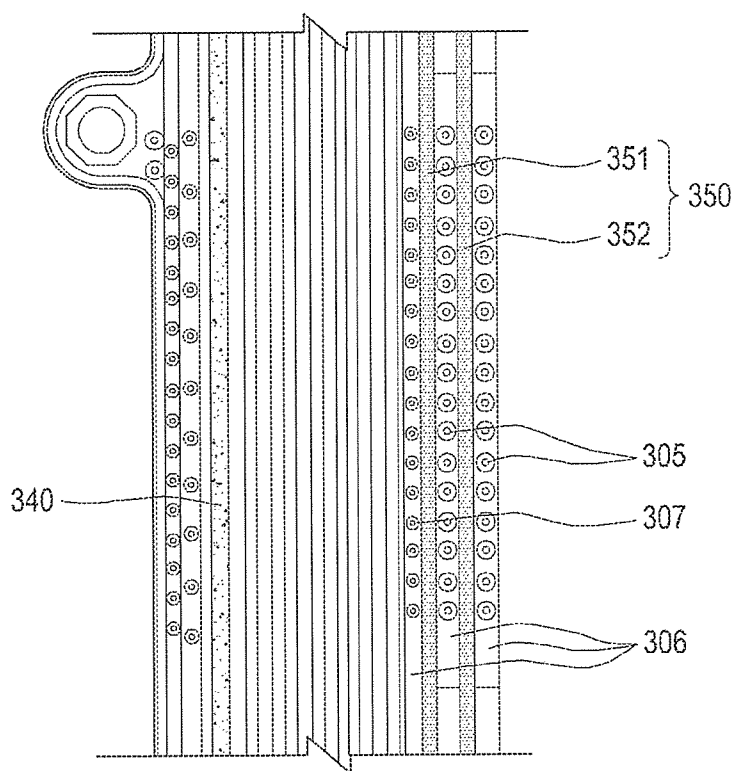


FIG. 13

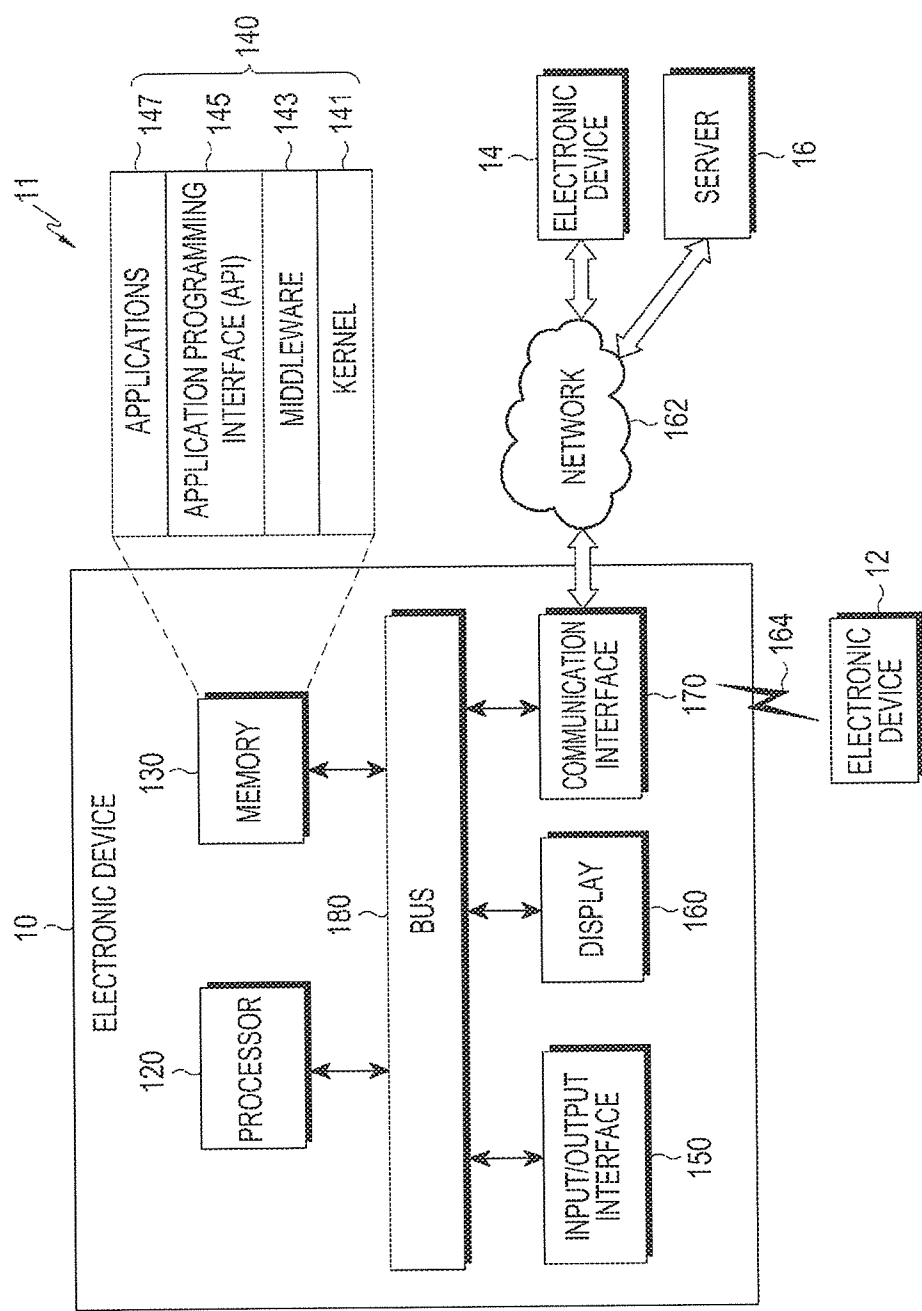


FIG.14

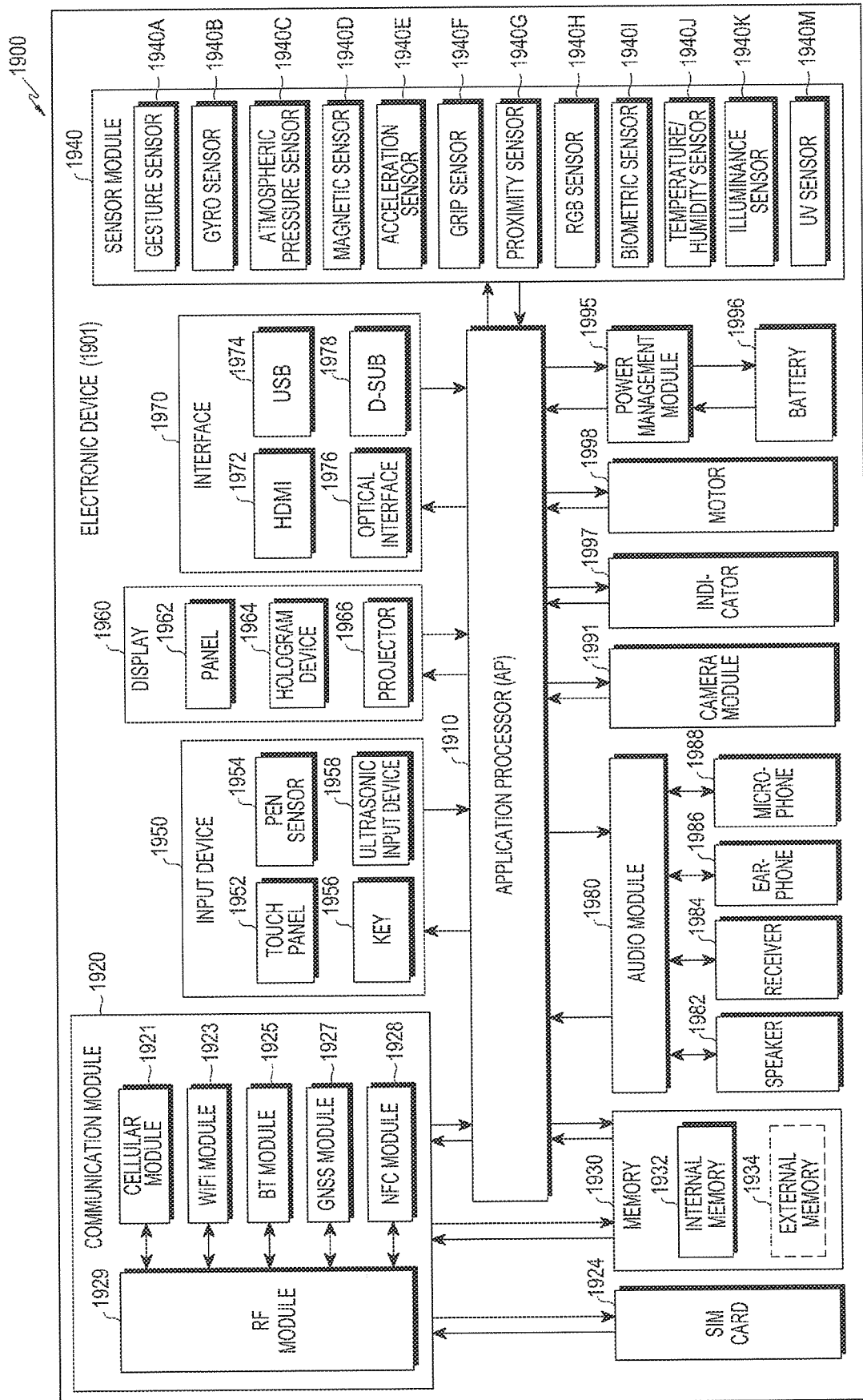


FIG.15

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ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT

PRIORITY

This application is a Continuation Application of U.S. patent application Ser. No. 15/621,687, filed on Jun. 13, 2017, and claims priority under 35 U.S.C. § 119(a) to a Korean Patent Application filed on Jun. 13, 2016 in the Korean Intellectual Property Office and assigned Serial No. 10-2016-0073355, the content of each of which is incorporated herein by reference.

BACKGROUND

1. Field of the Disclosure

The present disclosure relates generally to a high-frequency transmission circuit included in an electronic device, and more particularly, to a transmission line capable of effectively transmitting a noiseless high-frequency signal in a narrow space without affecting other modules.

2. Description of the Related Art

Recently, portable terminals having new functions have been rapidly developed and have accounted for a greater part of people's lives with the common use thereof.

Internal circuits of portable terminals are generally implemented on printed circuit boards (PCBs). PCB technologies have been rapidly developing, and flexible PCBs (FPCBs) that can be bent easily are currently used widely, as well as conventional rigid PCBs.

In addition, coaxial cables are typically used for high-frequency transmission lines (especially, radio frequency (RF) lines) that are applied to wireless terminals. A large amount of additional material cost may be required to produce coaxial cables used for electronic devices as high-frequency transmission lines, and additional labor cost may be incurred by an increase in the number of assembly steps. In another example, electronic devices have narrow internal spaces available on account of various types of circuit modules mounted therein, and due to the narrow internal spaces, a defect rate may increase when small connectors used for coaxial cables are assembled.

SUMMARY

An aspect of the present disclosure provides a transmission line capable of effectively transmitting a noiseless high-frequency signal in a narrow space without affecting other modules.

Another aspect of the present disclosure provides that a coaxial cable used as a transmission line in a portable terminal can be replaced by a circuit board that includes an FPCB, and thus it is possible to eliminate material costs associated with the manufacture of a coaxial cable.

Another aspect of the present disclosure provides a rigid PCB that is disposed between transmission lines of an FPCB, whereby it is possible to reduce line loss occurring in the transmission lines of the FPCB and to provide a high-frequency signal without loss.

Another aspect of the present disclosure provides that it is possible to efficiently use a mounting space in a wireless

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terminal, which enables diversification of the external design of the wireless terminal to improve customer satisfaction.

In accordance with an aspect of the present disclosure, an electronic device is provided. The electronic device includes a housing; a PCB disposed in the housing, wherein the PCB includes a plurality of layers that include one or more conductive layers and one or more insulation layers; a first electrical component formed to be at least a part of the housing or disposed in the housing; a second electrical component disposed above or near the PCB in the housing, wherein the second electrical component is separated from the first electrical component; and at least one electrical path extending from the first electrical component to the second electrical component, wherein at least a portion of the electrical path runs on or inside the PCB, wherein the PCB includes a region including a pattern of conductive vias, wherein each of the conductive vias extends through at least a part of the plurality of layers to contact at least one of the one or more conductive layers, and wherein the electrical path may run through the region without making contact with the conductive vias.

In accordance with another aspect of the present disclosure, an electronic device is provided. The electronic device includes a first PCB; a second PCB connected to the first PCB through a connector and including at least one FPCB and at least one rigid PCB that are integrated with each other, wherein the at least one FPCB flexible and the at least one rigid PCB each include a plurality of layers; an electrical path that transmits a high-frequency signal through the connector and passes over or through the second PCB; and at least one conductive via disposed in the at least one rigid PCB to connect grounds between the plurality of layers.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other aspects, features, and advantages of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings, in which:

FIG. 1 is a front view of a high-frequency transmission line according to an embodiment of the present disclosure, in which a PCB in an electronic device is used for the high-frequency transmission line;

FIG. 2 is a partial sectional view of a plurality of layers of the PCB of FIG. 1, according to an embodiment of the present disclosure, where FIG. 2 is taken along line B-B' of FIG. 1;

FIG. 3 is a sectional view of an interlayer configuration of a plurality of layers of the PCB of FIG. 1, according to an embodiment of the present disclosure, where FIG. 3 is taken along line B-B' of FIG. 1;

FIG. 4 is a front view of a high-frequency transmission line according to an embodiment of the present disclosure, where a PCB in an electronic device is used for the high-frequency transmission line;

FIG. 5 is a front view of a high-frequency transmission line according to an embodiment of the present disclosure, where a PCB in an electronic device is used for the high-frequency transmission line;

FIGS. 6A, 6B, 6C, 7A, 7B, and 7C are front views of conductive vias according to an embodiment of the present disclosure, the conductive vias being arranged in PCBs in electronic devices;

FIG. 8 is a perspective view of an electronic device according to an embodiment of the present disclosure;

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FIG. 9 is a schematic of an antenna arrangement in an electronic device according to an embodiment of the present disclosure;

FIG. 10 illustrates an antenna device and a PCB in an electronic device according to an embodiment of the present disclosure;

FIG. 11 is an illustration of a signal transmission using a second PCB according to an embodiment of the present disclosure;

FIGS. 12 and 13 are enlarged views of a first FPCB and a first rigid PCB;

FIG. 14 is a block diagram of an electronic device according to an embodiment of the present disclosure in a network environment; and

FIG. 15 is a block diagram of an electronic device according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE PRESENT DISCLOSURE

Hereinafter, various embodiments of the present disclosure are described with reference to the accompanying drawings. However, it should be understood that there is no intent to limit the present disclosure to the particular forms disclosed herein; rather, the present disclosure is intended to be construed to cover various modifications, equivalents, and/or alternatives of embodiments of the present disclosure. In the description of the accompanying drawings, similar reference numerals may be used to designate similar elements.

As used herein, the expressions “have”, “may have”, “include”, or “may include” refer to the existence of a corresponding feature (e.g., numeral, function, operation, or element such as a component), and do not exclude one or more additional features.

In the present disclosure, the expressions “A or B”, “at least one of A and/or B”, and “one or more of A and/or B” may include all possible combinations of the items listed. For example, the expressions “A or B”, “at least one of A and B”, and “at least one of A or B” refer to all of (1) including at least one A, (2) including at least one B, and (3) including all of at least one A and at least one B.

The expressions “a first”, “a second”, “the first”, or “the second” used in various embodiments of the present disclosure may modify various components regardless of the order and/or the importance of the components but are not intended to limit the corresponding components. For example, a first user device and a second user device indicate different user devices although both of them are user devices. For example, a first element may be referred to as a second element, and similarly, a second element may be referred to as a first element without departing from the scope of the present disclosure.

It should be understood that when an element (e.g., a first element) is referred to as being (operatively or communicatively) “connected,” or “coupled,” to another element (e.g., a second element), the element may be directly connected or directly coupled to the other element or any other element (e.g., a third element) may be interposer between them. In contrast, it may be understood that when an element (e.g., a first element) is referred to as being “directly connected,” or “directly coupled” to another element (e.g., a second element), there is no element (e.g., a third element) interposed between them.

The expression “configured to” used in the present disclosure may be used interchangeably with the expressions, for example, “suitable for”, “having the capacity to”,

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“designed to”, “adapted to”, “made to”, and “capable of” according to the situation. The term “configured to” may not necessarily imply “specifically designed to” in hardware. Alternatively, in some situations, the expression “device configured to” may indicate that the device, together with other devices or components, “is able to”. For example, the expression “processor adapted (or configured) to perform A, B, and C” may indicate a dedicated processor (e.g., an embedded processor) only for performing the corresponding operations or a general purpose processor (e.g., a central processing unit (CPU) or an application processor (AP)) that can perform the corresponding operations by executing one or more software programs stored in a memory device.

The terms used herein are merely used for the purpose of describing particular embodiments but are not intended to limit the scope of the present disclosure. A singular expression may include a plural expression unless they are definitely different in a context. Unless defined otherwise, all terms used herein, have the same meanings as those commonly understood by a person skilled in the art to which the present disclosure pertains. Such terms as those defined in a generally used dictionary may be interpreted to have the meanings equal to the contextual meanings in the relevant field of art, and are not intended to be interpreted to have ideal or excessively formal meanings unless clearly defined in the present disclosure. In some cases, even the terms defined in the present disclosure are not intended to be interpreted to exclude embodiments of the present disclosure.

An electronic device according to various embodiments of the present disclosure may include at least one of, for example, a smart phone, a tablet personal computer (PC), a mobile phone, a video phone, an electronic book reader (e-book reader), a desktop PC, a laptop PC, a netbook computer, a workstation, a server, a personal digital assistant (PDA), a portable multimedia player (PMP), a moving picture experts group (MPEG-1) audio layer-3 (MP3) player, a mobile medical device, a camera, and a wearable device. According to various embodiments of the present disclosure, the wearable device may include at least one of an accessory type (e.g., a watch, a ring, a bracelet, an anklet, a necklace, glasses, a contact lens, or a head-mounted device (HMD)), a fabric or clothing integrated type (e.g., electronic clothing), a body-mounted type (e.g., a skin pad, or tattoo), and a bio-implantable type (e.g., an implantable circuit).

According to an embodiment of the present disclosure, the electronic device may be a home appliance. A home appliance may include at least one of, for example, a television, a digital video disk (DVD) player, an audio player, a refrigerator, an air conditioner, a vacuum cleaner, an oven, a microwave oven, a washing machine, an air cleaner, a set-top box, a home automation control panel, a security control panel, a TV box (e.g., Samsung Home-Sync®, Apple TV®, or Google TV™), a game console (e.g., Xbox® and PlayStation®), an electronic dictionary, an electronic key, a camcorder, and an electronic photo frame.

According to an embodiment of the present disclosure, the electronic device may include at least one of various medical devices (e.g., various portable medical measuring devices (a blood glucose monitoring device, a heart rate monitoring device, a blood pressure measuring device, a body temperature measuring device, etc.), a magnetic resonance angiography (MRA), a magnetic resonance imaging (MRI), a computed tomography (CT) machine, and an ultrasonic machine), a navigation device, a global positioning system (GPS) receiver, an event data recorder (EDR), a flight data recorder (FDR), a vehicle infotainment device, an

electronic device for a ship (e.g., a navigation device for a ship, and a gyro-compass), avionics, security devices, an automotive head unit, a robot for home or industry, an automated teller machine (ATM), point of sales (POS) device in a shop, or an Internet of Things (IoT) device (e.g., a light bulb, various sensors, electric or gas meter, a sprinkler device, a fire alarm, a thermostat, a streetlamp, a toaster, sporting goods, a hot water tank, a heater, a boiler, etc.).

According to an embodiment of the present disclosure, an electronic device may include at least one of a part of furniture or a building/structure, an electronic board, an electronic signature receiving device, a projector, and various types of measuring instruments (e.g., a water meter, an electric meter, a gas meter, and a radio wave meter). An electronic device may be a combination of one or more of the aforementioned various devices. An electronic device may also be a flexible device. Further, an electronic device is not intended to be limited to the aforementioned devices, but may include an electronic device to be developed in the future.

Hereinafter, an electronic device according to various embodiments of the present disclosure is described with reference to the accompanying drawings. In the present disclosure, the term “user” may indicate a person using an electronic device or a device (e.g., an artificial intelligence electronic device) using an electronic device.

FIG. 1 is a front view of a high-frequency transmission line according to an embodiment of the present disclosure, in which a PCB 300 in an electronic device is used for the high-frequency transmission line. The electronic device may be a smart phone or a wearable device.

Referring to FIG. 1, the electronic device may include a housing corresponding to a case; a printed circuit unit including at least one PCB 300 in the housing; and an electrical path 350 passing over or through the printed circuit board 300.

The PCB 300 according to an embodiment of the present disclosure may be a first PCB 310 and a second PCB 320. For example, the first PCB 310 may be a main board on which electronic components included in the electronic device are mounted, the main board being capable of transmitting signals between the electronic components, and the second PCB 320 may be a sub-board that transmits signals generated by the electronic components. The sub-board may further include the electronic components.

The PCB 300 according to an embodiment of the present disclosure may be provided with a circuit of a terminal, which may be, for example, at least one of an AP, a communication processor (CP), a memory, and an RF transceiver, and may include a signal line (for example, an RF signal line).

The first and second PCBs 310 and 320 according to an embodiment of the present disclosure may be formed of a plurality of layers, and the plurality of layers may include one or more conductive layers 306 in FIG. 2 described below and one or more insulation layers 303.

The first PCB 310 in FIG. 1 according to an embodiment of the present disclosure may be a rigid PCB. The first PCB 310 may be electrically connected to the second PCB 320 through a connector 370. The connector 370 may include a ground pad 390 and a signal pad 391 of the first PCB 310 and a ground pad 392 and a signal pad 393 of the second PCB 320. The ground pad 390 of the first PCB 310 and the ground pad 392 of the second PCB 320 may be connected together by soldering. The signal pad 391 of the first PCB 310 and the signal pad 393 of the second PCB 320 may be connected together by soldering. The second PCB 320 may

include an interconnection pad 394. An antenna contact member (for example, a C-clip) or an RF component (for example, a matching circuit) may be mounted on the interconnection pad 394.

The second PCB 320 according to an embodiment of the present disclosure may be connected to the first PCB 310 and may include an FPCB 330 or a rigid PCB 340. The FPCB 330 and the rigid PCB 340 of the second PCB 320 may be integrated with each other. For example, a first FPCB 331, a first rigid printed circuit board 341, a second flexible printed circuit board 332, and a second rigid PCB 342 may be arranged in sequence. The second rigid PCB 342 may include an antenna connector and a universal serial bus (USB) connector, for example.

The second PCB 320 according to an embodiment of the present disclosure may have the electrical path 350 therein through which electrical signals can be transmitted. The electrical path 350 may be connected with the first PCB 310 through the FPCB 330 and the rigid PCB 340, which are integrated with each other.

The electrical path 350 according to an embodiment of the present disclosure may extend from a first electrical component 230 in FIG. 10 described below to a second electrical component 240 and may be disposed on or in the PCB 300 in FIG. 1. The electrical path 350 may be implemented as a transmission line of a side-guarded micro-strip-line type or a strip-line type, where the transmission line is shielded by a ground line 360 in order to transmit a high-frequency signal, which may be, for example, an RF signal.

The electrical path 350 according to an embodiment of the present disclosure, which is disposed on or in the second PCB 320, may receive a high-frequency signal through the connector 370 and may forward the received signal to antenna radiators 231 and 232 in FIG. 10 described below. Furthermore, a plurality of electrical paths 350 in FIG. 1 may be provided, and a ground line 360 may be disposed between the electrical paths 350 so that the electrical paths 350 can independently transmit the high-frequency signal without mutually affecting the characteristics thereof.

The ground line 360 according to an embodiment of the present disclosure may be disposed adjacent to the electrical path 350, and conductive vias 305 may be arranged in a portion of the area of a ground line 360(a). Accordingly, the electrical path 350 can pass through the PCB 300 without making contact with the conductive vias 305.

According to an embodiment of the present disclosure, the connector 370 for connecting the electrical path 350 may be disposed between the first and second PCBs 310 and 320 in order to transmit a high-frequency signal therebetween. The connector 370 may include a soldering pad, a zip-type connector, or a B-to-B-type connector. A rigid PCB may be employed for the connector 370, and, for example, a B-to-B-type connector may be mounted on the rigid printed circuit board, or a soldering pad may be located on the rigid PCB. Furthermore, for example, the pin map of a transmission line included in the connector 370 may be implemented in the sequence of a ground, an RF signal, and a ground.

Hereinafter, the plurality of layers and the conductive vias 305 of the PCB 300 is described in detail.

FIG. 2 is a partial sectional view of the plurality of layers of the PCB of FIG. 1, according to an embodiment of the present disclosure, where FIG. 2 is taken along line B-B' of FIG. 1. Hereinafter, the PCB is described as the second PCB 320. However, the following description may be applied to second printed circuit boards 420, 520, 620, and 720 in other embodiments, which will be described below.

Referring to FIG. 2, the second printed circuit board 320 according to an embodiment of the present disclosure may be formed of a plurality of layers, each including at least one conductive layer 306, at least one first insulation layer 302, and at least one second insulation layer 303. The conductive layer 306 and the first insulation layer 302 may be included in a flexible copper clad laminated (FCCL) layer. Further, the first insulation layer 302 may include polyamide.

The conductive layers 306 according to an embodiment of the present disclosure are arranged in the stacking direction, and the first insulation layer 302 is disposed on the rear or front surface of each conductive layer 306. The individual layers may be alternately arranged. One or more conductive vias 305 may be formed to electrically connect the conductive layers 306.

According to an embodiment of the present disclosure, the second insulation layer 303 is provided between the conductive layers 306 to prevent the conductive layers 306 from being electrically connected together through contact therebetween. For example, the plurality of second insulation layers 303 disposed between the plurality of conductive layers 306 may function to insulate the layers and may include a resin and a glass fabric.

According to an embodiment of the present disclosure, a plurality of electronic components may be disposed on and electrically connected to the conductive layers 306. The electronic components may be electrically connected to the antenna radiators 231 and 232 in FIG. 10 described below provided in the electronic device. For example, the electronic components may include an antenna clip, a connector, and a switch device.

Without being limited thereto, however, various materials capable of passing electricity therethrough, other than metal, may be applied to the conductive layers 306 in FIG. 2.

According to an embodiment of the present disclosure, the conductive layers 306 may be electrically connected to any of a plurality of signal lines and ground lines.

The conductive layers 306 according to an embodiment of the present disclosure may be provided, on the top and bottom thereof, with a protective layer 301 capable of protecting the conductive layers 306. The protective layers 301 may be formed of an insulating coating material, such as solder resist, and, for example, a photo imageable solder resist ink (PSR ink) may be used for the solder resist. The PSR ink may be left on a product because it provides insulation and protection even after a component is mounted. Accordingly, the protective layers 301 coated on the exterior of the conductive layers 306 can prevent a short circuit, a connection, corrosion, and contamination of a circuit during a manufacturing process and can protect the conductive layers 306 from external shock, humidity, and chemical substances after the manufacturing process.

The conductive vias 305 according to an embodiment of the present disclosure, which electrically connect the conductive layers 306, may include a first conductive via 305(a) capable of electrically connecting all the layers and a second conductive via 305(b) capable of electrically connecting adjacent layers. For example, the first conductive via 305(a) may be formed by computerized numerical control (CNC) hole machining and copper plating, and the second conductive via 305(b) may be formed by laser hole machining and copper plating.

The conductive vias 305 according to an embodiment of the present disclosure may include a region that forms a predetermined pattern within the PCB 300. The conductive

vias 305 may be configured to make contact with at least one of the conductive layers 306 and to pass through at least one of the multiple layers.

A pattern cover layer 304 according to an embodiment of the present disclosure may be disposed in a single layer on the bottom or top of the conductive layers 306 while surrounding a pattern and thus can protect the internal pattern. For example, the pattern cover layer 304 may be formed of polyimide (PI) and an adhesive.

According to an embodiment of the present disclosure, rigid PCBs 380 and 382 and a FPCB 381 may be integrated with one another. The FPCB 381 may not include the second insulation layer 303, but may include the pattern cover layer 304 instead of the protective layer 301.

FIG. 3 is a sectional view of an interlayer configuration of a plurality of layers of the PCB of FIG. 1, according to an embodiment of the present disclosure, where FIG. 3 is taken along line B-B' of FIG. 1.

Referring to FIG. 3, the second PCB 320 may include the first FPCB 331, the first rigid PCB 341, and the second FPCB 332, which are integrally connected together.

Further, the second PCB 320 may include first to fourth ground pattern layers 371, 372, 373, and 374 that are connected by the conductive vias 305 in order to form a ground. The entirety of the first and second ground pattern layers 371 and 372 forms a ground. Only a part of the second and third ground pattern layers 372 and 373 forms a ground. According to an embodiment of the present disclosure, there may be a layer having no ground layer.

Signal line patterns 364 not connected to the ground may be provided. These signal line patterns 364 may be used as RF signal lines.

The ground pattern layers according to an embodiment of the present disclosure may be connected by the first conductive via 305(a) capable of electrically connecting all of the layers and the second conductive via 305(b) capable of electrically connecting adjacent layers. For example, all or some of the ground pattern layers may be connected using the first conductive via 305(a) passing through all of the layers. For example, the first to fourth ground pattern layers 371, 372, 373, and 374 may be connected by the first conductive via 305(a). The adjacent ground pattern layers may be connected to each other by the second conductive via 305(b). For example, the first and second ground pattern layers 371 and 372 may be connected to each other by the second conductive via 305(b).

According to an embodiment of the present disclosure, a PCB (for example, the PCB 300 of FIG. 1) may include the first PCB 310 and the second PCB 320 including a rigid PCB and a FPCB. In this case, the PCB is described as the PCB 300 of FIG. 1 as an example, but may be applied to the PCBs 400, 500, 600, and 700 of FIGS. 4, 5, 6A, 6B, 6C, 7A, 7B, and 7C.

According to an embodiment of the present disclosure, the high-frequency transmission line provided by the PCB in the electronic device may be configured such that the FPCB 330 having advantages of reducing a material cost and ensuring a mounting space can substitute for a coaxial cable.

For example, the first and second FPCBs 331 and 332 may be used between the second rigid PCB 342 and the first PCB 310 in order to transmit a high-frequency signal from the second rigid PCB 342 to the first PCB 310.

In another example, in order to eliminate loss that occurs when a high-frequency signal is transmitted using the FPCB 330, the first rigid PCB 341 may be disposed between the first and second FPCB 331 and 332, and the ground pattern layers may be connected by the conductive vias 305.

According to an embodiment of the present disclosure, in order to mitigate an increase in line loss, the electronic device may further include a section containing the first rigid PCB 341 in the middle of the FPCB 330. If the FPCB 330 has a length of 10 mm or more, the transmission line in the section may cause a great line loss. For example, the transmission line may have a change in the strip line characteristics due to the inductance generated by the ground lines 360 disposed on the opposite sides thereof, and thus the line loss generated by the electrical path 350 may increase markedly.

According to an embodiment of the present disclosure, the electronic device may have the one or more vias 305 arranged on the first rigid PCB 341. When the grounds are connected by the conductive vias 305 arranged on the first rigid PCB 341, the inductance generated by the ground lines 360 can be reduced, for example, by virtue of the interlayer ground connection on the ground lines 360, thereby reducing adverse effects, such as line loss. Furthermore, an all-ground layer (ground line) is disposed between the layers providing the second PCB 320, and the entire ground line 360 may be connected to the all-ground layer through the conductive vias 305, whereby it is possible to have a much greater line loss reduction effect.

According to an embodiment of the present disclosure, a rigid PCB may be disposed between the FPCBs, thereby preventing the layers from being separated or twisted as described above. In the case where the existing FPCBs are installed on a drive unit, the layers of the FPCBs may be separated or twisted. Even in this case, there may be a change in the characteristics of a strip line, which may cause line loss in the transmission line or other adverse property changes.

However, an electronic device having the aforementioned arrangement is advantageous in that the side effects mentioned above are not exhibited even when the FPCBs move, whereby it is possible to reduce line loss in the transmission line and to alleviate other adverse characteristics.

In addition, in the case where a transmission line has a greater length, it is possible to solve shortcomings, such as line loss, by additionally placing rigid PCBs in several positions on the FPCBs and using a connection between the ground line 360, other ground pattern layers 363, and the conductive vias 305. This is described below in greater detail with reference to FIG. 4.

FIG. 4 is a front view of a high-frequency transmission line according to an embodiment of the present disclosure, in which a PCB in an electronic device is used for the high-frequency transmission line. An electronic device may be a smart phone or a wearable device.

According to an embodiment of the present disclosure, an electronic device may include a housing corresponding to a case; a printed circuit unit including at least one PCB 400 in the housing; and an electrical path 450 passing over or through the PCB 400.

Referring to FIG. 4, the PCB 400 according to an embodiment of the present disclosure may include a second PCB 420, a third PCB 411, and a fourth PCB 412. For example, the third PCB 411 may be a main board on which electronic components included in an electronic device are mounted, the main board being capable of transmitting signals between the electronic components. Furthermore, for example, the fourth PCB 412 may be a main board on which electronic components included in an electronic device are mounted, the main board being capable of transmitting signals between the electronic components. Moreover, for example, the second PCB 420 may be a sub-board that

transmits signals generated by electronic components. The sub-board may further include the electronic components.

The PCB 400 according to an embodiment of the present disclosure may be provided with a circuit of a terminal, which may be, for example, at least one of an AP, a CP, a memory, and an RF transceiver, and may include a signal line (for example, an RF signal line).

The second to fourth PCBs 420, 411, and 412 according to an embodiment of the present disclosure may be formed of a plurality of layers, and the plurality of layers may include one or more conductive layers and one or more insulation layers.

The third and fourth PCBs 411 and 412 according to an embodiment of the present disclosure may be a rigid PCB. The third or fourth PCB 411 and 412 may be electrically connected to the second PCB 410 through a connector 470.

The third PCB 411 may be disposed on one end of the second PCB 420, and the fourth PCB 412 may be disposed on an opposite end of the second PCB 420. Hereinafter, the third and fourth printed circuit boards 411 and 412 are described as third and fourth rigid PCBs 411 and 412, respectively.

The third rigid PCB 411 according to an embodiment of the present disclosure may be connected with the second PCB 420, and a first connector 471 may be disposed to transmit a high-frequency signal through the electrical path 450. The first connector 471 may include a soldering pad, a zip-type connector, or a B-to-B-type connector. A rigid PCB may be employed for the first connector 471, and, for example, a B-to-B-type connector may be mounted on the rigid PCB, or a soldering pad may be located on the rigid PCB. Furthermore, for example, the pin map of a transmission line included in the first connector 471 may be implemented in the sequence of a ground, an RF signal, and a ground.

The first connector 471 according to an embodiment of the present disclosure may include a ground pad 490 and a signal pad 491 of the third rigid PCB 411 and a ground pad 492 and a signal pad 493 of the second PCB 420. The ground pad 490 of the third rigid PCB 411 and the ground pad 492 of the second PCB 420 may be connected together by soldering. The signal pad 491 of the third rigid PCB 411 and the signal pad 493 of the second PCB 420 may be connected by soldering.

The fourth rigid PCB 412 according to an embodiment of the present disclosure may be connected to the second PCB 420, and a second connector 472 may be disposed to transmit a high-frequency signal through the electrical path 450. The second connector 472 may include a soldering pad, a zip-type connector, or a B-to-B-type connector. A rigid PCB may be employed for the second connector 472, and, for example, a B-to-B-type connector may be mounted on the rigid PCB, or a soldering pad may be located on the rigid PCB. Furthermore, for example, the pin map of a transmission line included in the first connector 472 may be implemented in the sequence of a ground, an RF signal, and a ground.

The second connector 472 according to an embodiment of the present disclosure may include a ground pad 495 and a signal pad 496 of the fourth rigid PCB 412 and a ground pad 497 and a signal pad 498 of the second PCB 420. The ground pad 495 of the fourth rigid PCB 412 and the ground pad 497 of the second PCB 420 may be connected by soldering. The signal pad 496 of the fourth rigid PCB 412 and the signal pad 498 of the second PCB 420 may be connected by soldering. The second PCB 420 may include an interconnection pad 494. An antenna contact member (for example, a C-clip) or

an RF component (for example, a matching circuit) may be mounted on the interconnection pad 494.

The second PCB 420 according to an embodiment of the present disclosure may be disposed between the third rigid PCB 411 and the fourth rigid PCB 412, and may include an FPCB 430 or a rigid PCB 440. The flexible printed circuit board 430 and the rigid PCB 440 of the second PCB 420 may be integrated with each other. For example, a first FPCB 431, the first rigid PCB 440, and a second flexible PCB 432 may be arranged in sequence. The first rigid PCB 440 may include an antenna connector and a USB connector, for example.

The electrical path 450 according to an embodiment of the present disclosure may extend from a first electrical component including an antenna radiator to a second electrical component including a communication circuit, and may be disposed on or in the PCB 400. The electrical path 450 may be implemented to be a transmission line of a side-guarded micro-strip-line type or a strip-line type, the transmission line being shielded by a ground line in order to transmit a high-frequency signal, which may be, for example, an RF signal.

The electrical path 450 according to an embodiment of the present disclosure, which is disposed on or in the second PCB 420, may receive a high-frequency signal through the connector 470 and may forward the received signal to antenna radiators 231 and 232 in FIG. 10 described below. Furthermore, a plurality of electrical paths 450 in FIG. 4 may be provided, and a ground line 460 may be disposed between the electrical paths 450 so that the electrical paths 450 can independently transmit a high-frequency signal without mutually affecting the characteristics thereof.

The ground line 460 according to an embodiment of the present disclosure may be disposed adjacent to the electrical path 450, and conductive vias 405 may be arranged in a portion of the area of the ground line 460. Accordingly, the electrical path 450 can pass through the PCB 400 without making contact with the conductive vias 405.

A plurality of layers of the PCB 400 and the conductive vias 405 are the same as those in the above-described embodiment with reference to FIGS. 1 to 3, and therefore descriptions thereof are omitted here.

FIG. 5 is a front view of a high-frequency transmission line according to an embodiment of the present disclosure, where a PCB in an electronic device is used for the high-frequency transmission line. The electronic device may be a smart phone or a wearable device.

Referring to FIG. 5, an electronic device may include a housing corresponding to a case; a printed circuit unit including at least one PCB 500 in the housing; and an electrical path 550 passing over or through the PCB 500.

The PCB 500 according to an embodiment of the present disclosure may include a first PCB 510 and a second PCB 520. For example, the first PCB 510 may be a main board on which electronic components included in an electronic device are mounted, the main board being capable of transmitting signals between the electronic components, and the second PCB 520 may be a sub-board on which the electronic components are mounted, the sub-board being capable of transmitting signals generated by the electronic components. The sub-board may further include the electronic components.

The PCB 500 according to an embodiment of the present disclosure may be provided with a circuit of a terminal, which may be, for example, at least one of an AP, a CP, a memory, and an RF transceiver, and may include a signal line (for example, an RF signal line).

The first and second PCBs 510 and 520 according to an embodiment of the present disclosure may be formed of a plurality of layers, and the plurality of layers may include one or more conductive layers and one or more insulation layers.

The first PCB 510 according to an embodiment of the present disclosure may be a rigid PCB. The first PCB 520 may be electrically connected to the second PCB 510 through a connection unit 570. The first PCB 510 may be connected to the second PCB 520, and the connector 570 may be disposed to transmit a high-frequency signal through the electrical path 550. The connector 570 may include a soldering pad, a zip-type connector, or a B-to-B-type connector. A rigid PCB may be employed for the connector 570, and, for example, a B-to-B-type connector may be mounted on the rigid PCB, or a soldering pad may be located on the rigid PCB.

The connector 570 according to an embodiment of the present disclosure may include a ground pad 590 and a signal pad 591 of the first PCB 510 and a ground pad 592 and a signal pad 593 of the second PCB 520. The ground pad 590 of the first PCB 510 and the ground pad 592 of the second PCB 520 may be connected by soldering. The signal pad 591 of the first PCB 510 and the signal pad 593 of the second PCB 520 may be connected together by soldering. The second PCB 520 may include an interconnection pad 594. An antenna contact member (for example, a C-clip) or an RF component (for example, a matching circuit) may be mounted on the interconnection pad 594.

The second PCB 520 according to an embodiment of the present disclosure may include an FPCB 530 and a rigid PCB 540 that are integrated with each other. For example, the second PCB 520 may include a first FPCB 531, a first rigid PCB 541, and a second FPCB 532, which are arranged in sequence. The second FPCB 532 may include an antenna connector and a USB connector. Unlike in an embodiment described above with reference to FIG. 1, the second FPCB 532 in FIG. 5 is configured to perform the function of the second rigid PCB 342 in FIG. 1 as well.

The electrical path 550 in FIG. 5 according to an embodiment of the present disclosure may extend from a first electrical component including an antenna radiator to a second electrical component including a communication circuit, and may be disposed on or in the PCB 500. The electrical path 550 may be implemented to be a transmission line of a side-guarded micro-strip-line type or a strip-line type, the transmission line being shielded by a ground line in order to transmit a high-frequency signal, which may be, for example, an RF signal.

The electrical path 550 according to an embodiment of the present disclosure, which is disposed on or in the second PCB 520, may receive a high-frequency signal through the connector 570 and may forward the received signal to an antenna radiator. Furthermore, a plurality of electrical paths 550 may be provided, and a ground line 560 may be disposed between the electrical paths 550 so that the electrical paths 550 can independently transmit the high-frequency signal without mutually affecting the characteristics thereof.

The ground line 560 according to an embodiment of the present disclosure may be disposed adjacent to the electrical path 550, and conductive vias 505 may be arranged in a portion of the area of the ground line 560. Accordingly, the electrical path 550 can pass through the PCB 500 without making contact with the conductive vias 505.

A plurality of layers of the PCB 500 and the conductive vias 505 are the same as those in an above-described

embodiment with reference to FIGS. 1 to 3, and therefore descriptions thereof are omitted here.

FIGS. 6A, 6B, and 6C are front views of conductive vias according to an embodiment of the present disclosure, the conductive vias being arranged in electronic device PCBs. FIGS. 6A, 6B, and 6C illustrate various forms of PCBs.

Referring to FIG. 6A, a second PCB 620 according to an embodiment of the present disclosure may include an FPCB 630 and a rigid PCB 640, which are integrated with each other. For example, the second PCB 620 may include a first FPCB 631, the first rigid PCB 640, and a second FPCB 632, which are arranged in sequence.

An electrical path 650 according to an embodiment of the present disclosure may be disposed on or in the second PCB 620 and may be implemented to be a transmission line of a side-guarded micro-strip-line type or a strip-line type in order to transmit a high-frequency signal.

The electrical path 650 according to an embodiment of the present disclosure, which is disposed on or in the second PCB 620, may receive a high-frequency signal through a connector and may forward the received signal to an antenna radiator. Furthermore, a plurality of electrical paths 650 may be provided, and a ground line 660 may be disposed between the electrical paths 650 so that the electrical paths 650 can independently transmit the high-frequency signal without mutually affecting the characteristics thereof.

The ground lines 660 according to an embodiment of the present disclosure may be disposed adjacent to the electrical path 650 and may be connected with one or more conductive layers through conductive vias 605(c) in the PCB. Since the conductive vias 605(c) are arranged in one ground line 660, the electrical path 650 may pass through the PCB without making contact with the conductive vias 605(c).

The electrical path 650 according to an embodiment of the present disclosure is disposed to pass through the flexible PCB 630 and the rigid PCB 640, and thus the ground lines 660 disposed on opposite sides of the electrical path 650 may also be disposed to pass through the FPCB 630 and the rigid PCB 640. While two ground lines 660 are provided in FIG. 6A, the number of ground lines 660 may be diversely varied so as to avoid contact with the electrical path 650, without being limited thereto.

For example, the first ground line 660(a) disposed on the left side may have a greater width than the second ground line 660(b) disposed on the right side, and the plurality of conductive vias 605(c) may be formed in the region where the rigid PCB 640 and the ground line 660(a) cross each other. The second ground line 660(b) may not include the conductive vias 605(c) since the second ground line 660(b) has a lesser width than the first ground line 660(a).

The PCB is illustrated as having the conductive vias 605(c) arranged in the first ground line 660(a) having a great width. The conductive vias 605(c) arranged in the first ground line 660(a) may include vias passing through all layers. Ground patterns in all or some of the layers and the ground line on a side of the electrical path may be connected by the conductive vias 605(c) passing through all the layers.

Referring to FIG. 6B, ground lines 660 may be separately disposed on opposite sides of an electrical path 650 in the second PCB 620 and may be formed to pass through the flexible PCB and the rigid PCB along the lengthwise direction of the electrical path 650. While the two ground lines 660 are provided in FIG. 6B, the number of ground lines 660 may be diversely varied so as to avoid contact with the electrical path 650, without being limited thereto.

According to an embodiment of the present disclosure, the first ground line 660(a) disposed on the left side may

have a greater width than the second ground line 660(b) disposed on the right side, and a plurality of conductive vias 605(c) may be formed in the region where the first ground line 660(a) and the rigid PCB 640 cross each other. The second ground line 660(b) on the right side may not include the conductive vias 605(c) having the size since the second ground line 660(b) has a lesser width than the first ground line 660(a). Instead, the second ground line 660(b) on the right side may have conductive vias 605(d) having a lesser size than the conductive vias 605(c).

For example, the rigid PCB 640 formed of a plurality of layers may include the third conductive vias 605(c) passing through all of the layers in the PCB and the fourth conductive vias 605(d) passing through some of the layers. The third conductive vias 605(c) may have a greater hole size than the fourth conductive vias 605(d) passing through some of the layers, and may be arranged in the ground line 660(a) having a great width. However, the fourth conductive vias 605(d) may be arranged in the second ground line 660(b) having a lesser width and disposed on the right side, since the fourth conductive vias 605(d) have a lesser hole size than the third conductive vias 605(c). Even when the ground lines 660 are connected with a ground in a different layer through some conductive vias 605, it is possible to reduce line loss.

Referring to FIG. 6C, ground lines 660 may be separately disposed on opposite sides of an electrical path 650 and may be formed in the lengthwise direction of the electrical path 650. While two ground lines 660 are provided in FIG. 6C, the number of ground lines 660 may be diversely varied so as to avoid contact with the electrical path 650, without being limited thereto.

According to an embodiment of the present disclosure, the first ground line 660(a) disposed on the left side may have a lesser width than the second ground line 660(b) disposed on the right side, and a plurality of conductive vias 605(d) may be formed in the region where the second ground line 660(b) and the rigid PCB 640 cross each other. The second ground line 660(b) on the right side has a greater width than the first ground line 660(a), but is too narrow to have the great conductive vias 605(c). Instead, the conductive vias 605(d) having a lesser width may be arranged in the second ground line 660(b) on the right side.

Therefore, according to an embodiment of the present disclosure, the second ground line 660(b) having a lesser width and disposed on the right side may electrically connect the PCB using the fourth conductive vias 605(d) passing through some layers. Even when the ground lines 660 are connected with a ground in a different layer through some of the fourth conductive vias 605(d), it is possible to reduce line loss.

FIGS. 7A, 7B, and 7C are top views illustrating conductive vias arranged in PCBs according to an embodiment of the present disclosure. FIGS. 7A, 7B, and 7C illustrate various forms of PCBs.

Referring to FIG. 7A, a second PCB 720 may include a plurality of FPCBs and/or a plurality of rigid PCBs. For example, the second PCB 720 may be configured such that FPCBs 730 and rigid PCBs 740 are alternately disposed and are integrated with each other.

The FPCBs 730 according to an embodiment of the present disclosure may include a first FPCB 731, a second FPCB 732, and a third FPCB 733. The rigid PCBs 740 may include a first rigid PCB 741 and a fifth rigid PCB 742.

According to an embodiment of the present disclosure, at least two rigid PCBs 740 may be provided between the FPCBs 730. For example, the second PCB 720 may include

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the first FPCB 731, the first rigid PCB 741, the second FPCB 732, the fifth rigid PCB 742, and the third FPCB 733, which are arranged in sequence.

The first and fifth rigid PCBs 741 and 742 may be disposed to be spaced apart from each other and may include a plurality of conductive vias 705(c) and ground lines 760.

An electrical path 750 according to an embodiment of the present disclosure may be disposed on or in the second PCB 720 and may be implemented to be a transmission line of a side-guarded micro-strip-line type or a strip-line type in order to transmit a high-frequency signal.

The electrical path 750 according to an embodiment of the present disclosure, which is disposed on or in the second PCB 720, may receive a high-frequency signal through a connector and may forward the received signal to an antenna radiator. Furthermore, a plurality of electrical paths 750 may be provided, and the ground line 760 may be disposed between the electrical paths 750 so that the electrical paths 750 can independently transmit the high-frequency signal without mutually affecting the characteristics thereof.

The ground lines 760 according to an embodiment of the present disclosure may be disposed adjacent to the electrical path 750 and may be connected with one or more conductive layers through the conductive vias 705(c) in the PCB 720.

The ground lines 760 according to an embodiment of the present disclosure may be formed to pass through the FPCB 730 and the rigid PCBs 740 along the lengthwise direction of the electrical path 750. While two ground lines 760 are provided in the present disclosure, the number of ground lines 760 may be diversely varied so as to avoid contact with the electrical path 750, without being limited thereto.

For example, the first ground line 760(a) disposed on the left side of the first and fifth rigid PCBs 741 and 742 may have a greater width than the second ground line 760(b) disposed on the right side, and the plurality of conductive vias 705(c) may be formed in the regions where the first ground line 760(a) crosses the first and fifth rigid PCBs 741 and 742. In another example, the second ground line 760(b) on the right side may not include the conductive vias 705(c) having the size since the second ground line 760(b) has a lesser width than the first ground line 760(a). The PCB is illustrated as having the conductive vias 705(c) arranged in the first ground line 760(a) having a great width.

Referring to FIG. 7B, the second PCB 720 may include the FPCBs 730 and the rigid PCBs 740 that are integrated with each other. For example, the second PCB 720 may be configured such that the FPCBs 730 and the rigid PCBs 740 are alternately disposed and are integrated with each other.

The second PCB 720 according to an embodiment of the present disclosure may include the first FPCB 731, the first rigid PCB 741, the second FPCB 732, the fifth rigid PCB 742, and the third FPCB 733, which are arranged in sequence.

Ground lines 760 according to an embodiment of the present disclosure may be separately disposed on opposite sides of the electrical path 750 and may be formed along the lengthwise direction of the electrical path 750. While two ground lines 760 are provided in the FIG. 7B, the number of ground lines 760 may be diversely varied so as to avoid contact with the electrical path 750, without being limited thereto.

According to an embodiment of the present disclosure, the first ground line 760(a) disposed on the left side of the first and fifth rigid PCBs 741 and 742 may have a greater width than the second ground line 760(b) disposed on the right side, and a plurality of conductive vias 705(c) may be formed in the regions where the first ground line 760(a)

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crosses the rigid PCBs 740. However, the second ground line 760(b) on the right side may not include the conductive vias 705(c) having the size since the second ground line 760(b) has a lesser width than the first ground line 760(a). Instead, the second ground line 760(b) on the right side may have conductive vias 705(d) having a lesser size than the conductive vias 705(c).

For example, the rigid PCBs 740 formed of a plurality of layers may include the third conductive vias 705(c) passing through all the layers in the PCB and the fourth conductive vias 705(d) passing through some of the layers. The third conductive vias 705(c) may have a greater hole size than the fourth conductive vias 705(d) passing through some of the layers, and may be arranged in the ground line 760(a) having a great width. However, the fourth conductive vias 705(d) may be arranged in the second ground line 760(b) having a lesser width and disposed on the right side, since the fourth conductive vias 705(d) have a lesser hole size than the third conductive vias 705(c). Even when the ground lines 760 are connected with a ground in a different layer through some conductive vias 705, it is possible to reduce a line loss.

Referring to FIG. 7C, the second PCB 720 may include the FPCBs 730 and the rigid PCBs 740 that are alternately disposed and are integrated with each other. In this embodiment, at least two rigid PCBs 740 may be provided between the FPCBs 730. For example, the second PCB 720 may include the first FPCB 731, the second FPCB 732, the first rigid PCB 741, the fifth rigid PCB 742, and the third FPCB 733, which are arranged in sequence.

Ground lines 760 according to an embodiment of the present disclosure may be separately disposed on opposite sides of the electrical path 750 and may be formed along the lengthwise direction of the electrical path 750. While two ground lines 760 are provided in the present disclosure, the number of ground lines 760 may be diversely varied so as to avoid contact with the electrical path 750, without being limited thereto.

According to an embodiment of the present disclosure, the first ground line 760(a) disposed on the left side of the first and fifth rigid PCBs 741 and 742 may have a lesser width than the second ground line 760(b) disposed on the right side, and a plurality of conductive vias 705(d) may be formed in the regions where the second ground line 760(b) crosses the rigid PCBs 740. The second ground line 760(b) on the right side has a greater width than the first ground line 760(a), but is too narrow to have the great conductive vias 705(c). Instead, the conductive vias 705(d) having a lesser width may be arranged in the second ground line 760(b) on the right side.

Therefore, according to an embodiment of the present disclosure, even when the ground lines 760 are connected to a ground in a different layer through some of the fourth conductive vias 705(d) arranged in the second ground line 760(b), it is possible to reduce a line loss.

FIG. 8 is a perspective view of an electronic device according to an embodiment of the present disclosure. The electronic device 10 may be a smart phone or a wearable device. The electronic device 10 may be an electronic device that includes the PCBs described above with reference to FIGS. 1 to 7.

Referring to FIG. 8, a display 101 may be mounted on the front 107 of the electronic device 10. A speaker device 102 for receiving a counterpart's speech may be provided on the upper side of the display 101. A microphone device 103 for transmitting the speech of a user of the electronic device 10 to the counterpart may be provided on the lower side of the display 101.

According to an embodiment of the present disclosure, one or more components for performing various functions of the electronic device 10 may be arranged around the speaker device 102. For example, the one or more arranged components may include at least one sensor module 104. The sensor module 104 may include, for example, at least one of an illuminance sensor (e.g., an optical sensor), a proximity sensor, an infrared sensor, and an ultrasonic sensor. The components may also include a camera device 105. The components may also include a light emitting diode (LED) indicator 106 for informing a user of status information of the electronic device 10.

According to an embodiment of the present disclosure, the electronic device 10 may include a metal bezel 110 (for example, capable of serving as at least a part of a metal housing). The metal bezel 110 may be disposed along the outer periphery of the electronic device 10 and may extend to at least one area of the back of the electronic device 10, the at least one area being connected to the outer periphery. The metal bezel 110 may define the thickness of the electronic device 10 along the outer periphery of the electronic device 10 and may be formed in a loop shape. Without being limited thereto, however, the metal bezel 110 may also be formed in such a manner that the metal bezel contributes to at least part of the thickness of the electronic device 10. The metal bezel 110 may also be disposed only in at least one area of the outer periphery of the electronic device 10. The metal bezel 110 may include one or more cut-off portions 115 and 116. Unit bezel parts separated by the cut-off portions 115 and 116 may be used as an antenna radiator that operates in at least one frequency band.

According to an embodiment of the present disclosure, the metal bezel 110 may have a loop shape along the outer periphery of the electronic device and may be disposed to contribute to the entirety or a part of the thickness of the electronic device 10. When the electronic device 10 is viewed from the front, the metal bezel 110 may include a right bezel part 111, a left bezel part 112, an upper bezel part 113, and a lower bezel part 114. In this case, the lower bezel part 114 described above may serve as a unit bezel part formed by a pair of cut-off portions 116.

According to an embodiment of the present disclosure, a main antenna device may be disposed in the lower region (an antenna region of a main part) of the electronic device 10. The lower bezel part 114 may be used as a main antenna radiator by virtue of the pair of cut-off portions 116. The lower bezel part 114 may serve as an antenna radiator that operates in at least two operating frequency bands depending on the feeding position. For example, the lower bezel part 114 may be a part of an antenna that supports a low band (LB) and a high/middle band (H/MB), or a part of an antenna that supports an H/MB.

According to an embodiment of the present disclosure, the antenna device of the present disclosure is simply for illustrative purpose, and the aforementioned functions of the lower bezel part 114 may be performed by the upper bezel part 113 separated by the other cut-off portions 115, or may be performed by the upper and lower bezel parts.

According to an embodiment of the present disclosure, an antenna region of a diversity part may be used as an antenna for diversity/multiple input multiple output (MIMO). For example, the upper bezel part 113 may be a part of a diversity antenna that supports LB and H/MB.

According to an embodiment of the present disclosure, at least one of the right and left bezel parts 111 and 112 may be supplied with electrical power to operate as an antenna. For example, the right or left bezel part 111 or 112 may be

a part of an antenna that supports an H/MB band or LB and H/MB bands. For example, an antenna including the right or left bezel part 111 or 112 included in the antenna region of the main part may operate as a main antenna. An antenna including the right or left bezel part 111 or 112 included in the antenna region of the diversity part may operate as a diversity antenna. Hereinafter, the configuration of an antenna is described in detail.

FIG. 9 is a schematic of an antenna arrangement in an electronic device 10 according to an embodiment of the present disclosure. The electronic device 10 may be a smart phone or a wearable device. The electronic device 10 may be an electronic device that includes the PCBs described above with reference to FIGS. 1 to 7.

The electronic device 10 according to an embodiment of the present disclosure may include a plurality of antenna radiators 230. For example, the electronic device 10 may include first and second antennas 231 and 232 of a main part 210 and third and fourth antennas 233 and 234 of a diversity part 220.

The first antenna 231 may support LB and H/MB bands, and the second antenna 232 may support an H/MB band. According to an embodiment of the present disclosure, an LB band has a relatively great wavelength so that an antenna may have a relatively great size, and there may be a mounting space limitation by adding an antenna supporting multiple LBs to the main part 210 of the terminal. Therefore, only one antenna may be configured to support an LB band.

In the case where the electronic device 10 additionally has the diversity part 220 according to an embodiment of the present disclosure, the electronic device 10 may include the third and fourth antennas 233 and 234. In order to enhance correlation/isolation characteristics, the antenna radiators 230 of the main part 210 may be included in the lower end portion of the terminal, and the antenna radiators 230 of the diversity part 220 may be included in the upper end portion of the terminal. For example, in the case where the antenna radiators 230 of the main part 210 are included in the lower end portion of the terminal, and the antenna radiators 230 of the diversity part 220 are included in the upper end portion of the terminal, it is possible to ensure an antenna separation distance available in the terminal even when a plurality of LB band antennas are included.

By virtue of the structure described above, the electronic device 10 can receive signals using four antennas in an H/MB band and can receive signals using two antennas in an LB band. That is, 4th order diversity/MIMO can be performed in an H/MB band, and 2nd order diversity/MIMO can be performed in an LB band.

FIG. 10 illustrates an antenna device and a PCB of an electronic device according to an embodiment of the present disclosure. Referring to FIG. 10, a main RF circuit in the main part 210 may supply electrical power to the first and second antennas 231 and 232 connected thereto through the connector 370 and the second PCB 320 using RF signal lines. The second PCB 320 may include the first FPCB 331, the first rigid PCB 341, the second FPCB 332, and the second rigid PCB 342. A ground may be removed from an antenna feeding part or from the surroundings of an antenna in order to enhance antenna characteristics.

According to an embodiment of the present disclosure, the metal bezel 110 may include the right bezel part 111, the left bezel part 112, the upper bezel part 113, and the lower bezel part 114 when viewed from the front. The upper bezel part 113 may be maintained separately from the right and left bezel parts 111 and 112 by a pair of cut-off portions that are formed with a predetermined interval therebetween. The

lower bezel part 114 may be maintained separately from the right and left bezel parts 111 and 112 by a pair of cut-off portions that are formed with a predetermined interval therebetween. The pair of cut-off portions may be formed of a dielectric material. The pair of cut-off portions may be formed by double-injection molding or insert molding a synthetic resin into the metal bezel. Without being limited thereto, however, various types of electrical insulating materials may be applied to the pair of cut-off portions.

According to an embodiment of the present disclosure, a first feeding piece may be integrally formed with the lower bezel part 114 and may be supplied with electrical power by a first feeding part of the PCB. The first feeding piece of the lower bezel part 114 may be connected to the first feeding part of the PCB merely by installing the PCB in the electronic device, or may be electrically connected to the first feeding part by a separate electrical connection member (e.g., a C-clip, etc.).

According to an embodiment of the present disclosure, a first feeding pad may be disposed on the PCB and may be electrically connected to the first feeding piece of the lower bezel part 114. A first electrical path (e.g., an interconnection wire line) may be formed from the first feeding pad to the first feeding part. The first feeding piece may be integrally formed with the lower bezel part 114 and may be supplied with electrical power by the first feeding part of the PCB. The first feeding piece of the lower bezel part 114 may be connected to the first feeding part of the PCB merely by installing the PCB in the electronic device, or may be electrically connected to the first feeding part by a separate electrical connection member (e.g., a C-clip, etc.).

According to an embodiment of the present disclosure, the first feeding pad may be disposed on the PCB and may be electrically connected to the first feeding piece of the lower bezel part 114. The first electrical path (e.g., an interconnection wire line) may be formed from the first feeding pad to the first feeding part.

The lower bezel part may be a portion of the first antenna 231 of the main part 210 that supports an LB or H/MB band. The left and right bezel parts 112 and 111 may also be supplied with electrical power by the same method. The left and right bezel parts 112 and 111 may be portions of the second antenna 231 of the main part 210 that supports an LB or H/MB band.

According to an embodiment of the present disclosure, a first electrical connection piece may be integrally formed with the lower bezel part 114 so as to be located at a position separate from the first feeding piece by a predetermined distance, and may be grounded to a first ground part of the PCB. The first electrical connection piece of the lower bezel part 114 may be grounded to the first ground part of the PCB merely by installing the PCB in the electronic device, or may be electrically connected to the first ground part by a separate electrical connection member (e.g., a C-clip, etc.).

According to an embodiment of the present disclosure, a first ground pad may be disposed on the PCB and may be electrically connected to the first electrical connection piece of the lower bezel part 114. A second electrical path (e.g., an interconnection wire line) may be formed from the first ground pad to the first ground part.

According to an embodiment of the present disclosure, a second feeding piece may be integrally formed with the lower bezel part 114 and may be supplied with electrical power by a first feeding part of the PCB. The second feeding piece of the lower bezel part 114 may be connected to a second feeding part of the PCB merely by installing the PCB

in the electronic device, or may be electrically connected to the second feeding part by a separate electrical connection member (e.g., a C-clip, etc.).

According to an embodiment of the present disclosure, a second feeding pad may be disposed on the PCB and may be electrically connected to the second feeding piece of the lower bezel part 114. A third electrical path (e.g., an interconnection wire line) may be formed from the second feeding pad to the second feeding part. The lower bezel part 114 may be a portion of the second antenna 232 of the main part 210 that supports an LB or H/MB band.

According to an embodiment of the present disclosure, a second electrical connection piece may be integrally formed with the right bezel part 111 so as to be located at a position separate from the cut-off portions by a predetermined distance, and may be grounded to a second ground part of the PCB. The second electrical connection piece of the right bezel part 111 may be grounded to a second ground part of the PCB merely by installing the PCB in the electronic device, or may be electrically connected to the second ground part by a separate electrical connection member (e.g., a C-clip, etc.).

According to an embodiment of the present disclosure, a second ground pad may be disposed on the PCB and may be electrically connected to the second electrical connection piece of the right bezel part 111. A fourth electrical path (e.g., an interconnection wire line) may be formed from the second ground pad to the second ground part.

For example, the feeding parts, the feeding pads, the ground parts, and the ground pads of the main part 210 may be disposed on the second PCB 320. The main part 210 and an RF circuit of the diversity part 220 may be disposed on the first PCB (main PCB) 310. The first and second PCBs may be connected by an FPCB. The first and second PCBs may be integrated with each other.

According to an embodiment of the present disclosure, the second rigid PCB 342 included in the second PCB 320 may be disposed at a lower vertical position than the first PCB (main PCB) 310. Accordingly, components included in the second rigid PCB 342 may be additionally spaced apart from the antenna. Further, relatively thick components, such as a USB connector and a speaker, can also be disposed on the second rigid PCB 342.

According to an embodiment of the present disclosure, a transmitted or received signal of the RF circuit of the main part 210 may be forwarded to the first and second feeding parts of the second PCB 320.

The diversity part 220 may include the third and fourth antennas 233 and 234. The third antenna 233 may include a part of the upper bezel part 113, and the fourth antenna 234 may include the left or right bezel part 112 or 111.

The third antenna 233 may support LB and H/MB bands, and the fourth antenna 234 may support an H/MB band. Feeding parts, feeding pads, ground parts, and ground pads of the diversity part 220 may be disposed on the first PCB (main PCB) 310. Electrical paths connecting the feeding parts and the feeding pads and electrical paths connecting the ground parts and the ground pads in the diversity part 220 may be disposed on the first PCB (main PCB) 310.

According to an embodiment of the present disclosure, when the main part 210 uses the right bezel part 111 as the second antenna 232 for the purpose of signal separation between the antennas, the diversity part 220 may use the opposite left bezel part 112 as the fourth antenna 234. Alternately, when the main part 210 uses the left bezel part

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112 as the second antenna 232, the diversity part 220 may use the opposite right bezel part 111 as the fourth antenna 234.

FIG. 11 is an illustration of a signal transmission using the second PCB 320 according to an embodiment of the present disclosure. The second PCB 320 may be the same as the second PCBs 320, 420, and 520 described above with reference to FIGS. 1 to 7.

Referring to FIG. 11, the second PCB 320 may include the FPCB 330 and the rigid PCB 340, which are integrated with each other. The second PCB 320 may include the first FPCB 331, the second FPCB 332, the first rigid PCB 341, and the second rigid PCB 342.

According to an embodiment of the present disclosure, the electrical path 350 for transmitting a high-frequency signal may also transmit signals other than RF signals using the second PCB 320. The other signals may include, for example, a USB signal, an ear-jack signal, a microphone (MIC) signal, a speaker signal, and a touch key signal.

According to an embodiment of the present disclosure, an antenna contact to which an RF signal is connected, and a USB connector, an ear-jack connector, a microphone (MIC) connector, and a speaker contact to which the other signals are connected may be disposed on the second rigid PCB 342. For example, RF signals may be transmitted through two electrical paths 350: the first electrical path 351 and the second electrical path 352. The ground line 360 may be disposed between the first and second electrical paths 351 and 352 to prevent the first and second electrical paths 351 and 352 from mutually affecting the characteristics thereof. The conductive vias 305 may be arranged in the region where the ground line 360 and the first rigid PCB 341 cross each other, and may be connected to a ground in a different layer. A rigid PCB connected to the rear end of the second rigid PCB 342 may be various types of modules of the electronic device, and may function as a touch key, for example.

According to an embodiment of the present disclosure, a first connector 342(a) connected to the first electrical path 351 may be a first antenna contact, a first USB connector, a first ear-jack connector, a first MIC connector, or a first speaker contact, and a second connector 342(b) connected to the second electrical path 352 may be a second antenna contact, a second USB connector, a second ear-jack connector, a second MIC connector, and a second speaker contact.

FIGS. 12 and 13 are enlarged views of a first FPCB and a first rigid PCB. The second PCB 320 may be one of the second PCBs 320, 420, 520, 620, and 720 described above with reference to FIGS. 1 to 7.

Referring to FIGS. 12 and 13, a high-frequency signal may be connected from a B-to-B-type connector A of the connector 370 to antenna (ANT) feeding B and C by a micro strip line or a strip line on the FPCB. Further, the ground lines 306 may exist on the left and right sides of the electrical path 350 through which an RF signal is transmitted on the connector, and may be formed parallel to the electrical path 350 along the lengthwise direction of the electrical path 350 so as to be connected to the ANT feeding B and C. In addition, the conductive vias 305 may be formed in the ground lines 306. Conductive vias 307 lesser in size than the conductive vias 305 may be formed on the right side of the electrical path 351 so as to prevent contact between the ground lines and a signal line on one side.

An electronic device according to an embodiment of the present disclosure includes a housing; a PCB disposed in the housing, the PCB including a plurality of layers that include

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one or more conductive layers and one or more insulation layers; a first electrical component formed to be at least a part of the housing or disposed in the housing; a second electrical component disposed above or near the PCB in the housing, the second electrical component being separated from the first electronic component; and at least one electrical path extending from the first electrical component to the second electronic component, wherein at least a portion of the electrical path runs on or inside the PCB, wherein the PCB may include a region including a pattern of conductive vias, each of the vias extending through at least a part of the plurality of layers to contact at least one of the one or more conductive layers, and wherein the electrical path may run through the region without making contact with the conductive vias.

According to an embodiment of the present disclosure, the first electrical component may include an antenna radiator and the second electrical component may include a communication circuit.

According to an embodiment of the present disclosure, the antenna radiator may form at least one part of the housing.

According to an embodiment of the present disclosure, the pattern of the conductive vias may be aligned in one or more rows or columns along the electrical path when viewed from above the region of the PCB.

According to an embodiment of the present disclosure, the PCB may include an FPCB as at least a part thereof, where the FPCB may transmit a signal through the electrical path.

According to an embodiment of the present disclosure, the PCB may include a first PCB; and a second PCB connected to the first PCB through a connector, where the second PCB includes at least one FPCB and at least one rigid PCB, and where the electrical path may transmit a high-frequency signal through the connector and may pass over or through the second PCB.

According to an embodiment of the present disclosure, the electronic device may further include a ground line disposed around the electrical path so as to be adjacent thereto, where the ground line may be connected to the one or more conductive layers through the conductive vias in the rigid PCB.

According to an embodiment of the present disclosure, the second PCB may include a first FPCB connected to the first PCB through a communication circuit; a second PCB disposed to face the first flexible PCB with a separation distance therebetween; and a first rigid PCB disposed between the first and second FPCBs, where the first rigid PCB includes the pattern of the conductive vias.

According to an embodiment of the present disclosure, the electronic device may further include a second rigid PCB connected to the second FPCB, wherein a part of the antenna radiator that forms at least one portion of the housing may be disposed on the second rigid PCB.

According to an embodiment of the present disclosure, at least one of the plurality of layers may form a ground pattern in the ground line of the second PCB, where the entire line of the ground pattern includes a ground.

According to an embodiment of the present disclosure, the conductive vias of the first rigid PCB may include a first conductive via passing through all of the plurality of layers; and a second conductive via passing through some of the plurality of layers, and where the first conductive via may connect the ground pattern disposed on all or some of the layers and the ground line.

According to an embodiment of the present disclosure, a plurality of ground lines may be formed on the rigid PCB so as to be spaced apart from the electrical path, and may have different widths.

According to an embodiment of the present disclosure, the conductive vias formed along the width of the ground line may include a third conductive via having a first hole size; and a fourth via having a second hole size that differs from the first hole size.

According to an embodiment of the present disclosure, a space for mounting an electronic component may be formed on or in the rigid PCB of the second PCB.

According to an embodiment of the present disclosure, the connector may be disposed on one side of the first or second PCB, where the electrical path passing over or through the second PCB may be connected to the first PCB.

According to an embodiment of the present disclosure, the connector may include a first connector disposed on one end of the first FPCB of the second PCB; and a second connector disposed on one end of the second FPCB of the second PCB.

According to an embodiment of the present disclosure, the electronic device may include a plurality of electrical paths and a plurality of antenna radiators, wherein the electrical paths may be connected to the respective antenna radiators corresponding thereto, and the ground line may be disposed between the plurality of electrical paths.

According to an embodiment of the present disclosure, the first PCB may include at least a part of the first electrical component and may be connected to the antenna radiator, and the second PCB may have the second electrical component disposed thereon.

According to an embodiment of the present disclosure, the FPCB and the rigid PCB of the second PCB may be integrated with each other.

According to an embodiment of the present disclosure, the region of the PCB may be a rigid PCB region, and the electronic device may include a first FPCB region disposed between the first electrical component and the region; and a second FPCB region disposed between the second electrical component and the region.

An electronic device according to an embodiment of the present disclosure includes a first PCB; a second PCB connected with the first PCB through a connector and including at least one FPCB and at least one rigid PCB that are integrated with each other, where the flexible and rigid PCBs include a plurality of layers; an electrical path that transmits a high-frequency signal through the connector and passes over or through the second PCB; and at least one conductive via disposed in the rigid PCB to connect grounds between the plurality of layers.

FIG. 14 is a block diagram of an electronic device 10, according to an embodiment of the present disclosure, in a network environment 11.

Referring to FIG. 14, the electronic device 10 may include a bus 180, a processor 120, a memory 130, an input/output interface 150, a display 160, and a communication interface 170. The electronic device 10 may omit at least one of the elements, or may further include other elements.

The bus 180 may include, for example, a circuit that interconnects the elements 120 to 170 and delivers communication (for example, a control message and/or data) between the elements.

The processor 120 may include one or more of a CPU, an AP, and a CP. For example, the processor 120 may carry out

operations or data processing relating to the control and/or communication of at least one other element of the electronic device 10.

The memory 130 may include a volatile and/or non-volatile memory. For example, the memory 130 may store instructions or data relevant to at least one other element 120 to 170 of the electronic device 10. The memory 130 may store software and/or a program 140. The program 140 may include, for example, a kernel 141, middleware 143, an application programming interface (API) 145, and/or application programs (or applications) 147. At least some of the kernel 141, the middleware 143, and the API 145 may be referred to as an operating system (OS).

For example, the kernel 141 may control or manage system resources (for example, the bus 180, the processor 120, and the memory 130) that are used to execute operations or functions implemented by the other programs (for example, the middleware 143, the API 145, and the applications 147). Furthermore, the kernel 141 may provide an interface by which the middleware 143, the API 145, or the applications 147 access the individual elements of the electronic device 10 to control or manage the system resources.

For example, the middleware 143 may serve as an intermediary to allow the API 145 or the applications 147 to exchange data with the kernel 141 through communication.

Furthermore, the middleware 143 may process one or more task requests received from the applications 147 according to the priorities of the requests. For example, the middleware 143 may assign, to at least one of the applications 147, a priority to use the system resources (for example, the bus 180, the processor 120, and the memory 130) of the electronic device 10. For example, the middleware 143 may perform scheduling or loading balancing on the one or more task requests by processing the one or more task requests according to the priority assigned to the at least one of the applications.

The API 145 is, for example, an interface by which the applications 147 control functions provided by the kernel 141 or the middleware 143, and the API 145 may include at least one interface or function (for example, an instruction) for file control, window control, image processing, or text control, for example.

The input/output interface 150 may serve as, for example, an interface that can forward instructions or data input from a user or an external device to the other element(s) 120 to 170 of the electronic device 10. Furthermore, the input/output interface 150 may output instructions or data received from the other element(s) of the electronic device 10 to the user or the external device.

Examples of the display 160 may include a liquid crystal display (LCD), an LED display, an organic LED (OLED) display, a microelectromechanical systems (MEMS) display, and an electronic paper display. For example, the display 160 may display various types of content (for example, text, images, videos, icons, or symbols) for a user. The display 160 may include a touch screen, and may receive, for example, a touch, gesture, a proximity input, or a hovering input using an electronic pen or a part of a user's body.

For example, the communication interface 170 may establish communication between the electronic device 10 and a first external electronic device 12, a second external electronic device 14, or a server 16. For example, the communication interface 170 may be connected to a network 162 through wireless or wired communication to communicate with the second external electronic device 14 or the server 16.

The wireless communication may use, for example, at least one of long term evolution (LTE), LTE-advanced (LTE-A), code division multiple access (CDMA), wideband CDMA (WCDMA), universal mobile telecommunications system (UMTS), wireless broadband (WiBro), and global system for mobile communications (GSM), for example, as a cellular communication protocol. Furthermore, the wireless communication may include short range communication 164, for example. The short range communication 164 may include, for example, at least one of wireless fidelity (WiFi), Bluetooth (BT), near field communication (NFC), and global navigation satellite system (GNSS). Depending on the service area and the bandwidth, the GNSS may include, for example, at least one of a global positioning system (GPS), a global navigation satellite system (Glonass), a Beidou navigation satellite system (Beidou), and a European global satellite-based navigation system (Galileo). Hereinafter, "GPS" may be interchangeably used with "GNSS." The wired communication may include, for example, at least one of a USB, a high definition multimedia interface (HDMI), recommended standard 232 (RS-232), and a plain old telephone service (POTS). The network 162 may include at least one of a communication network such as a computer network (e.g., a local area network (LAN) or a wide area network (WAN)), the Internet, and a telephone network.

Each of the first and second external electronic devices 12 and 14 may be of the same or a different type from the electronic device 10. According to an embodiment of the present disclosure, the server 16 may include a group of one or more servers. All or some of the operations executed in the electronic device 10 may be executed in the electronic device 102, the electronic device 104, or the server 16. When the electronic device 10 must perform a function or service automatically or in response to a request, the electronic device 10 may request the electronic device 12, the electronic device 14, or the server 16 to perform at least some functions relating thereto, instead of or in addition to performing the function or service by itself. The electronic device 102, the electronic device 104, or the server 106 may execute the requested function or additional functions and then deliver the result to the electronic apparatus 10. The electronic device 10 may provide a received result as is, or may further process the received result to provide the requested function or service. To this end, for example, cloud computing, distributed computing, or client-server computing technology may be used.

The electronic device 10, which is described below in greater detail, may be one of a wearable device, a notebook computer, a net-book computer, a smart phone, a tablet PC, a Galaxy tab, an i-Pad, and a wireless charging device mentioned above. In this embodiment, the electronic device 10 may be a smart phone.

The wireless charging device, according to an embodiment of the present disclosure, refers to a device that recharges the electronic device by wirelessly transmitting and receiving power in a short range.

In addition, the display 160 of the electronic device 10 may increase in size and may implement a luxurious design by minimizing the bezel area thereof, or may be implemented to be flexible, convex, or concave.

Namely, the peripheral portion of the display 160 may be bent to allow a view area to extend to the lateral portion. As the view area of the display 160 is bent to extend to the lateral, the view area may be enlarged, a separate screen may be used on the lateral portion, or a luxurious design may be

implemented. In other words, the display 160 may include a first view area and second view areas on opposite sides of the first view area.

FIG. 15 is a block diagram of an electronic device 1901 according to an embodiment of the present disclosure. The electronic device 1901 may include, for example, the entirety or a part of the electronic device 10 described above with reference to FIG. 14. The electronic device 1901 in FIG. 15 may include at least one processor (e.g., an AP) 1910, a communication module 1920, a subscriber identification module card 1924, a memory 1930, a sensor module 1940, an input device 1950, a display 1960, an interface 1970, an audio module 1980, a camera module 1991, a power management module 1995, a battery 1996, an indicator 1997, and a motor 1998.

The processor 1910 may operate, for example, an OS or an application program to control a plurality of hardware or software elements connected to the processor 1910 and to perform various types of data processing and operations. The processor 1910 may be implemented to be a system on chip (SoC), for example. According to an embodiment of the present disclosure, the processor 1910 may further include a graphics processing unit (GPU) and/or an image signal processor. The processor 1910 may also include at least some (for example, a cellular module 1921) of the elements illustrated in FIG. 15. The processor 1910 may load, in a volatile memory, instructions or data received from at least one of the other elements (for example, a non-volatile memory) to process the loaded instructions or data, and may store various types of data in the non-volatile memory.

The configuration of the communication module 1920 may be the same as or similar to that of the communication interface 170 of FIG. 14. The communication module 1920 may include, for example, a cellular module 1921, a WiFi module 1923, a BT module 1925, a GNSS module 1927 (for example, a GPS module, a Glonass module, a Beidou module, or a Galileo module), an NFC module 1928, and an RF module 1929.

The cellular module 1921 may provide, for example, a voice call, a video call, a text message service, or an Internet service through a communication network. According to an embodiment of the present disclosure, the cellular module 1921 may identify and authenticate the electronic device 1901 in a communication network using the subscriber identification module card (SIM card) 1924. The cellular module 1921 may perform at least some of the functions that the processor 1910 can provide. The cellular module 1921 may include a CP.

The WiFi module 1923, the BT module 1925, the GNSS module 1927, and the NFC module 1928 may each include, for example, a processor for processing data transmitted and received through the corresponding module. According to an embodiment of the present disclosure, at least some (for example, two or more) of the cellular module 1921, the WiFi module 1923, the BT module 1925, the GNSS module 1927, and the NFC module 1928 may be included in one integrated circuit (IC) or IC package.

The RF module 1929 may transmit and receive, for example, a communication signal (for example, an RF signal). The RF module 1929 may include, for example, a transceiver, a power amplifier module (PAM), a frequency filter, a low noise amplifier (LNA), or an antenna. At least one of the cellular module 1921, the WiFi module 1923, the BT module 1925, the GNSS module 1927, and the NFC module 1928 may transmit and receive an RF signal through a separate RF module.

The subscriber identification module card **1924** may include, for example, an embedded SIM, and may include unique identification information (for example, an integrated circuit card identifier (ICCID)) or subscriber information (for example, an international mobile subscriber identity (IMSI)).

The memory **1930** (for example, the memory **130** in FIG. **14**) may include, for example, an internal memory **1932** or an external memory **1934**. The internal memory **1932** may include, for example, at least one of a volatile memory (for example, a dynamic random access memory (DRAM), a static RAM (SRAM), or a synchronous DRAM (SDRAM)) and a non-volatile memory (for example, a one-time programmable read only memory (OTPROM), a programmable ROM (PROM), an erasable and programmable ROM (EPROM), an electrically erasable PROM (EEPROM), a mask ROM, a flash ROM, a flash memory (for example, a NAND flash memory or a NOR flash memory), a hard disc drive, or a solid state drive (SSD)).

The external memory **1934** may further include a flash drive, which may be, for example, a compact flash (CF) drive, a secure digital (SD) memory card, a micro SD (Micro-SD) memory card, a mini SD (Mini-SD) memory card, an extreme digital (xD) memory card, a multi-media card (MMC), or a memory stick. The external memory **1934** may be functionally and/or physically connected to the electronic device **1901** through various interfaces.

For example, the sensor module **1940** may measure a physical quantity, or may sense an operating state of the electronic device **1901**, to convert the measured or sensed information into an electrical signal. The sensor module **1940** may include, for example, at least one of a gesture sensor **1940A**, a gyro sensor **1940B**, an atmospheric pressure sensor **1940C**, a magnetic sensor **1940D**, an acceleration sensor **1940E**, a grip sensor **1940F**, a proximity sensor **1940G**, a color sensor **1940H** (for example, a red, green, blue (RGB) sensor), a biometric sensor **1940I**, a temperature/humidity sensor **1940J**, an illuminance sensor **1940K**, and a ultraviolet (UV) light sensor **1940M**. Additionally or alternatively, the sensor module **1940** may include, for example, an electronic nose (E-nose) sensor, an electromyography (EMG) sensor, an electroencephalogram (EEG) sensor, an electrocardiogram (ECG) sensor, an infrared (IR) sensor, an iris sensor, and/or a fingerprint sensor. The sensor module **1940** may further include a control circuit for controlling one or more sensors included therein. In an embodiment of the present disclosure, the electronic device **1901** may further include a processor configured to control the sensor module **1940** as a part of the processor **1910** or separately from the processor **1910** to control the sensor module **1940** while the processor **1910** is in a reduced power or sleep state.

The input device **1950** may include, for example, a touch panel **1952**, a (digital) pen sensor **1954**, a key **1956**, or an ultrasonic input device **1958**. The touch panel **1952** may use, for example, at least one of a capacitive type panel, a resistive type panel, an infrared type panel, and an ultrasonic type panel. Furthermore, the touch panel **1952** may further include a control circuit. The touch panel **1952** may further include a tactile layer to provide a tactile reaction to a user.

The (digital) pen sensor **1954** may include, for example, a recognition sheet that is a part of, or separate from, the touch panel. The key **1956** may include, for example, a physical button, an optical key, or a keypad. The ultrasonic input device **1958** may sense ultrasonic waves generated by an input tool through a microphone **1988** and may identify data corresponding to the sensed ultrasonic waves.

The display **1960** (for example, the display **160** in FIG. **14**) may include a panel **1962**, a hologram device **1964**, or a projector **1966**. The panel **1962** may have a configuration that is the same as, or similar to, that of the display **160** illustrated in FIG. **1**. The panel **1962** in FIG. **15** may be implemented to be, for example, flexible, transparent, or wearable. The panel **1962**, together with the touch panel **1952**, may be implemented to be one module. The hologram device **1964** may show a three-dimensional image in the air by using an interference of light. The projector **1966** may project light onto a screen to display an image. The screen may be located, for example, inside or outside of the electronic device **1901**. According to an embodiment of the present disclosure, the display **1960** may further include a control circuit for controlling the panel **1962**, the hologram device **1964**, and the projector **1966**.

The interface **1970** may include, for example, an HDMI **1972**, a USB **1974**, an optical interface **1976**, or a D-subminiature (D-sub) connector **1978**. The interface **1970** may be included, for example, in the communication interface **170** illustrated in FIG. **3**. Additionally or alternatively, the interface **1970** in FIG. **15** may include, for example, a mobile high-definition link (MHL) interface, an SD card/MMC interface, or an Infrared Data Association (IrDA) standard interface.

For example, the audio module **1980** may convert a sound into an electrical signal, and vice versa. At least some elements of the audio module **1980** may be included, for example, in the input/output interface **145** illustrated in FIG. **1**. The audio module **1980** in FIG. **15** may process sound information that is input or output, for example, through a speaker **1982**, a receiver **1984**, an earphone **1986**, or the microphone **1988**.

The camera module **1991** is a device that can photograph a still image and a dynamic image. According to an embodiment of the present disclosure, the camera module **1991** may include one or more image sensors (for example, a front sensor or a rear sensor), a lens, an image sensor processor (ISP), or a flash (for example, an LED or a xenon lamp).

The power management module **1995** may manage, for example, power of the electronic device **1901**. According to an embodiment of the present disclosure, the power management module **1995** may include a power management IC (PMIC), a charger IC, or a battery gauge. The PMIC may use a wired and/or wireless charging method. Examples of the wireless charging method may include a magnetic resonance method, a magnetic induction method, and an electromagnetic wave method. An additional circuit (for example, a coil loop, a resonance circuit, or a rectifier) for wireless charging may be further included. The battery gauge may measure, for example, a residual amount of the battery **1996** and a voltage, current, or temperature while charging. The battery **1996** may include, for example, a rechargeable battery and/or a solar battery.

The indicator **1997** may indicate a particular state (for example, a booting state, a message state, or a charging state) of the electronic device **1901** or a part thereof (for example, the processor **1910**). The motor **1998** may convert an electrical signal into a mechanical vibration and may generate a vibration or a haptic effect. The electronic device **1901** may include a processing unit (for example, a GPU) for supporting mobile TV. The processing unit for supporting mobile TV may process media data according to a standard, such as digital multimedia broadcasting (DMB), digital video broadcasting (DVB), or MediaFlo™.

Each of the above-described component elements of hardware according to the present disclosure may be con-

figured with one or more components, and the names of the corresponding component elements may vary based on the type of electronic device. The electronic device according to various embodiments of the present disclosure may include at least one of the aforementioned elements. Some elements may be omitted or other additional elements may be further included in the electronic device. In addition, some of the hardware components according to various embodiments may be combined into one entity, which may perform functions identical to those of the relevant components before the combination.

While the present disclosure has been shown and described with reference to certain embodiments thereof, it will be apparent to those skilled in the art that the camera lens module according to the present disclosure is not limited to these embodiments, and various changes in form and details may be made therein without departing from the scope of the present disclosure as defined by the appended claims and their equivalents.

What is claimed is:

1. A portable communication device comprising:

a housing including a first conduction portion forming a first portion of a side surface of the portable communication device and a second conduction portion forming a second portion of the side surface, the first conduction portion operating as a first antenna and the second conduction portion operating as a second antenna;

a wireless communication circuitry; and

a first printed circuit board (PCB) including:

a first PCB portion;

a second PCB portion in contact with the first PCB portion and being more flexible than the first PCB portion;

a third PCB portion in contact with the second PCB portion and being less flexible than the second PCB portion;

a fourth PCB portion in contact with the third PCB portion and being more flexible than the third PCB portion; and

a plurality of lines located from the first PCB portion to the fourth PCB portion through the second PCB portion and the third PCB portion, the plurality of lines including a first ground line, a second ground line at least partially spaced apart from the first ground line, a first signal line located between the first ground line and the second ground line, and a second signal line located between the first signal line and the second ground line, the first signal line electrically connected with the first antenna and the wireless communication circuitry, and the second signal line electrically connected with the second antenna and the wireless communication circuitry.

2. The portable communication device of claim 1, wherein the plurality of lines includes a third ground line located between the first signal line and the second signal line, and

wherein the first PCB further includes an insulating layer on which the first ground line, the second ground line, the third ground line, the first signal line, and the second signal line are located, and a ground plate located below the insulating layer.

3. The portable communication device of claim 2, wherein the ground plate is electrically connected with the first ground line through a first conduction via formed in a first portion of the insulating layer, the second ground line through a second conduction via formed in a second portion of the insulating layer, and the third

ground line through a third conduction via formed in a third portion of the insulating layer.

4. The portable communication device of claim 2, wherein the first PCB further includes a solder resist ink forming at least a part of an exterior surface of the first PCB portion or the third PCB portion.

5. The portable communication device of claim 1, wherein the first PCB further includes a fifth PCB portion in contact with the fourth PCB portion and being less flexible than the fourth PCB portion, wherein the first PCB portion further includes a first connector for a connection with the antenna, wherein the fifth PCB portion includes a second connector for a connection with the wireless communication circuitry, and

wherein a first ending portion of the first signal line and a second ending portion of the second signal line are in contact with the second connector.

6. The portable communication device of claim 1, further comprising:

a second PCB on which the wireless communication circuitry is located,

wherein the first ground line and the second ground line are electrically connected with a ground member located in the second PCB, or

wherein the first ground line and the second ground line are electrically connected with a ground member located in the first PCB portion.

7. A portable communication device comprising:

a first electronic component;

a second electronic component; and

a printed circuit board (PCB) electrically connected with the first electronic component and the second electronic component, the PCB including:

a first rigid PCB portion;

a first flexible PCB portion in contact with the first rigid PCB portion;

a second rigid PCB portion in contact with the first flexible PCB portion; and

a second flexible PCB portion in contact with the second rigid PCB portion,

wherein each of the first rigid PCB portion and the second rigid PCB portion includes:

a first insulating layer including a first material and forming a first exterior surface of each of the first rigid PCB portion and the second rigid PCB portion;

a first conduction layer located on or above the first insulating layer;

a second insulating layer located on or above the first conduction layer and having a first thickness;

a second conduction layer located on or above the second insulating layer; and

a third insulating layer including the first material and located on or above the second conduction layer, the third insulating layer forming a second exterior surface of each of the first rigid PCB portion and the second rigid PCB portion, and

wherein each of the first flexible PCB portion and the second flexible PCB portion includes:

a first insulating layer including a second material different from the first material and forming a first exterior surface of each of the first flexible PCB portion and the second flexible PCB portion;

a first conduction layer located on or above the first insulating layer and being electrically connected with

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the first conduction layer of first rigid PCB portion and the first conduction layer of the second rigid PCB portion;

a second insulating layer located on or above the first conduction layer and having a second thickness thinner than the first thickness;

a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the first rigid PCB portion and the second conduction layer of the second rigid PCB portion; and

a third insulating layer including the second material and located on or above the second conduction layer, the third insulating layer forming a second exterior surface of each of the first flexible PCB portion and the second flexible PCB portion.

8. The portable communication device of claim 7, wherein the second insulating layer of the first flexible PCB portion or the second insulating layer of the second flexible PCB portion includes the second material.

9. The portable communication device of claim 7, wherein each of the first flexible PCB portion and the second flexible PCB portion includes a fourth insulating layer located between the second conduction layer and the third insulating layer, and a third conduction layer located between the third insulating layer and the fourth insulating layer.

10. The portable communication device of claim 9, wherein the fourth insulating layer includes the second material.

11. The portable communication device of claim 9, wherein the first conduction layer of each of the first flexible PCB portion and the second flexible PCB portion includes a first ground portion, wherein the second conduction layer of each of the first flexible PCB portion and the second flexible PCB portion includes a second ground portion electrically connected with the first ground portion through a first conduction via, and wherein the third conduction layer includes a third ground portion electrically connected with the second ground portion through a second conduction via.

12. The portable communication device of claim 7, wherein the first material is a solder resist ink and the second material is a polyimide, and wherein the second insulating layer of the first rigid PCB portion or the second insulating layer of the second rigid PCB portion includes a third material.

13. The portable communication device of claim 12, wherein the third material includes a resin or a glass fabric.

14. A portable communication device comprising:

a first electronic component;

a second electronic component; and

a printed circuit board (PCB) electrically connected with the first electronic component and the second electronic component, the PCB including:

a first flexible PCB portion including:

a first insulating layer forming a first exterior surface of the first flexible PCB portion and including a first material;

a first conduction layer located on or above the first insulating layer;

a second insulating layer located on or above the first conduction layer having a first thickness;

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a second conduction layer located on or above the second insulating layer; and

a third insulating layer located on or above the second conduction layer and forming a second exterior surface of the first flexible PCB portion, the third insulating layer including the first material;

a rigid PCB portion in contact with the first flexible PCB portion, the rigid PCB portion including:

a first insulating layer forming a first exterior surface of the rigid PCB portion and including a second material different from the first material;

a first conduction layer located on or above the first insulating layer and being electrically connected with the first conduction layer of the first flexible PCB portion;

a second insulating layer located on or above the first conduction layer and having a second thickness thicker than the first thickness;

a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the first flexible PCB portion; and

a third insulating layer located on or above the second conduction layer and forming a second exterior surface of the rigid PCB portion, the third insulating layer including the second material;

a second flexible PCB portion in contact with the rigid PCB portion, the second flexible PCB portion including:

a first insulating layer forming a first exterior surface of the second flexible PCB portion and including the first material;

a first conduction layer located on or above the first insulating layer and being electrically connected with the first conduction layer of the rigid PCB portion;

a second insulating layer located on or above the first conduction layer and having the first thickness;

a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the rigid PCB portion;

a third insulating layer located on or above the second conduction layer and forming a second exterior surface of the second flexible PCB portion, the third insulating layer including the first material; and

a contact pad located on the third insulating layer.

15. The portable communication device of claim 14, wherein the second insulating layer of the first flexible PCB portion or the second insulating layer of the second flexible PCB portion includes the second material.

16. The portable communication device of claim 14, wherein each of the first flexible PCB portion and the second flexible PCB portion includes a fourth insulating layer located between the second conduction layer and the third insulating layer, and a third conduction layer located between the third insulating layer and the fourth insulating layer.

17. The portable communication device of claim 16, wherein the fourth insulating layer includes the second material.

18. The portable communication device of claim 16, wherein the first conduction layer of each of the first flexible PCB portion and the second flexible PCB portion includes a first ground portion, wherein the second conduction layer of each of the first flexible PCB portion and the second flexible PCB

portion includes a second ground portion electrically connected with the first ground portion through a first conduction via, and

wherein the third conduction layer includes a third ground portion electrically connected with the second ground portion through a second conduction via.

19. The portable communication device of claim 14, wherein the first material is a solder resist ink and the second material is a polyimide, and

wherein the second insulating layer of the rigid PCB portion includes a third material.

20. The portable communication device of claim 19, wherein the third material includes a resin or a glass fabric.

* * * * *

APPENDIX A

8660338



THE UNITED STATES OF AMERICA

TO ALL TO WHOM THESE PRESENTS SHALL COME:

UNITED STATES DEPARTMENT OF COMMERCE

United States Patent and Trademark Office

December 08, 2025

**THIS IS TO CERTIFY THAT ANNEXED IS A TRUE COPY FROM THE
RECORDS OF THIS OFFICE OF THE FILE WRAPPER AND CONTENTS
OF:**

APPLICATION NUMBER: 16/939,708

FILING DATE: *July 27, 2020*

PATENT NUMBER: 10978789

ISSUE DATE: *April 13, 2021*



Certified by

DIRECTOR OF THE UNITED STATES PATENT AND TRADEMARK OFFICE

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FEE TRANSMITTAL		Complete if known	
		Application Number	
		Filing Date	July 27, 2020
☐ Applicant asserts small entity status. See 37 CFR 1.27.		First Named Inventor	Han-Min CHO et al.
☐ Applicant certifies micro entity status. See 37 CFR 1.29. Form PTO/SB/15A or B or equivalent must either be enclosed or have been submitted previously.		Examiner Name	
		Art Unit	
TOTAL AMOUNT OF PAYMENT	(\$ 5,860.00	Practitioner Docket No.	678-5477 CON (P23235-US1-CON)

METHOD OF PAYMENT (check all that apply)
☐ Check ☐ Credit Card ☐ Money Order ☐ None ☐ Other (please identify): _____

☒ Deposit Account Deposit Account Number: 50-4053 Deposit Account Name: The Farrell Law Firm, PC

For the above-identified deposit account, the Director is hereby authorized to (check all that apply):

☐ Charge fee(s) indicated below☐ Charge fee(s) indicated below, **except for the filing fee**
☒ Charge any additional fee(s) or underpayment of fee(s) ☒ Credit any overpayment of fee(s)
under 37 CFR 1.16 and 1.17

WARNING: Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038.

FEE CALCULATION**1. BASIC FILING, SEARCH, AND EXAMINATION FEES (U = undiscounted fee; S = small entity fee; M = micro entity fee)**

Application Type	FILING FEES			SEARCH FEES			EXAMINATION FEES			Fees Paid (\$)
	U (\$)	S (\$)	M (\$)	U (\$)	S (\$)	M (\$)	U (\$)	S (\$)	M (\$)	
Utility	300	150*	75	660	330	165	760	380	190	1,720.00
Design	200	100	50	160	80	40	600	300	150	
Plant	200	100	50	420	210	105	620	310	155	
Reissue	300	150	75	660	330	165	2,200	1,100	550	
Provisional	280	140	70	0	0	0	0	0	0	

* The \$150 small entity status filing fee for a utility application is further reduced to \$75 for a small entity status applicant who files the application via EFS-Web.

2. EXCESS CLAIM FEES

Fee Description	Undiscounted Fee (\$)	Small Entity Fee (\$)	Micro Entity Fee (\$)
Each claim over 20 (including Reissues)	100	50	25
Each independent claim over 3 (including Reissues)	460	230	115
Multiple dependent claims	820	410	205
Total Claims			
20 _____ -20 or HP = _____ x _____	100		
HP = highest number of total claims paid for, if greater than 20.			
Indep. Claims			
2 _____ -3 or HP = _____ x _____	460		
HP = highest number of independent claims paid for, if greater than 3.			
		Multiple Dependent Claims	
		Fee (\$)	Fee Paid (\$)
		820	

3. APPLICATION SIZE FEE

If the specification and drawings exceed 100 sheets of paper (excluding electronically filed sequence or computer listings under 37 CFR 1.52(e)), the application size fee due is \$400 (\$200 for small entity) (\$100 for micro entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).

Total Sheets	Extra Sheets	Number of each additional 50 or fraction thereof	Fee (\$)	Fee Paid (\$)
65	- 100 = _____ / 50 = _____ (round up to a whole number)	x	400	

4. OTHER FEE(S)

Non-English specification, \$130 fee (no small or micro entity discount)

Non-electronic filing fee under 37 CFR 1.16(t) for a utility application, \$400 fee (\$200 small or micro entity)

Other (e.g., late filing surcharge): Request for Prioritized Examination & Processing Fee 4,140

SUBMITTED BY			
Signature	/Paul J. Farrell/	Registration No. (Attorney/Agent)	33,494
Name (Print/Type)	Paul J. Farrell	Telephone	(516) 228-3565
		Date	July 27, 2020

This collection of information is required by 37 CFR 1.136. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

(Only for new nonprovisional applications under 37 CFR 1.53(b))

Attorney Docket No.	678-5477 CON (P23235-US1-CON)
First Named Inventor	Han-Min CHO et al.
Title	ELECTRONIC DEVICE...
Priority Mail Express® Label No.	

See MPEP chapter 600 concerning utility patent application contents.

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

1. ☒ **Fee Transmittal Form**
(PTO/SB/17 or equivalent)
2. ☐ **Applicant asserts small entity status.**
See 37 CFR 1.27
3. ☐ **Applicant certifies micro entity status.** See 37 CFR 1.29.
Applicant must attach form PTO/SB/15A or B or equivalent.
4. ☒ **Specification** [Total Pages 51]
Both the claims and abstract must start on a new page.
(See MPEP § 608.01(a) for information on the preferred arrangement)
5. ☒ **Drawing(s)** (35 U.S.C. 113) [Total Sheets 14]
6. **Inventor's Oath or Declaration** [Total Pages 7]
(including substitute statements under 37 CFR 1.64 and assignments
serving as an oath or declaration under 37 CFR 1.63(e))
 - a. ☐ Newly executed (original or copy)
 - b. ☒ A copy from a prior application (37 CFR 1.63(d))
7. ☒ **Application Data Sheet** * See note below.
See 37 CFR 1.76 (PTO/AIA/14 or equivalent)
8. **CD-ROM or CD-R**
in duplicate, large table, or Computer Program (Appendix)
☐ Landscape Table on CD
9. **Nucleotide and/or Amino Acid Sequence Submission**
(if applicable, items a. -- c. are required)
 - a. ☐ Computer Readable Form (CRF)
 - b. ☐ Specification Sequence Listing on:
 - i. ☐ CD-ROM or CD-R (2 copies); or
 - ii. ☐ Paper
 - c. ☐ Statements verifying identity of above copies

10. ☐ **Assignment Papers**
(cover sheet & document(s))
Name of Assignee Samsung Electronics Co., Ltd.

11. ☐ **37 CFR 3.73(c) Statement** ☒ **Power of Attorney**
(when there is an assignee)

12. ☐ **English Translation Document**
(if applicable)

13. ☒ **Information Disclosure Statement**
(PTO/SB/08 or PTO-1449)
☐ Copies of citations attached

14. ☐ **Preliminary Amendment**

15. ☐ **Return Receipt Postcard**
(MPEP § 503) (Should be specifically itemized)

16. ☐ **Certified Copy of Priority Document(s)**
(if foreign priority is claimed)

17. ☐ **Nonpublication Request**
Under 35 U.S.C. 122(b)(2)(B)(i). Applicant must attach form PTO/SB/35 or equivalent.

18. ☒ **Other:** Request for Prioritized Examination

***Note:** (1) Benefit claims under 37 CFR 1.78 and foreign priority claims under 1.55 **must** be included in an Application Data Sheet (ADS).
(2) For applications filed under 35 U.S.C. 111, the application must contain an ADS specifying the applicant if the applicant is an assignee, person to whom the inventor is under an obligation to assign, or person who otherwise shows sufficient proprietary interest in the matter. See 37 CFR 1.46(b).

19. CORRESPONDENCE ADDRESS

☒ The address associated with Customer Number: 66547 OR ☐ Correspondence address below

Name					
Address					
City		State		Zip Code	
Country		Telephone		Email	
Signature	/Paul J. Farrell/			Date	July 27, 2020
Name (Print/Type)	Paul J. Farrell			Registration No. (Attorney/Agent)	33,494

This collection of information is required by 37 CFR 1.53(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	678-5477 CON
		Application Number	
Title of Invention	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Inventor Information:

Inventor 1					Remove
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Inventor 2					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
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Postal Code			Country ⁱ	KR	
Inventor 3					Remove
Legal Name					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	678-5477 CON
		Application Number	
Title of Invention	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT		

Prefix	Given Name	Middle Name	Family Name	Suffix
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City	Gyeonggi-do	State/Province		
Postal Code		Country ⁱ	KR	
Inventor 4				Remove
Legal Name				

Prefix	Given Name	Middle Name	Family Name	Suffix
	Tae-Wook		HAM	
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Postal Code		Country ⁱ	KR	
Inventor 5				Remove
Legal Name				

Prefix	Given Name	Middle Name	Family Name	Suffix
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	678-5477 CON
		Application Number	
Title of Invention	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT		

Inventor 6					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
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All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the **Add** button.

[Add](#)

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence Information of this application.			
Customer Number	66547		
Email Address	genmail@farrelliplaw.com	Add Email	Remove Email

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	678-5477 CON
		Application Number	
Title of Invention	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT		

Application Information:

Title of the Invention	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT		
Attorney Docket Number	678-5477 CON	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	14	Suggested Figure for Publication (if any)	

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☐ Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.			
Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	66547		

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	678-5477 CON
		Application Number	
Title of Invention	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending		
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
	Continuation of	15/621687	2017-06-13
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

			
Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
10-2016-0073355	KR	2016-06-13	
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	678-5477 CON
		Application Number	
Title of Invention	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	678-5477 CON
		Application Number	
Title of Invention	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.			
Applicant 1			
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.			
Clear			
☒ Assignee	☐ Legal Representative under 35 U.S.C. 117	☐ Joint Inventor	
☐ Person to whom the inventor is obligated to assign.		☐ Person who shows sufficient proprietary interest	
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:			
Name of the Deceased or Legally Incapacitated Inventor:			
If the Applicant is an Organization check here. ☒			
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Mailing Address Information For Applicant:			
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Address 2			
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Country	KR	Postal Code	16677
Phone Number		Fax Number	
Email Address			
Additional Applicant Data may be generated within this form by selecting the Add button.			

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	678-5477 CON
		Application Number	
Title of Invention	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT		

Assignee 1

Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.

If the Assignee or Non-Applicant Assignee is an Organization check here. ☐

Prefix	Given Name	Middle Name	Family Name	Suffix

Mailing Address Information For Assignee including Non-Applicant Assignee:

Address 1			
Address 2			
City		State/Province	
Country ⁱ		Postal Code	
Phone Number		Fax Number	
Email Address			

Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.

Signature:

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the **INITIAL** filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Paul J. Farrell/		Date (YYYY-MM-DD)	2020-07-27
First Name	Paul	Last Name	Farrell	Registration Number
33,494				

Additional Signature may be generated within this form by selecting the Add button.

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	678-5477 CON
		Application Number	
Title of Invention	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

- 1 The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
- 2 A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
- 4 A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
- 5 A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
- 8 A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION
CIRCUIT

PRIORITY

5

This application is a Continuation Application of U.S. Patent Application No. 15/621,687, filed on June 13, 2017, and claims priority under 35 U.S.C. §119(a) to a Korean Patent Application filed on June 13, 2016 in the Korean Intellectual Property Office and assigned Serial No. 10-2016-0073355, the content of each of which is
10 incorporated herein by reference.

BACKGROUND

1. Field of the Disclosure

15 The present disclosure relates generally to a high-frequency transmission circuit included in an electronic device, and more particularly, to a transmission line capable of effectively transmitting a noiseless high-frequency signal in a narrow space without affecting other modules.

20

2. Description of the Related Art

Recently, portable terminals having new functions have been rapidly developed and have accounted for a greater part of people's lives with the common use thereof.

Internal circuits of portable terminals are generally implemented on printed circuit
25 boards (PCBs). PCB technologies have been rapidly developing, and flexible PCBs (FPCBs) that can be bent easily are currently used widely, as well as conventional rigid PCBs.

In addition, coaxial cables are typically used for high-frequency transmission lines (especially, radio frequency (RF) lines) that are applied to wireless terminals.

A large amount of additional material cost may be required to produce coaxial cables used for electronic devices as high-frequency transmission lines, and additional labor cost may be incurred by an increase in the number of assembly steps. In another example, electronic devices have narrow internal spaces
5 available on account of various types of circuit modules mounted therein, and due to the narrow internal spaces, a defect rate may increase when small connectors used for coaxial cables are assembled.

SUMMARY

10

An aspect of the present disclosure provides a transmission line capable of effectively transmitting a noiseless high-frequency signal in a narrow space without affecting other modules.

Another aspect of the present disclosure provides that a coaxial cable used as a
15 transmission line in a portable terminal can be replaced by a circuit board that includes an FPCB, and thus it is possible to eliminate material costs associated with the manufacture of a coaxial cable.

Another aspect of the present disclosure provides a rigid PCB that is disposed between transmission lines of an FPCB, whereby it is possible to reduce line loss
20 occurring in the transmission lines of the FPCB and to provide a high-frequency signal without loss.

Another aspect of the present disclosure provides that it is possible to efficiently use a mounting space in a wireless terminal, which enables diversification of the external design of the wireless terminal to improve customer satisfaction.

25 In accordance with an aspect of the present disclosure, an electronic device is provided. The electronic device includes a housing; a PCB disposed in the housing, wherein the PCB includes a plurality of layers that include one or more conductive layers and one or more insulation layers; a first electrical component formed to be at least a part

of the housing or disposed in the housing; a second electrical component disposed above or near the PCB in the housing, wherein the second electrical component is separated from the first electrical component; and at least one electrical path extending from the first electrical component to the second electrical component, wherein at least a portion
 5 of the electrical path runs on or inside the PCB, wherein the PCB includes a region including a pattern of conductive vias, wherein each of the conductive vias extends through at least a part of the plurality of layers to contact at least one of the one or more conductive layers, and wherein the electrical path may run through the region without making contact with the conductive vias.

10 In accordance with another aspect of the present disclosure, an electronic device is provided. The electronic device includes a first PCB; a second PCB connected to the first PCB through a connector and including at least one FPCB and at least one rigid PCB that are integrated with each other, wherein the at least one FPCB flexible and the at least one rigid PCB each include a plurality of layers; an electrical path that transmits a high-
 15 frequency signal through the connector and passes over or through the second PCB; and at least one conductive via disposed in the at least one rigid PCB to connect grounds between the plurality of layers.

BRIEF DESCRIPTION OF THE DRAWINGS

20

The above and other aspects, features, and advantages of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings, in which:

FIG. 1 is a front view of a high-frequency transmission line according to an
 25 embodiment of the present disclosure, in which a PCB in an electronic device is used for the high-frequency transmission line;

FIG. 2 is a partial sectional view of a plurality of layers of the PCB of FIG. 1, according to an embodiment of the present disclosure, where FIG. 2 is taken along line B-B' of FIG. 1;

FIG. 3 is a sectional view of an interlayer configuration of a plurality of layers of the PCB of FIG. 1, according to an embodiment of the present disclosure, where FIG. 3 is taken along line B-B' of FIG. 1;

FIG. 4 is a front view of a high-frequency transmission line according to an embodiment of the present disclosure, where a PCB in an electronic device is used for the high-frequency transmission line;

FIG. 5 is a front view of a high-frequency transmission line according to an embodiment of the present disclosure, where a PCB in an electronic device is used for the high-frequency transmission line;

FIGS. 6A, 6B, 6C, 7A, 7B, and 7C are front views of conductive vias according to an embodiment of the present disclosure, the conductive vias being arranged in PCBs in electronic devices;

FIG. 8 is a perspective view of an electronic device according to an embodiment of the present disclosure;

FIG. 9 is a schematic of an antenna arrangement in an electronic device according to an embodiment of the present disclosure;

FIG. 10 illustrates an antenna device and a PCB in an electronic device according to an embodiment of the present disclosure;

FIG. 11 is an illustration of a signal transmission using a second PCB according to an embodiment of the present disclosure;

FIGS. 12 and 13 are enlarged views of a first FPCB and a first rigid PCB;

FIG. 14 is a block diagram of an electronic device according to an embodiment of the present disclosure in a network environment; and

FIG. 15 is a block diagram of an electronic device according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE PRESENT DISCLOSURE

Hereinafter, various embodiments of the present disclosure are described with reference to the accompanying drawings. However, it should be understood that there is no intent to limit the present disclosure to the particular forms disclosed herein; rather, the present disclosure is intended to be construed to cover various modifications, 5 equivalents, and/or alternatives of embodiments of the present disclosure. In the description of the accompanying drawings, similar reference numerals may be used to designate similar elements.

As used herein, the expressions “have”, “may have”, “include”, or “may include” refer to the existence of a corresponding feature (e.g., numeral, function, operation, or 10 element such as a component), and do not exclude one or more additional features.

In the present disclosure, the expressions “A or B”, “at least one of A and/or B”, and “one or more of A and/or B” may include all possible combinations of the items listed. For example, the expressions “A or B”, “at least one of A and B”, and “at least one of A or B” refer to all of (1) including at least one A, (2) including at least one B, and 15 (3) including all of at least one A and at least one B.

The expressions “a first”, “a second”, “the first”, or “the second” used in various embodiments of the present disclosure may modify various components regardless of the order and/or the importance of the components but are not intended to limit the corresponding components. For example, a first user device and a second user device 20 indicate different user devices although both of them are user devices. For example, a first element may be referred to as a second element, and similarly, a second element may be referred to as a first element without departing from the scope of the present disclosure.

It should be understood that when an element (e.g., a first element) is referred to as being (operatively or communicatively) “connected,” or “coupled,” to another element 25 (e.g., a second element), the element may be directly connected or directly coupled to the other element or any other element (e.g., a third element) may be interposer between them. In contrast, it may be understood that when an element (e.g., a first element) is referred to as being “directly connected,” or “directly coupled” to another element (e.g., a second element), there is no element (e.g., a third element) interposed between them.

The expression “configured to” used in the present disclosure may be used interchangeably with the expressions, for example, “suitable for”, “having the capacity to”, “designed to”, “adapted to”, “made to”, and “capable of” according to the situation. The term “configured to” may not necessarily imply “specifically designed to” in hardware. Alternatively, in some situations, the expression “device configured to” may indicate that the device, together with other devices or components, “is able to”. For example, the expression “processor adapted (or configured) to perform A, B, and C” may indicate a dedicated processor (e.g., an embedded processor) only for performing the corresponding operations or a general purpose processor (e.g., a central processing unit (CPU) or an application processor (AP)) that can perform the corresponding operations by executing one or more software programs stored in a memory device.

The terms used herein are merely used for the purpose of describing particular embodiments but are not intended to limit the scope of the present disclosure. A singular expression may include a plural expression unless they are definitely different in a context. Unless defined otherwise, all terms used herein, have the same meanings as those commonly understood by a person skilled in the art to which the present disclosure pertains. Such terms as those defined in a generally used dictionary may be interpreted to have the meanings equal to the contextual meanings in the relevant field of art, and are not intended to be interpreted to have ideal or excessively formal meanings unless clearly defined in the present disclosure. In some cases, even the terms defined in the present disclosure are not intended to be interpreted to exclude embodiments of the present disclosure.

An electronic device according to various embodiments of the present disclosure may include at least one of, for example, a smart phone, a tablet personal computer (PC), a mobile phone, a video phone, an electronic book reader (e-book reader), a desktop PC, a laptop PC, a netbook computer, a workstation, a server, a personal digital assistant (PDA), a portable multimedia player (PMP), a moving picture experts group (MPEG-1) audio layer-3 (MP3) player, a mobile medical device, a camera, and a wearable device. According to various embodiments of the present disclosure, the wearable device may

include at least one of an accessory type (e.g., a watch, a ring, a bracelet, an anklet, a necklace, glasses, a contact lens, or a head-mounted device (HMD)), a fabric or clothing integrated type (e.g., electronic clothing), a body-mounted type (e.g., a skin pad, or tattoo), and a bio-implantable type (e.g., an implantable circuit).

5 According to an embodiment of the present disclosure, the electronic device may be a home appliance. A home appliance may include at least one of, for example, a television, a digital video disk (DVD) player, an audio player, a refrigerator, an air conditioner, a vacuum cleaner, an oven, a microwave oven, a washing machine, an air cleaner, a set-top box, a home automation control panel, a security control panel, a TV
10 box (e.g., Samsung HomeSync®, Apple TV®, or Google TV™), a game console (e.g., Xbox® and PlayStation®), an electronic dictionary, an electronic key, a camcorder, and an electronic photo frame.

 According to an embodiment of the present disclosure, the electronic device may include at least one of various medical devices (e.g., various portable medical measuring
15 devices (a blood glucose monitoring device, a heart rate monitoring device, a blood pressure measuring device, a body temperature measuring device, etc.), a magnetic resonance angiography (MRA), a magnetic resonance imaging (MRI), a computed tomography (CT) machine, and an ultrasonic machine), a navigation device, a global positioning system (GPS) receiver, an event data recorder (EDR), a flight data recorder
20 (FDR), a vehicle infotainment device, an electronic device for a ship (e.g., a navigation device for a ship, and a gyro-compass), avionics, security devices, an automotive head unit, a robot for home or industry, an automated teller machine (ATM), point of sales (POS) device in a shop, or an Internet of Things (IoT) device (e.g., a light bulb, various sensors, electric or gas meter, a sprinkler device, a fire alarm, a thermostat, a streetlamp, a
25 toaster, sporting goods, a hot water tank, a heater, a boiler, etc.).

 According to an embodiment of the present disclosure, an electronic device may include at least one of a part of furniture or a building/structure, an electronic board, an electronic signature receiving device, a projector, and various types of measuring instruments (e.g., a water meter, an electric meter, a gas meter, and a radio wave meter).

An electronic device may be a combination of one or more of the aforementioned various devices. An electronic device may also be a flexible device. Further, an electronic device is not intended to be limited to the aforementioned devices, but may include an electronic device to be developed in the future.

5 Hereinafter, an electronic device according to various embodiments of the present disclosure is described with reference to the accompanying drawings. In the present disclosure, the term “user” may indicate a person using an electronic device or a device (e.g., an artificial intelligence electronic device) using an electronic device.

FIG. 1 is a front view of a high-frequency transmission line according to an
10 embodiment of the present disclosure, in which a PCB 300 in an electronic device is used for the high-frequency transmission line. The electronic device may be a smart phone or a wearable device.

Referring to FIG. 1, the electronic device may include a housing corresponding to a case; a printed circuit unit including at least one PCB 300 in the housing; and an
15 electrical path 350 passing over or through the printed circuit board 300.

The PCB 300 according to an embodiment of the present disclosure may be a first PCB 310 and a second PCB 320. For example, the first PCB 310 may be a main board on which electronic components included in the electronic device are mounted, the main board being capable of transmitting signals between the electronic components, and
20 the second PCB 320 may be a sub-board that transmits signals generated by the electronic components. The sub-board may further include the electronic components.

The PCB 300 according to an embodiment of the present disclosure may be provided with a circuit of a terminal, which may be, for example, at least one of an AP, a communication processor (CP), a memory, and an RF transceiver, and may include a
25 signal line (for example, an RF signal line).

The first and second PCBs 310 and 320 according to an embodiment of the present disclosure may be formed of a plurality of layers, and the plurality of layers may include one or more conductive layers 306 in FIG. 2 described below and one or more insulation layers 303.

The first PCB 310 in FIG. 1 according to an embodiment of the present disclosure may be a rigid PCB. The first PCB 310 may be electrically connected to the second PCB 320 through a connector 370. The connector 370 may include a ground pad 390 and a signal pad 391 of the first PCB 310 and a ground pad 392 and a signal pad 393 of the second PCB 320. The ground pad 390 of the first PCB 310 and the ground pad 392 of the second PCB 320 may be connected together by soldering. The signal pad 391 of the first PCB 310 and the signal pad 393 of the second PCB 320 may be connected together by soldering. The second PCB 320 may include an interconnection pad 394. An antenna contact member (for example, a C-clip) or an RF component (for example, a matching circuit) may be mounted on the interconnection pad 394.

The second PCB 320 according to an embodiment of the present disclosure may be connected to the first PCB 310 and may include an FPCB 330 or a rigid PCB 340. The FPCB 330 and the rigid PCB 340 of the second PCB 320 may be integrated with each other. For example, a first FPCB 331, a first rigid printed circuit board 341, a second flexible printed circuit board 332, and a second rigid PCB 342 may be arranged in sequence. The second rigid PCB 342 may include an antenna connector and a universal serial bus (USB) connector, for example.

The second PCB 320 according to an embodiment of the present disclosure may have the electrical path 350 therein through which electrical signals can be transmitted. The electrical path 350 may be connected with the first PCB 310 through the FPCB 330 and the rigid PCB 340, which are integrated with each other.

The electrical path 350 according to an embodiment of the present disclosure may extend from a first electrical component 230 in FIG. 10 described below to a second electrical component 240 and may be disposed on or in the PCB 300 in FIG. 1. The electrical path 350 may be implemented as a transmission line of a side-guarded micro-strip-line type or a strip-line type, where the transmission line is shielded by a ground line 360 in order to transmit a high-frequency signal, which may be, for example, an RF signal.

The electrical path 350 according to an embodiment of the present disclosure,

which is disposed on or in the second PCB 320, may receive a high-frequency signal through the connector 370 and may forward the received signal to antenna radiators 231 and 232 in FIG. 10 described below. Furthermore, a plurality of electrical paths 350 in FIG. 1 may be provided, and a ground line 360 may be disposed between the electrical paths 350 so that the electrical paths 350 can independently transmit the high-frequency signal without mutually affecting the characteristics thereof.

The ground line 360 according to an embodiment of the present disclosure may be disposed adjacent to the electrical path 350, and conductive vias 305 may be arranged in a portion of the area of a ground line 360(a). Accordingly, the electrical path 350 can pass through the PCB 300 without making contact with the conductive vias 305.

According to an embodiment of the present disclosure, the connector 370 for connecting the electrical path 350 may be disposed between the first and second PCBs 310 and 320 in order to transmit a high-frequency signal therebetween. The connector 370 may include a soldering pad, a zip-type connector, or a B-to-B-type connector. A rigid PCB may be employed for the connector 370, and, for example, a B-to-B-type connector may be mounted on the rigid printed circuit board, or a soldering pad may be located on the rigid PCB. Furthermore, for example, the pin map of a transmission line included in the connector 370 may be implemented in the sequence of a ground, an RF signal, and a ground.

Hereinafter, the plurality of layers and the conductive vias 305 of the PCB 300 is described in detail.

FIG. 2 is a partial sectional view of the plurality of layers of the PCB of FIG. 1, according to an embodiment of the present disclosure, where FIG. 2 is taken along line B-B' of FIG. 1. Hereinafter, the PCB is described as the second PCB 320. However, the following description may be applied to second printed circuit boards 420, 520, 620, and 720 in other embodiments, which will be described below.

Referring to FIG. 2, the second printed circuit board 320 according to an embodiment of the present disclosure may be formed of a plurality of layers, each including at least one conductive layer 306, at least one first insulation layer 302, and at

least one second insulation layer 303. The conductive layer 306 and the first insulation layer 302 may be included in a flexible copper clad laminated (FCCL) layer. Further, the first insulation layer 302 may include polyamide.

The conductive layers 306 according to an embodiment of the present disclosure
5 are arranged in the stacking direction, and the first insulation layer 302 is disposed on the rear or front surface of each conductive layer 306. The individual layers may be alternately arranged. One or more conductive vias 305 may be formed to electrically connect the conductive layers 306.

According to an embodiment of the present disclosure, the second insulation layer
10 303 is provided between the conductive layers 306 to prevent the conductive layers 306 from being electrically connected together through contact therebetween. For example, the plurality of second insulation layers 303 disposed between the plurality of conductive layers 306 may function to insulate the layers and may include a resin and a glass fabric.

According to an embodiment of the present disclosure, a plurality of electronic
15 components may be disposed on and electrically connected to the conductive layers 306. The electronic components may be electrically connected to the antenna radiators 231 and 232 in FIG. 10 described below provided in the electronic device. For example, the electronic components may include an antenna clip, a connector, and a switch device.

Without being limited thereto, however, various materials capable of passing
20 electricity therethrough, other than metal, may be applied to the conductive layers 306 in FIG. 2.

According to an embodiment of the present disclosure, the conductive layers 306 may be electrically connected to any of a plurality of signal lines and ground lines.

The conductive layers 306 according to an embodiment of the present disclosure
25 may be provided, on the top and bottom thereof, with a protective layer 301 capable of protecting the conductive layers 306. The protective layers 301 may be formed of an insulating coating material, such as solder resist, and, for example, a photo imageable solder resist ink (PSR ink) may be used for the solder resist. The PSR ink may be left on a product because it provides insulation and protection even after a component is

mounted. Accordingly, the protective layers 301 coated on the exterior of the conductive layers 306 can prevent a short circuit, a connection, corrosion, and contamination of a circuit during a manufacturing process and can protect the conductive layers 306 from external shock, humidity, and chemical substances after the
5 manufacturing process.

The conductive vias 305 according to an embodiment of the present disclosure, which electrically connect the conductive layers 306, may include a first conductive via 305(a) capable of electrically connecting all the layers and a second conductive via 305(b) capable of electrically connecting adjacent layers. For example, the first
10 conductive via 305(a) may be formed by computerized numerical control (CNC) hole machining and copper plating, and the second conductive via 305(b) may be formed by laser hole machining and copper plating.

The conductive vias 305 according to an embodiment of the present disclosure may include a region that forms a predetermined pattern within the PCB 300. The
15 conductive vias 305 may be configured to make contact with at least one of the conductive layers 306 and to pass through at least one of the multiple layers.

A pattern cover layer 304 according to an embodiment of the present disclosure may be disposed in a single layer on the bottom or top of the conductive layers 306 while surrounding a pattern and thus can protect the internal pattern. For example, the pattern
20 cover layer 304 may be formed of polyimide (PI) and an adhesive.

According to an embodiment of the present disclosure, rigid PCBs 380 and 382 and a FPCB 381 may be integrated with one another. The FPCB 381 may not include the second insulation layer 303, but may include the pattern cover layer 304 instead of the protective layer 301.

25 FIG. 3 is a sectional view of an interlayer configuration of a plurality of layers of the PCB of FIG. 1, according to an embodiment of the present disclosure, where FIG. 3 is taken along line B-B' of FIG. 1.

Referring to FIG. 3, the second PCB 320 may include the first FPCB 331, the first rigid PCB 341, and the second FPCB 332, which are integrally connected together.

Further, the second PCB 320 may include first to fourth ground pattern layers 371, 372, 373, and 374 that are connected by the conductive vias 305 in order to form a ground. The entirety of the first and second ground pattern layers 371 and 372 forms a ground. Only a part of the second and third ground pattern layers 372 and 373 forms a
 5 ground. According to an embodiment of the present disclosure, there may be a layer having no ground layer.

Signal line patterns 364 not connected to the ground may be provided. These signal line patterns 364 may be used as RF signal lines.

The ground pattern layers according to an embodiment of the present disclosure
 10 may be connected by the first conductive via 305(a) capable of electrically connecting all of the layers and the second conductive via 305(b) capable of electrically connecting adjacent layers. For example, all or some of the ground pattern layers may be connected using the first conductive via 305(a) passing through all of the layers. For example, the first to fourth ground pattern layers 371, 372, 373, and 374 may be connected by the first
 15 conductive via 305(a). The adjacent ground pattern layers may be connected to each other by the second conductive via 305(b). For example, the first and second ground pattern layers 371 and 372 may be connected to each other by the second conductive via 305(b).

According to an embodiment of the present disclosure, a PCB (for example, the
 20 PCB 300 of FIG. 1) may include the first PCB 310 and the second PCB 320 including a rigid PCB and a FPCB. In this case, the PCB is described as the PCB 300 of FIG. 1 as an example, but may be applied to the PCBs 400, 500, 600, and 700 of FIGS. 4, 5, 6A, 6B, 6C, 7A, 7B, and 7C.

According to an embodiment of the present disclosure, the high-frequency
 25 transmission line provided by the PCB in the electronic device may be configured such that the FPCB 330 having advantages of reducing a material cost and ensuring a mounting space can substitute for a coaxial cable.

For example, the first and second FPCBs 331 and 332 may be used between the second rigid PCB 342 and the first PCB 310 in order to transmit a high-frequency signal

from the second rigid PCB 342 to the first PCB 310.

In another example, in order to eliminate loss that occurs when a high-frequency signal is transmitted using the FPCB 330, the first rigid PCB 341 may be disposed between the first and second FPCB 331 and 332, and the ground pattern layers may be
5 connected by the conductive vias 305.

According to an embodiment of the present disclosure, in order to mitigate an increase in line loss, the electronic device may further include a section containing the first rigid PCB 341 in the middle of the FPCB 330. If the FPCB 330 has a length of 10 mm or more, the transmission line in the section may cause a great line loss. For
10 example, the transmission line may have a change in the strip line characteristics due to the inductance generated by the ground lines 360 disposed on the opposite sides thereof, and thus the line loss generated by the electrical path 350 may increase markedly.

According to an embodiment of the present disclosure, the electronic device may have the one or more vias 305 arranged on the first rigid PCB 341. When the grounds
15 are connected by the conductive vias 305 arranged on the first rigid PCB 341, the inductance generated by the ground lines 360 can be reduced, for example, by virtue of the interlayer ground connection on the ground lines 360, thereby reducing adverse effects, such as line loss. Furthermore, an all-ground layer (ground line) is disposed between the layers providing the second PCB 320, and the entire ground line 360 may be
20 connected to the all-ground layer through the conductive vias 305, whereby it is possible to have a much greater line loss reduction effect.

According to an embodiment of the present disclosure, a rigid PCB may be disposed between the FPCBs, thereby preventing the layers from being separated or twisted as described above. In the case where the existing FPCBs are installed on a
25 drive unit, the layers of the FPCBs may be separated or twisted. Even in this case, there may be a change in the characteristics of a strip line, which may cause line loss in the transmission line or other adverse property changes.

However, an electronic device having the aforementioned arrangement is advantageous in that the side effects mentioned above are not exhibited even when the

FPCBs move, whereby it is possible to reduce line loss in the transmission line and to alleviate other adverse characteristics.

In addition, in the case where a transmission line has a greater length, it is possible to solve shortcomings, such as line loss, by additionally placing rigid PCBs in several
5 positions on the FPCBs and using a connection between the ground line 360, other ground pattern layers 363, and the conductive vias 305. This is described below in greater detail with reference to FIG. 4.

FIG. 4 is a front view of a high-frequency transmission line according to an embodiment of the present disclosure, in which a PCB in an electronic device is used for
10 the high-frequency transmission line. An electronic device may be a smart phone or a wearable device.

According to an embodiment of the present disclosure, an electronic device may include a housing corresponding to a case; a printed circuit unit including at least one PCB 400 in the housing; and an electrical path 450 passing over or through the PCB 400.

Referring to FIG. 4, the PCB 400 according to an embodiment of the present disclosure may include a second PCB 420, a third PCB 411, and a fourth PCB 412. For example, the third PCB 411 may be a main board on which electronic components included in an electronic device are mounted, the main board being capable of transmitting signals between the electronic components. Furthermore, for example, the
15 fourth PCB 412 may be a main board on which electronic components included in an electronic device are mounted, the main board being capable of transmitting signals between the electronic components. Moreover, for example, the second PCB 420 may be a sub-board that transmits signals generated by electronic components. The sub-board may further include the electronic components.

The PCB 400 according to an embodiment of the present disclosure may be provided with a circuit of a terminal, which may be, for example, at least one of an AP, a CP, a memory, and an RF transceiver, and may include a signal line (for example, an RF signal line).
25

The second to fourth PCBs 420, 411, and 412 according to an embodiment of the

present disclosure may be formed of a plurality of layers, and the plurality of layers may include one or more conductive layers and one or more insulation layers.

The third and fourth PCBs 411 and 412 according to an embodiment of the present disclosure may be a rigid PCB. The third or fourth PCB 411 and 412 may be
5 electrically connected to the second PCB 410 through a connector 470.

The third PCB 411 may be disposed on one end of the second PCB 420, and the fourth PCB 412 may be disposed on an opposite end of the second PCB 420. Hereinafter, the third and fourth printed circuit boards 411 and 412 are described as third and fourth rigid PCBs 411 and 412, respectively.

10 The third rigid PCB 411 according to an embodiment of the present disclosure may be connected with the second PCB 420, and a first connector 471 may be disposed to transmit a high-frequency signal through the electrical path 450. The first connector 471 may include a soldering pad, a zip-type connector, or a B-to-B-type connector. A rigid PCB may be employed for the first connector 471, and, for example, a B-to-B-type
15 connector may be mounted on the rigid PCB, or a soldering pad may be located on the rigid PCB. Furthermore, for example, the pin map of a transmission line included in the first connector 471 may be implemented in the sequence of a ground, an RF signal, and a ground.

The first connector 471 according to an embodiment of the present disclosure may
20 include a ground pad 490 and a signal pad 491 of the third rigid PCB 411 and a ground pad 492 and a signal pad 493 of the second PCB 420. The ground pad 490 of the third rigid PCB 411 and the ground pad 492 of the second PCB 420 may be connected together by soldering. The signal pad 491 of the third rigid PCB 411 and the signal pad 493 of the second PCB 420 may be connected by soldering.

25 The fourth rigid PCB 412 according to an embodiment of the present disclosure may be connected to the second PCB 420, and a second connector 472 may be disposed to transmit a high-frequency signal through the electrical path 450. The second connector 472 may include a soldering pad, a zip-type connector, or a B-to-B-type connector. A rigid PCB may be employed for the second connector 472, and, for

example, a B-to-B-type connector may be mounted on the rigid PCB, or a soldering pad may be located on the rigid PCB. Furthermore, for example, the pin map of a transmission line included in the first connector 472 may be implemented in the sequence of a ground, an RF signal, and a ground.

5 The second connector 472 according to an embodiment of the present disclosure may include a ground pad 495 and a signal pad 496 of the fourth rigid PCB 412 and a ground pad 497 and a signal pad 498 of the second PCB 420. The ground pad 495 of the fourth rigid PCB 412 and the ground pad 497 of the second PCB 420 may be connected by soldering. The signal pad 496 of the fourth rigid PCB 412 and the signal
10 pad 498 of the second PCB 420 may be connected by soldering. The second PCB 420 may include an interconnection pad 494. An antenna contact member (for example, a C-clip) or an RF component (for example, a matching circuit) may be mounted on the interconnection pad 494.

 The second PCB 420 according to an embodiment of the present disclosure may
15 be disposed between the third rigid PCB 411 and the fourth rigid PCB 412, and may include an FPCB 430 or a rigid PCB 440. The flexible printed circuit board 430 and the rigid PCB 440 of the second PCB 420 may be integrated with each other. For example, a first FPCB 431, the first rigid PCB 440, and a second flexible PCB 432 may be arranged in sequence. The first rigid PCB 440 may include an antenna connector and a
20 USB connector, for example.

 The electrical path 450 according to an embodiment of the present disclosure may extend from a first electrical component including an antenna radiator to a second electrical component including a communication circuit, and may be disposed on or in the PCB 400. The electrical path 450 may be implemented to be a transmission line of a
25 side-guarded micro-strip-line type or a strip-line type, the transmission line being shielded by a ground line in order to transmit a high-frequency signal, which may be, for example, an RF signal.

 The electrical path 450 according to an embodiment of the present disclosure, which is disposed on or in the second PCB 420, may receive a high-frequency signal

through the connector 470 and may forward the received signal to antenna radiators 231 and 232 in FIG. 10 described below. Furthermore, a plurality of electrical paths 450 in FIG. 4 may be provided, and a ground line 460 may be disposed between the electrical paths 450 so that the electrical paths 450 can independently transmit a high-frequency
5 signal without mutually affecting the characteristics thereof.

The ground line 460 according to an embodiment of the present disclosure may be disposed adjacent to the electrical path 450, and conductive vias 405 may be arranged in a portion of the area of the ground line 460. Accordingly, the electrical path 450 can pass through the PCB 400 without making contact with the conductive vias 405.

10 A plurality of layers of the PCB 400 and the conductive vias 405 are the same as those in the above-described embodiment with reference to FIGS. 1 to 3, and therefore descriptions thereof are omitted here.

FIG. 5 is a front view of a high-frequency transmission line according to an embodiment of the present disclosure, where a PCB in an electronic device is used for the
15 high-frequency transmission line. The electronic device may be a smart phone or a wearable device.

Referring to FIG. 5, an electronic device may include a housing corresponding to a case; a printed circuit unit including at least one PCB 500 in the housing; and an electrical path 550 passing over or through the PCB 500.

20 The PCB 500 according to an embodiment of the present disclosure may include a first PCB 510 and a second PCB 520. For example, the first PCB 510 may be a main board on which electronic components included in an electronic device are mounted, the main board being capable of transmitting signals between the electronic components, and the second PCB 520 may be a sub-board on which the electronic components are
25 mounted, the sub-board being capable of transmitting signals generated by the electronic components. The sub-board may further include the electronic components.

The PCB 500 according to an embodiment of the present disclosure may be provided with a circuit of a terminal, which may be, for example, at least one of an AP, a CP, a memory, and an RF transceiver, and may include a signal line (for example, an RF

signal line).

The first and second PCBs 510 and 520 according to an embodiment of the present disclosure may be formed of a plurality of layers, and the plurality of layers may include one or more conductive layers and one or more insulation layers.

5 The first PCB 510 according to an embodiment of the present disclosure may be a rigid PCB. The first PCB 520 may be electrically connected to the second PCB 510 through a connection unit 570. The first PCB 510 may be connected to the second PCB 520, and the connector 570 may be disposed to transmit a high-frequency signal through the electrical path 550. The connector 570 may include a soldering pad, a zip-type
10 connector, or a B-to-B-type connector. A rigid PCB may be employed for the connector 570, and, for example, a B-to-B-type connector may be mounted on the rigid PCB, or a soldering pad may be located on the rigid PCB.

The connector 570 according to an embodiment of the present disclosure may include a ground pad 590 and a signal pad 591 of the first PCB 510 and a ground pad 592
15 and a signal pad 593 of the second PCB 520. The ground pad 590 of the first PCB 510 and the ground pad 592 of the second PCB 520 may be connected by soldering. The signal pad 591 of the first PCB 510 and the signal pad 593 of the second PCB 520 may be connected together by soldering. The second PCB 520 may include an interconnection pad 594. An antenna contact member (for example, a C-clip) or an RF
20 component (for example, a matching circuit) may be mounted on the interconnection pad 594.

The second PCB 520 according to an embodiment of the present disclosure may include an FPCB 530 and a rigid PCB 540 that are integrated with each other. For example, the second PCB 520 may include a first FPCB 531, a first rigid PCB 541, and a
25 second FPCB 532, which are arranged in sequence. The second FPCB 532 may include an antenna connector and a USB connector. Unlike in an embodiment described above with reference to FIG. 1, the second FPCB 532 in FIG. 5 is configured to perform the function of the second rigid PCB 342 in FIG. 1 as well.

The electrical path 550 in FIG. 5 according to an embodiment of the present

disclosure may extend from a first electrical component including an antenna radiator to a second electrical component including a communication circuit, and may be disposed on or in the PCB 500. The electrical path 550 may be implemented to be a transmission line of a side-guarded micro-strip-line type or a strip-line type, the transmission line
 5 being shielded by a ground line in order to transmit a high-frequency signal, which may be, for example, an RF signal.

The electrical path 550 according to an embodiment of the present disclosure, which is disposed on or in the second PCB 520, may receive a high-frequency signal through the connector 570 and may forward the received signal to an antenna radiator.
 10 Furthermore, a plurality of electrical paths 550 may be provided, and a ground line 560 may be disposed between the electrical paths 550 so that the electrical paths 550 can independently transmit the high-frequency signal without mutually affecting the characteristics thereof.

The ground line 560 according to an embodiment of the present disclosure may be
 15 disposed adjacent to the electrical path 550, and conductive vias 505 may be arranged in a portion of the area of the ground line 560. Accordingly, the electrical path 550 can pass through the PCB 500 without making contact with the conductive vias 505.

A plurality of layers of the PCB 500 and the conductive vias 505 are the same as those in an above-described embodiment with reference to FIGS. 1 to 3, and therefore
 20 descriptions thereof are omitted here.

FIGS. 6A, 6B, and 6C are front views of conductive vias according to an embodiment of the present disclosure, the conductive vias being arranged in electronic device PCBs. FIGS. 6A, 6B, and 6C illustrate various forms of PCBs.

Referring to FIG. 6A, a second PCB 620 according to an embodiment of the
 25 present disclosure may include an FPCB 630 and a rigid PCB 640, which are integrated with each other. For example, the second PCB 620 may include a first FPCB 631, the first rigid PCB 640, and a second FPCB 632, which are arranged in sequence.

An electrical path 650 according to an embodiment of the present disclosure may be disposed on or in the second PCB 620 and may be implemented to be a transmission

line of a side-guarded micro-strip-line type or a strip-line type in order to transmit a high-frequency signal.

The electrical path 650 according to an embodiment of the present disclosure, which is disposed on or in the second PCB 620, may receive a high-frequency signal through a connector and may forward the received signal to an antenna radiator. Furthermore, a plurality of electrical paths 650 may be provided, and a ground line 660 may be disposed between the electrical paths 650 so that the electrical paths 650 can independently transmit the high-frequency signal without mutually affecting the characteristics thereof.

The ground lines 660 according to an embodiment of the present disclosure may be disposed adjacent to the electrical path 650 and may be connected with one or more conductive layers through conductive vias 605(c) in the PCB. Since the conductive vias 605(c) are arranged in one ground line 660, the electrical path 650 may pass through the PCB without making contact with the conductive vias 605(c).

The electrical path 650 according to an embodiment of the present disclosure is disposed to pass through the flexible PCB 630 and the rigid PCB 640, and thus the ground lines 660 disposed on opposite sides of the electrical path 650 may also be disposed to pass through the FPCB 630 and the rigid PCB 640. While two ground lines 660 are provided in FIG. 6A, the number of ground lines 660 may be diversely varied so as to avoid contact with the electrical path 650, without being limited thereto.

For example, the first ground line 660(a) disposed on the left side may have a greater width than the second ground line 660(b) disposed on the right side, and the plurality of conductive vias 605(c) may be formed in the region where the rigid PCB 640 and the ground line 660(a) cross each other. The second ground line 660(b) may not include the conductive vias 605(c) since the second ground line 660(b) has a lesser width than the first ground line 660(a).

The PCB is illustrated as having the conductive vias 605(c) arranged in the first ground line 660(a) having a great width. The conductive vias 605(c) arranged in the first ground line 660(a) may include vias passing through all layers. Ground patterns in

all or some of the layers and the ground line on a side of the electrical path may be connected by the conductive vias 605(c) passing through all the layers.

Referring to FIG. 6B, ground lines 660 may be separately disposed on opposite sides of an electrical path 650 in the second PCB 620 and may be formed to pass through the flexible PCB and the rigid PCB along the lengthwise direction of the electrical path 650. While the two ground lines 660 are provided in FIG. 6B, the number of ground lines 660 may be diversely varied so as to avoid contact with the electrical path 650, without being limited thereto.

According to an embodiment of the present disclosure, the first ground line 660(a) disposed on the left side may have a greater width than the second ground line 660(b) disposed on the right side, and a plurality of conductive vias 605(c) may be formed in the region where the first ground line 660(a) and the rigid PCB 640 cross each other. The second ground line 660(b) on the right side may not include the conductive vias 605(c) having the size since the second ground line 660(b) has a lesser width than the first ground line 660(a). Instead, the second ground line 660(b) on the right side may have conductive vias 605(d) having a lesser size than the conductive vias 605(c).

For example, the rigid PCB 640 formed of a plurality of layers may include the third conductive vias 605(c) passing through all of the layers in the PCB and the fourth conductive vias 605(d) passing through some of the layers. The third conductive vias 605(c) may have a greater hole size than the fourth conductive vias 605(d) passing through some of the layers, and may be arranged in the ground line 660(a) having a great width. However, the fourth conductive vias 605(d) may be arranged in the second ground line 660(b) having a lesser width and disposed on the right side, since the fourth conductive vias 605(d) have a lesser hole size than the third conductive vias 605(c). Even when the ground lines 660 are connected with a ground in a different layer through some conductive vias 605, it is possible to reduce line loss.

Referring to FIG. 6C, ground lines 660 may be separately disposed on opposite sides of an electrical path 650 and may be formed in the lengthwise direction of the electrical path 650. While two ground lines 660 are provided in FIG. 6C, the number of

ground lines 660 may be diversely varied so as to avoid contact with the electrical path 650, without being limited thereto.

According to an embodiment of the present disclosure, the first ground line 660(a) disposed on the left side may have a lesser width than the second ground line 660(b) disposed on the right side, and a plurality of conductive vias 605(d) may be formed in the region where the second ground line 660(b) and the rigid PCB 640 cross each other. The second ground line 660(b) on the right side has a greater width than the first ground line 660(a), but is too narrow to have the great conductive vias 605(c). Instead, the conductive vias 605(d) having a lesser width may be arranged in the second ground line 660(b) on the right side.

Therefore, according to an embodiment of the present disclosure, the second ground line 660(b) having a lesser width and disposed on the right side may electrically connect the PCB using the fourth conductive vias 605(d) passing through some layers. Even when the ground lines 660 are connected with a ground in a different layer through some of the fourth conductive vias 605(d), it is possible to reduce line loss.

FIGS. 7A, 7B, and 7C are top views illustrating conductive vias arranged in PCBs according to an embodiment of the present disclosure. FIGS. 7A, 7B, and 7C illustrate various forms of PCBs.

Referring to FIG. 7A, a second PCB 720 may include a plurality of FPCBs and/or a plurality of rigid PCBs. For example, the second PCB 720 may be configured such that FPCBs 730 and rigid PCBs 740 are alternately disposed and are integrated with each other.

The FPCBs 730 according to an embodiment of the present disclosure may include a first FPCB 731, a second FPCB 732, and a third FPCB 733. The rigid PCBs 740 may include a first rigid PCB 741 and a fifth rigid PCB 742.

According to an embodiment of the present disclosure, at least two rigid PCBs 740 may be provided between the FPCBs 730. For example, the second PCB 720 may include the first FPCB 731, the first rigid PCB 741, the second FPCB 732, the fifth rigid PCB 742, and the third FPCB 733, which are arranged in sequence.

The first and fifth rigid PCBs 741 and 742 may be disposed to be spaced apart from each other and may include a plurality of conductive vias 705(c) and ground lines 760.

An electrical path 750 according to an embodiment of the present disclosure may
 5 be disposed on or in the second PCB 720 and may be implemented to be a transmission line of a side-guarded micro-strip-line type or a strip-line type in order to transmit a high-frequency signal.

The electrical path 750 according to an embodiment of the present disclosure, which is disposed on or in the second PCB 720, may receive a high-frequency signal
 10 through a connector and may forward the received signal to an antenna radiator. Furthermore, a plurality of electrical paths 750 may be provided, and the ground line 760 may be disposed between the electrical paths 750 so that the electrical paths 750 can independently transmit the high-frequency signal without mutually affecting the characteristics thereof.

15 The ground lines 760 according to an embodiment of the present disclosure may be disposed adjacent to the electrical path 750 and may be connected with one or more conductive layers through the conductive vias 705(c) in the PCB 720.

The ground lines 760 according to an embodiment of the present disclosure may be formed to pass through the FPCB 730 and the rigid PCBs 740 along the lengthwise
 20 direction of the electrical path 750. While two ground lines 760 are provided in the present disclosure, the number of ground lines 760 may be diversely varied so as to avoid contact with the electrical path 750, without being limited thereto.

For example, the first ground line 760(a) disposed on the left side of the first and fifth rigid PCBs 741 and 742 may have a greater width than the second ground line
 25 760(b) disposed on the right side, and the plurality of conductive vias 705(c) may be formed in the regions where the first ground line 760(a) crosses the first and fifth rigid PCBs 741 and 742. In another example, the second ground line 760(b) on the right side may not include the conductive vias 705(c) having the size since the second ground line 760(b) has a lesser width than the first ground line 760(a). The PCB is illustrated as

having the conductive vias 705(c) arranged in the first ground line 760(a) having a great width.

Referring to FIG. 7B, the second PCB 720 may include the FPCBs 730 and the rigid PCBs 740 that are integrated with each other. For example, the second PCB 720
5 may be configured such that the FPCBs 730 and the rigid PCBs 740 are alternately disposed and are integrated with each other.

The second PCB 720 according to an embodiment of the present disclosure may include the first FPCB 731, the first rigid PCB 741, the second FPCB 732, the fifth rigid PCB 742, and the third FPCB 733, which are arranged in sequence.

10 Ground lines 760 according to an embodiment of the present disclosure may be separately disposed on opposite sides of the electrical path 750 and may be formed along the lengthwise direction of the electrical path 750. While two ground lines 760 are provided in the FIG. 7B, the number of ground lines 760 may be diversely varied so as to avoid contact with the electrical path 750, without being limited thereto.

15 According to an embodiment of the present disclosure, the first ground line 760(a) disposed on the left side of the first and fifth rigid PCBs 741 and 742 may have a greater width than the second ground line 760(b) disposed on the right side, and a plurality of conductive vias 705(c) may be formed in the regions where the first ground line 760(a) crosses the rigid PCBs 740. However, the second ground line 760(b) on the right side
20 may not include the conductive vias 705(c) having the size since the second ground line 760(b) has a lesser width than the first ground line 760(a). Instead, the second ground line 760(b) on the right side may have conductive vias 705(d) having a lesser size than the conductive vias 705(c).

For example, the rigid PCBs 740 formed of a plurality of layers may include the
25 third conductive vias 705(c) passing through all the layers in the PCB and the fourth conductive vias 705(d) passing through some of the layers. The third conductive vias 705(c) may have a greater hole size than the fourth conductive vias 705(d) passing through some of the layers, and may be arranged in the ground line 760(a) having a great width. However, the fourth conductive vias 705(d) may be arranged in the second

ground line 760(b) having a lesser width and disposed on the right side, since the fourth conductive vias 705(d) have a lesser hole size than the third conductive vias 705(c). Even when the ground lines 760 are connected with a ground in a different layer through some conductive vias 705, it is possible to reduce a line loss.

5 Referring to FIG. 7C, the second PCB 720 may include the FPCBs 730 and the rigid PCBs 740 that are alternately disposed and are integrated with each other. In this embodiment, at least two rigid PCBs 740 may be provided between the FPCBs 730. For example, the second PCB 720 may include the first FPCB 731, the second FPCB 732, the first rigid PCB 741, the fifth rigid PCB 742, and the third FPCB 733, which are arranged
10 in sequence.

Ground lines 760 according to an embodiment of the present disclosure may be separately disposed on opposite sides of the electrical path 750 and may be formed along the lengthwise direction of the electrical path 750. While two ground lines 760 are provided in the present disclosure, the number of ground lines 760 may be diversely
15 varied so as to avoid contact with the electrical path 750, without being limited thereto.

According to an embodiment of the present disclosure, the first ground line 760(a) disposed on the left side of the first and fifth rigid PCBs 741 and 742 may have a lesser width than the second ground line 760(b) disposed on the right side, and a plurality of conductive vias 705(d) may be formed in the regions where the second ground line
20 760(b) crosses the rigid PCBs 740. The second ground line 760(b) on the right side has a greater width than the first ground line 760(a), but is too narrow to have the great conductive vias 705(c). Instead, the conductive vias 705(d) having a lesser width may be arranged in the second ground line 760(b) on the right side.

Therefore, according to an embodiment of the present disclosure, even when the
25 ground lines 760 are connected to a ground in a different layer through some of the fourth conductive vias 705(d) arranged in the second ground line 760(b), it is possible to reduce a line loss.

FIG. 8 is a perspective view of an electronic device according to an embodiment of the present disclosure. The electronic device 10 may be a smart phone or a wearable

device. The electronic device 10 may be an electronic device that includes the PCBs described above with reference to FIGS. 1 to 7.

Referring to FIG. 8, a display 101 may be mounted on the front 107 of the electronic device 10. A speaker device 102 for receiving a counterpart's speech may be provided on the upper side of the display 101. A microphone device 103 for transmitting the speech of a user of the electronic device 10 to the counterpart may be provided on the lower side of the display 101.

According to an embodiment of the present disclosure, one or more components for performing various functions of the electronic device 10 may be arranged around the speaker device 102. For example, the one or more arranged components may include at least one sensor module 104. The sensor module 104 may include, for example, at least one of an illuminance sensor (e.g., an optical sensor), a proximity sensor, an infrared sensor, and an ultrasonic sensor. The components may also include a camera device 105. The components may also include a light emitting diode (LED) indicator 106 for informing a user of status information of the electronic device 10.

According to an embodiment of the present disclosure, the electronic device 10 may include a metal bezel 110 (for example, capable of serving as at least a part of a metal housing). The metal bezel 110 may be disposed along the outer periphery of the electronic device 10 and may extend to at least one area of the back of the electronic device 10, the at least one area being connected to the outer periphery. The metal bezel 110 may define the thickness of the electronic device 10 along the outer periphery of the electronic device 10 and may be formed in a loop shape. Without being limited thereto, however, the metal bezel 110 may also be formed in such a manner that the metal bezel contributes to at least part of the thickness of the electronic device 10. The metal bezel 110 may also be disposed only in at least one area of the outer periphery of the electronic device 10. The metal bezel 110 may include one or more cut-off portions 115 and 116. Unit bezel parts separated by the cut-off portions 115 and 116 may be used as an antenna radiator that operates in at least one frequency band.

According to an embodiment of the present disclosure, the metal bezel 110 may

have a loop shape along the outer periphery of the electronic device and may be disposed to contribute to the entirety or a part of the thickness of the electronic device 10. When the electronic device 10 is viewed from the front, the metal bezel 110 may include a right bezel part 111, a left bezel part 112, an upper bezel part 113, and a lower bezel part 114.

5 In this case, the lower bezel part 114 described above may serve as a unit bezel part formed by a pair of cut-off portions 116.

According to an embodiment of the present disclosure, a main antenna device may be disposed in the lower region (an antenna region of a main part) of the electronic device 10. The lower bezel part 114 may be used as a main antenna radiator by virtue of the
10 pair of cut-off portions 116. The lower bezel part 114 may serve as an antenna radiator that operates in at least two operating frequency bands depending on the feeding position. For example, the lower bezel part 114 may be a part of an antenna that supports a low band (LB) and a high/middle band (H/MB), or a part of an antenna that supports an H/MB.

15 According to an embodiment of the present disclosure, the antenna device of the present disclosure is simply for illustrative purpose, and the aforementioned functions of the lower bezel part 114 may be performed by the upper bezel part 113 separated by the other cut-off portions 115, or may be performed by the upper and lower bezel parts.

According to an embodiment of the present disclosure, an antenna region of a
20 diversity part may be used as an antenna for diversity/multiple input multiple output (MIMO). For example, the upper bezel part 113 may be a part of a diversity antenna that supports LB and H/MB.

According to an embodiment of the present disclosure, at least one of the right and left bezel parts 111 and 112 may be supplied with electrical power to operate as an
25 antenna. For example, the right or left bezel part 111 or 112 may be a part of an antenna that supports an H/MB band or LB and H/MB bands. For example, an antenna including the right or left bezel part 111 or 112 included in the antenna region of the main part may operate as a main antenna. An antenna including the right or left bezel part 111 or 112 included in the antenna region of the diversity part may operate as a diversity

antenna. Hereinafter, the configuration of an antenna is described in detail.

FIG. 9 is a schematic of an antenna arrangement in an electronic device 10 according to an embodiment of the present disclosure. The electronic device 10 may be a smart phone or a wearable device. The electronic device 10 may be an electronic
5 device that includes the PCBs described above with reference to FIGS. 1 to 7.

The electronic device 10 according to an embodiment of the present disclosure may include a plurality of antenna radiators 230. For example, the electronic device 10 may include first and second antennas 231 and 232 of a main part 210 and third and fourth antennas 233 and 234 of a diversity part 220.

10 The first antenna 231 may support LB and H/MB bands, and the second antenna 232 may support an H/MB band. According to an embodiment of the present disclosure, an LB band has a relatively great wavelength so that an antenna may have a relatively great size, and there may be a mounting space limitation by adding an antenna supporting multiple LBs to the main part 210 of the terminal. Therefore, only one antenna may be
15 configured to support an LB band.

In the case where the electronic device 10 additionally has the diversity part 220 according to an embodiment of the present disclosure, the electronic device 10 may include the third and fourth antennas 233 and 234. In order to enhance correlation/isolation characteristics, the antenna radiators 230 of the main part 210 may
20 be included in the lower end portion of the terminal, and the antenna radiators 230 of the diversity part 220 may be included in the upper end portion of the terminal. For example, in the case where the antenna radiators 230 of the main part 210 are included in the lower end portion of the terminal, and the antenna radiators 230 of the diversity part 220 are included in the upper end portion of the terminal, it is possible to ensure an
25 antenna separation distance available in the terminal even when a plurality of LB band antennas are included.

By virtue of the structure described above, the electronic device 10 can receive signals using four antennas in an H/MB band and can receive signals using two antennas in an LB band. That is, 4th order diversity/MIMO can be performed in an H/MB band,

and 2nd order diversity/MIMO can be performed in an LB band.

FIG. 10 illustrates an antenna device and a PCB of an electronic device according to an embodiment of the present disclosure. Referring to FIG. 10, a main RF circuit in the main part 210 may supply electrical power to the first and second antennas 231 and 232 connected thereto through the connector 370 and the second PCB 320 using RF signal lines. The second PCB 320 may include the first FPCB 331, the first rigid PCB 341, the second FPCB 332, and the second rigid PCB 342. A ground may be removed from an antenna feeding part or from the surroundings of an antenna in order to enhance antenna characteristics.

According to an embodiment of the present disclosure, the metal bezel 110 may include the right bezel part 111, the left bezel part 112, the upper bezel part 113, and the lower bezel part 114 when viewed from the front. The upper bezel part 113 may be maintained separately from the right and left bezel parts 111 and 112 by a pair of cut-off portions that are formed with a predetermined interval therebetween. The lower bezel part 114 may be maintained separately from the right and left bezel parts 111 and 112 by a pair of cut-off portions that are formed with a predetermined interval therebetween. The pair of cut-off portions may be formed of a dielectric material. The pair of cut-off portions may be formed by double-injection molding or insert molding a synthetic resin into the metal bezel. Without being limited thereto, however, various types of electrical insulating materials may be applied to the pair of cut-off portions.

According to an embodiment of the present disclosure, a first feeding piece may be integrally formed with the lower bezel part 114 and may be supplied with electrical power by a first feeding part of the PCB. The first feeding piece of the lower bezel part 114 may be connected to the first feeding part of the PCB merely by installing the PCB in the electronic device, or may be electrically connected to the first feeding part by a separate electrical connection member (e.g., a C-clip, etc.).

According to an embodiment of the present disclosure, a first feeding pad may be disposed on the PCB and may be electrically connected to the first feeding piece of the lower bezel part 114. A first electrical path (e.g., an interconnection wire line) may be

formed from the first feeding pad to the first feeding part. The first feeding piece may be integrally formed with the lower bezel part 114 and may be supplied with electrical power by the first feeding part of the PCB. The first feeding piece of the lower bezel part 114 may be connected to the first feeding part of the PCB merely by installing the
5 PCB in the electronic device, or may be electrically connected to the first feeding part by a separate electrical connection member (e.g., a C-clip, etc.).

According to an embodiment of the present disclosure, the first feeding pad may be disposed on the PCB and may be electrically connected to the first feeding piece of the lower bezel part 114. The first electrical path (e.g., an interconnection wire line) may be
10 formed from the first feeding pad to the first feeding part.

The lower bezel part may be a portion of the first antenna 231 of the main part 210 that supports an LB or H/MB band. The left and right bezel parts 112 and 111 may also be supplied with electrical power by the same method. The left and right bezel parts 112 and 111 may be portions of the second antenna 231 of the main part 210 that supports
15 an LB or H/MB band.

According to an embodiment of the present disclosure, a first electrical connection piece may be integrally formed with the lower bezel part 114 so as to be located at a position separate from the first feeding piece by a predetermined distance, and may be grounded to a first ground part of the PCB. The first electrical connection piece of the
20 lower bezel part 114 may be grounded to the first ground part of the PCB merely by installing the PCB in the electronic device, or may be electrically connected to the first ground part by a separate electrical connection member (e.g., a C-clip, etc.).

According to an embodiment of the present disclosure, a first ground pad may be disposed on the PCB and may be electrically connected to the first electrical connection
25 piece of the lower bezel part 114. A second electrical path (e.g., an interconnection wire line) may be formed from the first ground pad to the first ground part.

According to an embodiment of the present disclosure, a second feeding piece may be integrally formed with the lower bezel part 114 and may be supplied with electrical power by a first feeding part of the PCB. The second feeding piece of the

lower bezel part 114 may be connected to a second feeding part of the PCB merely by installing the PCB in the electronic device, or may be electrically connected to the second feeding part by a separate electrical connection member (e.g., a C-clip, etc.).

According to an embodiment of the present disclosure, a second feeding pad may be disposed on the PCB and may be electrically connected to the second feeding piece of the lower bezel part 114. A third electrical path (e.g., an interconnection wire line) may be formed from the second feeding pad to the second feeding part. The lower bezel part 114 may be a portion of the second antenna 232 of the main part 210 that supports an LB or H/MB band.

According to an embodiment of the present disclosure, a second electrical connection piece may be integrally formed with the right bezel part 111 so as to be located at a position separate from the cut-off portions by a predetermined distance, and may be grounded to a second ground part of the PCB. The second electrical connection piece of the right bezel part 111 may be grounded to a second ground part of the PCB merely by installing the PCB in the electronic device, or may be electrically connected to the second ground part by a separate electrical connection member (e.g., a C-clip, etc.).

According to an embodiment of the present disclosure, a second ground pad may be disposed on the PCB and may be electrically connected to the second electrical connection piece of the right bezel part 111. A fourth electrical path (e.g., an interconnection wire line) may be formed from the second ground pad to the second ground part.

For example, the feeding parts, the feeding pads, the ground parts, and the ground pads of the main part 210 may be disposed on the second PCB 320. The main part 210 and an RF circuit of the diversity part 220 may be disposed on the first PCB (main PCB) 310. The first and second PCBs may be connected by an FPCB. The first and second PCBs may be integrated with each other.

According to an embodiment of the present disclosure, the second rigid PCB 342 included in the second PCB 320 may be disposed at a lower vertical position than the first PCB (main PCB) 310. Accordingly, components included in the second rigid PCB 342

may be additionally spaced apart from the antenna. Further, relatively thick components, such as a USB connector and a speaker, can also be disposed on the second rigid PCB 342.

According to an embodiment of the present disclosure, a transmitted or received
5 signal of the RF circuit of the main part 210 may be forwarded to the first and second feeding parts of the second PCB 320.

The diversity part 220 may include the third and fourth antennas 233 and 234. The third antenna 233 may include a part of the upper bezel part 113, and the fourth antenna 234 may include the left or right bezel part 112 or 111.

10 The third antenna 233 may support LB and H/MB bands, and the fourth antenna 234 may support an H/MB band. Feeding parts, feeding pads, ground parts, and ground pads of the diversity part 220 may be disposed on the first PCB (main PCB) 310. Electrical paths connecting the feeding parts and the feeding pads and electrical paths connecting the ground parts and the ground pads in the diversity part 220 may be
15 disposed on the first PCB (main PCB) 310.

According to an embodiment of the present disclosure, when the main part 210 uses the right bezel part 111 as the second antenna 232 for the purpose of signal separation between the antennas, the diversity part 220 may use the opposite left bezel part 112 as the fourth antenna 234. Alternately, when the main part 210 uses the left
20 bezel part 112 as the second antenna 232, the diversity part 220 may use the opposite right bezel part 111 as the fourth antenna 234.

FIG. 11 is an illustration of a signal transmission using the second PCB 320 according to an embodiment of the present disclosure. The second PCB 320 may be the same as the second PCBs 320, 420, and 520 described above with reference to FIGS. 1 to
25 7.

Referring to FIG. 11, the second PCB 320 may include the FPCB 330 and the rigid PCB 340, which are integrated with each other. The second PCB 320 may include the first FPCB 331, the second FPCB 332, the first rigid PCB 341, and the second rigid PCB 342.

According to an embodiment of the present disclosure, the electrical path 350 for transmitting a high-frequency signal may also transmit signals other than RF signals using the second PCB 320. The other signals may include, for example, a USB signal, an ear-jack signal, a microphone (MIC) signal, a speaker signal, and a touch key signal.

5 According to an embodiment of the present disclosure, an antenna contact to which an RF signal is connected, and a USB connector, an ear-jack connector, a microphone (MIC) connector, and a speaker contact to which the other signals are connected may be disposed on the second rigid PCB 342. For example, RF signals may be transmitted through two electrical paths 350: the first electrical path 351 and the
10 second electrical path 352. The ground line 360 may be disposed between the first and second electrical paths 351 and 352 to prevent the first and second electrical paths 351 and 352 from mutually affecting the characteristics thereof. The conductive vias 305 may be arranged in the region where the ground line 360 and the first rigid PCB 341 cross each other, and may be connected to a ground in a different layer. A rigid PCB
15 connected to the rear end of the second rigid PCB 342 may be various types of modules of the electronic device, and may function as a touch key, for example.

According to an embodiment of the present disclosure, a first connector 342(a) connected to the first electrical path 351 may be a first antenna contact, a first USB connector, a first ear-jack connector, a first MIC connector, or a first speaker contact, and
20 a second connector 342(b) connected to the second electrical path 352 may be a second antenna contact, a second USB connector, a second ear-jack connector, a second MIC connector, and a second speaker contact.

FIGS. 12 and 13 are enlarged views of a first FPCB and a first rigid PCB. The second PCB 320 may be one of the second PCBs 320, 420, 520, 620, and 720 described
25 above with reference to FIGS. 1 to 7.

Referring to FIGS. 12 and 13, a high-frequency signal may be connected from a B-to-B-type connector A of the connector 370 to antenna (ANT) feeding B and C by a micro strip line or a strip line on the FPCB. Further, the ground lines 306 may exist on the left and right sides of the electrical path 350 through which an RF signal is

transmitted on the connector, and may be formed parallel to the electrical path 350 along the lengthwise direction of the electrical path 350 so as to be connected to the ANT feeding B and C. In addition, the conductive vias 305 may be formed in the ground lines 306. Conductive vias 307 lesser in size than the conductive vias 305 may be
5 formed on the right side of the electrical path 351 so as to prevent contact between the ground lines and a signal line on one side.

An electronic device according to an embodiment of the present disclosure includes a housing; a PCB disposed in the housing, the PCB including a plurality of layers that include one or more conductive layers and one or more insulation layers; a
10 first electrical component formed to be at least a part of the housing or disposed in the housing; a second electrical component disposed above or near the PCB in the housing, the second electrical component being separated from the first electronic component; and at least one electrical path extending from the first electrical component to the second electronic component, wherein at least a portion of the electrical path runs on or inside
15 the PCB, wherein the PCB may include a region including a pattern of conductive vias, each of the vias extending through at least a part of the plurality of layers to contact at least one of the one or more conductive layers, and wherein the electrical path may run through the region without making contact with the conductive vias.

According to an embodiment of the present disclosure, the first electrical
20 component may include an antenna radiator and the second electrical component may include a communication circuit.

According to an embodiment of the present disclosure, the antenna radiator may form at least one part of the housing.

According to an embodiment of the present disclosure, the pattern of the
25 conductive vias may be aligned in one or more rows or columns along the electrical path when viewed from above the region of the PCB.

According to an embodiment of the present disclosure, the PCB may include an FPCB as at least a part thereof, where the FPCB may transmit a signal through the electrical path.

According to an embodiment of the present disclosure, the PCB may include a first PCB; and a second PCB connected to the first PCB through a connector, where the second PCB includes at least one FPCB and at least one rigid PCB, and where the electrical path may transmit a high-frequency signal through the connector and may pass
5 over or through the second PCB.

According to an embodiment of the present disclosure, the electronic device may further include a ground line disposed around the electrical path so as to be adjacent thereto, where the ground line may be connected to the one or more conductive layers through the conductive vias in the rigid PCB.

10 According to an embodiment of the present disclosure, the second PCB may include a first FPCB connected to the first PCB through a communication circuit; a second PCB disposed to face the first flexible PCB with a separation distance therebetween; and a first rigid PCB disposed between the first and second FPCBs, where the first rigid PCB includes the pattern of the conductive vias.

15 According to an embodiment of the present disclosure, the electronic device may further include a second rigid PCB connected to the second FPCB, wherein a part of the antenna radiator that forms at least one portion of the housing may be disposed on the second rigid PCB.

According to an embodiment of the present disclosure, at least one of the plurality
20 of layers may form a ground pattern in the ground line of the second PCB, where the entire line of the ground pattern includes a ground.

According to an embodiment of the present disclosure, the conductive vias of the first rigid PCB may include a first conductive via passing through all of the plurality of layers; and a second conductive via passing through some of the plurality of layers, and
25 where the first conductive via may connect the ground pattern disposed on all or some of the layers and the ground line.

According to an embodiment of the present disclosure, a plurality of ground lines may be formed on the rigid PCB so as to be spaced apart from the electrical path, and may have different widths.

According to an embodiment of the present disclosure, the conductive vias formed along the width of the ground line may include a third conductive via having a first hole size; and a fourth via having a second hole size that differs from the first hole size.

According to an embodiment of the present disclosure, a space for mounting an
5 electronic component may be formed on or in the rigid PCB of the second PCB.

According to an embodiment of the present disclosure, the connector may be disposed on one side of the first or second PCB, where the electrical path passing over or through the second PCB may be connected to the first PCB.

According to an embodiment of the present disclosure, the connector may include
10 a first connector disposed on one end of the first FPCB of the second PCB; and a second connector disposed on one end of the second FPCB of the second PCB.

According to an embodiment of the present disclosure, the electronic device may include a plurality of electrical paths and a plurality of antenna radiators, wherein the electrical paths may be connected to the respective antenna radiators corresponding
15 thereto, and the ground line may be disposed between the plurality of electrical paths.

According to an embodiment of the present disclosure, the first PCB may include at least a part of the first electrical component and may be connected to the antenna radiator, and the second PCB may have the second electrical component disposed thereon.

According to an embodiment of the present disclosure, the FPCB and the rigid
20 PCB of the second PCB may be integrated with each other.

According to an embodiment of the present disclosure, the region of the PCB may be a rigid PCB region, and the electronic device may include a first FPCB region disposed between the first electrical component and the region; and a second FPCB region disposed between the second electrical component and the region.

25 An electronic device according to an embodiment of the present disclosure includes a first PCB; a second PCB connected with the first PCB through a connector and including at least one FPCB and at least one rigid PCB that are integrated with each other, where the flexible and rigid PCBs include a plurality of layers; an electrical path that transmits a high-frequency signal through the connector and passes over or through the

second PCB; and at least one conductive via disposed in the rigid PCB to connect grounds between the plurality of layers.

FIG. 14 is a block diagram of an electronic device 10, according to an embodiment of the present disclosure, in a network environment 11.

5 Referring to FIG. 14, the electronic device 10 may include a bus 180, a processor 120, a memory 130, an input/output interface 150, a display 160, and a communication interface 170. The electronic device 10 may omit at least one of the elements, or may further include other elements.

The bus 180 may include, for example, a circuit that interconnects the elements 10 120 to 170 and delivers communication (for example, a control message and/or data) between the elements.

The processor 120 may include one or more of a CPU, an AP, and a CP. For example, the processor 120 may carry out operations or data processing relating to the control and/or communication of at least one other element of the electronic device 10.

15 The memory 130 may include a volatile and/or non-volatile memory. For example, the memory 130 may store instructions or data relevant to at least one other element 120 to 170 of the electronic device 10. The memory 130 may store software and/or a program 140. The program 140 may include, for example, a kernel 141, middleware 143, an application programming interface (API) 145, and/or application 20 programs (or applications) 147. At least some of the kernel 141, the middleware 143, and the API 145 may be referred to as an operating system (OS).

For example, the kernel 141 may control or manage system resources (for example, the bus 180, the processor 120, and the memory 130) that are used to execute operations or functions implemented by the other programs (for example, the middleware 143, the 25 API 145, and the applications 147). Furthermore, the kernel 141 may provide an interface by which the middleware 143, the API 145, or the applications 147 access the individual elements of the electronic device 10 to control or manage the system resources.

For example, the middleware 143 may serve as an intermediary to allow the API 145 or the applications 147 to exchange data with the kernel 141 through communication.

Furthermore, the middleware 143 may process one or more task requests received from the applications 147 according to the priorities of the requests. For example, the middleware 143 may assign, to at least one of the applications 147, a priority to use the system resources (for example, the bus 180, the processor 120, and the memory 130) of the electronic device 10. For example, the middleware 143 may perform scheduling or loading balancing on the one or more task requests by processing the one or more task requests according to the priority assigned to the at least one of the applications.

The API 145 is, for example, an interface by which the applications 147 control functions provided by the kernel 141 or the middleware 143, and the API 145 may include at least one interface or function (for example, an instruction) for file control, window control, image processing, or text control, for example.

The input/output interface 150 may serve as, for example, an interface that can forward instructions or data input from a user or an external device to the other element(s) 120 to 170 of the electronic device 10. Furthermore, the input/output interface 150 may output instructions or data received from the other element(s) of the electronic device 10 to the user or the external device.

Examples of the display 160 may include a liquid crystal display (LCD), an LED display, an organic LED (OLED) display, a microelectromechanical systems (MEMS) display, and an electronic paper display. For example, the display 160 may display various types of content (for example, text, images, videos, icons, or symbols) for a user. The display 160 may include a touch screen, and may receive, for example, a touch, gesture, a proximity input, or a hovering input using an electronic pen or a part of a user's body.

For example, the communication interface 170 may establish communication between the electronic device 10 and a first external electronic device 12, a second external electronic device 14, or a server 16. For example, the communication interface 170 may be connected to a network 162 through wireless or wired communication to communicate with the second external electronic device 14 or the server 16.

The wireless communication may use, for example, at least one of long term

evolution (LTE), LTE-advanced (LTE-A), code division multiple access (CDMA), wideband CDMA (WCDMA), universal mobile telecommunications system (UMTS), wireless broadband (WiBro), and global system for mobile communications (GSM), for example, as a cellular communication protocol. Furthermore, the wireless
 5 communication may include short range communication 164, for example. The short range communication 164 may include, for example, at least one of wireless fidelity (WiFi), Bluetooth (BT), near field communication (NFC), and global navigation satellite system (GNSS). Depending on the service area and the bandwidth, the GNSS may include, for example, at least one of a global positioning system (GPS), a global
 10 navigation satellite system (Glonass), a Beidou navigation satellite system (Beidou), and a European global satellite-based navigation system (Galileo). Hereinafter, “GPS” may be interchangeably used with “GNSS.” The wired communication may include, for example, at least one of a USB, a high definition multimedia interface (HDMI), recommended standard 232 (RS-232), and a plain old telephone service (POTS). The
 15 network 162 may include at least one of a communication network such as a computer network (e.g., a local area network (LAN) or a wide area network (WAN)), the Internet, and a telephone network.

Each of the first and second external electronic devices 12 and 14 may be of the same or a different type from the electronic device 10. According to an embodiment of
 20 the present disclosure, the server 16 may include a group of one or more servers. All or some of the operations executed in the electronic device 10 may be executed in the electronic device 102, the electronic device 104, or the server 16. When the electronic device 10 must perform a function or service automatically or in response to a request, the electronic device 10 may request the electronic device 12, the electronic device 14, or
 25 the server 16 to perform at least some functions relating thereto, instead of or in addition to performing the function or service by itself. The electronic device 102, the electronic device 104, or the server 106 may execute the requested function or additional functions and then deliver the result to the electronic apparatus 10. The electronic device 10 may provide a received result as is, or may further process the received result to provide the

requested function or service. To this end, for example, cloud computing, distributed computing, or client-server computing technology may be used.

The electronic device 10, which is described below in greater detail, may be one of a wearable device, a notebook computer, a net-book computer, a smart phone, a tablet PC,
 5 a Galaxy tab, an i-Pad, and a wireless charging device mentioned above. In this embodiment, the electronic device 10 may be a smart phone.

The wireless charging device, according to an embodiment of the present disclosure, refers to a device that recharges the electronic device by wirelessly transmitting and receiving power in a short range.

10 In addition, the display 160 of the electronic device 10 may increase in size and may implement a luxurious design by minimizing the bezel area thereof, or may be implemented to be flexible, convex, or concave.

Namely, the peripheral portion of the display 160 may be bent to allow a view area to extend to the lateral portion. As the view area of the display 160 is bent to extend to
 15 the lateral, the view area may be enlarged, a separate screen may be used on the lateral portion, or a luxurious design may be implemented. In other words, the display 160 may include a first view area and second view areas on opposite sides of the first view area.

FIG. 15 is a block diagram of an electronic device 1901 according to an
 20 embodiment of the present disclosure. The electronic device 1901 may include, for example, the entirety or a part of the electronic device 10 described above with reference to FIG. 14. The electronic device 1901 in FIG. 15 may include at least one processor (e.g., an AP) 1910, a communication module 1920, a subscriber identification module card 1924, a memory 1930, a sensor module 1940, an input device 1950, a display 1960,
 25 an interface 1970, an audio module 1980, a camera module 1991, a power management module 1995, a battery 1996, an indicator 1997, and a motor 1998.

The processor 1910 may operate, for example, an OS or an application program to control a plurality of hardware or software elements connected to the processor 1910 and to perform various types of data processing and operations. The processor 1910 may be

implemented to be a system on chip (SoC), for example. According to an embodiment of the present disclosure, the processor 1910 may further include a graphics processing unit (GPU) and/or an image signal processor. The processor 1910 may also include at least some (for example, a cellular module 1921) of the elements illustrated in FIG. 15.

5 The processor 1910 may load, in a volatile memory, instructions or data received from at least one of the other elements (for example, a non-volatile memory) to process the loaded instructions or data, and may store various types of data in the non-volatile memory.

The configuration of the communication module 1920 may be the same as or
10 similar to that of the communication interface 170 of FIG. 14. The communication module 1920 may include, for example, a cellular module 1921, a WiFi module 1923, a BT module 1925, a GNSS module 1927 (for example, a GPS module, a Glonass module, a Beidou module, or a Galileo module), an NFC module 1928, and an RF module 1929.

The cellular module 1921 may provide, for example, a voice call, a video call, a
15 text message service, or an Internet service through a communication network. According to an embodiment of the present disclosure, the cellular module 1921 may identify and authenticate the electronic device 1901 in a communication network using the subscriber identification module card (SIM card) 1924. The cellular module 1921 may perform at least some of the functions that the processor 1910 can provide. The
20 cellular module 1921 may include a CP.

The WiFi module 1923, the BT module 1925, the GNSS module 1927, and the NFC module 1928 may each include, for example, a processor for processing data transmitted and received through the corresponding module. According to an embodiment of the present disclosure, at least some (for example, two or more) of the
25 cellular module 1921, the WiFi module 1923, the BT module 1925, the GNSS module 1927, and the NFC module 1928 may be included in one integrated circuit (IC) or IC package.

The RF module 1929 may transmit and receive, for example, a communication signal (for example, an RF signal). The RF module 1929 may include, for example, a

transceiver, a power amplifier module (PAM), a frequency filter, a low noise amplifier (LNA), or an antenna. At least one of the cellular module 1921, the WiFi module 1923, the BT module 1925, the GNSS module 1927, and the NFC module 1928 may transmit and receive an RF signal through a separate RF module.

5 The subscriber identification module card 1924 may include, for example, an embedded SIM, and may include unique identification information (for example, an integrated circuit card identifier (ICCID)) or subscriber information (for example, an international mobile subscriber identity (IMSI)).

10 The memory 1930 (for example, the memory 130 in FIG. 14) may include, for example, an internal memory 1932 or an external memory 1934. The internal memory 1932 may include, for example, at least one of a volatile memory (for example, a dynamic random access memory (DRAM), a static RAM (SRAM), or a synchronous DRAM (SDRAM)) and a non-volatile memory (for example, a one-time programmable read only memory (OTPROM), a programmable ROM (PROM), an erasable and
15 programmable ROM (EPROM), an electrically erasable PROM (EEPROM), a mask ROM, a flash ROM, a flash memory (for example, a NAND flash memory or a NOR flash memory), a hard disc drive, or a solid state drive (SSD)).

20 The external memory 1934 may further include a flash drive, which may be, for example, a compact flash (CF) drive, a secure digital (SD) memory card, a micro SD (Micro-SD) memory card, a mini SD (Mini-SD) memory card, an extreme digital (xD) memory card, a multi-media card (MMC), or a memory stick. The external memory 1934 may be functionally and/or physically connected to the electronic device 1901 through various interfaces.

25 For example, the sensor module 1940 may measure a physical quantity, or may sense an operating state of the electronic device 1901, to convert the measured or sensed information into an electrical signal. The sensor module 1940 may include, for example, at least one of a gesture sensor 1940A, a gyro sensor 1940B, an atmospheric pressure sensor 1940C, a magnetic sensor 1940D, an acceleration sensor 1940E, a grip sensor 1940F, a proximity sensor 1940G, a color sensor 1940H (for example, a red, green, blue

(RGB) sensor), a biometric sensor 1940I, a temperature/humidity sensor 1940J, an illuminance sensor 1940K, and a ultraviolet (UV) light sensor 1940M. Additionally or alternatively, the sensor module 1940 may include, for example, an electronic nose (E-nose) sensor, an electromyography (EMG) sensor, an electroencephalogram (EEG) sensor, an electrocardiogram (ECG) sensor, an infrared (IR) sensor, an iris sensor, and/or a fingerprint sensor. The sensor module 1940 may further include a control circuit for controlling one or more sensors included therein. In an embodiment of the present disclosure, the electronic device 1901 may further include a processor configured to control the sensor module 1940 as a part of the processor 1910 or separately from the processor 1910 to control the sensor module 1940 while the processor 1910 is in a reduced power or sleep state.

The input device 1950 may include, for example, a touch panel 1952, a (digital) pen sensor 1954, a key 1956, or an ultrasonic input device 1958. The touch panel 1952 may use, for example, at least one of a capacitive type panel, a resistive type panel, an infrared type panel, and an ultrasonic type panel. Furthermore, the touch panel 1952 may further include a control circuit. The touch panel 1952 may further include a tactile layer to provide a tactile reaction to a user.

The (digital) pen sensor 1954 may include, for example, a recognition sheet that is a part of, or separate from, the touch panel. The key 1956 may include, for example, a physical button, an optical key, or a keypad. The ultrasonic input device 1958 may sense ultrasonic waves generated by an input tool through a microphone 1988 and may identify data corresponding to the sensed ultrasonic waves.

The display 1960 (for example, the display 160 in FIG. 14) may include a panel 1962, a hologram device 1964, or a projector 1966. The panel 1962 may have a configuration that is the same as, or similar to, that of the display 160 illustrated in FIG. 1. The panel 1962 in FIG. 15 may be implemented to be, for example, flexible, transparent, or wearable. The panel 1962, together with the touch panel 1952, may be implemented to be one module. The hologram device 1964 may show a three-dimensional image in the air by using an interference of light. The projector 1966 may project light onto a

screen to display an image. The screen may be located, for example, inside or outside of the electronic device 1901. According to an embodiment of the present disclosure, the display 1960 may further include a control circuit for controlling the panel 1962, the hologram device 1964, and the projector 1966.

5 The interface 1970 may include, for example, an HDMI 1972, a USB 1974, an optical interface 1976, or a D-subminiature (D-sub) connector 1978. The interface 1970 may be included, for example, in the communication interface 170 illustrated in FIG. 3. Additionally or alternatively, the interface 1970 in FIG. 15 may include, for example, a mobile high-definition link (MHL) interface, an SD card/ MMC interface, or an Infrared
10 Data Association (IrDA) standard interface.

For example, the audio module 1980 may convert a sound into an electrical signal, and vice versa. At least some elements of the audio module 1980 may be included, for example, in the input/output interface 145 illustrated in FIG. 1. The audio module 1980 in FIG. 15 may process sound information that is input or output, for example, through a
15 speaker 1982, a receiver 1984, an earphone 1986, or the microphone 1988.

The camera module 1991 is a device that can photograph a still image and a dynamic image. According to an embodiment of the present disclosure, the camera module 1991 may include one or more image sensors (for example, a front sensor or a rear sensor), a lens, an image sensor processor (ISP), or a flash (for example, an LED or a
20 xenon lamp).

The power management module 1995 may manage, for example, power of the electronic device 1901. According to an embodiment of the present disclosure, the power management module 1995 may include a power management IC (PMIC), a charger IC, or a battery gauge. The PMIC may use a wired and/or wireless charging
25 method. Examples of the wireless charging method may include a magnetic resonance method, a magnetic induction method, and an electromagnetic wave method. An additional circuit (for example, a coil loop, a resonance circuit, or a rectifier) for wireless charging may be further included. The battery gauge may measure, for example, a residual amount of the battery 1996 and a voltage, current, or temperature while charging.

The battery 1996 may include, for example, a rechargeable battery and/or a solar battery.

The indicator 1997 may indicate a particular state (for example, a booting state, a message state, or a charging state) of the electronic device 1901 or a part thereof (for example, the processor 1910). The motor 1998 may convert an electrical signal into a mechanical vibration and may generate a vibration or a haptic effect. The electronic device 1901 may include a processing unit (for example, a GPU) for supporting mobile TV. The processing unit for supporting mobile TV may process media data according to a standard, such as digital multimedia broadcasting (DMB), digital video broadcasting (DVB), or MediaFlo™.

10 Each of the above-described component elements of hardware according to the present disclosure may be configured with one or more components, and the names of the corresponding component elements may vary based on the type of electronic device. The electronic device according to various embodiments of the present disclosure may include at least one of the aforementioned elements. Some elements may be omitted or
15 other additional elements may be further included in the electronic device. In addition, some of the hardware components according to various embodiments may be combined into one entity, which may perform functions identical to those of the relevant components before the combination.

While the present disclosure has been shown and described with reference to
20 certain embodiments thereof, it will be apparent to those skilled in the art that the camera lens module according to the present disclosure is not limited to these embodiments, and various changes in form and details may be made therein without departing from the scope of the present disclosure as defined by the appended claims and their equivalents.

WHAT IS CLAIMED IS:

1. An electronic device, comprising:
a housing;
5 a printed circuit board (PCB) disposed in the housing, wherein the PCB comprises a plurality of layers that comprise one or more conductive layers and one or more insulation layers;
a first electrical component formed to be at least a part of the housing or disposed in the housing;
10 a second electrical component disposed above or near the PCB in the housing, wherein the second electrical component is separated from the first electrical component; and
at least one electrical path extending from the first electrical component to the second electrical component, wherein at least a portion of the electrical path runs on or
15 inside the PCB,
wherein the PCB comprises a region including a pattern of conductive vias,
wherein each of the conductive vias extends through at least a part of the plurality of layers to contact at least one of the one or more conductive layers, and
wherein the electrical path runs through the region without making contact with
20 the conductive vias.
2. The electronic device of claim 1, wherein the first electrical component comprises an antenna radiator, and the second electrical component comprises a communication circuit.
- 25 3. The electronic device of claim 1, wherein the antenna radiator forms at least a part of the housing.
4. The electronic device of claim 1, wherein the pattern of the conductive

vias is aligned in one or more rows or columns along the electrical path when viewed from above the region of the PCB.

5 5. The electronic device of claim 1, wherein the PCB comprises a flexible PCB (FPCB) as at least a part thereof, and the FPCB transmits a signal through the electrical path.

 6. The electronic device of claim 1, wherein the PCB comprises:
a first PCB; and
10 a second PCB connected to the first PCB through a connector, wherein the second PCB comprises at least one flexible PCB (FPCB) and at least one rigid PCB,
 wherein the electrical path transmits a high-frequency signal through the connector and passes over or through the second PCB.

15 7. The electronic device of claim 6, further comprising:
a ground line disposed around the electrical path so as to be adjacent thereto,
 wherein the ground line is connected to the one or more conductive layers through the conductive vias in the at least one rigid PCB.

20 8. The electronic device of claim 6, wherein the second PCB comprises:
a first FPCB connected to the first PCB through a communication circuit;
a second PCB disposed to face the first FPCB with a separation distance therebetween; and
a first rigid PCB disposed between the first and second FPCBs and comprising the
25 pattern of the conductive vias.

 9. The electronic device of claim 8, further comprising:
a second rigid PCB connected to the second FPCB, wherein a part of an antenna radiator that forms at least one portion of the housing is disposed on the second rigid PCB.

10. The electronic device of claim 7, wherein at least one of the plurality of layers forms a ground pattern in the ground line of the second PCB, wherein an entire line of the ground pattern includes a ground.

5

11. The electronic device of claim 10, wherein the conductive vias of the first rigid PCB comprise:

a first conductive via passing through all of the plurality of layers and a second conductive via passing through some of the plurality of layers,

10 wherein the first conductive via connects the ground pattern disposed on all or some of the layers and the ground line.

12. The electronic device of claim 10, wherein a plurality of ground lines are formed on the rigid PCB so as to be spaced apart from the electrical path, wherein the
15 ground lines having different widths.

13. The electronic device of claim 10, wherein the conductive vias formed along a width of the ground line comprise a third conductive via having a first hole size and a fourth via having a second hole size that differs from the first hole size.

20

14. The electronic device of claim 6, wherein the connector is disposed on one side of the first or second PCB, and the electrical path passing over or through the second PCB is connected to the first PCB.

25

15. The electronic device of claim 14, wherein the connector comprises: a first connector disposed on one end of a first of the at least one FPCB of the second PCB; and a second connector disposed on one end of the second FPCB of the second PCB.

16. The electronic device of claim 6, wherein the electronic device comprises a plurality of electrical paths and a plurality of antenna radiators,

wherein the electrical paths are connected to the antenna radiators corresponding thereto, respectively, and

5 a ground line is disposed between the plurality of electrical paths.

17. The electronic device of claim 6, wherein the first PCB comprises at least a part of the first electrical component and is connected to an antenna radiator, and wherein the second PCB has the second electrical component disposed thereon.

10

18. The electronic device of claim 6, wherein the at least one FPCB and the at least one rigid PCB of the second PCB are integrated with each other.

19. The electronic device of claim 1, wherein the region of the PCB is a rigid
15 PCB region, and

the electronic device comprises a first FPCB region disposed between the first electrical component and the region and a second FPCB region disposed between the second electrical component and the region.

20 20. An electronic device, comprising:

a first printed circuit board (PCB);

a second PCB connected to the first PCB through a connector and comprising at least one flexible PCB (FPCB) and at least one rigid PCB that are integrated with each other, wherein the at least one FPCB and the at least one rigid PCB each include a
25 plurality of layers;

an electrical path that transmits a high-frequency signal through the connector and passes over or through the second PCB; and

at least one conductive via disposed in the at least one rigid PCB to connect grounds between the plurality of layers.

ABSTRACT

Disclosed is an electronic device including a housing, a printed circuit board (PCB) in the housing, wherein the PCB includes a plurality of layers with one or more
5 conductive and insulation layers, a first electrical component formed as at least a part of or in the housing, a second electrical component above or near the PCB in the housing, wherein the first and second electrical components are separated, and at least one electrical path extending from the first electrical component to the second electrical component, wherein at least a portion of the electrical path runs on or inside the PCB,
10 wherein the PCB includes a region including a pattern of conductive vias, wherein each of the vias extends through at least part of the plurality of layers to contact at least one of the one or more conductive layers, and wherein the electrical path runs through the region without contacting the vias.

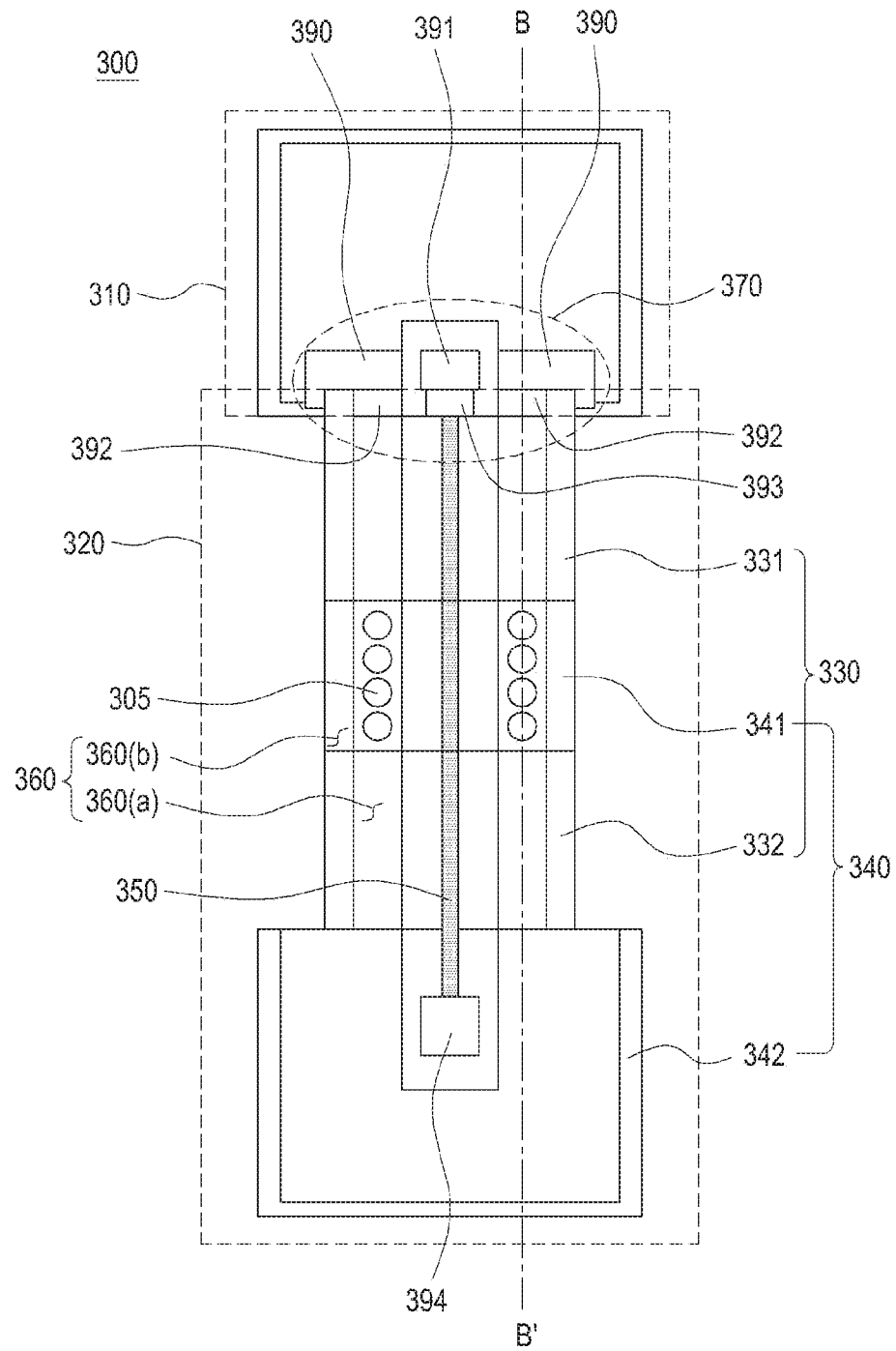


FIG.1

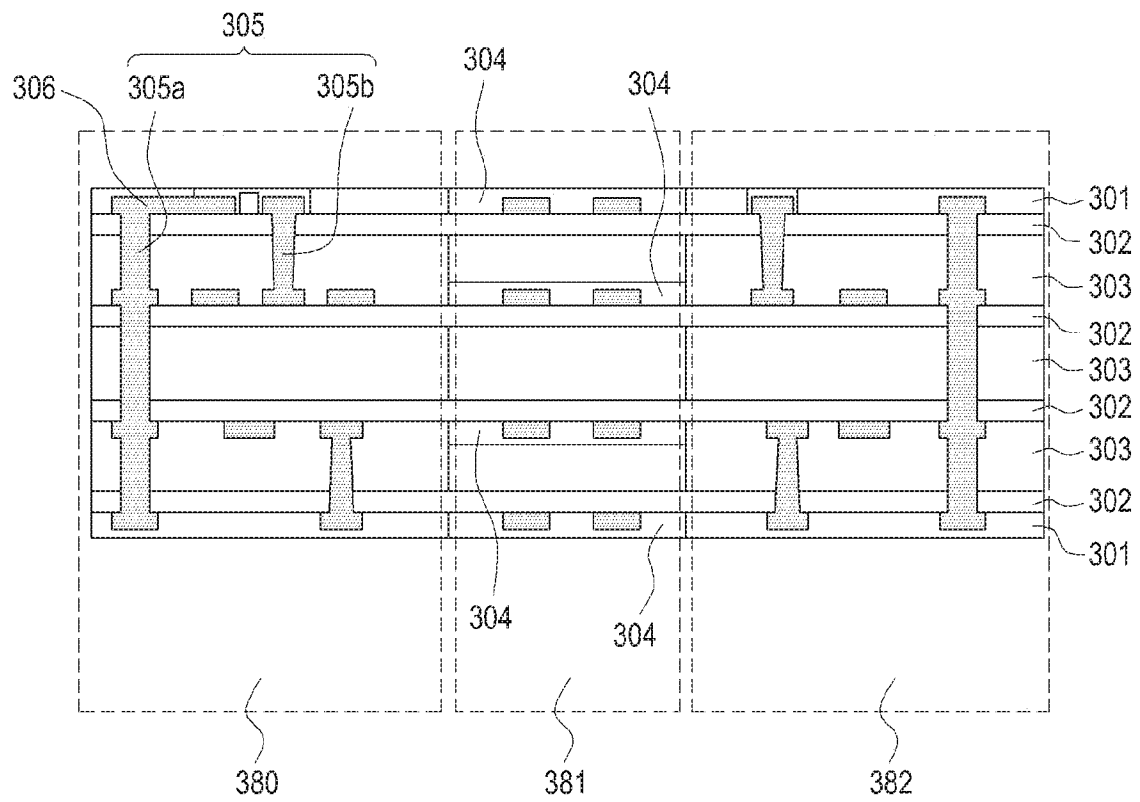


FIG.2

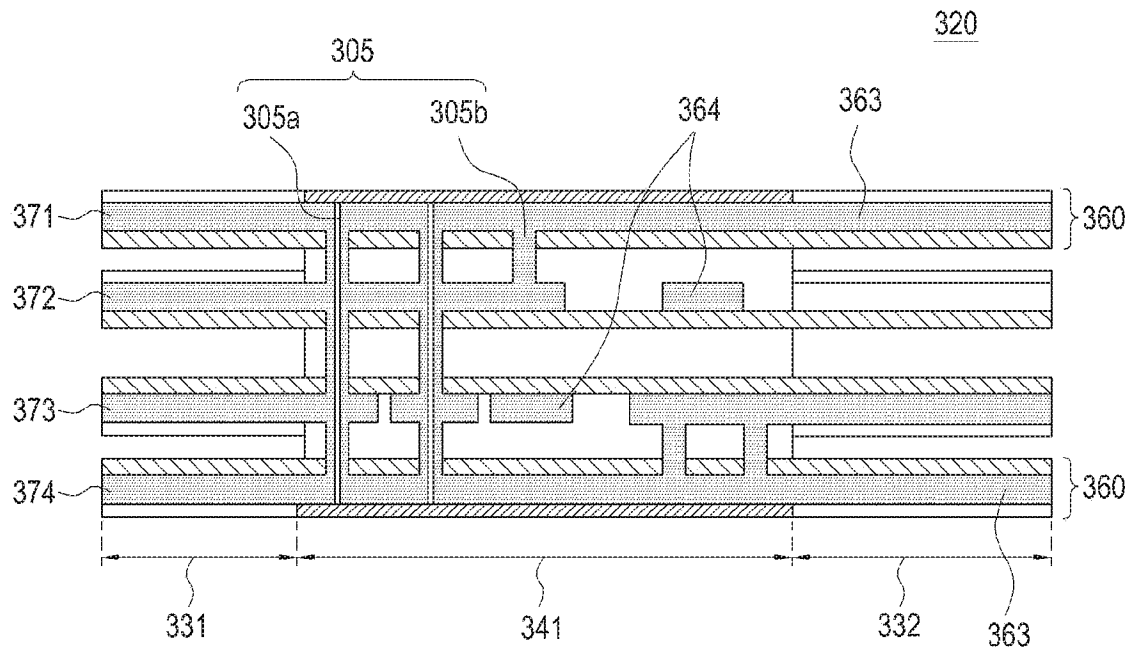


FIG.3

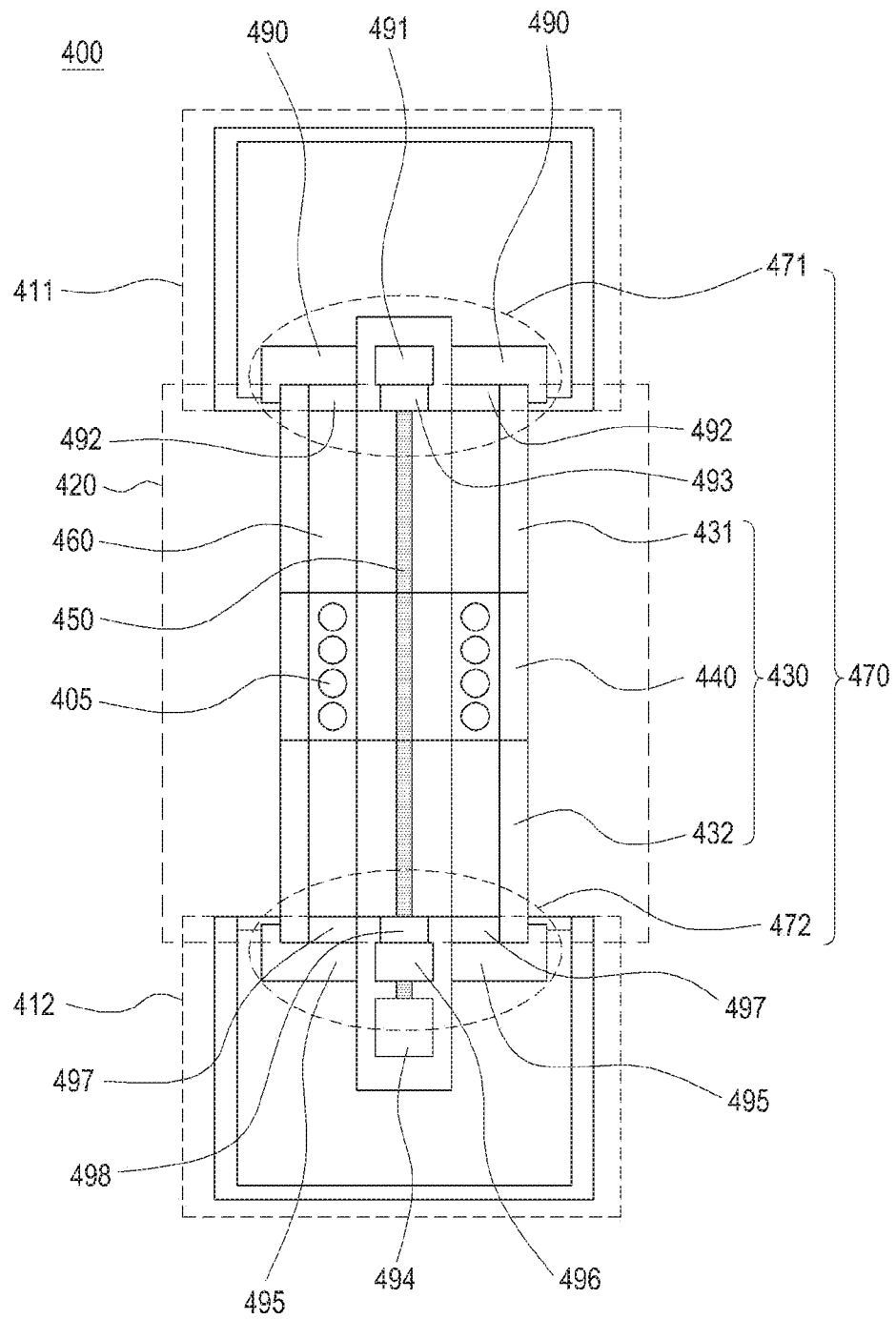


FIG.4

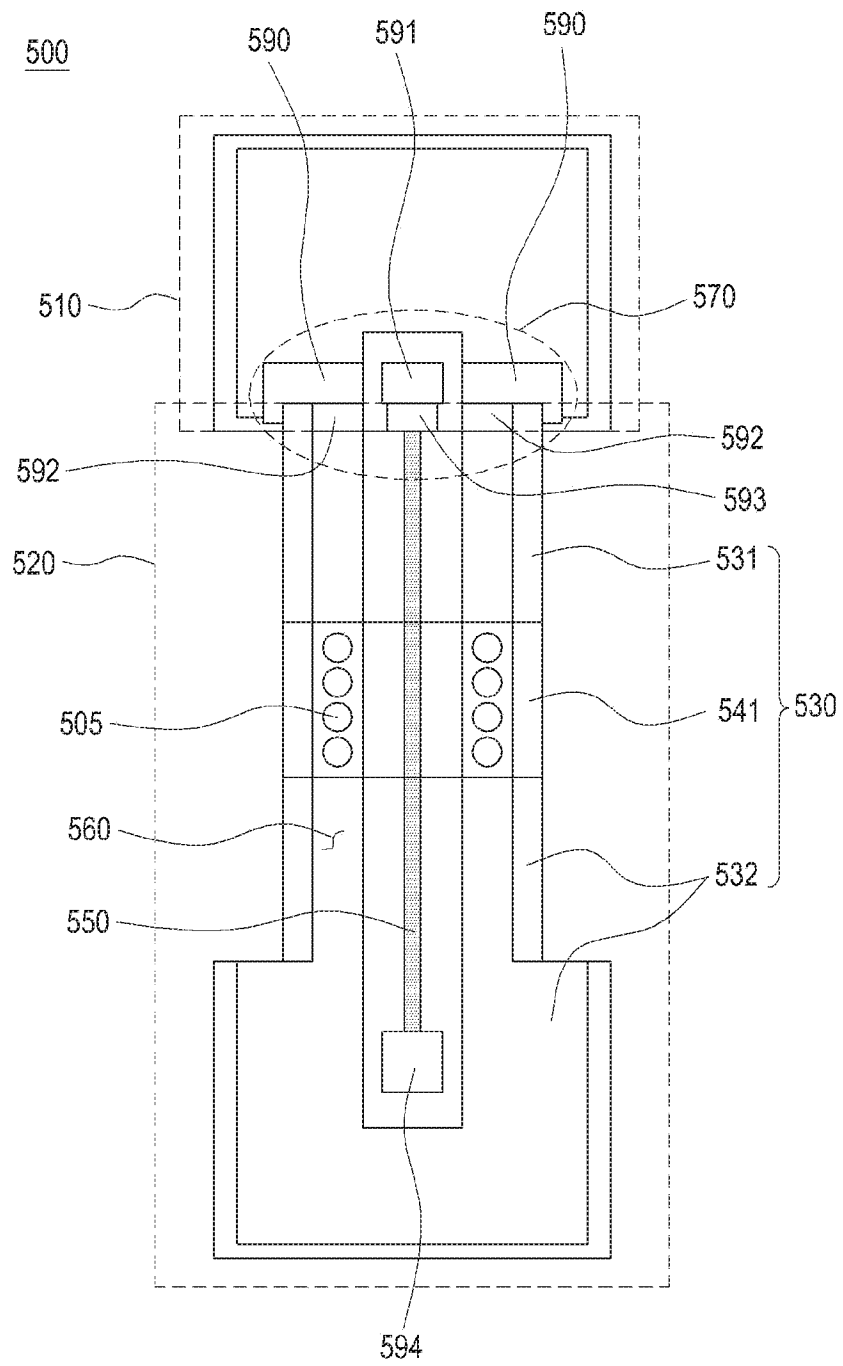


FIG.5

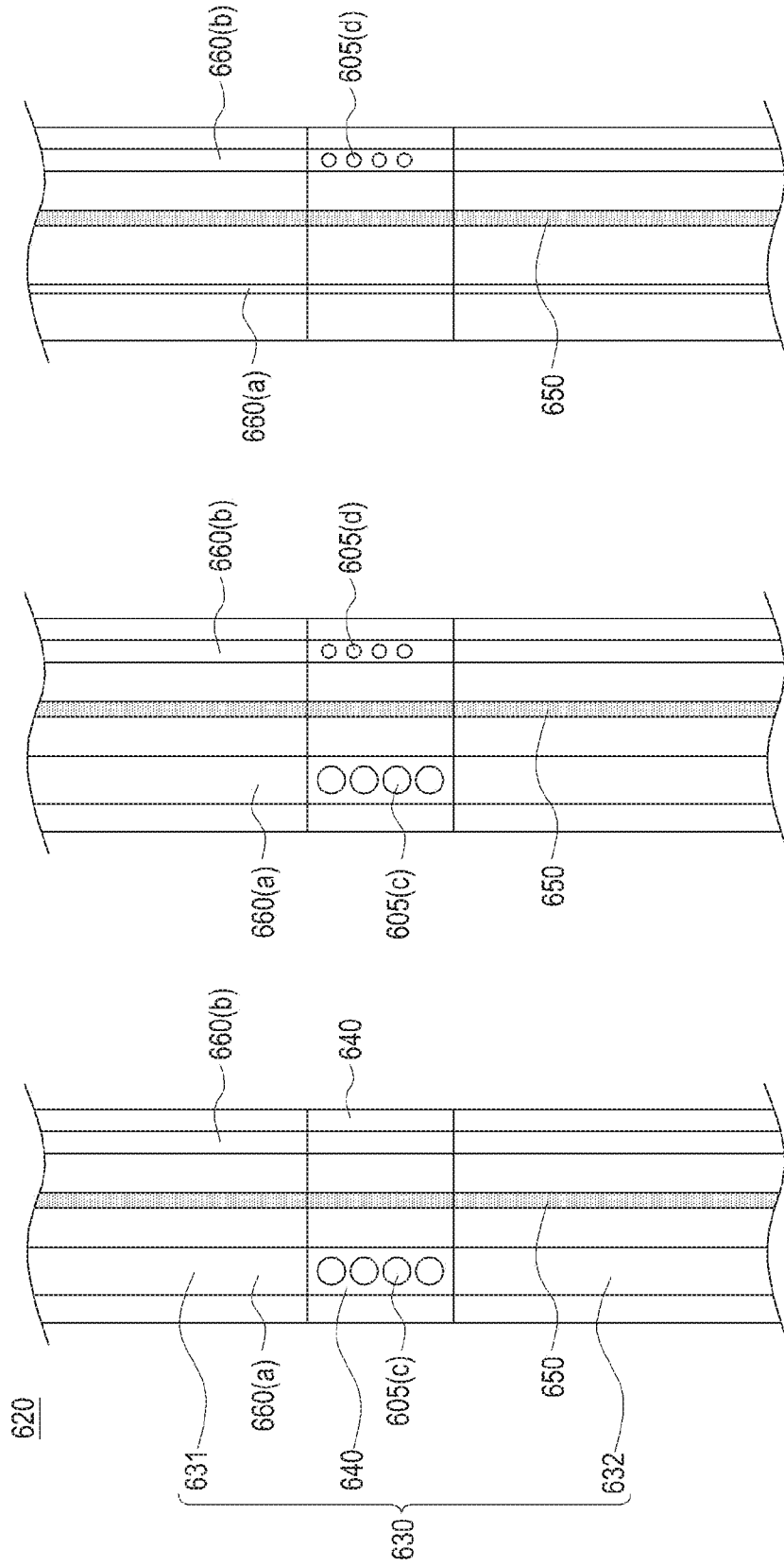


FIG. 6A

FIG. 6B

FIG. 6C

720

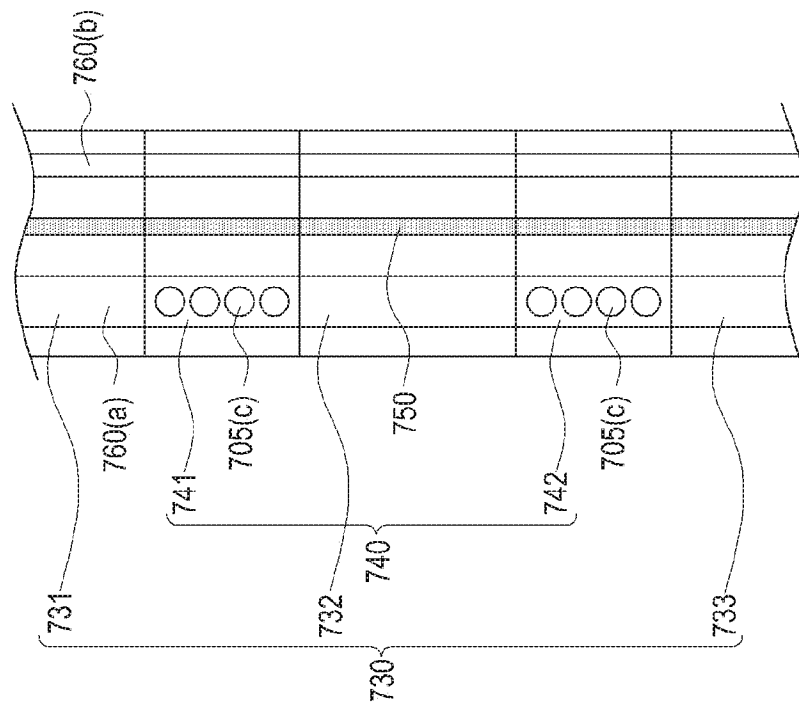


FIG. 7A

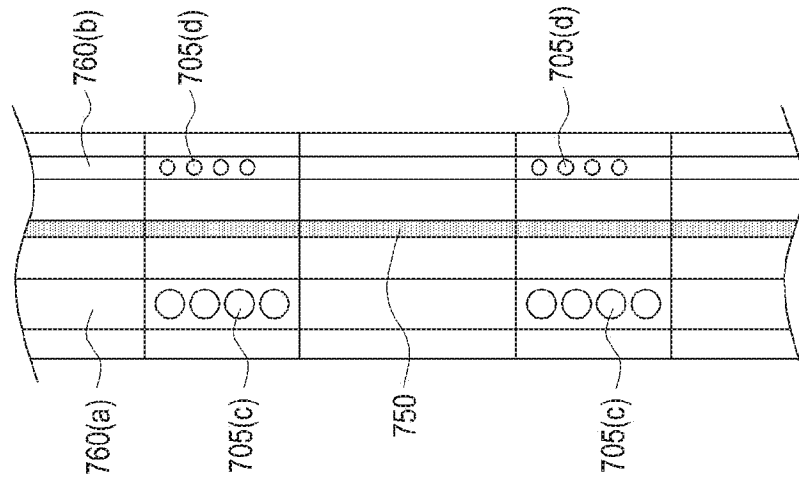


FIG. 7B

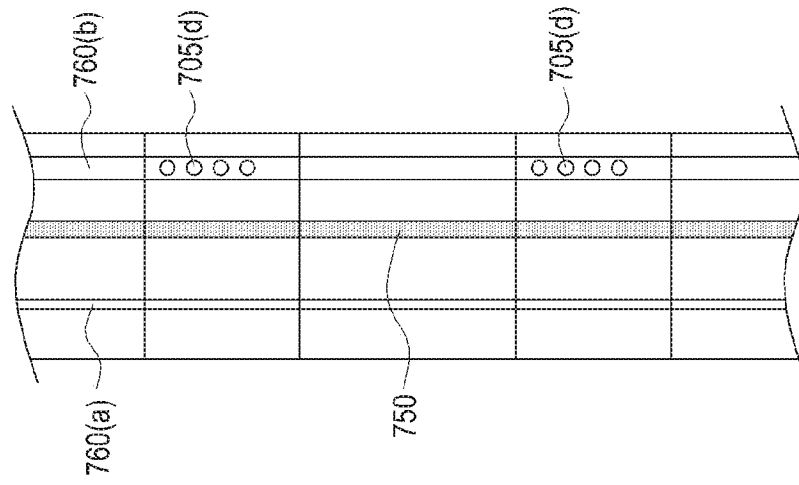


FIG. 7C

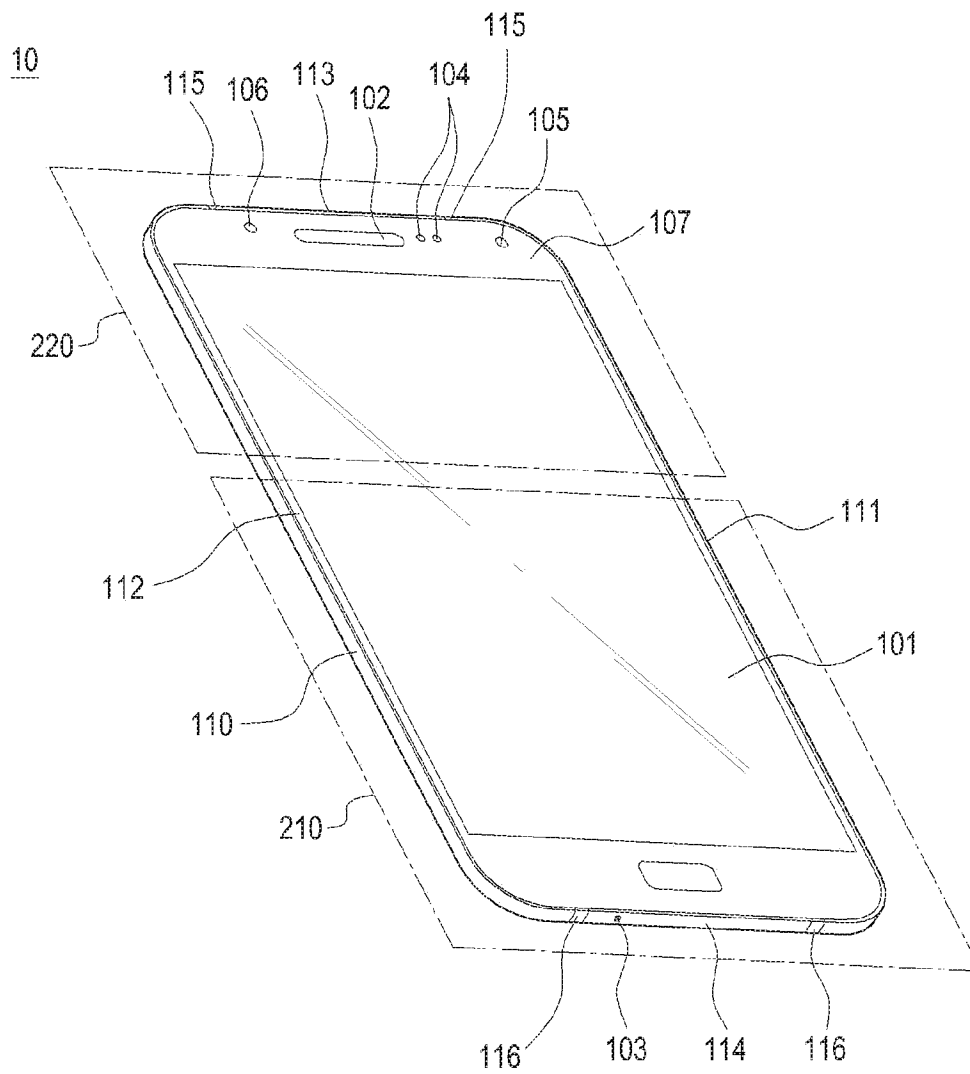


FIG. 8

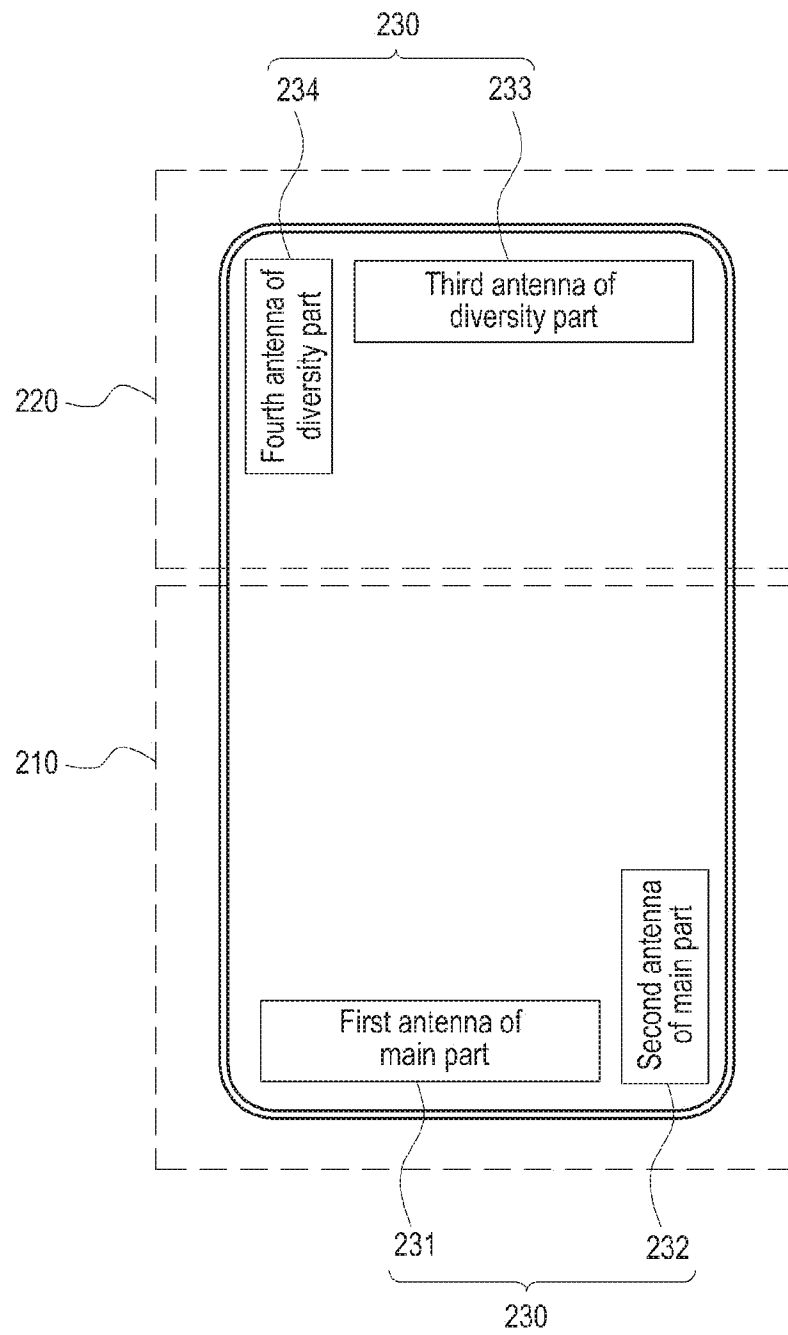


FIG.9

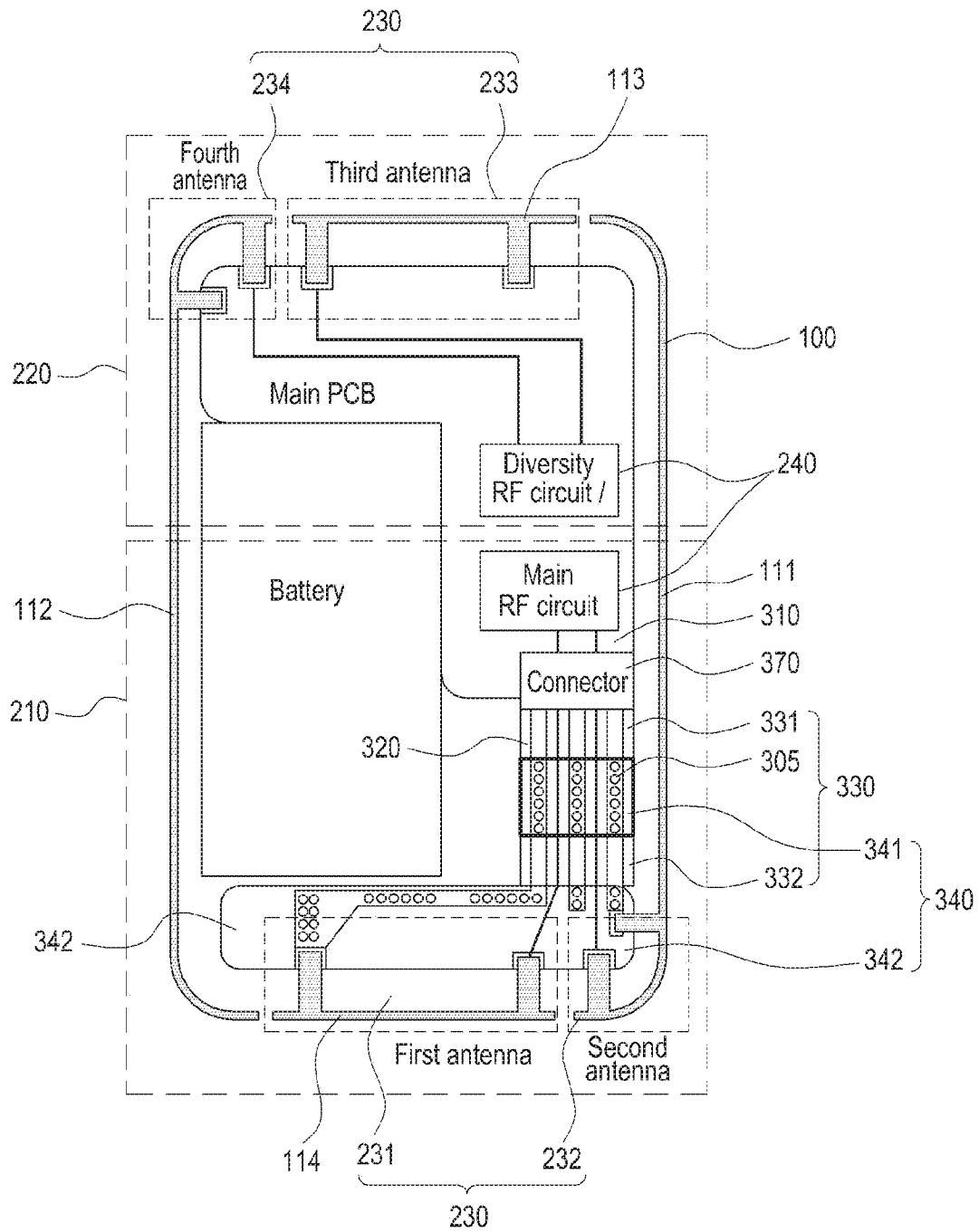


FIG.10

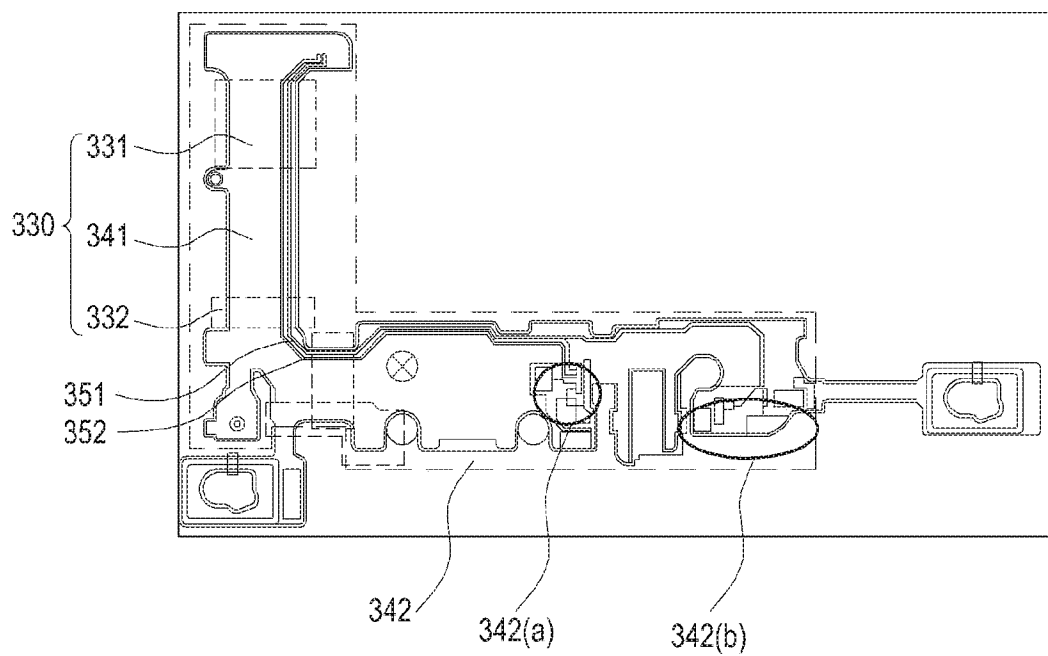


FIG. 11

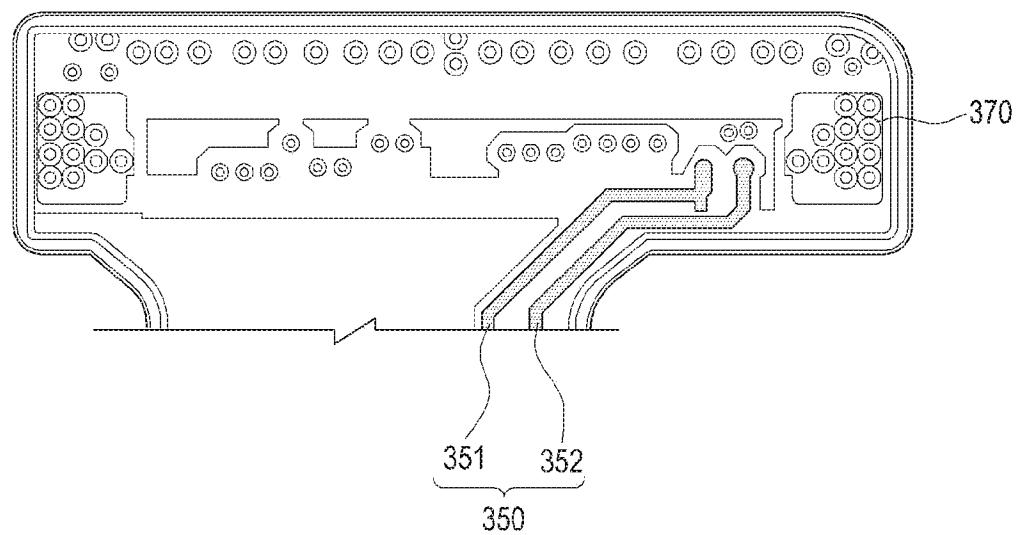


FIG. 12

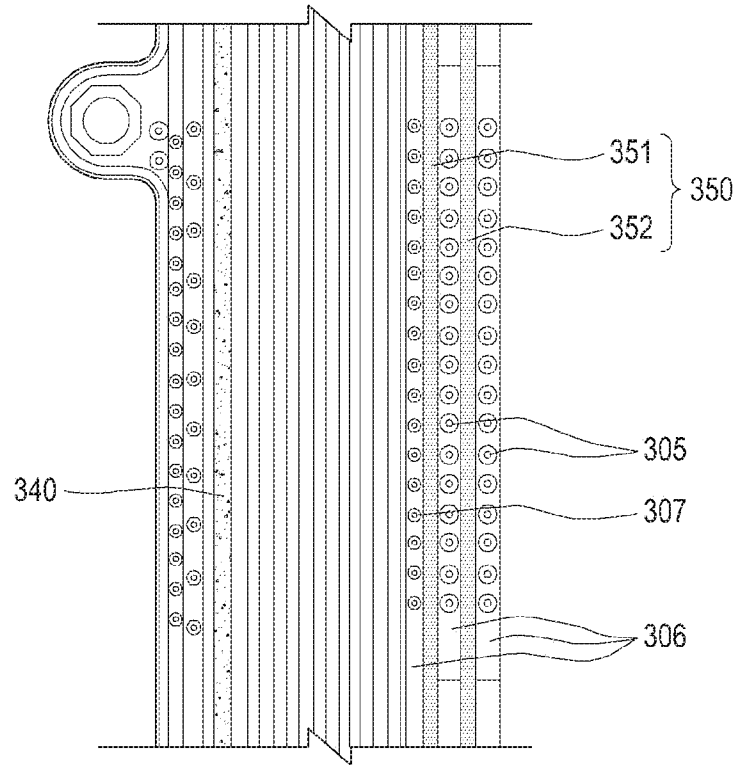


FIG.13

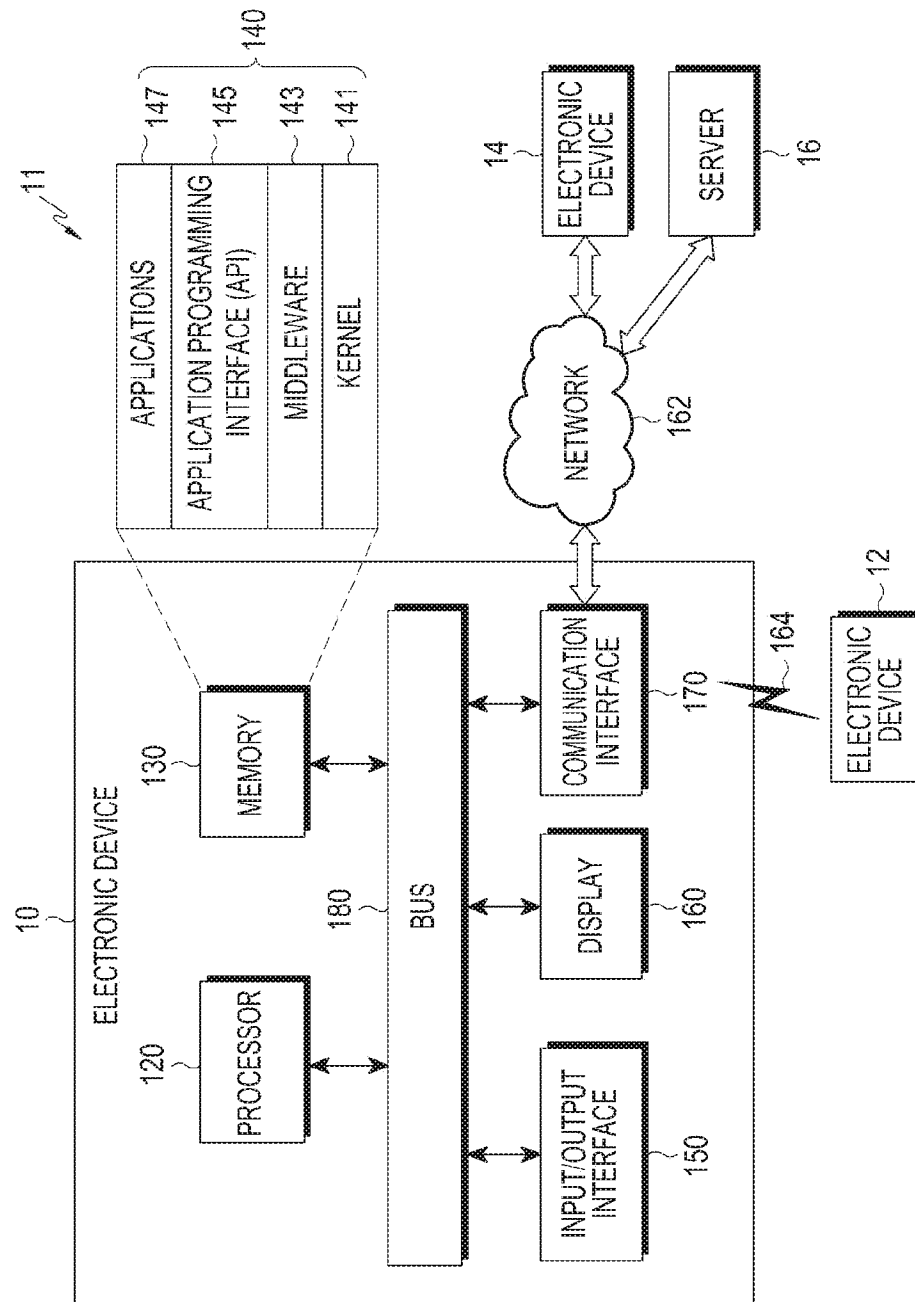


FIG. 14

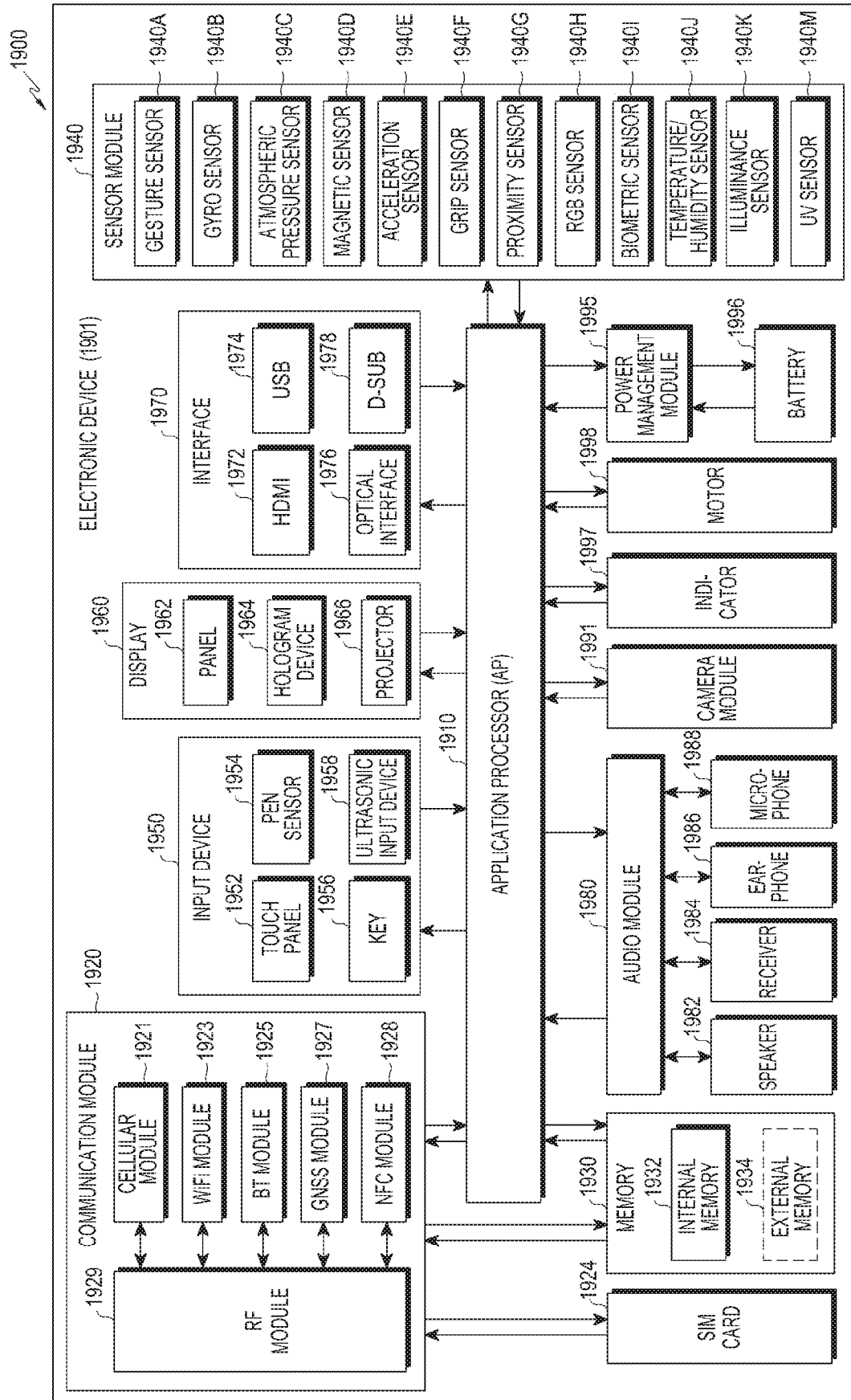


FIG. 15

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DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

Title of Invention	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT
-----------------------	---

As the below named inventor, I hereby declare that:

This declaration ☒ The attached application, or
☐ United States application or PCT international application number _____
 filed on _____

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

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LEGAL NAME OF INVENTOR

Inventor: Han-Min CHO

Date (Optional): May 17, 2017

Signature: Han-Min Cho

(Client Ref.: P23235-US (GM-201507-095-1-US0))

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DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

Title of
Invention

ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT

As the below named inventor, I hereby declare that:

This declaration
is directed to:



The attached application, or



United States application or PCT international application number _____

filed on _____

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

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LEGAL NAME OF INVENTOR

Inventor: Chan-Gi PARK

Date (Optional): May 17 2017

Signature: Chan-Gi PARK

(Client Ref.: P23235-US (GM-201507-095-1-US0))

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**DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN
APPLICATION DATA SHEET (37 CFR 1.76)**

Title of Invention	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT
-----------------------	---

As the below named inventor, I hereby declare that:

This declaration is directed to: ☒ The attached application, or
☐ United States application or PCT International application number _____
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LEGAL NAME OF INVENTOR

Inventor: Yeon-Sang YUN Date (Optional): May, 17, 2017
Signature: Yeon-Sang YUN (Client Ref.: P23235-US (GM-201507-095-1-USO))

Note: An application data sheet (PTO/AIA/14 or equivalent), including naming the entire inventive entity, must accompany this form. Use an additional PTO/SB/AIA01 form for each additional inventor.

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1 minute to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

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**DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN
APPLICATION DATA SHEET (37 CFR 1.76)**

Title of Invention **ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT**

As the below named inventor, I hereby declare that:

This declaration is directed to: ☒ The attached application, or
☐ United States application or PCT international application number _____
filed on _____

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

WARNING:

Petitioner/applicant is cautioned to avoid submitting personal information in documents filed in a patent application that may contribute to identity theft. Personal information such as social security numbers, bank account numbers, or credit card numbers (other than a check or credit card authorization form PTO-2038 submitted for payment purposes) is never required by the USPTO to support a petition or an application. If this type of personal information is included in documents submitted to the USPTO, petitioners/applicants should consider redacting such personal information from the documents before submitting them to the USPTO. Petitioner/applicant is advised that the record of a patent application is available to the public after publication of the application (unless a non-publication request in compliance with 37 CFR 1.213(a) is made in the application) or issuance of a patent. Furthermore, the record from an abandoned application may also be available to the public if the application is referenced in a published application or an issued patent (see 37 CFR 1.14). Checks and credit card authorization forms PTO-2038 submitted for payment purposes are not retained in the application file and therefore are not publicly available.

LEGAL NAME OF INVENTOR

Inventor: **Tae-Wook HAM**

Date (Optional): **MAY 17, 2017**

Signature: **Tae-wook HAM**

(Client Ref.: P23235-US (GM-201507-095-1-US0))

Note: An application data sheet (PTO/AIA/14 or equivalent), including naming the entire inventive entity, must accompany this form. Use an additional PTO/SB/AIA01 form for each additional inventor.

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DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT

Title of
Invention

As the below named inventor, I hereby declare that:

This declaration
is directed to:

☒

The attached application, or

☐

United States application or PCT international application number _____

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LEGAL NAME OF INVENTOR

Inventor: Hei-Seong KWAK

Date (Optional): May 17, 2017

Signature: Hei-Seong Kwak

(Client Ref.: P23235-US (GM-201507-095-1-USO))

Note: An application data sheet (PTO/AIA/14 or equivalent), including naming the entire inventive entity, must accompany this form. Use an additional PTO/SB/AIA01 form for each additional inventor.

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**DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN
APPLICATION DATA SHEET (37 CFR 1.76)**

Title of Invention	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT
-----------------------	---

As the below named inventor, I hereby declare that:

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 filed on _____.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

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LEGAL NAME OF INVENTOR

Inventor: Byoung-II SON

Date (Optional): May 17, 2017

Signature: Byoung-II SON

(Client Ref.: P23235-US (GM-201507-095-1-US0))

Note: An application data sheet (PTO/AIA/14 or equivalent), including naming the entire inventive entity, must accompany this form.
 Use an additional PTO/SB/AIA01 form for each additional inventor.

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DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR 1.76)

Title of Invention	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT
--------------------	---

As the below named inventor, I hereby declare that:

This declaration is directed to: ☒ The attached application, or
☐ United States application or PCT international application number _____
 filed on _____

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

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LEGAL NAME OF INVENTOR

Inventor: Sung-Chul PARK Date (Optional): May 11, 2012
 Signature: [Signature] (Client Ref.: P23235-US (CM-201507-095-1-US0))

Note: An application data sheet (PTO/AIA/14 or equivalent), including naming the entire inventive entity, must accompany this form. Use an additional PTO/SB/AIA01 form for each additional inventor.

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 1 minute to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

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TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS

NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B or equivalent) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5. If the Power of Attorney by Applicant form is not accompanied by this transmittal form or an equivalent, the Power of Attorney will not be recognized in the application.

Application Number	not yet assigned
Filing Date	herewith
First Named Inventor	Han-Min CHO et al.
Title	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT
Art Unit	not yet assigned
Examiner Name	not yet assigned
Attorney Docket Number	678-5477 CON (P23235-US1-CON)

SIGNATURE of Applicant or Patent Practitioner

Signature	/Paul J. Farrell/	Date	July 27, 2020
Name	Paul J. Farrell	Telephone	(516) 228-3565
Registration Number	33,494		

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications.

☐ *Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below:

Application Number	Filing Date

(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)

- ☒ I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above:
- OR
- ☐ I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)

Please recognize or change the correspondence address for the application identified in the attached transmittal letter or the boxes above to:

- ☒ The address associated with the above-mentioned Customer Number
- OR
- ☐ The address associated with Customer Number:
- OR

☐ Firm or Individual Name			
Address			
City	State	Zip	
Country			
Telephone	Email		

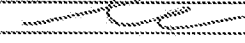
I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

--

- ☐ Inventor or Joint Inventor (title not required below)
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity):

Signature		Date (Optional)	
Name	SUNGHYUK SHIN		
Title	Corporate SVP, Samsung Electronics Co., Ltd.		

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.

☐ Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-8199 and select option 2.

PATENT

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicants: Han-Min CHO et al.

Serial No.: not yet assigned

Docket: 678-5477 CON

Filed: herewith

Dated: July 27, 2020

For: ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION
CIRCUIT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Sir:

Pursuant to Applicants' duty of disclosure, it is respectfully requested that the items of information listed in the attached form PTO-1449 be considered by the Examiner and made of record in the above-identified application.

The listed items were previously cited and submitted in the parent application, namely, U.S. Patent Appln. No. 15/621,687, filed June 13, 2017. In accordance with 37 C.F.R. §1.98(d), copies of these items can be obtained in the file of the above-identified parent application.

The citation of the listed items is not a representation that a search has been made.

The filing of this Information Disclosure Statement is not an admission that the information cited herein is, or is considered to be, material to patentability as defined in 37 C.F.R. § 1.56(b).

- ☒ This Information Disclosure Statement is being filed within three (3) months of the filing date of a national application other than a continued prosecution application.
- ☐ This Information Disclosure Statement is being filed within three (3) months of the date of entry of the national stage in an international application.

- ☐ To the best of Applicant's knowledge, this Information Disclosure Statement is being filed before the mailing of a first Office Action on the merits.
- ☐ This Information Disclosure Statement is being filed before the mailing of a first Office Action after the filing of a Request for Continued Examination.
- ☐ The \$240.00 surcharge will be paid via credit card.
- ☒ Please charge any additional fees which may become due in connection with this filing to Deposit Account No. 50-4053.

The claims of the application as now presented are believed to patentably distinguish over the prior art and to be in condition for allowance. Early and favorable consideration of the case is respectfully requested.

Respectfully submitted,

/Paul J. Farrell/

Paul J. Farrell
Reg. No. 33,494
Attorney for Applicants

THE FARRELL LAW FIRM, P.C.
290 Broadhollow Road, Suite 210E
Melville, NY 11747
(516) 228-3565

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Substitute for form 1449/PTO

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**

(Use as many sheets as necessary)

Sheet 1 of 2

Complete if Known

Application Number	not yet assigned
Filing Date	herewith
First Named Inventor	Han-Min CHO et al.
Art Unit	not yet assigned
Examiner Name	not yet assigned
Attorney Docket Number	678-5477 CON

U. S. PATENT DOCUMENTS

Examiner Initials*	Cite No. ¹	Document Number	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1	US- 2014/0225806	08-2014	LEE et al	
	2	US- 2009/0260859	10-2009	PAI et al.	
	3	US- 8,633,394	01-2014	Lai et al.	
	4	US- 8,143,526	03-2012	Hsu	
	5	US- 8,022,309	09-2011	Pai et al.	
	6	US- 7,781,680	08-2010	Pai et al.	
	7	US- 8,263,873	09-2012	Choi et al.	
	8	US- 2009/0294155	12-2009	Suzuki et al.	
	9	US- 2013/0063914	03-2013	Lin et al.	
	10	US- 2011/0094784	04-2011	Muro et al.	
	11	US- 2015/0223323	08-2015	O'Flynn	
	12	US- 2013/0257659	10-2013	Darnell et al.	
	13	US- 2012/0229346	09-2012	Chen et al.	
	14	US- 2006/0270472	11-2006	Chen et al.	
	15	US- 4,845,311	07-1989	Schreiber et al.	
	16	US- 2015/0234426	08-2015	Bailey	
	17	US- 2014/0147128	05-2014	Han	
		US-			
		US-			

FOREIGN PATENT DOCUMENTS

Examiner Initials*	Cite No. ¹	Foreign Patent Document	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages Or Relevant Figures Appear	T ⁶
		Country Code ³ -Number ⁴ -Kind Code ⁵ (if known)				
	18	KR 101261983	05-08-2013			
	19	KR 20150062059	06-05-2015			
	20	EP 2 180 769	04-28-2010			
	21	KR 20140136634	12-01-2014			
	22	KR 101305518	09-06-2013			
	23	KR 1020130081450	07-17-2013			

Examiner
SignatureDate
Considered

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patent Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST.3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

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Substitute for form 1449/PTO

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT***(Use as many sheets as necessary)***Complete if Known**

Application Number	not yet assigned
Filing Date	herewith
First Named Inventor	Han-Min CHO et al.
Art Unit	not yet assigned
Examiner Name	not yet assigned
Attorney Docket Number	678-5477 CON

Sheet 2 of 2

NON PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	24	International Search Report dated September 19, 2017 issued in counterpart application No. PCT/KR2017/006070, 7 pages.	
	25	Samsung Galaxy et al: "Samsung Galaxy S7 Teardown Geschrieben von: Jeff Suovanen", IFIXIT Guide, XP055578568, March 8, 2016, 19 pages.	
	26	European Search Report dated April 17, 2019 issued in counterpart application No. 17813538.0-1203, 11 pages.	
	27	European Search Report dated February 5, 2020 issued in counterpart application No. 17813538.0-1203, 12 pages.	

Examiner
SignatureDate
Considered

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ Applicant's unique citation designation number (optional). ² Applicant is to place a check mark here if English language Translation is attached.This collection of information is required by 37 CFR 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 2 hours to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO:****Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.***If you need assistance in completing the form, call 1-800-PTO-9199 (1-800-786-9199) and select option 2.*

Electronic Patent Application Fee Transmittal

Application Number:				
Filing Date:				
Title of Invention:	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT			
First Named Inventor/Applicant Name:	Han-Min CHO			
Filer:	Paul J. Farrell/Deanna Marman			
Attorney Docket Number:	678-5477 CON			
Filed as Large Entity				
Filing Fees for Track I Prioritized Examination - Nonprovisional Application under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
UTILITY APPLICATION FILING	1011	1	300	300
UTILITY SEARCH FEE	1111	1	660	660
UTILITY EXAMINATION FEE	1311	1	760	760
REQUEST FOR PRIORITIZED EXAMINATION	1817	1	4000	4000
Pages:				
Claims:				
Miscellaneous-Filing:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
PUBL. FEE- EARLY, VOLUNTARY, OR NORMAL	1504	1	0	0
PROCESSING FEE, EXCEPT PROV. APPLS.	1830	1	140	140
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				5860

Electronic Acknowledgement Receipt

EFS ID:	40113856
Application Number:	16939708
International Application Number:	
Confirmation Number:	8134
Title of Invention:	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT
First Named Inventor/Applicant Name:	Han-Min CHO
Customer Number:	66547
Filer:	Paul J. Farrell/Deanna Marman
Filer Authorized By:	Paul J. Farrell
Attorney Docket Number:	678-5477 CON
Receipt Date:	27-JUL-2020
Filing Date:	
Time Stamp:	15:57:45
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$5860
RAM confirmation Number	E20207QF58026895
Deposit Account	504053
Authorized User	Deanna Marman

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

37 CFR 1.16 (National application filing, search, and examination fees)

37 CFR 1.17 (Patent application and reexamination processing fees)

37 CFR 1.19 (Document supply fees)
37 CFR 1.20 (Post Issuance fees)
37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	TrackOne Request	Track_One_Request.pdf	125617	no	1
			be4508cc61f1f9eeca23ad9cc532bc4a66180725		
Warnings:					
Information:					
2	Transmittal of New Application	Fee_Transmittal.pdf	335155	no	2
			0e1acb28a9f77c1e82b44c416d230145db3d2127		
Warnings:					
Information:					
3	Application Data Sheet	ADS.pdf	122153	no	10
			42d3d2c8a3cfe349088a881134f12481e1d5aabb		
Warnings:					
Information:					
This is not an USPTO supplied ADS fillable form					
4	Specification	Specification.pdf	265679	no	46
			6a519a8f76dc8cd524f286a587d60798a39cbe28		
Warnings:					
Information:					
5	Claims	Claims.pdf	95769	no	4
			5aa74d58bc35c2437632a817ec830faa982f2219		
Warnings:					
Information:					
6	Abstract	Abstract.pdf	84748	no	1
			6ccb9cf2fd09695ef43c7759f53e1fca886adcce		
Warnings:					

Information:					
7	Drawings-only black and white line drawings	Drawings.pdf	6391974	no	14
			2cd7bb5ea2b925969272eb801acfeceb19f843f9		
Warnings:					
Information:					
8	Oath or Declaration filed	Declaration.pdf	2294318	no	7
			02b63f18519c5248227ddda8bb30d8fc9ce3beb7		
Warnings:					
Information:					
9	Power of Attorney	POA.pdf	1248872	no	2
			6d5a99df752baae35422bf97422844bf86411f3f		
Warnings:					
Information:					
10	Information Disclosure Statement (IDS) Form (SB08)	IDS.pdf	306576	no	4
			121299306f867d25a415d8c68dc059b0b1f51d84		
Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
11	Fee Worksheet (SB06)	fee-info.pdf	39961	no	2
			615844bc141193f4236a77e41a4e0a9b8bc77793		
Warnings:					
Information:					
Total Files Size (in bytes):			11310822		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

**CERTIFICATION AND REQUEST FOR PRIORITIZED EXAMINATION
UNDER 37 CFR 1.102(e) (Page 1 of 1)**

First Named Inventor:	Han-Min CHO et al.	Nonprovisional Application Number (if known):	not yet assigned
Title of Invention:	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT		

APPLICANT HEREBY CERTIFIES THE FOLLOWING AND REQUESTS PRIORITIZED EXAMINATION FOR THE ABOVE-IDENTIFIED APPLICATION.

1. The processing fee set forth in 37 CFR 1.17(i)(1) and the prioritized examination fee set forth in 37 CFR 1.17(c) have been filed with the request. The publication fee requirement is met because that fee, set forth in 37 CFR 1.18(d), is currently \$0. The basic filing fee, search fee, and examination fee are filed with the request or have been already been paid. I understand that any required excess claims fees or application size fee must be paid for the application.
2. I understand that the application may not contain, or be amended to contain, more than four independent claims, more than thirty total claims, or any multiple dependent claims, and that any request for an extension of time will cause an outstanding Track I request to be dismissed.
3. The applicable box is checked below:

I. ☒ Original Application (Track One) - Prioritized Examination under § 1.102(e)(1)

- i. (a) The application is an original nonprovisional utility application filed under 35 U.S.C. 111(a). This certification and request is being filed with the utility application via EFS-Web.
---OR---
(b) The application is an original nonprovisional plant application filed under 35 U.S.C. 111(a). This certification and request is being filed with the plant application in paper.
- ii. An executed inventor's oath or declaration under 37 CFR 1.63 or 37 CFR 1.64 for each inventor, or the application data sheet meeting the conditions specified in 37 CFR 1.53(f)(3)(i) is filed with the application.

II. ☐ Request for Continued Examination - Prioritized Examination under § 1.102(e)(2)

- i. A request for continued examination has been filed with, or prior to, this form.
- ii. If the application is a utility application, this certification and request is being filed via EFS-Web.
- iii. The application is an original nonprovisional utility application filed under 35 U.S.C. 111(a), or is a national stage entry under 35 U.S.C. 371.
- iv. This certification and request is being filed prior to the mailing of a first Office action responsive to the request for continued examination.
- v. No prior request for continued examination has been granted prioritized examination status under 37 CFR 1.102(e)(2).

Signature /Paul J. Farrell/	Date July 27, 2020
Name (Print/Typed) Paul J. Farrell	Practitioner Registration Number 33,494

Note: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. Submit multiple forms if more than one signature is required.*

☐ *Total of _____ forms are submitted.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
16/939,708	07/27/2020	2833	1720	678-5477 CON	20	2

CONFIRMATION NO. 8134

FILING RECEIPT



0000000118984248

66547
THE FARRELL LAW FIRM, P.C.
Paul J Farrell
290 Broadhollow Road
Suite 210E
Melville, NY 11747

Date Mailed: 08/06/2020

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Han-Min CHO, Gyeonggi-do, KOREA, REPUBLIC OF;
Chan-Gi PARK, Gyeonggi-do, KOREA, REPUBLIC OF;
Yeon-Sang YUN, Gyeonggi-do, KOREA, REPUBLIC OF;
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Byoung-Il SON, Gyeonggi-do, KOREA, REPUBLIC OF;
Sung-Chul PARK, Seoul, KOREA, REPUBLIC OF;

Applicant(s)

Samsung Electronics Co., Ltd., Gyeonggi-do, KOREA, REPUBLIC OF;

Power of Attorney: The patent practitioners associated with Customer Number 66547

Domestic Priority data as claimed by applicant

This application is a CON of 15/621,687 06/13/2017 PAT 10727568

Foreign Applications (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)

REPUBLIC OF KOREA 10-2016-0073355 06/13/2016 No Access Code Provided

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 08/05/2020

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 16/939,708**

Projected Publication Date: 11/12/2020

Non-Publication Request: No

Early Publication Request: No
Title

ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT

Preliminary Class

439

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific

countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

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No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop

technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
16/939,708

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	300
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	660
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	760
TOTAL CLAIMS (37 CFR 1.16(j))	20 minus 20 =	*			OR	x 100 =	0.00
INDEPENDENT CLAIMS (37 CFR 1.16(h))	2 minus 3 =	*				x 460 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1720

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.

Electronic Patent Application Fee Transmittal

Application Number:	16939708			
Filing Date:	27-Jul-2020			
Title of Invention:	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT			
First Named Inventor/Applicant Name:	Han-Min CHO			
Filer:	Paul J. Farrell/Deanna Marman			
Attorney Docket Number:	678-5477 CON			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
CLAIMS IN EXCESS OF 20	1202	5	100	500
INDEPENDENT CLAIMS IN EXCESS OF 3	1201	1	460	460
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				960

Electronic Acknowledgement Receipt

EFS ID:	40202697
Application Number:	16939708
International Application Number:	
Confirmation Number:	8134
Title of Invention:	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT
First Named Inventor/Applicant Name:	Han-Min CHO
Customer Number:	66547
Filer:	Paul J. Farrell/Deanna Marman
Filer Authorized By:	Paul J. Farrell
Attorney Docket Number:	678-5477 CON
Receipt Date:	05-AUG-2020
Filing Date:	27-JUL-2020
Time Stamp:	15:30:56
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$960
RAM confirmation Number	E202085F31316640
Deposit Account	504053
Authorized User	Deanna Marman

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

37 CFR 1.16 (National application filing, search, and examination fees)

37 CFR 1.17 (Patent application and reexamination processing fees)

37 CFR 1.19 (Document supply fees)
37 CFR 1.20 (Post Issuance fees)
37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Preliminary Amendment	Preliminary_Amendment.pdf	852871	no	12
			a7f426bb0641cdb9e53589bfb7f8346ccf9996a5		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	32085	no	2
			edc6a785a0b36d4e4c93ff71a26752a3c7cb8a8ac		

Warnings:

Information:

Total Files Size (in bytes):			884956
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New Applications Under 35 U.S.C. 111

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National Stage of an International Application under 35 U.S.C. 371

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New International Application Filed with the USPTO as a Receiving Office

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Inventor(s): Han-Min CHO et al.

Examiner: not yet assigned

Serial No.: 16/939,708

Art Unit: 2833

Filed: July 27, 2020

Date: August 5, 2020

For: **ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT**

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT TRANSMITTAL FORM

Sir:

Transmitted herewith is an amendment in the above-identified application.

- ☐ Small entity status of this application under 37 C.F.R. §§1.9 and 1.27 has been established by a verified statement previously submitted.
- ☐ A verified statement to establish small entity under 37 C.F.R. §§1.9 and 1.27 is enclosed.
- ☐ No additional fee is required.

For	Claims Remaining After Amendment	Highest No. Previously Paid For	Present Extra	Rate (Small Entity)	Addit. Fee	Rate	Addit. Fee
TOTAL CLAIMS*	25	20	5	x50 =	\$0	x100 =	\$500
INDEPENDENT CLAIMS	4	3	1	x230 =	\$0	X460 =	\$460
☐ First Presentation of Multiple Dep. Claim				410		820	\$0

* If the entry in Col. 1 is less than entry in Col. 2, write "0" in Col. 3.

** If the "Highest No. Previously Paid for" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest No. Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The Highest No. Previously Paid For" (Total or indep.) is the highest number found in the appropriate box in Col. 1 of a prior amendment or the number of claims originally filed.

- ☐ Please charge Deposit Account No. 50-4053 in the amount of \$____. Two (2) copies of this sheet are enclosed.
- ☐ A check in the amount of \$____ is enclosed.
- ☒ Please charge any deficiency as well as any other fee(s) which may become due under 37 C.F.R. §§1.16 and/or 1.17 at any time during the pendency of this application, or credit any overpayment of such fee(s) to Deposit Account No. 50-4053. Also, in the event any extensions of time for responding are required for the pending application(s), please treat this paper as a petition to extend the time as required and charge Deposit Account No. 50-4053 therefor.

Respectfully submitted,

/Douglas M. Owens III/

Douglas M. Owens III
Reg No. 51,314
Attorney for Applicant(s)

THE FARRELL LAW FIRM, PC
290 Broadhollow Road, Suite 210 E
Melville, New York 11747
(516) 228-3565

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

INVENTORS(s): Han-Min CHO et al. **EXAMINER:** not yet assigned

SERIAL NO.: 16/939,708 **ART UNIT:** not yet assigned

FILED: July 27, 2020 **DATE:** August 5, 2020

**FOR: ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY
TRANSMISSION CIRCUIT**

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

PRELIMINARY AMENDMENT

Before examination of the above-identified application, please enter the following amendments and consider the following Remarks.

AMENDMENTS TO THE CLAIMS

Please amend the claims as follows:

1-20. (Cancelled)

21. (New) An electronic device comprising:

a first component;

a second component; and

a signal path interface coupled between the first component and the second component, the signal path interface including a printed circuit board (PCB) having a rigid PCB portion and a flexible PCB portion,

wherein a first signal line and a second signal line extend through the rigid PCB portion and the flexible PCB portion for transmitting signals from the first component to the second components, and a plurality of ground lines extend through the rigid PCB portion and the flexible PCB portion, and

wherein each of the plurality of ground lines extending through the rigid PCB portion is connected to one or more conductive layers through conductive vias.

22. (New) The electronic device of claim 21,

wherein the first component comprises a communication module, and the second component comprises an antenna radiator, and

wherein the first signal line and the second signal line extending through the signal path interface are configured to transmit high-frequency signals from the communication module to the antenna radiator.

23. (New) The electronic device of claim 22,

wherein the plurality of ground lines include a first ground line, a second ground line at least partially spaced apart from the first ground line, and a third ground line located between the first signal line and the second signal line,

wherein first signal line is located between the first ground line and the second ground line, and

wherein the second signal line is located between the first signal line and the second ground line.

24. (New) The electronic device of claim 22,

wherein the PCB includes an insulating layer on which the first ground line, the second ground line, the third ground line, the first signal line, and the second signal line are located, and a ground plate located below the insulating layer.

25. (New) The electronic device of claim 24,

wherein the ground plate is electrically connected with the first ground line through a first conduction via formed in a first portion of the insulating layer, the second ground line through a second conduction via formed in a second portion of the insulating layer, and the third ground line through a third conduction via formed in a third portion of the insulating layer.

26. (New) A portable communication device comprising:

a housing including a first conduction portion forming a first portion of a side surface of the portable communication device and a second conduction portion forming a second portion of the side surface, the first conduction portion operating as a first antenna and the second conduction portion operating as a second antenna;

a wireless communication circuitry; and

a first printed circuit board (PCB) including:

a first PCB portion;

a second PCB portion in contact with the first PCB portion and being more flexible than the first PCB portion;

a third PCB portion in contact with the second PCB portion and being less flexible than the second PCB portion;

a fourth PCB portion in contact with the third PCB portion and being more flexible than the third PCB portion; and

a plurality of lines located from the first PCB portion to the fourth PCB portion through the second PCB portion and the third PCB portion, the plurality of lines including a first ground line, a second ground line at least partially spaced apart from the first ground line, a first signal line located between the first ground line and the second ground line, and a second signal line located between the first signal line and the second ground line, the first signal line electrically connected with the first antenna and the wireless communication circuitry, and the second signal line electrically connected with the second antenna and the wireless communication circuitry.

27. (New) The portable communication device of claim 26,
wherein the plurality of lines includes a third ground line located between the first signal line and the second signal line, and

wherein the first PCB further includes an insulating layer on which the first ground line, the second ground line, the third ground line, the first signal line, and the second signal line are located, and a ground plate located below the insulating layer.

28. (New) The portable communication device of claim 27,
wherein the ground plate is electrically connected with the first ground line through a first conduction via formed in a first portion of the insulating layer, the second ground line through a second conduction via formed in a second portion of the insulating layer, and the third ground line through a third conduction via formed in a third portion of the insulating layer.

29. (New) The portable communication device of claim 27,
wherein the first PCB further includes a solder resist ink forming at least a part of an exterior surface of the first PCB portion or the third PCB portion.

30. (New) The portable communication device of claim 26,
wherein the first PCB further includes a fifth PCB portion in contact with the fourth PCB portion and being less flexible than the fourth PCB portion,
wherein the first PCB portion further includes a first connector for a connection with the antenna,

wherein the fifth PCB portion includes a second connector for a connection with the wireless communication circuitry, and

wherein a first ending portion of the first signal line and a second ending portion of the second signal line are in contact with the second connector.

31. (New) The portable communication device of claim 26, further comprising:
a second PCB on which the wireless communication circuitry is located,
wherein the first ground line and the second ground line are electrically connected with a ground member located in the second PCB, or
wherein the first ground line and the second ground line are electrically connected with a ground member located in the first PCB portion.

32. (New) A portable communication device comprising:
a first electronic component;
a second electronic component; and
a printed circuit board (PCB) electrically connected with the first electronic component and the second electronic component, the PCB including:
a first rigid PCB portion;
a first flexible PCB portion in contact with the first rigid PCB portion;
a second rigid PCB portion in contact with the first flexible PCB portion; and
a second flexible PCB portion in contact with the second rigid PCB portion,
wherein each of the first rigid PCB portion and the second rigid PCB portion includes:
a first insulating layer including a first material and forming a first exterior surface of each of the first rigid PCB portion and the second rigid PCB portion;
a first conduction layer located on or above the first insulating layer;
a second insulating layer located on or above the first conduction layer and having a first thickness;
a second conduction layer located on or above the second insulating layer; and
a third insulating layer including the first material and located on or above the second conduction layer, the third insulating layer forming a second exterior surface of each of the first rigid PCB portion and the second rigid PCB portion, and

wherein each of the first flexible PCB portion and the second flexible PCB portion includes:

a first insulating layer including a second material different from the first material and forming a first exterior surface of each of the first flexible PCB portion and the second flexible PCB portion;

a first conduction layer located on or above the first insulating layer and being electrically connected with the first conduction layer of first rigid PCB portion and the first conduction layer of the second rigid PCB portion;

a second insulating layer located on or above the first conduction layer and having a second thickness thinner than the first thickness;

a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the first rigid PCB portion and the second conduction layer of the second rigid PCB portion; and

a third insulating layer including the second material and located on or above the second conduction layer, the third insulating layer forming a second exterior surface of each of the first flexible PCB portion and the second flexible PCB portion.

33. (New) The portable communication device of claim 32,
wherein the second insulating layer of the first flexible PCB portion or the second insulating layer of the second flexible PCB portion includes the second material.

34. (New) The portable communication device of claim 32,
wherein each of the first flexible PCB portion and the second flexible PCB portion includes a fourth insulating layer located between the second conduction layer and the third insulating layer, and a third conduction layer located between the third insulating layer and the fourth insulating layer.

35. (New) The portable communication device of claim 34,
wherein the fourth insulating layer includes the second material.

36. (New) The portable communication device of claim 34,

wherein the first conduction layer of each of the first flexible PCB portion and the second flexible PCB portion includes a first ground portion,

wherein the second conduction layer of each of the first flexible PCB portion and the second flexible PCB portion includes a second ground portion electrically connected with the first ground portion through a first conduction via, and

wherein the third conduction layer includes a third ground portion electrically connected with the second ground portion through a second conduction via.

37. (New) The portable communication device of claim 32,
wherein the first material is a solder resist ink and the second material is a polyimide, and
wherein the second insulating layer of the first rigid PCB portion or the second insulating layer of the second rigid PCB portion includes a third material.

38. (New) The portable communication device of claim 37,
wherein the third material includes a resin or a glass fabric.

39. (New) A portable communication device comprising:
a first electronic component;
a second electronic component; and
a printed circuit board (PCB) electrically connected with the first electronic component and the second electronic component, the PCB including:
a first flexible PCB portion including:
a first insulating layer forming a first exterior surface of the first flexible PCB portion and including a first material;
a first conduction layer located on or above the first insulating layer;
a second insulating layer located on or above the first conduction layer having a first thickness;
a second conduction layer located on or above the second insulating layer; and
a third insulating layer located on or above the second conduction layer and forming a second exterior surface of the first flexible PCB portion, the third insulating layer including the first material;

a rigid PCB portion in contact with the first flexible PCB portion, the rigid PCB portion including:

- a first insulating layer forming a first exterior surface of the rigid PCB portion and including a second material different from the first material;

- a first conduction layer located on or above the first insulating layer and being electrically connected with the first conduction layer of the first flexible PCB portion;

- a second insulating layer located on or above the first conduction layer and having a second thickness thicker than the first thickness;

- a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the first flexible PCB portion; and

- a third insulating layer located on or above the second conduction layer and forming a second exterior surface of the rigid PCB portion, the third insulating layer including the second material;

a second flexible PCB portion in contact with the rigid PCB portion, the second flexible PCB portion including:

- a first insulating layer forming a first exterior surface of the second flexible PCB portion and including the first material;

- a first conduction layer located on or above the first insulating layer and being electrically connected with the first conduction layer of the rigid PCB portion;

- a second insulating layer located on or above the first conduction layer and having the first thickness;

- a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the rigid PCB portion;

- a third insulating layer located on or above the second conduction layer and forming a second exterior surface of the second flexible PCB portion, the third insulating layer including the first material; and

- a contact pad located on the third insulating layer.

40. (New) The portable communication device of claim 39,
wherein the second insulating layer of the first flexible PCB portion or the second insulating layer of the second flexible PCB portion includes the second material.

41. (New) The portable communication device of claim 39,
wherein each of the first flexible PCB portion and the second flexible PCB portion includes a fourth insulating layer located between the second conduction layer and the third insulating layer, and a third conduction layer located between the third insulating layer and the fourth insulating layer.

42. (New) The portable communication device of claim 41,
wherein the fourth insulating layer includes the second material.

43. (New) The portable communication device of claim 41,
wherein the first conduction layer of each of the first flexible PCB portion and the second flexible PCB portion includes a first ground portion,
wherein the second conduction layer of each of the first flexible PCB portion and the second flexible PCB portion includes a second ground portion electrically connected with the first ground portion through a first conduction via, and
wherein the third conduction layer includes a third ground portion electrically connected with the second ground portion through a second conduction via.

44. (New) The portable communication device of claim 39,
wherein the first material is a solder resist ink and the second material is a polyimide, and
wherein the second insulating layer of the rigid PCB portion includes a third material.

45. (New) The portable communication device of claim 44, wherein the third material includes a resin or a glass fabric.

REMARKS

Prior to an examination on the merits, please enter the amendments shown above.

Specifically, please cancel Claims 1-20 without prejudice, and add new Claims 21-45 as shown. No new matter has been added.

Accordingly, Claims 21-45 are pending in the application, with Claims 21, 26, 32 and 39 being the independent claims.

Should the Examiner believe that a telephone conference or personal interview would facilitate resolution of any remaining matters, it is requested that the Examiner contact Applicants' attorney at the number below.

Respectfully Submitted,

/Douglas M. Owens III/

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Reg. No. 51,314
Attorneys for Applicants

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/939,708	07/27/2020	Han-Min CHO	678-5477 CON	8134

66547 7590 08/07/2020
THE FARRELL LAW FIRM, P.C.
Paul J Farrell
290 Broadhollow Road
Suite 210E
Melville, NY 11747

EXAMINER

ART UNIT	PAPER NUMBER
2848	

NOTIFICATION DATE	DELIVERY MODE
08/07/2020	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

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Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

pto@farrelliplaw.com

<i>Decision Granting Request for Prioritized Examination (Track I)</i>	Application No. 16/939,708	Applicant(s) CHO et al.	
	Examiner BRIAN W BROWN	Art Unit OPET	AIA (FITF) Status Yes
1. THE REQUEST FILED <u>27 July 2020</u> IS GRANTED . The above-identified application has met the requirements for prioritized examination A. ☒ for an original nonprovisional application (Track I). B. ☐ for an application undergoing continued examination (RCE). 2. The above-identified application will undergo prioritized examination. The application will be accorded special status throughout its entire course of prosecution until one of the following occurs: A. filing a <u>petition for extension of time</u> to extend the time period for filing a reply; B. filing an <u>amendment to amend the application to contain more than four independent claims, more than thirty total claims</u>, or a multiple dependent claim; C. filing a <u>request for continued examination</u> ; D. filing a notice of appeal; E. filing a request for suspension of action; F. mailing of a notice of allowance; G. mailing of a final Office action; H. completion of examination as defined in 37 CFR 41.102; or I. abandonment of the application. Telephone inquiries with regard to this decision should be directed to BRIAN BROWN at (571)272-5338. In his/her absence, calls may be directed to Petition Help Desk at (571) 272-3282.			
/BRIAN W BROWN/ Petitions Examiner, OPET			

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Application	Document	Mailroom Date	Attorney Docket No.
16939708	TRACK1.GRANT	08/07/2020	678-5477 CON

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/939,708	07/27/2020	Han-Min CHO	678-5477 CON	8134

66547 7590 08/31/2020
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Paul J Farrell
290 Broadhollow Road
Suite 210E
Melville, NY 11747

EXAMINER

AYCHILLHUM, ANDARGIE M

ART UNIT	PAPER NUMBER
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2848

NOTIFICATION DATE	DELIVERY MODE
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08/31/2020

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DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Information Disclosure Statement

2. The information disclosure statements (IDSs) submitted on 07/27/2020 is being considered by the examiner.

Claim Rejections - 35 USC § 102

3. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

4. Claim 21-25 are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Han et al. (US 2014/0147128 A1).

Pertaining to claim 21, Han et al. discloses An electronic device (see fig. 1) comprising: a first component (30, see fig. 1); a second component (40, see fig. 1); and a signal path interface (see paragraph [0037]) coupled between the first component (30) and the second component (40, see fig. 1), the signal path interface (see paragraph [0037]) including a printed circuit board (PCB) having a rigid PCB portion (20, see fig. 1) and a flexible PCB portion (10, see fig. 1), wherein a first signal line (12) and a second

signal line (22, see fig. 1) extend through the rigid PCB (20) portion and the flexible PCB portion (10) for transmitting signals from the first component to the second components (40), and a plurality of ground lines (14, 24, see fig. 1) extend through the rigid PCB portion (20) and the flexible PCB portion (10), and wherein each of the plurality of ground lines (14, 24) extending through the rigid PCB portion (20) is connected to one or more conductive layers through conductive vias (70, 80, see fig. 1).

Pertaining to claim 22, Han et al. discloses, wherein the first component (20, see fig. 1) comprises a communication module (see paragraph [0034]), and the second component (40, see fig. 1) comprises an antenna radiator (see paragraph [0035]), and wherein the first signal line (12) and the second signal line (22) extending through the signal path interface (see paragraph [0037]) are configured to transmit high-frequency signals (see paragraph [0036]) from the communication module (see paragraph [0034]) to the antenna radiator (see paragraph [0035]).

Pertaining to claim 23, Han et al. discloses, wherein the plurality of ground lines (14, 24) include a first ground line (14), a second ground line (24) at least partially spaced apart from the first ground line (14), and a third ground line (16, see fig. 6) located between the first signal line (12) and the second signal line (22), wherein first signal line (12) is located between the first ground line (14, see fig. 2) and the second ground line (24, see fig. 2), and wherein the second signal line (22, see fig. 2) is located between the first signal line (12, see fig. 2) and the second ground line (24, see fig. 2).

Pertaining to claim 24, Han et al. discloses, wherein the PCB (10, 20) includes an insulating layer (see paragraph [0041]) on which the first ground line (14, see fig. 6), the second ground line (24, see fig. 6), the third ground line (16, see fig. 6), the first signal line (12), and the second signal line (22) are located, and a ground plate (18, see fig. 6) located below the insulating layer (11, see fig. 6).

Pertaining to claim 25, Han et al. discloses, wherein the ground plate (18) is electrically connected with the first ground line (14, see fig. 6) through a first conduction via (70) formed in a first portion of the insulating layer (11), the second ground line (24) through a second conduction via (80) formed in a second portion of the insulating layer (21, see fig. 2), and the third ground line (16) through a third conduction via formed in a third portion of the insulating layer (19, see fig. 6).

Allowable Subject Matter

5. Claims 26-31, 32-38 and 39-45 are allowed.

6. The following is a statement of reasons for the indication of allowable subject matter: a third PCB portion in contact with the second PCB portion and being less flexible than the second PCB portion; a fourth PCB portion in contact with the third PCB portion and being more flexible than the third PCB portion; and a plurality of lines located from the first PCB portion to the fourth PCB portion through the second PCB portion and the third PCB portion, the plurality of lines including a first ground line, a second ground line at least partially spaced apart from the first ground line, a first signal line located between the first ground line and the second ground line, and a second

signal line located between the first signal line and the second ground line, the first signal line electrically connected with the first antenna and the wireless communication circuitry, and the second signal line electrically connected with the second antenna and the wireless communication circuitry.

Conclusion

7. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ANDARGIE M AYCHILLHUM whose telephone number is (571)270-1607. The examiner can normally be reached on M-F 9-5.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jenny L Wagner can be reached on (571) 272-5359. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <https://ppair-my.uspto.gov/pair/PrivatePair>. Should you have questions on access to the Private

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/ANDARGIE M AYCHILLHUM/
Primary Examiner, Art Unit 2848

<i>Notice of References Cited</i>	Application/Control No. 16/939,708		Applicant(s)/Patent Under Reexamination CHO et al.	
	Examiner ANDARGIE M AYCHILLHUM		Art Unit 2848	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20140147128-A1	05-2014	Han; Young-Tak	H05K1/142	398/135
	B					
	C					
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
	L					
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FOREIGN PATENT DOCUMENTS

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	N					
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NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
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*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Search Notes</i> 	Application/Control No. 16/939,708	Applicant(s)/Patent Under Reexamination CHO et al.
	Examiner ANDARGIE M AYCHILLHUM	Art Unit 2848

CPC - Searched*		
Symbol	Date	Examiner

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
361	790		

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
East (US-PGPUB, USPAT, SUOCR, FPRS, EPO, JPO, DERWENT AND IBM-TDB) See Search Printout	08/26/2020	A.A.

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
361	785	08/26/2020	A.A.

/ANDARGIE M AYCHILLHUM/ Primary Examiner, Art Unit 2848	08/26/2020
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PATENT**IN THE UNITED STATES PATENT AND TRADEMARK OFFICE**

Applicants: Han-Min CHO et al.

Serial No.: not yet assigned

Docket: 678-5477 CON

Filed: herewith

Dated: July 27, 2020

For: ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION
CIRCUIT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

INFORMATION DISCLOSURE STATEMENT

Sir:

Pursuant to Applicants' duty of disclosure, it is respectfully requested that the items of information listed in the attached form PTO-1449 be considered by the Examiner and made of record in the above-identified application.

The listed items were previously cited and submitted in the parent application, namely, U.S. Patent Appln. No. 15/621,687, filed June 13, 2017. In accordance with 37 C.F.R. §1.98(d), copies of these items can be obtained in the file of the above-identified parent application.

The citation of the listed items is not a representation that a search has been made.

The filing of this Information Disclosure Statement is not an admission that the information cited herein is, or is considered to be, material to patentability as defined in 37 C.F.R. § 1.56(b).

- ☒ This Information Disclosure Statement is being filed within three (3) months of the filing date of a national application other than a continued prosecution application.
- ☐ This Information Disclosure Statement is being filed within three (3) months of the date of entry of the national stage in an international application.

- ☐ To the best of Applicant's knowledge, this Information Disclosure Statement is being filed before the mailing of a first Office Action on the merits.
- ☐ This Information Disclosure Statement is being filed before the mailing of a first Office Action after the filing of a Request for Continued Examination.
- ☐ The \$240.00 surcharge will be paid via credit card.
- ☒ Please charge any additional fees which may become due in connection with this filing to Deposit Account No. 50-4053.

The claims of the application as now presented are believed to patentably distinguish over the prior art and to be in condition for allowance. Early and favorable consideration of the case is respectfully requested.

Respectfully submitted,

/Paul J. Farrell/

Paul J. Farrell
Reg. No. 33,494
Attorney for Applicants

THE FARRELL LAW FIRM, P.C.
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PTO/SB/08a (07-09)

Approved for use through 07/31/2016. OMB 0651-0031

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Substitute for form 1449/PTO

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**

(Use as many sheets as necessary)

Sheet 1 of 2

Complete if Known

Application Number	not yet assigned - GAU: 2848
Filing Date	herewith
First Named Inventor	Han-Min CHO et al.
Art Unit	not yet assigned 16/939,708 -
Examiner Name	not yet assigned /ANDARGIE M
Attorney Docket Number	678-5477 CON

GAU: 284
MAYCHILLH**U. S. PATENT DOCUMENTS**

Examiner Initials*	Cite No. ¹	Document Number	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1	US- 2014/0225806	08-2014	LEE et al	
	2	US- 2009/0260859	10-2009	PAI et al.	
	3	US- 8,633,394	01-2014	Lai et al.	
	4	US- 8,143,526	03-2012	Hsu	
	5	US- 8,022,309	09-2011	Pai et al.	
	6	US- 7,781,680	08-2010	Pai et al.	
	7	US- 8,263,873	09-2012	Choi et al.	
	8	US- 2009/0294155	12-2009	Suzuki et al.	
	9	US- 2013/0063914	03-2013	Lin et al.	
	10	US- 2011/0094784	04-2011	Muro et al.	
	11	US- 2015/0223323	08-2015	O'Flynn	
	12	US- 2013/0257659	10-2013	Darnell et al.	
	13	US- 2012/0229346	09-2012	Chen et al.	
	14	US- 2006/0270472	11-2006	Chen et al.	
	15	US- 4,845,311	07-1989	Schreiber et al.	
	16	US- 2015/0234426	08-2015	Bailey	
	17	US- 2014/0147128	05-2014	Han	
		US-			
		US-			

FOREIGN PATENT DOCUMENTS

Examiner Initials*	Cite No. ¹	Foreign Patent Document	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages Or Relevant Figures Appear	T ⁶
		Country Code ³ -Number ⁴ -Kind Code ⁵ (if known)				
	18	KR 101261983	05-08-2013			
	19	KR 20150062059	06-05-2015			
	20	EP 2 180 769	04-28-2010			
	21	KR 20140136634	12-01-2014			
	22	KR 101305518	09-06-2013			
	23	KR 1020130081450	07-17-2013			

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>		Complete if Known	
		Application Number	not yet assigned 16/939,708 - GAU: 2848
		Filing Date	herewith
		First Named Inventor	Han-Min CHO et al.
		Art Unit	not yet assigned 16/939,708 - GAU: 2848
		Examiner Name	not yet assigned/ANDARGIE M AYCHILLHUM/
Sheet 2	of 2	Attorney Docket Number	678-5477 CON

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	24	International Search Report dated September 19, 2017 issued in counterpart application No. PCT/KR2017/006070, 7 pages.	
	25	Samsung Galaxy et al: "Samsung Galaxy S7 Teardown Geschrieben von: Jeff Suovanen", IFIXIT Guide, XP055578568, March 8, 2016, 19 pages.	
	26	European Search Report dated April 17, 2019 issued in counterpart application No. 17813538.0-1203, 11 pages.	
	27	European Search Report dated February 5, 2020 issued in counterpart application No. 17813538.0-1203, 12 pages.	

Examiner Signature	/ANDARGIE M AYCHILLHUM/	Date Considered	08/26/2020
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¹ Applicant's unique citation designation number (optional). ² Applicant is to place a check mark here if English language Translation is attached.

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ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /A.M.A/

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	2	"20140147128"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 12:56
L2	142,544	portable with communication with device	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 12:58
L3	7,792	2 and (housing case chassis enclosure) with conduct\$3	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 12:59
L4	4,475	3 and second with conduct\$3	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 12:59
L5	2,214	4 and (antennas)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 13:00
L6	0	5 and wireless with communicat\$3 with circuitry	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 13:00
L7	177	5 and wireless with communicat\$3 with circuitry	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 13:01
L8	42	7 and (first with printed with circuit with board PCB)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 13:08
L9	42	8 and (second with printed with circuit with board PCB)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 13:08
L10	1	8 and (second with printed with circuit with board PCB) with portion with contact\$3 with (first with printed with circuit with board PCB) with (flexible with first with PCB with portion)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 13:11
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L12	0	"16939708"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 13:15
S1	1	"15621687"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 11:53

S2	1	"20090294155"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 15:18
S3	7	("2009/0294155").URPN.	USPAT	OR	OFF	2019/09/20 15:21
S4	117	("20030174529" "20030176023" "20030176053" "20030180011" "20030214860" "20040057220" "20050018409" "20050024885" "20050045374" "20050061542" "20050122627" "20050190006" "20060200977" "20070102830" "20070253176" "20080029293" "20090008131" "20090044968" "20090078452" "20090101399" "20090126970" "20090133906" "20090188699" "20090188711" "20090191377" "20090242259" "20090255717" "20090260859" "20090260860" "20090294155" "20100015822" "20100116590" "20100118443" "20100244871" "20100258338" "20100276183" "20100307798" "20100314159" "20100315752" "20100326706" "20110019371" "20110048771" "20110088842" "20110100694" "20110121922" "20110132641" "20110259632" "20110308840" "20120025924" "20120061129" "20120067626" "20120138340" "20120138344" "20120175151" "20120224281" "20120247826" "20120292085" "20130020112" "20130048344" "20130081868" "20130099876" "20130105208" "20130133939" "20130147581" "20130319743" "20130319748" "20140027165" "20140158406" "20140266505" "20140333329" "20150305142" "2703854" "3271214" "5262590" "5274401" "6057600" "6313912" "6326553" "6373740" "6380493" "6414820" "6531346" "6535083" "6631551" "6700748" "6846991" "7006331" "7142395" "7414814" "7439449" "7532438" "7550842" "7554829" "7724475" "7737365" "7813082" "7826177" "8045296" "8106408" "8119921" "8169746").PN. OR ("8208227" "8361896" "8390958"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 15:51

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S5	2	("2015/0305142").URPN.	USPAT	OR	OFF	2019/09/20 15:57
S6	9	("2011/0121922").URPN.	USPAT	OR	OFF	2019/09/20 16:14
S7	2	"20130063914"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:42
S8	1	"20110094784"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:43
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S11	2	"20140225806"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:47
S12	1,456,772	electronic with device	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:48
S13	3,148	S12 and printed with circuit with board with conduct\$3 with insulat\$3	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:49
S14	62	S13 and first with component with housing	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:50
S15	34	S14 and second with component with (printed with circuit with board PCB)	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:50
S16	7	S15 and (electric with path transmission with line signal with line) with first with component with second with component	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:51
S17	0	S16 and conduct\$3 with vias with conduct\$3 with layer	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:52
S18	3	S16 and (conduct\$3 with vias hole)	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:52

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S21	2	"20140147128"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 17:32
S22	1,460,177	electronic with device housing with printed with circuit with board with electrical with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:03
S23	112	electronic with device with housing with printed with circuit with board with electrical with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:04
S24	0	electronic with device with housing with printed with circuit with board with electrical with connector with conductor with vias	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:04
S25	112	electronic with device with housing with printed with circuit with board with electrical with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:05
S26	13	electronic with device with housing with flexible with printed with circuit with board with electrical with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:20
S27	8	electronic with device with housing with first with flexible with printed with circuit with board with electrical with first with connector with second with flexible with printed with circuit with board with second with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:37
S28	8	electronic with device with housing with first with flexible with printed with circuit with board with electrical with first with connector with second with flexible with printed with circuit with board with second with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:38
S29	271	first with flexible with printed with circuit with board with electrical with first with connector with second with flexible with printed with circuit with board with second with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:39
S30	79	(electronic with device) and first with flexible with printed with circuit with board with electrical with first with connector with second with flexible	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:42

		with printed with circuit with board with second with connector				
S31	109	(electronic with device) and first with flexible with printed with circuit with board with electrical with first with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 12:31
S32	83	S31 and second with flexible with printed with circuit with board with second with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 12:31
S33	83	S31 and second with flexible with printed with circuit with board with second with connector	US-PGPUB; USPAT; USOCR; DERWENT	OR	OFF	2019/09/23 12:50
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S35	2	"20140147128"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 11:24
S36	1	"15621687"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 11:51
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S38	0	elecontronic with device	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 14:49
S39	1,528,162	electronic with device	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 14:50
S40	358,341	S39 and (housing chassis)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 14:51
S41	1,147,311	S39 and (housing chassis case)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 14:51
S42	941,012	S41 and printed circuit with board	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 14:58
S43	0	S42 and componant with housing	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 14:59
S44	40,521	S42 and component with housing	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 14:59

S45	6,479	S44 and (printed with circuit with board PCB) with (first with printed with circuit with board PCB) with (second with printed with circuit with board PCB) with connect\$3 with (first with printed with circuit with board PCB)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 15:51
S46	1	S45 and (electrical with path with transmits with high with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 15:52
S47	1	S45 and (electrical same path same transmits same high same frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 15:53
S48	940	S45 and (electrical with path)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 15:54
S49	1	S48 and transmits with high with frequency	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 15:54
S50	244	S48 and high with frequency	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 15:54
S51	4	S50 and component with antenna with radiator	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 15:55
S52	0	elecontronic with device	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:45
S53	0	elecontronic with device	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:45
S54	1,528,162	electronic with device	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:46
S55	1,147,311	S54 and (housing chassis case)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:46
S56	35,988	S55 and component with housing	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:47
S57	3,353	S56 and (printed with circuit with board PCB) with (first with printed with circuit with board PCB) with (second with printed with circuit with board PCB) with connect\$3 with (first with printed with circuit with board PCB)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:48

S58	1	S57 and (electrical with path with transmits with high with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:49
S59	1	S57 and (electrical same path same transmits same high same frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:50
S60	883	S57 and (electrical same path)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:50
S61	237	S60 and high with frequency	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:50
S62	4	S61 and component with antenna with radiator	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:51
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S67	4	"8143526"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:30
S68	4	"8022309"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:34
S69	1	"7781680"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:53
S70	1	"7781680"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:54
S71	1	"8263873"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:54
S72	267,406	(printed with circuit with board PCB) with (first with printed with circuit with board PCB) with (second with printed with circuit with board PCB)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:58

S73	169,320	(printed with circuit with board PCB) near5 (first with printed with circuit with board PCB) near5 (second with printed with circuit with board PCB)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:58
S74	21,245	S73 and second near5 Printed with circuit with board PCB near5 comprises near5 flexible with printed with circuit with board FPCB near5 rigid with printed with circuit with board rigid with PCB	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:16
S75	1	S74 and electrical near5 path with transmit\$3 with high with frequency with signal near5 connect\$3	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:17
S76	1	S74 and electrical near5 path with transmit\$3 with high with frequency	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:17
S77	4	S74 and electrical with path with transmit\$3 with high with frequency	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:18
S78	1,688	S74 and electrical near5 path	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:21
S79	44,650	second near5 Printed with circuit with board PCB near5 comprises near5 flexible with printed with circuit with board FPCB near5 rigid with printed with circuit with board rigid with PCB	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:26
S80	1	S79 and electrical near5 path with transmit\$3 with high with frequency with signal near5 connect\$3	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:26
S81	4,395	24and electrical with path with transmit\$3 with high with frequency	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:28
S82	5	S79 and electrical with path with transmit\$3 with high with frequency	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:28
S83	359	"30" and (electrical same path same transmits same high same frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:36
S84	1	"38" and (electrical with path with transmits with high with frequency).clm.	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:56
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S98	70	"4845311"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/04/21 09:19
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S108	2	"20140147128"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/25 11:14

8/26/2020 1:32:19 PM

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From: PAIR_eOfficeAction@uspto.gov
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Subject: Private PAIR Correspondence Notification for Customer Number 66547

Aug 31, 2020 03:30:36 AM

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Suite 210E
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Application	Document	Mailroom Date	Attorney Docket No.
16939708	CTNF	08/31/2020	678-5477 CON
	892	08/31/2020	678-5477 CON
	1449	08/31/2020	678-5477 CON

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
16/939,708	07/27/2020	Han-Min CHO	678-5477 CON

CONFIRMATION NO. 8134

PUBLICATION NOTICE



OC000000121298122

66547
THE FARRELL LAW FIRM, P.C.
Paul J Farrell
290 Broadhollow Road
Suite 210E
Melville, NY 11747

Title:ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT

Publication No.US-2020-0358166-A1

Publication Date:11/12/2020

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Application	Document	Mailroom Date	Attorney Docket No.
16939708	NTC.PUB	11/12/2020	678-5477 CON

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Electronic Acknowledgement Receipt

EFS ID:	41244705
Application Number:	16939708
International Application Number:	
Confirmation Number:	8134
Title of Invention:	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT
First Named Inventor/Applicant Name:	Han-Min CHO
Customer Number:	66547
Filer:	Paul J. Farrell/Deanna Marman
Filer Authorized By:	Paul J. Farrell
Attorney Docket Number:	678-5477 CON
Receipt Date:	30-NOV-2020
Filing Date:	27-JUL-2020
Time Stamp:	13:57:10
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Amendment/Req. Reconsideration-After Non-Final Reject	Amendment.pdf	249715	no	12
			c209e7214c3a25fd62fac07f61d3c1fd49af48e4		

Warnings:

Information:	
Total Files Size (in bytes):	249715
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 16/939,708		Filing Date 07/27/2020		☐ To be Mailed	
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO									
APPLICATION AS FILED - PART I									
		(Column 1)	(Column 2)						
FOR		NUMBER FILED	NUMBER EXTRA		RATE (\$)		FEE (\$)		
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A	N/A		N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A	N/A		N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A	N/A		N/A				
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *		x \$100 =					
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *		x \$460 =					
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).							
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))									
* If the difference in column 1 is less than zero, enter "0" in column 2.					TOTAL				
APPLICATION AS AMENDED - PART II									
		(Column 1)		(Column 2)	(Column 3)				
AMENDMENT	11/30/2020	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	x \$100 =		0	
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	x \$480 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE		0	
		(Column 1)		(Column 2)	(Column 3)				
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =			
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE			
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The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.									

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Inventor(s): Han-Min CHO et al.

Examiner: AYCHILLHUM, Andargie M.

Serial No.: 16/939,708

Art Unit: 2848

Filed: July 27, 2020

Date: November 30, 2020

For: **ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT**

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT TRANSMITTAL FORM

Sir:

Transmitted herewith is an amendment in the above-identified application.

☐ Small entity status of this application under 37 C.F.R. §§1.9 and 1.27 has been established by a verified statement previously submitted.

☐ A verified statement to establish small entity under 37 C.F.R. §§1.9 and 1.27 is enclosed.

☒ No additional fee is required.

For	Claims Remaining After Amendment	Highest No. Previously Paid For	Present Extra	Rate (Small Entity)	Addit. Fee	Rate	Addit. Fee
TOTAL CLAIMS*	20	25	0	x50 =	\$0	x100 =	\$0
INDEPENDENT CLAIMS	3	4	0	x240 =	\$0	X480 =	\$0
☐ First Presentation of Multiple Dep. Claim				410		820	\$0

* If the entry in Col. 1 is less than entry in Col. 2, write "0" in Col. 3.

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- ☐ A check in the amount of \$____ is enclosed.
- ☒ Please charge any deficiency as well as any other fee(s) which may become due under 37 C.F.R. §§1.16 and/or 1.17 at any time during the pendency of this application, or credit any overpayment of such fee(s) to Deposit Account No. 50-4053. Also, in the event any extensions of time for responding are required for the pending application(s), please treat this paper as a petition to extend the time as required and charge Deposit Account No. 50-4053 therefor.

Respectfully submitted,

/Paul J. Farrell/

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

INVENTORS(s): Han-Min CHO et al. **EXAMINER:** Aychillhum, Andargie M.

SERIAL NO.: 16/939,708 **ART UNIT:** 2848

FILED: July 27, 2020 **DATE:** November 30, 2020

**FOR: ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY
TRANSMISSION CIRCUIT**

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT

In response to the Office Action of the United States Patent and Trademark Office dated August 31, 2020, please consider the following amendments and remarks.

AMENDMENTS TO THE CLAIMS

Please amend the claims as follows:

1-25. (Cancelled)

26. (Previously Presented) A portable communication device comprising:

a housing including a first conduction portion forming a first portion of a side surface of the portable communication device and a second conduction portion forming a second portion of the side surface, the first conduction portion operating as a first antenna and the second conduction portion operating as a second antenna;

a wireless communication circuitry; and

a first printed circuit board (PCB) including:

a first PCB portion;

a second PCB portion in contact with the first PCB portion and being more flexible than the first PCB portion;

a third PCB portion in contact with the second PCB portion and being less flexible than the second PCB portion;

a fourth PCB portion in contact with the third PCB portion and being more flexible than the third PCB portion; and

a plurality of lines located from the first PCB portion to the fourth PCB portion through the second PCB portion and the third PCB portion, the plurality of lines including a first ground line, a second ground line at least partially spaced apart from the first ground line, a first signal line located between the first ground line and the second ground line, and a second signal line located between the first signal line and the second ground line, the first signal line electrically connected with the first antenna and the wireless communication circuitry, and the second signal line electrically connected with the second antenna and the wireless communication circuitry.

27. (Previously Presented) The portable communication device of claim 26,
wherein the plurality of lines includes a third ground line located between the first signal
line and the second signal line, and

wherein the first PCB further includes an insulating layer on which the first ground line,
the second ground line, the third ground line, the first signal line, and the second signal line are
located, and a ground plate located below the insulating layer.

28. (Previously Presented) The portable communication device of claim 27,
wherein the ground plate is electrically connected with the first ground line through a first
conduction via formed in a first portion of the insulating layer, the second ground line through a
second conduction via formed in a second portion of the insulating layer, and the third ground
line through a third conduction via formed in a third portion of the insulating layer.

29. (Previously Presented) The portable communication device of claim 27,
wherein the first PCB further includes a solder resist ink forming at least a part of an
exterior surface of the first PCB portion or the third PCB portion.

30. (Previously Presented) The portable communication device of claim 26,
wherein the first PCB further includes a fifth PCB portion in contact with the fourth PCB
portion and being less flexible than the fourth PCB portion,

wherein the first PCB portion further includes a first connector for a connection with the
antenna,

wherein the fifth PCB portion includes a second connector for a connection with the
wireless communication circuitry, and

wherein a first ending portion of the first signal line and a second ending portion of the
second signal line are in contact with the second connector.

31. (Previously Presented) The portable communication device of claim 26, further
comprising:

a second PCB on which the wireless communication circuitry is located,

wherein the first ground line and the second ground line are electrically connected with a ground member located in the second PCB, or

wherein the first ground line and the second ground line are electrically connected with a ground member located in the first PCB portion.

32. (Previously Presented) A portable communication device comprising:
- a first electronic component;
 - a second electronic component; and
 - a printed circuit board (PCB) electrically connected with the first electronic component and the second electronic component, the PCB including:
 - a first rigid PCB portion;
 - a first flexible PCB portion in contact with the first rigid PCB portion;
 - a second rigid PCB portion in contact with the first flexible PCB portion; and
 - a second flexible PCB portion in contact with the second rigid PCB portion,wherein each of the first rigid PCB portion and the second rigid PCB portion includes:
 - a first insulating layer including a first material and forming a first exterior surface of each of the first rigid PCB portion and the second rigid PCB portion;
 - a first conduction layer located on or above the first insulating layer;
 - a second insulating layer located on or above the first conduction layer and having a first thickness;
 - a second conduction layer located on or above the second insulating layer; and
 - a third insulating layer including the first material and located on or above the second conduction layer, the third insulating layer forming a second exterior surface of each of the first rigid PCB portion and the second rigid PCB portion, andwherein each of the first flexible PCB portion and the second flexible PCB portion includes:
 - a first insulating layer including a second material different from the first material and forming a first exterior surface of each of the first flexible PCB portion and the second flexible PCB portion;

a first conduction layer located on or above the first insulating layer and being electrically connected with the first conduction layer of first rigid PCB portion and the first conduction layer of the second rigid PCB portion;

a second insulating layer located on or above the first conduction layer and having a second thickness thinner than the first thickness;

a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the first rigid PCB portion and the second conduction layer of the second rigid PCB portion; and

a third insulating layer including the second material and located on or above the second conduction layer, the third insulating layer forming a second exterior surface of each of the first flexible PCB portion and the second flexible PCB portion.

33. (Previously Presented) The portable communication device of claim 32, wherein the second insulating layer of the first flexible PCB portion or the second insulating layer of the second flexible PCB portion includes the second material.

34. (Previously Presented) The portable communication device of claim 32, wherein each of the first flexible PCB portion and the second flexible PCB portion includes a fourth insulating layer located between the second conduction layer and the third insulating layer, and a third conduction layer located between the third insulating layer and the fourth insulating layer.

35. (Previously Presented) The portable communication device of claim 34, wherein the fourth insulating layer includes the second material.

36. (Previously Presented) The portable communication device of claim 34, wherein the first conduction layer of each of the first flexible PCB portion and the second flexible PCB portion includes a first ground portion,

wherein the second conduction layer of each of the first flexible PCB portion and the second flexible PCB portion includes a second ground portion electrically connected with the first ground portion through a first conduction via, and

wherein the third conduction layer includes a third ground portion electrically connected with the second ground portion through a second conduction via.

37. (Previously Presented) The portable communication device of claim 32, wherein the first material is a solder resist ink and the second material is a polyimide, and wherein the second insulating layer of the first rigid PCB portion or the second insulating layer of the second rigid PCB portion includes a third material.

38. (Previously Presented) The portable communication device of claim 37, wherein the third material includes a resin or a glass fabric.

39. (Previously Presented) A portable communication device comprising:
a first electronic component;
a second electronic component; and
a printed circuit board (PCB) electrically connected with the first electronic component and the second electronic component, the PCB including:
a first flexible PCB portion including:
a first insulating layer forming a first exterior surface of the first flexible PCB portion and including a first material;
a first conduction layer located on or above the first insulating layer;
a second insulating layer located on or above the first conduction layer having a first thickness;
a second conduction layer located on or above the second insulating layer; and
a third insulating layer located on or above the second conduction layer and forming a second exterior surface of the first flexible PCB portion, the third insulating layer including the first material;
a rigid PCB portion in contact with the first flexible PCB portion, the rigid PCB portion including:
a first insulating layer forming a first exterior surface of the rigid PCB portion and including a second material different from the first material;

a first conduction layer located on or above the first insulating layer and being electrically connected with the first conduction layer of the first flexible PCB portion;

a second insulating layer located on or above the first conduction layer and having a second thickness thicker than the first thickness;

a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the first flexible PCB portion; and

a third insulating layer located on or above the second conduction layer and forming a second exterior surface of the rigid PCB portion, the third insulating layer including the second material;

a second flexible PCB portion in contact with the rigid PCB portion, the second flexible PCB portion including:

a first insulating layer forming a first exterior surface of the second flexible PCB portion and including the first material;

a first conduction layer located on or above the first insulating layer and being electrically connected with the first conduction layer of the rigid PCB portion;

a second insulating layer located on or above the first conduction layer and having the first thickness;

a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the rigid PCB portion;

a third insulating layer located on or above the second conduction layer and forming a second exterior surface of the second flexible PCB portion, the third insulating layer including the first material; and

a contact pad located on the third insulating layer.

40. (Previously Presented) The portable communication device of claim 39, wherein the second insulating layer of the first flexible PCB portion or the second insulating layer of the second flexible PCB portion includes the second material.

41. (Previously Presented) The portable communication device of claim 39, wherein each of the first flexible PCB portion and the second flexible PCB portion includes a fourth insulating layer located between the second conduction layer and the third

insulating layer, and a third conduction layer located between the third insulating layer and the fourth insulating layer.

42. (Previously Presented) The portable communication device of claim 41, wherein the fourth insulating layer includes the second material.

43. (Previously Presented) The portable communication device of claim 41, wherein the first conduction layer of each of the first flexible PCB portion and the second flexible PCB portion includes a first ground portion,

wherein the second conduction layer of each of the first flexible PCB portion and the second flexible PCB portion includes a second ground portion electrically connected with the first ground portion through a first conduction via, and

wherein the third conduction layer includes a third ground portion electrically connected with the second ground portion through a second conduction via.

44. (Previously Presented) The portable communication device of claim 39, wherein the first material is a solder resist ink and the second material is a polyimide, and wherein the second insulating layer of the rigid PCB portion includes a third material.

45. (Previously Presented) The portable communication device of claim 44, wherein the third material includes a resin or a glass fabric.

REMARKS

Prior to this Amendment, Claims 21-45 were pending in the application, with Claims 21, 26, 32 and 39 being recited in independent form. It is gratefully acknowledged that the Examiner found allowable subject matter in Claims 26-45.

In the Office Action:

Claims 21-25 are rejected under 35 U.S.C. § 102(a)(1) as allegedly being anticipated by U.S. Patent Application Publication No. 2014/0147128 to Han et al. (hereinafter Han).

Please cancel Claims 21-25, without prejudice. No new matter has been added. Accordingly, Claims 26-45 are will be pending upon entry of this Amendment.

Regarding the §102(a)(1) rejection of Claims 21-25, it is respectfully asserted that this rejection is rendered moot in view of the cancellation of Claims 21-25. Therefore, it is respectfully submitted that the rejection of Claims 21-25 under 35 U.S.C. §102(a)(1) should be withdrawn.

Accordingly, all of the claims pending in the Application, namely, Claims 26-45, are believed to be in condition for allowance. Should the Examiner believe that a telephone conference or personal interview would facilitate resolution of any remaining matters, the Examiner may contact Applicants' attorney at the number given below.

Respectfully Submitted,

/Paul J. Farrell/

Paul J. Farrell
Reg. No. 33,494
Attorneys for Applicants

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66547 7590 12/11/2020
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Paul J Farrell
290 Broadhollow Road
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EXAMINER	
AYCHILLHUM, ANDARGIE M	
ART UNIT	PAPER NUMBER

2848

DATE MAILED: 12/11/2020

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/939,708	07/27/2020	Han-Min CHO	678-5477 CON	8134

TITLE OF INVENTION: ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	03/11/2021

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66547 7590 12/11/2020
THE FARRELL LAW FIRM, P.C.
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	(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/939,708	07/27/2020	Han-Min CHO	678-5477 CON	8134

TITLE OF INVENTION: ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	03/11/2021

EXAMINER	ART UNIT	CLASS-SUBCLASS
AYCHILLHUM, ANDARGIE M	2848	361-790000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. **Change in Entity Status** (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/939,708	07/27/2020	Han-Min CHO	678-5477 CON	8134
66547	7590	12/11/2020	EXAMINER	
THE FARRELL LAW FIRM, P.C. Paul J Farrell 290 Broadhollow Road Suite 210E Melville, NY 11747			AYCHILLHUM, ANDARGIE M	
			ART UNIT	PAPER NUMBER
			2848	
DATE MAILED: 12/11/2020				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 16/939,708	Applicant(s) CHO et al.	
	Examiner ANDARGIE M AYCHILLHUM	Art Unit 2848	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 11/30/2020.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 26-45. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892)	5. ☐ Examiner's Amendment/Comment
2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____.	6. ☒ Examiner's Statement of Reasons for Allowance
3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____.	7. ☐ Other _____.
4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	

/ANDARGIE M AYCHILLHUM/ Primary Examiner, Art Unit 2848	12/04/2020
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DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Allowable Subject Matter

2. Claims 26-45 are allowed over the prior art of record.

Reasons for Allowance

3. The following is an examiner's statement of reasons for allowance:

The prior art of record neither teach nor render obvious the instant application claimed invention as a whole, in particular, the prior art fails to teach "A portable communication device comprising: the first conduction portion operating as a first antenna and the second conduction portion operating as a second antenna; a wireless communication circuitry; and a first printed circuit board (PCB) including: a first PCB portion; a second PCB portion in contact with the first PCB portion and being more flexible than the first PCB portion; a third PCB portion in contact with the second PCB portion and being less flexible than the second PCB portion; a fourth PCB portion in contact with the third PCB portion and being more flexible than the third PCB portion; and a plurality of lines located from the first PCB portion to the fourth PCB portion through the second PCB portion and the third PCB portion, the plurality of lines including a first ground line, a second ground line at least partially spaced apart from the first ground line, a first signal line located between the first ground line and the second ground line, and a second signal line located between the first signal line and the

second ground line, the first signal line electrically connected with the first antenna and the wireless communication circuitry, and the second signal line electrically connected with the second antenna and the wireless communication circuitry.” as recited claim 26,

“A portable communication device comprising: a first electronic component; a second electronic component; and a printed circuit board (PCB) electrically connected with the first electronic component and the second electronic component, the PCB including: a first rigid PCB portion; a first flexible PCB portion in contact with the first rigid PCB portion; a second rigid PCB portion in contact with the first flexible PCB portion; and a second flexible PCB portion in contact with the second rigid PCB portion, wherein each of the first rigid PCB portion and the second rigid PCB portion includes: a first insulating layer including a first material and forming a first exterior surface of each of the first rigid PCB portion and the second rigid PCB portion; a first conduction layer located on or above the first insulating layer; a second insulating layer located on or above the first conduction layer and having a first thickness; a second conduction layer located on or above the second insulating layer; and a third insulating layer including the first material and located on or above the second conduction layer, the third insulating layer forming a second exterior surface of each of the first rigid PCB portion and the second rigid PCB portion, and wherein each of the first flexible PCB portion and the second flexible PCB portion includes: a first insulating layer including a second material different from the first material and forming a first exterior surface of each of the first flexible PCB portion and the second flexible PCB portion; a first conduction layer located on or above the first insulating layer and being electrically connected with the first conduction layer of first rigid PCB portion and the first conduction layer of the second rigid PCB portion; a

second insulating layer located on or above the first conduction layer and having a second thickness thinner than the first thickness; a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the first rigid PCB portion and the second conduction layer of the second rigid PCB portion; and a third insulating layer including the second material and located on or above the second conduction layer, the third insulating layer forming a second exterior surface of each of the first flexible PCB portion and the second flexible PCB portion.” as recited claim 32, “A portable communication device comprising: a first electronic component; a second electronic component; and a printed circuit board (PCB) electrically connected with the first electronic component and the second electronic component, the PCB including: a first flexible PCB portion including: a first insulating layer forming a first exterior surface of the first flexible PCB portion and including a first material; a first conduction layer located on or above the first insulating layer; a second insulating layer located on or above the first conduction layer having a first thickness; a second conduction layer located on or above the second insulating layer; and a third insulating layer located on or above the second conduction layer and forming a second exterior surface of the first flexible PCB portion, the third insulating layer including the first material; a rigid PCB portion in contact with the first flexible PCB portion, the rigid PCB portion including: a first insulating layer forming a first exterior surface of the rigid PCB portion and including a second material different from the first material; a first conduction layer located on or above the first insulating layer and being electrically connected with the first conduction layer of the first flexible PCB portion; a second insulating layer located on or above the first conduction layer and having a

second thickness thicker than the first thickness; a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the first flexible PCB portion; and a third insulating layer located on or above the second conduction layer and forming a second exterior surface of the rigid PCB portion, the third insulating layer including the second material; a second flexible PCB portion in contact with the rigid PCB portion, the second flexible PCB portion including: a first insulating layer forming a first exterior surface of the second flexible PCB portion and including the first material; a first conduction layer located on or above the first insulating layer and being electrically connected with the first conduction layer of the rigid PCB portion; a second insulating layer located on or above the first conduction layer and having the first thickness; a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the rigid PCB portion; a third insulating layer located on or above the second conduction layer and forming a second exterior surface of the second flexible PCB portion, the third insulating layer including the first material; and a contact pad located on the third insulating layer.” as recited claim 39.

Claims 27-31, 33-38 and 40-45 are also allowed as being directly or indirectly dependent of the allowed base claims 26, 32 and 39.

Conclusion

4. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ANDARGIE M AYCHILLHUM whose telephone number is (571)270-1607. The examiner can normally be reached on M-F 9-5.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jinhee Lee can be reached on (571) 272-1977. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <https://ppair-my.uspto.gov/pair/PrivatePair>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ANDARGIE M AYCHILLHUM/
Primary Examiner, Art Unit 2848

<i>Search Notes</i> 	Application/Control No. 16/939,708	Applicant(s)/Patent Under Reexamination CHO et al.
	Examiner ANDARGIE M AYCHILLHUM	Art Unit 2848

CPC - Searched*		
Symbol	Date	Examiner

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
361	790	12/04/2020	A.A.

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
East (US-PGPUB, USPAT, SUOCR, FPRS, EPO, JPO, DERWENT AND IBM-TDB) See Search Printout	12/04/2020	A.A.
Text, Class and Inventor search is completed.	12/04/2020	A.A.
Interference search is completed.	12/04/2020	A.A.

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
361	785	12/04/2020	A.A.

/ANDARGIE M AYCHILLHUM/ Primary Examiner, Art Unit 2848	08/26/2020
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Issue Classification 	Application/Control No. 16/939,708	Applicant(s)/Patent Under Reexamination CHO et al.	
	Examiner ANDARGIE M AYCHILLHUM	Art Unit 2848	

CPC						
Symbol					Type	Version
H01Q	/	1	/	243	F	2013-01-01
H01Q	/	5	/	35	I	2013-01-01
H05K	/	1	/	0218	I	2013-01-01
H01Q	/	21	/	28	I	2013-01-01
H01Q	/	1	/	38	I	2013-01-01
H01R	/	12	/	79	I	2013-01-01
H01R	/	13	/	6691	I	2013-01-01
H05K	/	1	/	0243	I	2013-01-01
H05K	/	1	/	028	I	2013-01-01
H05K	/	1	/	115	I	2013-01-01
H05K	/	1	/	142	I	2013-01-01
H05K	/	1	/	148	I	2013-01-01
H05K	/	1	/	18	I	2013-01-01
H05K	/	7	/	1427	I	2013-01-01
H05K	/	2201	/	0999	A	2013-01-01
H05K	/	2201	/	09609	A	2013-01-01
H05K	/	2201	/	10098	A	2013-01-01

CPC Combination Sets					
Symbol				Type	Set
	/		/		

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 20	
/ANDARGIE M AYCHILLHUM/ Primary Examiner, Art Unit 2848 (Primary Examiner) _____ (Date) _____		04 December 2020 O.G. Print Claim(s) 26	O.G. Print Figure 1

Issue Classification 	Application/Control No. 16/939,708	Applicant(s)/Patent Under Reexamination CHO et al.
	Examiner ANDARGIE M AYCHILLHUM	Art Unit 2848

INTERNATIONAL CLASSIFICATION			
CLAIMED			
H05K	/	7	/ 00
NON-CLAIMED			
/	/		

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
361	790

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					
361	785	735				

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	20	
/ANDARGIE M AYCHILLHUM/ Primary Examiner, Art Unit 2848	04 December 2020	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	26	1

<i>Issue Classification</i> 	Application/Control No. 16/939,708	Applicant(s)/Patent Under Reexamination CHO et al.
	Examiner ANDARGIE M AYCHILLHUM	Art Unit 2848

☐ Claims renumbered in the same order as presented by applicant
 ☐ CPA
 ☐ T.D.
 ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	26	10	35	19	44										
2	27	11	36	20	45										
3	28	12	37												
4	29	13	38												
5	30	14	39												
6	31	15	40												
7	32	16	41												
8	33	17	42												
9	34	18	43												

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	20	
/ANDARGIE M AYCHILLHUM/ Primary Examiner, Art Unit 2848	04 December 2020	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	26	1

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S1	1	"15621687"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 11:53
S2	1	"20090294155"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 15:18
S3	7	("2009/0294155").URPN.	USPAT	OR	OFF	2019/09/20 15:21
S4	117	("20030174529" "20030176023" "20030176053" "20030180011" "20030214860" "20040057220" "20050018409" "20050024885" "20050045374" "20050061542" "20050122627" "20050190006" "20060200977" "20070102830" "20070253176" "20080029293" "20090008131" "20090044968" "20090078452" "20090101399" "20090126970" "20090133906" "20090188699" "20090188711" "20090191377" "20090242259" "20090255717" "20090260859" "20090260860" "20090294155" "20100015822" "20100116590" "20100118443" "20100244871" "20100258338" "20100276183" "20100307798" "20100314159" "20100315752" "20100326706" "20110019371" "20110048771" "20110088842" "20110100694" "20110121922" "20110132641" "20110259632" "20110308840" "20120025924" "20120061129" "20120067626" "20120138340" "20120138344" "20120175151" "20120224281" "20120247826" "20120292085" "20130020112" "20130048344" "20130081868" "20130099876" "20130105208" "20130133939" "20130147581" "20130319743" "20130319748" "20140027165" "20140158406" "20140266505" "20140333329" "20150305142" "2703854" "3271214" "5262590" "5274401"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 15:51

		"6057600" "6313912" "6326553" "6373740" "6380493" "6414820" "6531346" "6535083" "6631551" "6700748" "6846991" "7006331" "7142395" "7414814" "7439449" "7532438" "7550842" "7554829" "7724475" "7737365" "7813082" "7826177" "8045296" "8106408" "8119921" "8169746").PN. OR ("8208227" "8361896" "8390958" "8395055" "8514522" "8547663" "8658903" "8760815" "8847696" "8878560" "9064513" "9077168" "9295156").PN. OR ("9532446").URPN.				
S5	2	("2015/0305142").URPN.	USPAT	OR	OFF	2019/09/20 15:57
S6	9	("2011/0121922").URPN.	USPAT	OR	OFF	2019/09/20 16:14
S7	2	"20130063914"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:42
S8	1	"20110094784"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:43
S9	1	"20150223323"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:44
S10	0	"2014225806"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:47
S11	2	"20140225806"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:47
S12	1,456,772	electronic with device	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:48
S13	3,148	S12 and printed with circuit with board with conduct\$3 with insulat\$3	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:49
S14	62	S13 and first with component with housing	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:50
S15	34	S14 and second with component with (printed with circuit with board PCB)	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:50

S16	7	S15 and (electric with path transmission with line signal with line) with first with component with second with component	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:51
S17	0	S16 and conduct\$3 with vias with conduct\$3 with layer	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:52
S18	3	S16 and (conduct\$3 with vias hole)	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:52
S19	2	"20110121922"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:54
S20	2	"20140147128"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 16:55
S21	2	"20140147128"	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/20 17:32
S22	1,460,177	electronic with device housing with printed with circuit with board with electrical with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:03
S23	112	electronic with device with housing with printed with circuit with board with electrical with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:04
S24	0	electronic with device with housing with printed with circuit with board with electrical with connector with conductor with vias	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:04
S25	112	electronic with device with housing with printed with circuit with board with electrical with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:05
S26	13	electronic with device with housing with flexible with printed with circuit with board with electrical with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:20
S27	8	electronic with device with housing with first with flexible with printed with circuit with board with electrical with first with connector with second with flexible with printed with circuit with board with second with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:37
S28	8	electronic with device with housing with first with flexible with printed with circuit with board with electrical with first with connector with second with	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:38

		flexible with printed with circuit with board with second with connector				
S29	271	first with flexible with printed with circuit with board with electrical with first with connector with second with flexible with printed with circuit with board with second with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:39
S30	79	(electronic with device) and first with flexible with printed with circuit with board with electrical with first with connector with second with flexible with printed with circuit with board with second with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 11:42
S31	109	(electronic with device) and first with flexible with printed with circuit with board with electrical with first with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 12:31
S32	83	S31 and second with flexible with printed with circuit with board with second with connector	US-PGPUB; USPAT; USOCR	OR	OFF	2019/09/23 12:31
S33	83	S31 and second with flexible with printed with circuit with board with second with connector	US-PGPUB; USPAT; USOCR; DERWENT	OR	OFF	2019/09/23 12:50
S34	3	"20150234426"	US-PGPUB; USPAT; USOCR; DERWENT	OR	OFF	2019/09/23 12:52
S35	2	"20140147128"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 11:24
S36	1	"15621687"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 11:51
S37	5	("2014/0147128").URPN.	USPAT	OR	OFF	2020/03/03 12:02
S38	0	elecontronic with device	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 14:49
S39	1,528,162	electronic with device	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 14:50
S40	358,341	S39 and (housing chassis)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 14:51

S41	1,147,311	S39 and (housing chassis case)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 14:51
S42	941,012	S41 and printed circuit with board	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 14:58
S43	0	S42 and component with housing	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 14:59
S44	40,521	S42 and component with housing	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 14:59
S45	6,479	S44 and (printed with circuit with board PCB) with (first with printed with circuit with board PCB) with (second with printed with circuit with board PCB) with connect\$3 with (first with printed with circuit with board PCB)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 15:51
S46	1	S45 and (electrical with path with transmits with high with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 15:52
S47	1	S45 and (electrical same path same transmits same high same frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 15:53
S48	940	S45 and (electrical with path)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 15:54
S49	1	S48 and transmits with high with frequency	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 15:54
S50	244	S48 and high with frequency	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 15:54
S51	4	S50 and component with antenna with radiator	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/03 15:55
S52	0	elecontronic with device	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:45
S53	0	elecontronic with device	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:45
S54	1,528,162	electronic with device	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:46

S55	1,147,311	S54 and (housing chassis case)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:46
S56	35,988	S55 and component with housing	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:47
S57	3,353	S56 and (printed with circuit with board PCB) with (first with printed with circuit with board PCB) with (second with printed with circuit with board PCB) with connect\$3 with (first with printed with circuit with board PCB)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:48
S58	1	S57 and (electrical with path with transmits with high with frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:49
S59	1	S57 and (electrical same path same transmits same high same frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:50
S60	883	S57 and (electrical same path)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:50
S61	237	S60 and high with frequency	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:50
S62	4	S61 and component with antenna with radiator	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 12:51
S63	2	"20140147128"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 14:46
S64	2	("20140147128" "20150234426").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:21
S65	1	"20090260859"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:28
S66	1	"8633394"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:29
S67	4	"8143526"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:30
S68	4	"8022309"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:34

S69	1	"7781680"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:53
S70	1	"7781680"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:54
S71	1	"8263873"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:54
S72	267,406	(printed with circuit with board PCB) with (first with printed with circuit with board PCB) with (second with printed with circuit with board PCB)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:58
S73	169,320	(printed with circuit with board PCB) near5 (first with printed with circuit with board PCB) near5 (second with printed with circuit with board PCB)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 15:58
S74	21,245	S73 and second near5 Printed with circuit with board PCB near5 comprises near5 flexible with printed with circuit with board FPCB near5 rigid with printed with circuit with board rigid with PCB	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:16
S75	1	S74 and electrical near5 path with transmit\$3 with high with frequency with signal near5 connect\$3	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:17
S76	1	S74 and electrical near5 path with transmit\$3 with high with frequency	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:17
S77	4	S74 and electrical with path with transmit\$3 with high with frequency	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:18
S78	1,688	S74 and electrical near5 path	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:21
S79	44,650	second near5 Printed with circuit with board PCB near5 comprises near5 flexible with printed with circuit with board FPCB near5 rigid with printed with circuit with board rigid with PCB	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:26
S80	1	S79 and electrical near5 path with transmit\$3 with high with frequency with signal near5 connect\$3	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:26
S81	4,395	24and electrical with path with transmit\$3 with high with frequency	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:28

S82	5	S79 and electrical with path with transmit\$3 with high with frequency	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:28
S83	359	"30" and (electrical same path same transmits same high same frequency)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:36
S84	1	"38" and (electrical with path with transmits with high with frequency).clm.	US-PGPUB; USPAT; USOCR	OR	OFF	2020/03/04 16:56
S93	1	"15621687"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/04/21 09:07
S94	2	("20140147128" "20150234426").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2020/04/21 09:10
S95	2	"20130257659"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/04/21 09:16
S96	2	"20120229346"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/04/21 09:18
S97	2	"20160270472"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/04/21 09:19
S98	70	"4845311"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/04/21 09:19
S99	2	"20130257659"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/04/28 14:50
S100	2	"20120229346"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/04/28 14:51
S101	1	"20060270472"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/04/28 14:52
S102	2	"20140147128"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/21 16:29
S103	0	"16939708"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/21 17:36
S104	2	"15621687"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/21 17:36

S105	2	"20140147128"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/24 15:56
S106	2	"20140147128"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/24 18:08
S107	2	"20140225806"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/24 18:08
S108	2	"20140147128"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/25 11:14
S109	2	"20140147128"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 12:56
S110	142,544	portable with communication with device	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 12:58
S111	7,792	S110 and (housing case chassis enclosure) with conduct\$3	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 12:59
S112	4,475	S111 and second with conduct\$3	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 12:59
S113	2,214	S112 and (antennas)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 13:00
S114	0	S113 and wireless with communicat\$3 with circuitry	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 13:00
S115	177	S113 and wireless with communicat\$3 with circuitry	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 13:01
S116	42	S115 and (first with printed with circuit with board PCB)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 13:08
S117	42	S116 and (second with printed with circuit with board PCB)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 13:08
S118	1	S116 and (second with printed with circuit with board PCB) with portion with contact\$3 with (first with printed with circuit with board PCB) with (flexible with first with PCB with portion)	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 13:11

S119	0	"16939,708"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 13:15
S120	0	"16939708"	US-PGPUB; USPAT; USOCR	OR	OFF	2020/08/26 13:15

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	20,893	portable with communication with device.clm.	US-PGPUB; USPAT	OR	OFF	2020/12/04 17:24
L2	16,644	1 and (housing case enclosure)	US-PGPUB; USPAT	OR	OFF	2020/12/04 17:24
L3	3,995	1 and (housing case enclosure).clm.	US-PGPUB; USPAT	OR	OFF	2020/12/04 17:24
L4	1	1 and (housing case enclosure) with conduction with portion.clm.	US-PGPUB; USPAT	OR	OFF	2020/12/04 17:25
S85	48,465	(printed with circuit with board PCB) with (first with printed with circuit with board PCB) with (second with printed with circuit with board PCB).clm.	US-PGPUB; USPAT	OR	OFF	2020/03/04 16:32
S86	462	S85 and second with flexible with printed with circuit with board with second with connector.clm.	US-PGPUB; USPAT	OR	OFF	2020/03/04 16:34
S87	53	S86 and electronic with device.clm.	US-PGPUB; USPAT	OR	OFF	2020/03/04 16:34
S88	70	"62" and (electrical same path same transmits same high same frequency)	US-PGPUB; USPAT	OR	OFF	2020/03/04 16:35
S89	358,869	electronic with device.clm.	US-PGPUB; USPAT	OR	OFF	2020/03/04 16:37
S90	21,175	"26" and (printed with circuit with board PCB) with (first with printed with circuit with board PCB) with (second with printed with circuit with board PCB).clm.	US-PGPUB; USPAT	OR	OFF	2020/03/04 16:53
S91	4,694	S89 and (printed with circuit with board PCB) with (first with printed with circuit with board PCB) with (second with printed with circuit with board PCB).clm.	US-PGPUB; USPAT	OR	OFF	2020/03/04 16:54

S92	1	S85 and (electrical with path with transmits with high with frequency).clm.	US- PGPUB; USPAT	OR	OFF	2020/03/04 16:58
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12/4/2020 5:26:44 PM

C:\Users\aychillhum\Documents\EAST\Workspaces\15621687.wsp



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BIB DATA SHEET

CONFIRMATION NO. 8134

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
16/939,708	07/27/2020	361	2848	678-5477 CON
RULE				
APPLICANTS Samsung Electronics Co., Ltd., Gyeonggi-do, KOREA, REPUBLIC OF;				
INVENTORS Han-Min CHO, Gyeonggi-do, KOREA, REPUBLIC OF; Chan-Gi PARK, Gyeonggi-do, KOREA, REPUBLIC OF; Yeon-Sang YUN, Gyeonggi-do, KOREA, REPUBLIC OF; Tae-Wook HAM, Gyeonggi-do, KOREA, REPUBLIC OF; Hei-Seong KWAK, Gyeonggi-do, KOREA, REPUBLIC OF; Byoung-II SON, Gyeonggi-do, KOREA, REPUBLIC OF; Sung-Chul PARK, Seoul, KOREA, REPUBLIC OF;				
** CONTINUING DATA ***** This application is a CON of 15/621,687 06/13/2017 PAT 10727568 NONE				
** FOREIGN APPLICATIONS ***** REPUBLIC OF KOREA 10-2016-0073355 06/13/2016				
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 08/05/2020				
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and /ANDARGIE M Aychillhum/ Acknowledged Examiner's Signature	☐ Met after Allowance A.A. Initials	STATE OR COUNTRY KOREA, REPUBLIC OF	SHEETS DRAWINGS 14	TOTAL CLAIMS 20
			INDEPENDENT CLAIMS 2 3	
ADDRESS THE FARRELL LAW FIRM, P.C. Paul J Farrell 290 Broadhollow Road Suite 210E Melville, NY 11747 UNITED STATES				
TITLE ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT				
FILING FEE RECEIVED 2680	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:			☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit

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16939708	NOA	12/11/2020	678-5477 CON

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PATENT

Atty. Docket No. 678-5477 CON (P23235-US1-CON/DMC)

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

APPLICANT(S): Han-Min CHO et al.

SERIAL NO.: 16/939,708

FILED: July 27, 2020

DATED: March 11, 2021

FOR: ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION
CIRCUIT

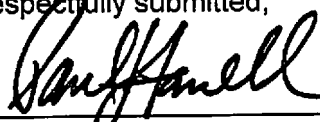
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

TRANSMITTAL OF PRIORITY DOCUMENT

Sir:

Enclosed is a certified copy of Korean Patent Appln. No. 10-2016-0073355 filed
on June 13, 2016, from which priority is claimed under 35 U.S.C. §119.

Respectfully submitted,



Paul J. Farrell
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Property Office.

출 원 번 호 : 10-2016-0073355

Application Number

출 원 년 월 일 : 2016년 06월 13일

Filing Date

Jun. 13, 2016

출 원 인 : 삼성전자주식회사

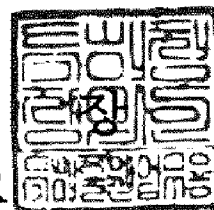
Applicant(s)

SAMSUNG ELECTRONICS CO., LTD.

2020년 04월 16일

특 허 청

COMMISSIONER



【서지사항】

【서류명】 특허출원서

【참조번호】 1

【출원구분】 특허출원

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【대리인】

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【대리인번호】 9-2004-000106-3

【포괄위임등록번호】 2013-008855-3

【발명의 국문명칭】 고주파 전송회로를 포함하는 전자 장치

【발명의 영문명칭】 ELECTRONIC DEVICE COMPRISING HIGH FREQUENCY
TRANSMITTING CIRCUIT

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【성명】 조한민

【성명의 영문표기】 CH0, Han Min

【주민등록번호】 710425-1XXXXXX

제출 일자 : 2016-06-13

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【성명의 영문표기】 HAM, Tae Wook
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【수수료】

【출원료】 0 면 46,000 원

【가산출원료】 89 면 0 원

【우선권주장료】 0 건 0 원

【심사청구료】 0 항 0 원

【합계】 46,000 원

【발명의 설명】

【발명의 명칭】

고주파 전송회로를 포함하는 전자 장치{ELECTRONIC DEVICE COMPRISING HIGH FREQUENCY TRANSMITTING CIRCUIT}

【기술분야】

【0001】 본 발명의 다양한 실시예들은 전자 장치에 포함된 고주파 전송회로에 관한 것이다.

【발명의 배경이 되는 기술】

【0003】 최근 새로운 기능을 가지는 휴대용 단말기의 개발이 빠른 속도로 이루어지고 있으며, 그 보급이 확대되어 감에 따라, 사람들의 생활 속에서 휴대용 단말기가 차지하는 비중이 점차 높아지고 있다.

【0004】 휴대용 단말기의 내부회로는 일반적으로 인쇄회로기판(printed circuit board:PCB)에 구현된다. 이러한 인쇄회로기판 기술은 비약적으로 발전하고 있으며, 현재에는 종래의 딱딱한 성질의 인쇄회로기판뿐만 아니라 자유로운 움직임이 가능한 플렉시블 회로기판(flexible PCB: FPCB)도 널리 사용되고 있다.

【0005】 더욱이, 무선단말기에 적용되는 고주파 선로, 특히 RF(Radio Frequency) 선로의 경우 일반적으로 동축케이블이 이용되고 있다.

【발명의 내용】

【해결하고자 하는 과제】

【0007】 전자 장치에 사용되는 동축 케이블은, 고주파 선로로 제작하기 위하여 추가 자재 비용이 많이 발생하며, 조립 공수 등으로 인한 추가 인건비가 발생할 수 있다. 또 다른 예로, 전자 장치의 내부 공간은 각종 회로 모듈들이 장착되어 공간이 협소하고, 협소한 공간으로 인해 동축케이블에 사용되는 작은 커넥터를 조립시 불량률이 높아질 수 있다.

【0008】 본 발명의 다양한 실시예에 따르면, 좁은 공간에서 다른 모듈에 영향을 미치지 않으면서도 잡음 없는 고주파 신호를 효과적으로 전송할 수 있는 전송 선로를 포함할 수 있다.

【과제의 해결 수단】

【0010】 본 발명에 따른 전자장치는, 하우징; 상기 하우징 내의 인쇄회로기판으로서, 복수개의 층을 포함하고 상기 복수개의 층은 하나 이상의 전도층과 하나 이상의 절연층을 포함하는 인쇄회로기판; 상기 하우징의 적어도 한 부분으로 형성되거나 상기 하우징 내에 배치되는 제 1전기 부품; 상기 하우징 내 상기 인쇄회로기판 상부 또는 근처에 배치되는 제 2전기 부품으로서, 상기 제 1전기 부품과 분리되는 제 2전기 부품; 및 상기 제 1부품으로부터 상기 제 2부품으로 연장하는 적어도 하나의 전기 패스로서, 적어도 한 부분은 상기 인쇄회로기판 상부 또는 내부로

지나가는 전기 패스를 포함하며, 상기 인쇄회로기판은 전도성 비아들의 패턴을 포함하는 영역을 포함하고, 상기 각각의 비아는 상기 복수의 층들의 적어도 일부를 통하여 상기 하나 이상의 도전층들의 적어도 하나와 접촉하고, 상기 전기 패스는 상기 전도성 비아들과 접촉하지 않으면서 상기 영역을 통해 지나갈 수 있다.

【0012】본 발명에 따른 전자장치에 있어서, 제 1 인쇄회로기판; 접속부를 통해 상기 제 1 인쇄회로기판과 연결되며, 복수 개의 층으로 구성된 적어도 하나의 연성 인쇄회로기판과 경성 인쇄회로기판을 일체형으로 구비된 제 2 인쇄회로기판; 상기 접속부를 통해 고주파 신호를 전달하며, 상기 제 2 인쇄회로기판의 상부 또는 내부를 통과하여 지나가는 전기 패스; 및 상기 적어도 하나의 경성 인쇄회로기판에 배치되어 상기 복수 개의 층간의 그라운드를 연결하는 적어도 하나의 전도성 비아를 포함할 수 있다.

【발명의 효과】

【0014】본 발명의 다양한 실시예에 따르면, 휴대용 단말기 내의 전송선로로서 사용하는 동축케이블을 연성 인쇄회로기판을 포함하는 회로기판으로 구현하여, 동축케이블 제작을 위해 사용되었던 재료비를 절감할 수 있다.

【0015】또한, 본 발명의 다양한 실시예에 따르면, 연성 인쇄회로기판의 전송선로 사이에 경성 인쇄회로기판을 추가 배치하여, 연성 인쇄회로기판의 전송선로

에서 발생하는 라인 로스(Line loss)를 감소시키고 고주파 신호를 손실없이 제공할 수 있다.

【0016】 또한, 본 발명의 다양한 실시예에 따르면, 무선 단말기 내의 실장 공간을 효율적으로 사용하여 무선 단말기 외부로 소비자 만족을 위한 다양한 디자인으로 제작할 수 있다.

【도면의 간단한 설명】

【0018】 도 1은 본 발명의 다양한 실시예에 따른, 인쇄회로기판을 이용한 고주파 전송 선로를 나타내는 정면도이다.

도 2는 본 발명의 다양한 실시예에 따른 도 1의 인쇄회로기판의 복수 개의 층을 B-B' 방향으로 절단하여 도시한 단면도의 일부이다.

도 3은 본 발명의 다양한 실시예에 따른 도 1의 인쇄회로기판의 복수 개의 층을 B-B' 방향으로 절단하여 층간 구성을 나타낸 단면도이다.

도 4는 본 발명의 다양한 실시예에 따른, 전자 장치 내의 인쇄회로기판을 이용한 고주파 전송 선로를 나타내는 정면도이다.

도 5는 본 발명의 다양한 실시예에 따른, 전자 장치 내의 인쇄회로기판을 이용한 고주파 전송 선로를 나타내는 정면도이다.

도 6, 7은 본 발명의 다양한 실시예에 따른 전자 장치의 인쇄회로기판에 배치된 전도성 비아를 나타낸 정면도이다.

도 8은 본 발명의 다양한 실시예에 따른 전자 장치의 사시도이다.

도 9는 본 발명의 다양한 실시예에 따른 전자 장치의 안테나의 배치 관계를 나타낸 간략도이다.

도 10은 본 발명의 다양한 실시예에 따른 전자 장치의 안테나 장치의 구성도 및 인쇄회로기판의 구성도이다.

도 11은 본 발명의 다양한 실시예에 따라 제 2인쇄회로기판을 이용하여 신호 전달을 나타내는 투시도이다.

도 12 및 13은 도 11의 제 1연성 인쇄회로기판 및 제 1경성 인쇄회로기판을 확대한 도면이다.

도 14는 본 발명의 다양한 실시예에서의, 네트워크 환경 내의 전자 장치를 나타낸다.

도 15는 다양한 실시예에 따른 전자 장치의 블록도이다.

【발명을 실시하기 위한 구체적인 내용】

【0019】 이하, 본 문서의 다양한 실시예가 첨부된 도면을 참조하여 기재된다. 그러나, 이는 본 문서에 기재된 기술을 특정한 실시 형태에 대해 한정하려는 것이 아니며, 본 문서의 실시예의 다양한 변경(modifications), 균등물(equivalents), 및/또는 대체물(alternatives)을 포함하는 것으로 이해되어야 한다. 도면의 설명과 관련하여, 유사한 구성요소에 대해서는 유사한 참조 부호가 사용될 수 있다.

【0020】 본 문서에서, "가진다," "가질 수 있다," "포함한다," 또는 "포함할 수 있다" 등의 표현은 해당 특징(예: 수치, 기능, 동작, 또는 부품 등의 구성요소)의 존재를 가리키며, 추가적인 특징의 존재를 배제하지 않는다.

【0021】 본 문서에서, "A 또는 B," "A 또는/및 B 중 적어도 하나," 또는 "A 또는/및 B 중 하나 또는 그 이상" 등의 표현은 함께 나열된 항목들의 모든 가능한 조합을 포함할 수 있다. 예를 들면, "A 또는 B," "A 및 B 중 적어도 하나," 또는 "A 또는 B 중 적어도 하나"는, (1) 적어도 하나의 A를 포함, (2) 적어도 하나의 B를 포함, 또는 (3) 적어도 하나의 A 및 적어도 하나의 B 모두를 포함하는 경우를 모두 지칭할 수 있다.

【0022】 본 문서에서 사용된 "제 1," "제 2," "첫째," 또는 "둘째," 등의 표현들은 다양한 구성요소들을, 순서 및/또는 중요도에 상관없이 수식할 수 있고, 한 구성요소를 다른 구성요소와 구분하기 위해 사용될 뿐 해당 구성요소들을 한정하지 않는다. 예를 들면, 제 1 사용자 기기와 제 2 사용자 기기는, 순서 또는 중요도와 무관하게, 서로 다른 사용자 기기를 나타낼 수 있다. 예를 들면, 본 문서에 기재된 권리 범위를 벗어나지 않으면서 제 1 구성요소는 제 2 구성요소로 명명될 수 있고, 유사하게 제 2 구성요소도 제 1 구성요소로 바꾸어 명명될 수 있다.

【0023】 어떤 구성요소(예: 제 1 구성요소)가 다른 구성요소(예: 제 2 구성요소)에 "(기능적으로 또는 통신적으로) 연결되어(operatively or communicatively) coupled with/to)" 있다거나 "접속되어(connected to)" 있다고 언급된 때에는, 상기 어떤 구성요소가 상기 다른 구성요소에 직접적으로 연결되거나

나, 다른 구성요소(예: 제 3 구성요소)를 통하여 연결될 수 있다고 이해되어야 할 것이다. 반면에, 어떤 구성요소(예: 제 1 구성요소)가 다른 구성요소(예: 제 2 구성요소)에 "직접 연결되어" 있다거나 "직접 접속되어" 있다고 언급된 때에는, 상기 어떤 구성요소와 상기 다른 구성요소 사이에 다른 구성요소(예: 제 3 구성요소)가 존재하지 않는 것으로 이해될 수 있다.

【0024】 본 문서에서 사용된 표현 "~하도록 구성된(또는 설정된)(configured to)"은 상황에 따라, 예를 들면, "~에 적합한(suitable for)," "~하는 능력을 가지는(having the capacity to)," "~하도록 설계된(designed to)," "~하도록 변경된(adapted to)," "~하도록 만들어진(made to)," 또는 "~를 할 수 있는(capable of)"과 바꾸어 사용될 수 있다. 용어 "~하도록 구성된(또는 설정된)"은 하드웨어적으로 "특별히 설계된(specifically designed to)" 것만을 반드시 의미하지 않을 수 있다. 대신, 어떤 상황에서는, "~하도록 구성된 장치"라는 표현은, 그 장치가 다른 장치 또는 부품들과 함께 "~할 수 있는" 것을 의미할 수 있다. 예를 들면, 문구 "A, B, 및 C를 수행하도록 구성된(또는 설정된) 프로세서"는 해당 동작을 수행하기 위한 전용 프로세서(예: 임베디드 프로세서), 또는 메모리 장치에 저장된 하나 이상의 소프트웨어 프로그램들을 실행함으로써, 해당 동작들을 수행할 수 있는 범용 프로세서(generic-purpose processor)(예: CPU 또는 application processor)를 의미할 수 있다.

【0025】 본 문서에서 사용된 용어들은 단지 특정한 실시예를 설명하기 위해 사용된 것으로, 다른 실시예의 범위를 한정하려는 의도가 아닐 수 있다. 단수의 표

현은 문맥상 명백하게 다르게 뜻하지 않는 한, 복수의 표현을 포함할 수 있다. 기술적이거나 과학적인 용어를 포함해서 여기서 사용되는 용어들은 본 문서에 기재된 기술 분야에서 통상의 지식을 가진 자에 의해 일반적으로 이해되는 것과 동일한 의미를 가질 수 있다. 본 문서에 사용된 용어들 중 일반적인 사전에 정의된 용어들은, 관련 기술의 문맥상 가지는 의미와 동일 또는 유사한 의미로 해석될 수 있으며, 본 문서에서 명백하게 정의되지 않는 한, 이상적이거나 과도하게 형식적인 의미로 해석되지 않는다. 경우에 따라서, 본 문서에서 정의된 용어일지라도 본 문서의 실시예들을 배제하도록 해석될 수 없다.

【0026】 본 문서의 다양한 실시예들에 따른 전자 장치는, 예를 들면, 스마트폰(smartphone), 태블릿 PC(tablet personal computer), 이동 전화기(mobile phone), 영상 전화기, 전자책 리더기(e-book reader), 데스크탑 PC(desktop personal computer), 랩탑 PC(laptop personal computer), 넷북 컴퓨터(netbook computer), 워크스테이션(workstation), 서버, PDA(personal digital assistant), PMP(portable multimedia player), MP3 플레이어, 모바일 의료기기, 카메라(camera), 또는 웨어러블 장치(wearable device) 중 적어도 하나를 포함할 수 있다. 다양한 실시예에 따르면, 웨어러블 장치는 액세서리형(예: 시계, 반지, 팔찌, 발찌, 목걸이, 안경, 콘택트 렌즈, 또는 머리 착용형 장치(head-mounted-device(HMD))), 직물 또는 의류 일체형(예: 전자 의복), 신체 부착형(예: 스킨 패드(skin pad) 또는 문신), 또는 생체 이식형(예: implantable circuit) 중 적어도 하나를 포함할 수 있다.

【0027】 어떤 실시예들에서, 전자 장치는 가전 제품(home appliance)일 수 있다. 가전 제품은, 예를 들면, 텔레비전, DVD(digital video disk) 플레이어, 오디오, 냉장고, 에어컨, 청소기, 오븐, 전자레인지, 세탁기, 공기 청정기, 셋톱 박스(set-top box), 홈 오토메이션 컨트롤 패널(home automation control panel), 보안 컨트롤 패널(security control panel), TV 박스(예: 삼성 HomeSync™, 애플TV™, 또는 구글 TV™), 게임 콘솔(예: Xbox™, PlayStation™), 전자 사전, 전자 키, 캠코더(camcorder), 또는 전자 액자 중 적어도 하나를 포함할 수 있다.

【0028】 다른 실시예에서, 전자 장치는, 각종 의료기기(예: 각종 휴대용 의료측정기기(혈당 측정기, 심박 측정기, 혈압 측정기, 또는 체온 측정기 등), MRA(magnetic resonance angiography), MRI(magnetic resonance imaging), CT(computed tomography), 촬영기, 또는 초음파기 등), 네비게이션(navigation) 장치, 위성 항법 시스템(GNSS(global navigation satellite system)), EDR(event data recorder), FDR(flight data recorder), 자동차 인포테인먼트(infotainment) 장치, 선박용 전자 장비(예: 선박용 항법 장치, 자이로 콤파스 등), 항공 전자기기(avionics), 보안 기기, 차량용 헤드 유닛(head unit), 산업용 또는 가정용 로봇, 금융 기관의 ATM(automatic teller's machine), 상점의 POS(point of sales), 또는 사물 인터넷 장치(internet of things)(예: 전구, 각종 센서, 전기 또는 가스 미터기, 스프링클러 장치, 화재경보기, 온도조절기(thermostat), 가로등, 토스터(toaster), 운동기구, 온수탱크, 히터, 보일러 등) 중 적어도 하나를 포함할 수 있다.

【0029】 어떤 실시예에 따르면, 전자 장치는 가구(furniture) 또는 건물/구조물의 일부, 전자 보드(electronic board), 전자 사인 수신 장치(electronic signature receiving device), 프로젝터(projector), 또는 각종 계측 기기(예: 수도, 전기, 가스, 또는 전파 계측 기기 등) 중 적어도 하나를 포함할 수 있다. 다양한 실시예에서, 전자 장치는 전술한 다양한 장치들 중 하나 또는 그 이상의 조합일 수 있다. 어떤 실시예에 따른 전자 장치는 플렉서블 전자 장치일 수 있다. 또한, 본 문서의 실시예에 따른 전자 장치는 전술한 기기들에 한정되지 않으며, 기술 발전에 따른 새로운 전자 장치를 포함할 수 있다.

【0030】 이하, 첨부 도면을 참조하여, 다양한 실시예에 따른 전자 장치가 설명된다. 본 문서에서, 사용자라는 용어는 전자 장치를 사용하는 사람 또는 전자 장치를 사용하는 장치(예: 인공지능 전자 장치)를 지칭할 수 있다.

【0032】 도 1은 본 발명의 다양한 실시예에 따른, 전자 장치 내의 인쇄회로기판을 이용한 고주파 전송 선로를 나타내는 정면도이다. 본 발명에 따른 전자 장치(10)는 전술한 스마트 폰이나 웨어러블 기기(wearable device)가 될 수 있다.

【0033】 본 발명의 다양한 실시예에 따라, 전자 장치(10)는 케이스에 해당하는 하우징, 상기 하우징 내의 적어도 하나의 인쇄회로기판(PCB)(300)을 포함하는 인쇄회로부, 상기 인쇄회로기판(300)의 상부 또는 내부로 지나가는 전기 패스(350)를 포함할 수 있다.

【0034】 도 1을 참고하면, 본 발명의 다양한 실시예에 따른, 인쇄회로기판(300)은 제 1 인쇄회로기판(310)과 제 2 인쇄회로기판(320)으로 구성될 수 있다. 예를 들어, 제 1 인쇄회로기판(310)은 전자 장치(10)에 포함되는 전자 부품들을 실장하고 상기 전자 부품들간 신호를 전달할 수 있는 메인보드 일 수 있으며, 제 2 인쇄회로기판(320)은 상기 전자 부품들로부터 발생한 신호를 전달하는 서브보드일 수 있다. 상기 서브보드는 상기 전자 부품들을 더 포함할 수 있다.

【0035】 본 발명의 다양한 실시예에 따른, 인쇄회로기판(300)에는 단말의 회로, 예를 들면, Application Processor(AP), Communication Processor(CP), 메모리, RF Transceiver, 중 적어도 하나를 실장할 수 있고, 신호라인(예: RF 신호라인(Signal Line))을 포함할 수 있다.

【0036】 본 발명의 다양한 실시예에 따른, 제 1 인쇄회로기판(310)과 제 2 인쇄회로기판(320)은 복수 개의 층으로 이루어져 있으며, 상기 복수 개의 층은 하나 이상의 도전층(306)과 하나 이상의 절연층(303)을 포함할 수 있다(도 2참조).

【0037】 본 발명의 다양한 실시예에 따른, 제 1 인쇄회로기판(310)은 경성 인쇄회로기판(Rigid PCB)일 수 있다. 상기 제 1 인쇄회로기판(310)은 접속부(370)를 통하여 상기 제 2 인쇄회로기판(320)과 전기적으로 연결될 수 있다. 상기 접속부(370)에서 제 1 인쇄회로기판(310)의 그라운드 패드(390)와 신호 패드(391)를 포함할 수 있고, 제 2 인쇄회로기판(320)의 그라운드 패드(392)와 신호 패드(393)를 포함할 수 있다. 제 1 인쇄회로기판(310)의 그라운드 패드(390)와 제 2 인쇄회로기판(320)의 그라운드 패드(392)는 납땜으로 연결될 수 있다. 제 1 인쇄회로

기판(310)의 신호 패드(391)와 제 2 인쇄회로기판(320)의 신호 패드(393)는 납땜으로 연결될 수 있다. 제 2 인쇄회로기판(320)은 실장 패드(394)를 포함할 수 있다. 실장 패드(394)에는 안테나 접촉용 부재(예를 들어 C-Clip) 또는 RF용 부품(예를 들어 정합 회로)이 실장될 수 있다.

【0038】 본 발명의 다양한 실시예에 따른, 제 2 인쇄회로기판(320)은 상기 제 1 인쇄회로기판(310)과 연결 배치되며, 연성 인쇄회로기판(FPCB)(330) 또는 경성 인쇄회로기판(Rigid PCB)(340)을 포함할 수 있다. 또한, 제 2 인쇄회로기판(320)은 상기 연성 인쇄회로기판(330)과 경성 인쇄회로기판(340)이 일체형으로 구성될 수 있다. 예를 들면, 순차적으로 제 1 연성 인쇄회로기판(331), 제 1 경성 인쇄회로기판(341), 제 2 연성 인쇄회로기판(332) 및 제 2 경성 인쇄회로기판(342)으로 배치될 수 있다. 상기 제 2 경성 인쇄회로기판(342)은 예를 들면, 안테나 접속부, USB 연결부 등을 포함할 수 있다.

【0039】 본 발명의 다양한 실시예에 따른, 제 2 인쇄회로기판(320)은 내측으로 전기적 신호를 전달할 수 있는 전기 패스(350)가 배치될 수 있다. 상기 전기 패스(350)는 일체로 구성된 적어도 하나의 연성 인쇄회로기판(330) 및 경성 인쇄회로기판(340)을 통과하여 제 1 인쇄회로기판(310)과 연결될 수 있다.

【0040】 본 발명의 다양한 실시예에 따른, 전기 패스(350)는 상기 제 1 전기 부품(230)으로부터 상기 제 2 전기 부품(240)으로 연장되며, 인쇄회로기판(300) 상부 또는 내부에 배치될 수 있다. (도 10참조) 전기 패스(350)는 고주파 신호를 전달하기 위한, 예를 들어, RF 신호를 전달하기 위해 그라운드 라인(360)으로 차폐

(Shielding)된, 사이드 가디드 마이크로 스트립라인(Side Guarded Micro strip line) 또는 스트립 라인 형태의 전송선로로 구현할 수 있다.

【0041】 본 발명의 다양한 실시예에 따른, 상기 제 2 인쇄회로기판(320)에 배치된 전기 패스(350)는 접속부(370)를 통해 고주파 신호를 전달받아 안테나 방사체(231,232)로 전달할 수 있다. 또한, 전기 패스(350)는 복수 개로 구성할 수 있으며, 각각의 전기 패스(350)가 상호 특성에 영향을 주지 않고 독립적으로 상기 고주파 신호를 전달할 수 있도록 상기 전기 패스(350) 사이에 그라운드 라인(360)을 배치할 수 있다.

【0042】 본 발명의 다양한 실시예에 따른, 상기 그라운드 라인(360)은 상기 전기 패스(350)와 인접하게 배치되며, 상기 전도성 비아(305)들은 그라운드 라인(360(a))의 일부 영역에 배치되기 때문에, 상기 전기 패스(350)는 전도성 비아(305)와 접촉하지 않으면서 인쇄회로기판(300) 상을 지나갈 수 있다.

【0043】 본 발명의 다양한 실시예에 따라, 제 1 인쇄회로기판(310)과 제 2 인쇄회로기판(320)은 상호간 고주파 신호를 전달하기 위해 전기 패스(350)를 연결하는 접속부(370)가 배치될 수 있다. 접속부(370)는 납땜패드, 쥘 타입(Zip Type) 커넥터 또는 B to B Type 커넥터를 포함할 수 있다. 접속부(370)는 경성 인쇄회로기판을 사용할 수 있으며, 예를 들어, B to B Type 커넥터가 경성 인쇄회로기판에 실장되거나 납땜패드가 위치할 수 있다. 또한, 예를 들어, 접속부(370)에 포함되는 전송선로의 핀맵은 그라운드 - RF 신호 - 그라운드 순서로 될 수 있다.

【0044】 이하, 상기 인쇄회로기판(300)의 복수개의 층과 전도성 비아(305)들

에 대하여 상세하게 설명하도록 한다.

【0046】 도 2는 본 발명의 다양한 실시예에 따른 도 1의 인쇄회로기판의 복수 개의 층을 B-B' 방향으로 절단하여 도시한 단면도의 일부이다. 여기서, 일 실시예에 따라, 인쇄회로기판을 제 2 인쇄회로기판(320)으로 설명하기로 한다. 다만, 하기 내용은 후술할 또 다른 실시예의 제 2 인쇄회로기판(420, 520, 620, 720)에 적용될 수 있다.

【0047】 도 2를 참조하면, 본 발명의 다양한 실시예에 따른, 제 2 인쇄회로기판(320)은 적어도 하나의 도전층(306)과 적어도 하나의 제 1절연층(302), 제 2절연층(303)을 포함한 복수개의 층으로 이루어질 수 있다. FCCL(Flexible Copper Clad Laminate)층에 상기 도전층(306)과 제 1절연층(302)이 포함될 수 있다. 또, 일 실시예에 따라, 제 1절연층(302)은 Polymaid를 포함할 수 있다.

【0048】 본 발명의 다양한 실시예에 따른, 상기 도전층(306)들은 적층 방향을 따라 배치되고, 제 1절연층(302)이 상기 도전층(306) 후면 또는 전면에 배치된다. 상기 각 층들은 번갈아 가면서 교대로 배치될 수 있다. 상기 도전층(306)들을 전기적으로 연결시킬 수 있도록 적어도 하나의 전도성 비아(305)를 형성할 수 있다.

【0049】 본 발명의 다양한 실시예에 따르면, 상기 제 2절연층(303)은 상기 도전층(306)들이 접촉하여 전기적으로 서로 연결되지 않도록, 도전층(306)들의 사

이에 구비되어 있다. 예를 들어, 복수 개의 도전층(306) 사이에 배치된 복수 개의 제 2절연층(303)은 각각의 층간을 절연시키는 역할을 하며, 수지(Resin) 및 글래스 천(Glass Fabric)을 포함하여 구성될 수 있다.

【0051】 본 발명의 다양한 실시예에 따르면, 상기 도전층(306)들의 상부에는 복수의 전자 부품이 배치됨과 동시에 전기적으로 연결될 수 있다. 전자 부품은 상기 전자 장치(10)에 구비된 안테나 방사체(231,232)와 전기적으로 연결될 수 있다. 예를 들어, 전자 부품은 안테나 클립, 커넥터 및 스위치 장치 등으로 이루어질 수 있다.

【0052】 그러나 이에 한정되는 것을 아니며, 상기 도전층(306)들은 금속 재질이외에 전기적으로 도통되는 재질이라면, 다양하게 적용될 수 있다.

【0053】 일 실시예에 따라, 상기 도전층(306)들은 복수의 신호선 및 복수의 그라운드 라인 중 어느 하나와 전기적으로 연결될 수 있다.

【0054】 본 발명의 다양한 실시예에 따른, 도전층(306)들은 상부와 하부에는 각각 도전층(306)들을 보호할 수 있는 보호층(301)이 구비될 수 있다. 보호층(301)은 솔더 레지스터와 같은 절연 코팅 물질로 구성되며, 솔더 레지스터를 할 때, 예를 들어, 포토 이미지저블 솔더 레지스트 잉크(Photo Imageable Solder Resist ink(PSR ink))를 사용할 수 있다. 상기 잉크는 부품의 실장 후에도 제품에 그대로 남아 절연 및 보호 작용을 할 수 있다. 따라서 보호층(301)이 도전층(306) 외부를

코팅하여, 제작과정 중 발생하는 회로의 단락, 합지, 부식, 오염 등을 방지하는 동시에, 제작 이후에도 외부의 충격, 습기, 화학 물질로부터 도전층(306)를 보호할 수 있다.

【0055】 본 발명의 다양한 실시예에 따른, 상기 도전층(306)들을 서로 전기적으로 연결하는 전도성 비아(305)는 전 층을 일괄 도통할 수 있는 제 1전도성 비아(305(a)) 및 인접한 층간 사이를 도통하는 제 2전도성 비아(305(b))를 포함할 수 있다. 예를 들어, 제 1전도성 비아(305(a))는 CNC(Computerized Numerical Control) 홀가공 및 동도금을 이용하여 제작하며, 제 2전도성 비아(305(b))는 레이저 홀가공 및 동도금을 이용하여 제작할 수 있다.

【0056】 본 발명의 다양한 실시예에 따른, 상기 전도성 비아(305)들은 인쇄 회로기판(300) 내에서 소정의 패턴을 형성하는 지역을 포함하며, 각각의 상기 전도성 비아(305)들은 하나 이상의 도전층(306)들의 적어도 하나와 접촉하도록 구성되는 동시에 상기 복수개의 층의 적어도 한 층을 통과할 수 있도록 구성할 수 있다.

【0057】 본 발명의 다양한 실시예에 따른, 패턴 커버층(304)은 상기 도전층(306)들의 하부 또는 상부에 하나의 층으로 패턴을 둘러싸면서 배치되어, 내부에 배치된 패턴을 보호할 수 있다. 예를 들어, 패턴 커버층(304)은 PI(Polyimide) 및 접착제(Adhesive)로 구성될 수 있다.

【0058】 본 발명의 다양한 실시예에 따르면 경성 인쇄회로기판(380,382)들과 연성 인쇄회로기판(381)이 일체형으로 구성될 수 있다. 연성 인쇄회로기판(381)은 제 2절연층(303)이 포함되지 않을 수 있고 상기 보호층(301) 대신 패턴 커버층

(304)이 포함될 수 있다.

【0060】 도 3은 본 발명의 다양한 실시예에 따른 도 1의 인쇄회로기판의 복수개의 층을 B-B' 방향으로 절단하여 층간 구성을 나타낸 단면도이다.

【0061】 도 3을 참조하면, 제 2 인쇄회로기판(320)은 제 1연성 인쇄회로기판(331), 제 1경성 인쇄회로기판(341) 및 제 2연성 인쇄회로기판(332)이 서로 연결되어 일체형으로 구성될 수 있다.

【0062】 그리고 제 2 인쇄회로기판(320)은 그라운드를 형성하기 위하여 제1층 내지 제4층 그라운드 패턴들(371,372,373,374)을 전도성 비아(305)로 연결할 수 있다. 제1층 그라운드 패턴(371) 과 제2층 그라운드 패턴(374)은 층 전체가 그라운드를 형성하고 있다. 제2층 그라운드 패턴(372) 과 제3층 그라운드 패턴(373)은 층의 일부만 그라운드를 형성하고 있다. 본 발명의 다양한 실시예에 따르면 그라운드 층이 형성되지 않은 층이 있을 수 있다.

【0063】 그라운드와 연결되지 않은 신호선 패턴(364)이 있을 수 있다. 상기 신호선 패턴(364)는 RF 신호선으로 이용될 수 있다.

【0064】

【0065】 본 발명의 다양한 실시예에 따른, 각 층의 그라운드 패턴 사이의 연결은 전 층을 일괄 도통할 수 있는 제 1전도성 비아(305(a)) 및 인접한 층간 사이를 도통하는 제 2전도성 비아(305(b))를 통해 이루어질 수 있다. 예를 들어, 진층

을 관통하는 제 1전도성 비아(305(a))를 이용하여, 전층 또는 일부층의 그라운드 패턴을 연결할 수 있다. 예를 들어 제 1층 내지 제 4층 그라운드 패턴(371,372,373,374)을 제1전도성 비아(305(a))로 연결할 수 있다. 제 2전도성 비아(305(b))로 인접한 층의 그라운드 패턴을 연결할 수 있다. 예를 들어, 제 1그라운드 패턴(371) 과 제 2그라운드 패턴(372) 를 제 2전도성 비아(305(b))로 연결 할 수 있다.

【0067】 본 발명의 다양한 실시예에 따르면, 인쇄회로기판(예: 도 1의 인쇄회로기판(300))은 제 1 인쇄회로기판(310) 및 경성-연성 인쇄회로기판을 포함한 제 2 인쇄회로기판(320)을 포함할 수 있다. 여기서, 일 예로 도 1의 인쇄회로기판(300)으로 설명하나, 그 외 도 4 내지 도 7의 인쇄회로기판(400,500,600,700)에 적용될 수 있다

【0068】 본 발명의 다양한 실시 예에 따르면, 전자 장치 내의 인쇄회로기판으로 구성된 고주파 전송 선로는 재료비 절감 및 실장 공간의 확보 등의 장점이 있는 연성 인쇄회로기판(330)이 동축케이블을 대체 적용할 수 있도록 구성할 수 있다..

【0069】 예를 들어, 제 2경성 인쇄회로기판(342)에서 제 1 인쇄회로기판(310)으로 고주파 신호를 전달하기 위하여, 제 2경성 인쇄회로기판(342)은 제 1 인쇄회로기판(310) 사이에 제 1연성 인쇄회로기판(331) 및 제 2연성 인쇄회로기판(332)을 사용할 수 있다.

【0070】 또한 일 예로, 연성 인쇄회로기판(330)을 이용하여 고주파 신호를 전달하는 경우, 발생하는 손실을 제거하기 위하여, 제 1연성 인쇄회로기판(331) 및 제 2연성 인쇄회로기판(332)사이에, 제 1경성 인쇄회로기판(341)을 배치하고 각 층의 그라운드 패턴을 전도성 비아(305)로 연결할 수 있다.

【0071】 본 발명의 다양한 실시예에 따르면, 상기 전자 장치는 증가하는 라인 로스를 개선하기 위해, 연성 인쇄회로기판(330) 중간에 제 1경성 인쇄회로기판(341)이 포함된 구간을 추가하여 배치할 수 있다. 만약, 10mm 이상의 긴 연성 인쇄회로기판(330)이 배치되는 경우, 상기 구간의 전송선로에서 큰 라인 로스(Line Loss)가 발생하는 부작용이 생길 수 있다. 예를 들어, 전송선로는 양측에 배치된 그라운드 라인(360)에서 발생하는 유도성 성분으로 인하여 스트립 라인(strip line)의 특성이 변화할 수 있으며, 그 결과 상기 전기 패스(350)에서 발생하는 라인 로스(Line Loss)가 가파르게 증가할 수 있다.

【0072】 본 발명의 다양한 실시예에 따르면, 상기 전자 장치는 제 1경성 인쇄회로기판(341) 상에 적어도 하나의 전도성 비아(305)를 배치할 수 있다. 제 1경성 인쇄회로기판(341)상에 배치된 상기 전도성 비아(305)를 통해 그라운드를 연결하게 되면, 예를 들어, 그라운드 라인(360) 상의 층간 그라운드 연결로, 그라운드 라인(360)에서 발생하는 유도성 성분을 줄일 수 있어 상기 라인 로스와 같은 부작용을 개선할 수 있다. 또한, 제 2 인쇄회로기판(320)을 구성하는 층 중에 전체가 그라운드인 층(그라운드 라인)이 있고, 상기 그라운드 라인(360) 전체가 그라운드가 존재하는 층과 전도성 비아(305)를 통해 연결되면 상기 라인 로스를 해결하는

개선 효과는 더 클 수 있다.

【0073】 본 발명의 다양한 실시예에 따르면, 상기 연성 인쇄회로기판 사이에 경성 인쇄회로기판을 배치하여 상기와 같은 층간 벌어진 또는 뒤뜰림을 방지할 수 있다. 만약, 기존 연성 인쇄회로기판이 구동부에 설치되는 경우, 연성 인쇄회로기판의 층간 간격이 벌어지거나 층 사이 뒤뜰림이 발생할 수 있다. 이 경우에도 스트립 라인의 특성이 변화하여 전송선로에서 라인 로스가 생기거나 다른 불리한 특성의 변화가 나타날 수 있다.

【0074】 그러나, 전술된 배치에 따른 전자 장치는, 연성인쇄회로기판이 움직이더라도 상기와 같은 부작용이 발생하지 않아 전송선로의 라인 로스가 개선되는 동시에 다른 불리한 특성도 완화할 수 있다.

【0075】 더욱이, 전송선로가 더 길어질 경우, 연성 인쇄회로기판 중간 중간에 각각 경성 인쇄회로기판을 더 추가하고 그라운드 라인(360)과 다른 층의 그라운드 패턴(363) 및 전도성 비아(305)의 연결을 이용하여 상기 라인 로스와 같은 단점을 방지할 수 있다. 이는 도 4에서 자세히 설명한다.

【0077】 도 4는 본 발명의 다양한 실시예에 따른, 전자 장치 내의 인쇄회로기판을 이용한 고주파 전송 선로를 나타내는 정면도이다. 본 발명에 따른 전자 장치(10)는 전술한 스마트 폰이나 웨어러블 기기(wearable device)가 될 수 있다.

【0078】 본 발명의 다양한 실시예에 따라, 전자 장치(10)는 케이스에 해당하는 하우징, 상기 하우징 내의 적어도 하나의 인쇄회로기판(PCB)(400)을 포함하는 인쇄회로부 및 상기 인쇄회로기판(400)의 상부 또는 내부로 지나가는 전기 패스(450)를 포함할 수 있다.

【0079】 도 4를 참조하면, 본 발명의 다양한 실시예에 따른, 인쇄회로기판(400)은 제 2 인쇄회로기판(420), 제 3 인쇄회로기판(411) 및 제 4 인쇄회로기판(412)으로 구성될 수 있다. 예를 들어, 제 3 인쇄회로기판(411)은 전자 장치(10)에 포함되는 전자 부품들을 실장하고 상기 전자 부품들간 신호를 전달할 수 있는 메인보드일 수 있다. 또한 일 예로, 제 4 인쇄회로기판(412)은 전자 장치(10)에 포함되는 전자 부품들을 실장하고 상기 전자 부품들간 신호를 전달할 수 있는 메인보드일 수 있다. 또한 일 예로, 제 2 인쇄회로기판(420)은 상기 전자 부품들로부터 발생한 신호를 전달하는 서브보드일 수 있다. 상기 서브보드는 상기 전자 부품들을 더 포함할 수 있다.

【0080】 본 발명의 다양한 실시예에 따른, 인쇄회로기판(400)에는 단말의 회로, 예를 들면, Application Processor(AP), Communication Processor(CP), 메모리, RF Transceiver 중 적어도 하나를 실장할 수 있고, 신호라인(예: RF 신호라인(Signal Line))을 포함할 수 있다.

【0081】 본 발명의 다양한 실시예에 따른, 제 2 인쇄회로기판(420), 제 3 인쇄회로기판(411) 및 제 4 인쇄회로기판(412)은 복수 개의 층으로 이루어져 있으며, 상기 복수개의 층은 하나 이상의 도전층과 하나 이상의 절연층을 포함할 수 있다.

【0082】 본 발명의 다양한 실시예에 따른, 제 3 인쇄회로기판(411) 및 제 4 인쇄회로기판(412)은 경성 인쇄회로기판(Rigid PCB)일 수 있다. 제 3 인쇄회로기판(411) 또는 제 4 인쇄회로기판(412)은 접속부(470)를 통하여 상기 제 2 인쇄회로기판(410)과 전기적으로 연결될 수 있다.

【0083】 제 3 인쇄회로기판(411)은 제 2 인쇄회로기판(420)의 일 단에 배치되며, 제 4 인쇄회로기판(412)은 제 2 인쇄회로기판(420)의 타단에 배치될 수 있다. 이하, 제 3 인쇄회로기판은 제 3경성 인쇄회로기판(411)으로 제 4 인쇄회로기판은 제 4경성 인쇄회로기판(412)으로 설명한다.

【0084】 본 발명의 다양한 실시예에 따른, 제 3경성 인쇄회로기판(411)은 제 2인쇄회로기판(320)과 연결되며, 전기 패스(450)를 통하여 고주파 신호를 전달하기 위하여 제 1접속부(471)가 배치될 수 있다. 제 1접속부(471)는 납땜패드, Zip Type 커넥터 또는 B to B Type 커넥터를 포함할 수 있다. 제 1접속부(471)는 경성 인쇄회로기판을 사용할 수 있으며, 예를 들어, B to B Type 커넥터가 경성 인쇄회로기판에 실장되거나 납땜패드가 위치할 수 있다. 또한, 예를 들어, 제 1접속부(471)에 포함되는 전송선로의 핀맵은 그라운드 - RF 신호 - 그라운드 순서로 될 수 있다.

【0085】 본 발명의 다양한 실시예에 따른, 상기 제 1접속부(471)에서 제 3경성 인쇄회로기판(411)의 그라운드 패드(490)와 신호 패드(491)를 포함할 수 있고, 제 2 인쇄회로기판(420)의 그라운드 패드(492)와 신호 패드(493)를 포함할 수 있다. 제 3경성 인쇄회로기판(411)의 그라운드 패드(490)와 제 2 인쇄회로기판(420)의 그라운드 패드(492)는 납땜으로 연결될 수 있다. 제 3경성 인쇄회로기판

(411)의 신호 패드(491)와 제 2 인쇄회로기판(420)의 신호 패드(493)는 납땜으로 연결될 수 있다.

【0086】본 발명의 다양한 실시예에 따른, 제 4경성 인쇄회로기판(412)은 제 2 인쇄회로기판(420)과 연결되며, 전기 패스(450)를 통하여 고주파 신호를 전달하기 위하여 제 2접속부(472)가 배치될 수 있다. 제 2접속부(472)는 납땜패드, Zip Type 커넥터 또는 B to B Type 커넥터를 포함할 수 있다. 제 2접속부(472)는 경성 인쇄회로기판을 사용할 수 있으며, 예를 들어, B to B Type 커넥터가 경성 인쇄회로기판에 실장되거나 납땜패드가 위치할 수 있다. 또한, 예를 들면, 제 2접속부(472)에 포함되는 전송선로의 편맵은 그라운드 - RF 신호 - 그라운드 순서로 될 수 있다.

【0087】본 발명의 다양한 실시예에 따른, 상기 제 2접속부(472)에서 제 4경성 인쇄회로기판(412)의 그라운드 패드(495)와 신호 패드(496)를 포함할 수 있고, 제 2 인쇄회로기판(420)의 그라운드 패드(497)와 신호 패드(498)를 포함할 수 있다. 제 4경성 인쇄회로기판(412)의 그라운드 패드(495)와 제 2 인쇄회로기판(420)의 그라운드 패드(497)는 납땜으로 연결될 수 있다. 제 4경성 인쇄회로기판(412)의 신호 패드(496)와 제 2 인쇄회로기판(420)의 신호 패드(498)은 납땜으로 연결될 수 있다. 제 2 인쇄회로기판(420)은 실장 패드(494)를 포함할 수 있다. 실장 패드(494)에는 안테나 접촉용 부재(예를 들어 C-Clip) 또는 RF용 부품(예를 들어 정합 회로)이 실장될 수 있다.

【0088】 본 발명의 다양한 실시예에 따른, 제 2 인쇄회로기판(420)은 상기 제 3경성 인쇄회로기판(411) 및 제 4경성 인쇄회로기판(412) 사이에 배치되며, 연성 인쇄회로기판(FPCB)(430) 또는 경성 인쇄회로기판(Rigid PCB)(440)을 포함할 수 있다. 또한, 제 2 인쇄회로기판(420)은 상기 연성 인쇄회로기판(430)과 경성 인쇄회로기판(440)이 일체형으로 구성될 수 있다. 예를 들면, 순차적으로 제 1 연성 인쇄회로기판(431), 제 1경성 인쇄회로기판(440), 제 2연성 인쇄회로기판(432)으로 배치될 수 있다. 상기 제 1경성 인쇄회로기판(440)은 예를 들면, 안테나 접속부, USB 연결부 등을 포함할 수 있다.

【0089】 본 발명의 다양한 실시예에 따른, 전기 패스(450)는 안테나 방사체를 포함하는 제 1전기 부품으로부터 통신회로를 포함하는 제 2전기 부품으로 확장 연결되며, 인쇄회로기판(400) 상부 또는 내부에 배치될 수 있다. 전기 패스(450)는 고주파 신호를 전달하기 위한, 예를 들어, RF 신호를 전달하기 위해 그라운드 라인으로 차폐(Shielding)된 사이드 가디드 마이크로 스트립라인(Side Guarded Micro strip line) 또는 스트립 라인 형태의 전송선로로 구현할 수 있다.

【0090】 본 발명의 다양한 실시예에 따른, 상기 제 2 인쇄회로기판(420)에 배치된 전기 패스(450)는 접속부(470)를 통해 고주파 신호를 전달받아 안테나 방사체(231,232)로 전달할 수 있다(도 10참조). 또한, 전기 패스(450)는 복수 개로 구성할 수 있으며, 각각의 전기 패스(450)가 상호 특성에 영향을 주지 않고 독립적으로 상기 고주파 신호를 전달할 수 있도록 상기 전기 패스(350) 사이에 그라운드 라인(460)을 배치할 수 있다.

【0091】 본 발명의 다양한 실시예에 따른, 상기 그라운드 라인(460)은 상기 전기 패스(450)와 인접하게 배치되며, 상기 전도성 비아(405)들은 상기 그라운드 라인(460)의 일부 영역에 배치되기 때문에, 전기 패스(450)는 전도성 비아(405)와 접촉하지 않으면서 인쇄회로기판(400) 상을 지나갈 수 있다.

【0092】 상기 인쇄회로기판(400)의 복수 개의 층과 전도성 비아(405)들에 대한 구체적인 설명은 상기 실시예(도 4,5,6)과 동일하므로 생략하도록 한다.

【0094】 도 5는 본 발명의 다양한 실시예에 따른, 전자 장치 내의 인쇄회로기판을 이용한 고주파 전송 선로를 나타내는 정면도이다. 본 발명에 따른 전자 장치(10)는 전술한 스마트 폰이나 웨어러블 기기(wearable device)가 될 수 있다.

【0095】 본 발명의 다양한 실시예에 따라, 전자 장치(10)는 케이스에 해당하는 하우징, 상기 하우징 내의 적어도 하나의 인쇄회로기판(PCB)(500)을 포함하는 인쇄회로부 및 상기 인쇄회로기판(500)의 상부 또는 내부로 지나가는 전기 패스(550)를 포함할 수 있다.

【0096】 도 5를 참조하면, 본 발명의 다양한 실시예에 따른, 인쇄회로기판(500)은 제 1 인쇄회로기판(510)과 제 2 인쇄회로기판(520)으로 구성될 수 있다. 예를 들어, 제 1 인쇄회로기판(510)은 전자 장치(10)에 포함되는 전자 부품들을 실장하고 상기 전자 부품들간 신호를 전달할 수 있는 메인보드일 수 있으며, 제 2 인쇄회로기판(520)은 상기 전자 부품들을 실장하고 상기 전자 부품들로부터 발생한

신호를 전달하는 서브보드일 수 있다. 상기 서브보드는 상기 전자 부품들을 더 포함할 수 있다.

【0097】 본 발명의 다양한 실시예에 따른, 인쇄회로기판(500)에는 단말의 회로, 예를 들면, Application Processor(AP), Communication Processor(CP), 메모리, RF Transceiver 중 적어도 하나를 실장할 수 있고, 신호라인(예를 들면, RF 신호라인(Signal Line))을 포함할 수 있다.

【0098】 본 발명의 다양한 실시예에 따른, 제 1 인쇄회로기판(510)과 제 2 인쇄회로기판(520)은 복수 개의 층으로 이루어져 있으며, 상기 복수 개의 층은 하나 이상의 도전층과 하나 이상의 절연층을 포함할 수 있다.

【0099】 본 발명의 다양한 실시예에 따른, 제 1 인쇄회로기판(510)은 경성 인쇄회로기판(Rigid PCB)일 수 있다. 제 1 인쇄회로기판(520)은 접속부(570)를 통하여 상기 제 2 인쇄회로기판(510)과 전기적으로 연결될 수 있다. 본 발명의 다양한 실시예에 따른, 제 1 인쇄회로기판(510)은 제 2 인쇄회로기판(520)과 연결되며, 전기 패스(550)를 통하여 고주파 신호를 전달하기 위하여 접속부(570)가 배치될 수 있다. 접속부(570)는 납땜패드, Zip Type 커넥터 또는 B to B Type 커넥터를 포함할 수 있다. 접속부(570)는 경성 인쇄회로기판을 사용할 수 있으며, 예를 들어, B to B Type 커넥터가 경성 인쇄회로기판에 실장되거나 납땜패드가 위치할 수 있다.

【0100】 본 발명의 다양한 실시예에 따른, 상기 접속부(570)에서 제 1 인쇄회로기판(510)의 그라운드 패드(590)와 신호 패드(591)를 포함할 수 있고, 제 2 인쇄회로기판(520)의 그라운드 패드(592)와 신호 패드(593)를 포함할 수 있다. 제

1 인쇄회로기판(510)의 그라운드 패드(590)와 제 2 인쇄회로기판(520)의 그라운드 패드(592)는 납땜으로 연결될 수 있다. 제 1 인쇄회로기판(510)의 신호 패드(591)와 제 2 인쇄회로기판(520)의 신호 패드(593)는 납땜으로 연결될 수 있다. 제 2 인쇄회로기판(520)은 실장 패드(594)를 포함할 수 있다. 실장 패드(594)에는 안테나 접촉용 부재(예를 들어 C-Clip) 또는 RF용 부품(예를 들어 정합 회로)이 실장될 수 있다.

【0101】본 발명의 다양한 실시예에 따른, 제 2 인쇄회로기판(520)은 상기 연성 인쇄회로기판(530)과 경성 인쇄회로기판(540)이 일체형으로 구성될 수 있다. 예를 들면, 제 2 인쇄회로기판(520)은 순차적으로 제 1 연성 인쇄회로기판(531), 제 1경성 인쇄회로기판(541), 제 2연성 인쇄회로기판(532)으로 배치될 수 있다. 상기 2연성 인쇄회로기판(532)은 안테나 접속부, USB 연결부 등을 포함할 수 있다. 본 발명에서 상기 실시예(도 1참조)와 달리, 제 2연성 인쇄회로기판(532)이 제 2경성 인쇄회로기판(542)(도 1참조)의 역할까지 수행할 수 있도록 구성하였다.

【0102】본 발명의 다양한 실시예에 따른, 전기 패스(550)는 안테나 방사체를 포함하는 제 1전기 부품으로부터 통신회로를 포함하는 제 2전기 부품으로 확장 연결되며, 인쇄회로기판(500) 상부 또는 내부에 배치될 수 있다. 전기 패스(550)는 고주파 신호를 전달하기 위한, 예를 들어, RF 신호를 전달하기 위해 그라운드 라인으로 차폐된 사이드 가디드 마이크로 스트립라인(Side Guarded Micro strip line) 또는 스트립 라인 형태의 전송선로로 구현할 수 있다.

【0103】 본 발명의 다양한 실시예에 따른, 상기 제 2 인쇄회로기판(520)에 배치된 전기 패스(550)는 접속부(570)를 통해 고주파 신호를 전달받아 안테나 방사체로 전달할 수 있다. 또한, 전기 패스(550)는 복수 개로 구성할 수 있으며, 각각의 전기 패스(550)가 상호 특성에 영향을 주지 않고 독립적으로 상기 고주파 신호를 전달할 수 있도록 상기 전기 패스(550) 사이에 그라운드 라인(560)을 배치할 수 있다.

【0104】 본 발명의 다양한 실시예에 따른, 상기 그라운드 라인(560)은 상기 전기 패스(550)와 인접하게 배치되며, 상기 전도성 비아(505)들은 그라운드 라인(560)의 일부 영역에 배치되기 때문에, 전기 패스(550)는 전도성 비아(505)들과 접촉하지 않으면서 인쇄회로기판(500) 상을 지나갈 수 있다.

【0105】 상기 인쇄회로기판(500)의 복수 개의 층과 전도성 비아(505)들에 대한 구체적인 설명은 상기 실시예(도 4,5,6)과 동일하므로 생략하도록 한다.

【0107】 도 6은 본 발명의 다양한 실시예에 따른 인쇄회로기판에 배치된 전도성 비아를 나타낸 정면도이다. 도 6(a) 내지 6(c)는 다양한 형태의 인쇄회로기판을 나타낸다.

【0108】 도 6(a)를 참조하면, 본 발명의 다양한 실시예에 따른, 제 2 인쇄회로기판(620)은 상기 연성 인쇄회로기판(630)과 경성 인쇄회로기판(640)이 연결된 일체형으로 구성될 수 있다. 예를 들면, 제 2 인쇄회로기판(620)은 순차적으로 제

1 연성 인쇄회로기판(631), 제 1경성 인쇄회로기판(640) 및 제 2연성 인쇄회로기판(632)으로 배치될 수 있다.

【0109】 본 발명의 다양한 실시예에 따른, 전기 패스(650)는 상기 제 2 인쇄회로기판(620) 상부 또는 내부에 배치되며, 고주파 신호를 전달하기 위하여 사이드 가이드 마이크로 스트립라인(Side Guarded Micro strip line) 또는 스트립 라인 형태의 전송선로로 구현할 수 있다.

【0110】 본 발명의 다양한 실시예에 따른, 상기 제 2 인쇄회로기판(620)에 배치된 전기 패스(650)는 접속부를 통해 고주파 신호를 전달받아 안테나 방사체로 전달할 수 있다. 또한, 전기 패스(650)는 복수 개로 구성할 수 있으며, 각각의 전기 패스(650)가 상호 특성에 영향을 주지 않고 독립적으로 상기 고주파 신호를 전달할 수 있도록 상기 전기 패스(650) 사이에 그라운드 라인(660)을 배치할 수 있다.

【0111】 본 발명의 다양한 실시예에 따른, 상기 그라운드 라인(660)은 상기 전기 패스(650)와 인접하게 배치되며, 인쇄회로기판에서 전도성 비아(605)들을 통해 하나의 이상의 도전층과 연결될 수 있다. 그라운드 라인(660)에 상기 전도성 비아(605)들이 배치되기 때문에, 전기 패스(650)는 전도성 비아(605)와 접촉하지 않으면서 인쇄회로기판 상을 지나갈 수 있다.

【0112】 본 발명의 다양한 실시예에 따른, 전기 패스(650)는 상기 연성회로기판(630)과 경성회로기판(640)을 가로질러 통과할 수 있도록 배치되기 때문에, 상기 전기 패스(650) 양 측에 이격 배치된 그라운드 라인(660)도 상기 연성회로기판

과 경성회로기판을 가로질러 통과할 수 있도록 배치될 수 있다. 본 발명에서 그라운드 라인(660)을 두 개로 구성하고 있으나, 이에 한정된 것은 아니며, 전기 패스(650)와 접촉하지 않도록 다양한 수로 형성할 수 있다.

【0113】 예를 들어, 좌측에 배치된 제 1그라운드 라인(660(a))은 우측에 배치된 제 2그라운드 라인(660(b))에 비하여 상대적으로 넓은 폭을 가지고 있으며, 경성 인쇄회로기판(640)과 그라운드 라인(660)이 교차되는 지역에는 복수개의 전도성 비아(605)들이 형성될 수 있다. 또 다른 예에 따른, 제 2그라운드 라인(660(b))은 제 1그라운드 라인(660(a))에 비하여 상대적으로 폭이 좁아 상기 전도성 비아(605(c))들을 구비하지 못 할 수 있다.

【0114】 본 발명의 실시예에 따르면, 넓은 폭을 가진 제 1그라운드 라인(660(a))에 배치된 전도성 비아(605(c))들로 구성된 인쇄회로기판을 나타내었다. 제 1그라운드 라인(660(a))에 배치된 전도성 비아(605(c))들은, 전 층을 관통하는 비아들로 구성될 수 있다. 상기 전층을 관통하는 전도성 비아(605(c))를 이용하여, 전층 또는 일부층의 그라운드 패턴과 전기 패스 측면의 그라운드 라인을 연결할 수 있다.

【0115】 도 6(b)에 따른 다른 실시예를 살펴보면, 제 2 인쇄회로기판(620)에서, 그라운드 라인(660)은 전기 패스(650) 양 측에 이격 배치되어 있으며, 전기 패스(650) 길이 방향에 따라 상기 연성회로기판과 경성회로기판을 가로질러 통과하도록 형성할 수 있다. 본 발명에서 그라운드 라인(660)을 두 개로 구성하고 있으나, 이에 한정된 것은 아니며, 전기 패스(650)와 접촉하지 않도록 다양한 수로 형성할

수 있다.

【0116】 본 발명의 다양한 실시예에 따른, 좌측에 배치된 제 1그라운드 라인(660(a))은 우측에 배치된 제 2그라운드 라인(660(b))에 비하여 상대적으로 넓은 폭을 가지고 있으며, 제 1그라운드 라인(660(a))과 경성 인쇄회로기판(640)과 교차되는 지역에 복수개의 전도성 비아(605)들을 구성할 수 있다. 또 다른 예에 따른, 우측의 제 2그라운드 라인(660(b))은 제 1그라운드 라인(660(a))에 비하여 상대적으로 폭이 좁아 상기 크기의 전도성 비아(605(c))들을 구비하지 못 할 수 있다. 대신, 본 발명의 실시예에 따르면, 우측의 제 2그라운드 라인(660(b))은 전도성 비아(605(c))보다 작은 크기를 가진 전도성 비아(605(d))들을 배치할 수 있다.

【0117】 예를 들어, 복수개의 층으로 이루어진 경성 인쇄회로기판(640)은 상기 회로기판 내부의 전 층을 관통하는 제 3전도성 비아(605(c))들과 일부 층을 관통하는 제 4전도성 비아(605(d))들을 포함할 수 있다. 제 3전도성 비아(605(c))는 일부 층을 관통하는 제 4전도성 비아(605(d))에 비하여 상대적으로 홀의 크기가 더 크며, 넓은 폭의 그라운드 라인(660(c)) 상에 배치될 수 있다. 그러나 제 4전도성 비아(605(d))들은 3전도성 비아(605(c))에 비하여 상대적으로 홀의 더 작기 때문에 우측의 제 2그라운드 라인(660(b))과 같은 상대적으로 좁은 폭의 그라운드 라인(660) 상에 배치될 수 있다. 일부 전도성 비아(605)를 통하여 그라운드 라인(660)과 다른 층의 그라운드를 연결하더라도 라인 로스 개선과 같은 성능은 향상될 수 있다.

【0118】 도 6(c)의 다른 실시예를 살펴보면, 그라운드는 전기 패스(650) 양측에 이격 배치되어 있으며, 전기 패스(650) 길이 방향에 따라 그라운드 라인(360)을 형성할 수 있다. 본 발명에서 그라운드 라인(660)을 두 개로 구성하고 있으나, 이에 한정된 것은 아니며, 전기 패스(650)와 접촉하지 않도록 다양한 수로 형성할 수 있다.

【0119】 본 발명의 다양한 실시예에 따른, 좌측에 배치된 제 1그라운드 라인(660(a))은 우측에 배치된 제 2그라운드 라인(660(b))에 비하여 상대적으로 좁은 폭을 가지고 있으며, 경성 인쇄회로기판(640)과 교차되는 지역에 복수개의 전도성 비아(605)들을 형성할 수 없다. 또 다른 예에 따른, 우측의 제 2그라운드 라인(660(b))은 제 1그라운드 라인(660(a))에 비하여 상대적으로 폭이 넓지만 큰 크기의 전도성 비아(605(c))들을 구비 하기엔 좁다. 대신, 본 발명의 실시예에 따르면, 우측의 제 2그라운드 라인(660(b))에서 상대적으로 좁은 폭의 전도성 비아(605(d))들을 배치할 수 있다.

【0120】 따라서 본 발명의 다양한 실시예에 따라, 우측의 제 2그라운드 라인(660(b))과 같은 상대적으로 좁은 폭의 그라운드 라인(660)의 경우, 일부 층을 관통하는 제 4전도성 비아(605(d))들을 이용하여 인쇄회로기판을 전기적으로 연결할 수 있다. 상기 일부 제 4전도성 비아(605(d))를 통하여 그라운드 라인(360)과 다른 층의 그라운드를 연결하더라도 라인 로스 개선과 같은 성능은 향상될 수 있다.

【0122】 도 7은 본 발명의 다양한 실시예에 따른 인쇄회로기판에 배치된 전

도성 비아를 나타낸 상면도이다. 도 7(a) 내지 7(c)는 다양한 형태의 인쇄회로기판을 나타낸다.

【0123】 도 7(a)를 살펴보면, 제 2 인쇄회로기판(720)은 연성 인쇄회로기판 또는 경성 인쇄회로기판 중 적어도 하나를, 복수 개 포함할 수 있다. 예를 들어 제 2 인쇄회로기판(720)은 연성 인쇄회로기판(730)과 경성 인쇄회로기판(740)이 일체형으로 연성 인쇄회로기판(730)과 경성 인쇄회로기판(740)이 교대로 배치되도록 구성할 수 있고, 일체형으로 구성될 수 있다.

【0124】 본 발명의 다양한 실시예에 따른, 상기 연성 인쇄회로기판(730)은 제 1 연성 인쇄회로기판(731), 제 2연성 인쇄회로기판(732), 제 3연성 인쇄회로기판(733)을 포함할 수 있다. 또한, 상기 경성 인쇄회로기판(740)은 제 1경성 인쇄회로기판(741), 제 5경성 인쇄회로기판(742)을 포함할 수 있다.

【0125】 본 발명의 다양한 실시예에 따른, 연성 회로기판(730) 사이에 적어도 두 개의 경성 인쇄회로기판(740)을 포함할 수 있다. 예를 들면, 상기 제 2 인쇄회로기판(720)은 순차적으로 제 1연성 인쇄회로기판(731), 제 1경성 인쇄회로기판(741), 제 2연성 인쇄회로기판(732), 제 5경성 인쇄회로기판(742) 및 제 3연성 인쇄회로기판(733)을 배치할 수 있다.

【0126】 각각의 제 1경성 인쇄회로기판(741) 및 제 5경성 인쇄회로기판(742)은 서로 이격되어 배치되며, 복수 개의 전도성 비아(705) 및 그라운드 라인(760)을 포함할 수 있다.

【0127】 본 발명의 다양한 실시예에 따른, 전기 패스(750)는 상기 제 2 인쇄 회로기판(720) 상부 또는 내부에 배치되며, 고주파 신호를 전달하기 위하여 사이드 가이드 마이크로 스트립라인(Side Guarded Micro strip line) 또는 스트립 라인 형태의 전송선로로 구현할 수 있다.

【0128】 본 발명의 다양한 실시예에 따른, 상기 제 2 인쇄회로기판(720)에 배치된 전기 패스(750)는 접속부를 통해 고주파 신호를 전달받아 안테나 방사체로 전달할 수 있다. 또한, 전기 패스(750)는 복수개로 구성할 수 있으며, 각각의 전기 패스(750)가 상호 특성에 영향을 주지 않고 독립적으로 상기 고주파 신호를 전달할 수 있도록 상기 전기 패스(750) 사이에 그라운드 라인(760)을 배치할 수 있다.

【0129】 본 발명의 다양한 실시예에 따른, 그라운드 라인(760)은 상기 전기 패스(750)와 인접하게 배치되며, 제 2인쇄회로기판(720)에서 전도성 비아(705)들을 통해 하나의 이상의 도전층과 연결될 수 있다.

【0130】 본 발명의 다양한 실시예에 따른, 그라운드 라인(760)은 전기 패스(750) 길이 방향에 따라 연성 인쇄회로기판(730) 및 경성 인쇄회로기판(740)을 가로질러 형성할 수 있다. 본 발명에서 그라운드 라인(760)을 두 개로 구성하고 있으나, 이에 한정된 것은 아니며, 전기 패스(750)와 접촉하지 않도록 다양한 수로 형성할 수 있다.

【0131】 예를 들어, 제 1경성 인쇄회로기판(741) 및 제 5경성 인쇄회로기판(742)의 좌측에 배치된 제 1그라운드 라인(760(a))은 우측에 배치된 제 2그라운드

라인(760(b))에 비하여 상대적으로 넓은 폭을 가지고 있으며, 제 1그라운드 라인(760(a))과 제 1경성 인쇄회로기판(741) 및 제 5경성 인쇄회로기판(742)과 교차되는 지역에 복수개의 전도성 비아(705)들을 구성할 수 있다. 또 다른 예에 따른, 우측의 제 2그라운드 라인(760(b))은 제 1그라운드 라인(660(b))에 비하여 상대적으로 폭이 좁아 상기 크기의 전도성 비아(705(c))들을 구비하지 못 할 수 있다. 본 발명의 실시예에 따르면, 넓은 폭을 가진 제 1그라운드 라인(760(a))에 배치된 전도성 비아(705(c))들로 구성된 인쇄회로기판을 나타내었다.

【0132】 도 7(b)의 다른 실시예를 살펴보면, 제 2 인쇄회로기판(720)은 상기 연성 인쇄회로기판(730)과 경성 인쇄회로기판(740)이 일체형으로 구성될 수 있다. 예를 들면, 제 2 인쇄회로기판(720)은 연성 인쇄회로기판(730) 과 경성 인쇄회로기판(740)이 일체형으로 연성 인쇄회로기판(730) 과 경성 인쇄회로기판(740)이 교대로 배치되도록 구성할 수 있고, 일체형으로 구성될 수 있다.

【0133】 본 발명의 다양한 실시예에 따른, 제 2인쇄회로기판(720)은 순차적으로 제 1 연성 인쇄회로기판(731), 제 1경성 인쇄회로기판(741), 제 2연성 인쇄회로기판(732), 및 제 5경성 인쇄회로기판(742) 및 제 3연성 인쇄회로기판(733)으로 배치될 수 있다.

【0134】 본 발명의 다양한 실시예에 따른, 그라운드(760)는 전기 패스(750) 양 측에 이격 배치되어 있으며, 전기 패스(750) 길이 방향에 따라 그라운드 라인(760)을 형성할 수 있다. 본 발명에서 그라운드 라인(760)을 두 개로 구성하고 있으나, 이에 한정된 것은 아니며, 전기 패스(350)와 접촉하지 않도록 다양한 수로

형성할 수 있다.

【0135】 본 발명의 다양한 실시예에 따른, 제 1경성 인쇄회로기판(741) 및 제 5경성 인쇄회로기판(742)의 좌측에 배치된 제 1그라운드 라인(760(a))은 우측에 배치된 제 2그라운드 라인(760(b))에 비하여 상대적으로 넓은 폭을 가지고 있으며, 제 1그라운드 라인(760(a))과 경성 인쇄회로기판(740)과 교차되는 지역에 복수개의 전도성 비아(705)들을 구성할 수 있다. 그러나 우측의 제 2그라운드 라인(760(b))은 제 1그라운드 라인(660(b))에 비하여 상대적으로 폭이 좁아 상기 크기의 전도성 비아(705(c))들을 구비할 수 없다. 대신, 우측의 제 2그라운드 라인(760(b))은 전도성 비아(705(c))보다 작은 크기를 가진 전도성 비아(705(d))들을 배치할 수 있다.

【0136】 예를 들어, 복수개의 층으로 이루어진 경성 인쇄회로기판(740)은 상기 회로기판 내부의 전 층을 관통하는 제 3전도성 비아(705(c))들과 일부 층을 관통하는 제 4전도성 비아(705(d))들을 포함할 수 있다. 제 3전도성 비아(705(c))는 일부 층을 관통하는 제 4전도성 비아(705(d))에 비하여 상대적으로 홀의 크기가 더 크며, 넓은 폭의 그라운드 라인(760(c)) 상에 배치될 수 있다. 그러나 제 4전도성 비아(705(d))들은 3전도성 비아(705(c))에 비하여 상대적으로 홀의 더 작기 때문에 우측의 제 2그라운드 라인(760(b))과 같은 상대적으로 좁은 폭의 그라운드 라인(660) 상에 배치될 수 있다. 일부 전도성 비아(605)를 통하여 그라운드 라인(660)과 다른 층의 그라운드를 연결하더라도 라인 로스 개선과 같은 성능은 향상될 수 있다.

【0137】 도 7(c)의 다른 실시예를 살펴보면, 제 2 인쇄회로기판(720)은 상기 연성 인쇄회로기판(730)과 경성 인쇄회로기판(740)이 교대로 배치되도록 구성할 수 있고, 일체형으로 구성될 수 있다. 본 실시예에서는 연성 회로기판(730) 사이에 적어도 두 개의 경성 인쇄회로기판(740)을 포함할 수 있다. 예를 들면, 제 2인쇄회로기판(720)은 순차적으로 제 1연성 인쇄회로기판(731), 제 2연성 인쇄회로기판(732), 제 1경성 인쇄회로기판(741), 제 5경성 인쇄회로기판(742) 및 제 3연성 인쇄회로기판(733)으로 배치될 수 있다.

【0138】 본 발명의 다양한 실시예에 따른, 그라운드(760)는 전기 패스(750) 양 측에 이격 배치되어 있으며, 전기 패스(750) 길이 방향에 따라 그라운드 라인(760)을 형성할 수 있다. 본 발명에서 그라운드 라인(760)을 두 개로 구성하고 있으나, 이에 한정된 것은 아니며, 전기 패스(750)와 접촉하지 않도록 다양한 수로 형성할 수 있다.

【0139】 본 발명의 다양한 실시예에 따른, 제 1경성 인쇄회로기판(741) 및 제 5경성 인쇄회로기판(742)의 좌측에 배치된 제 1그라운드 라인(760(a))은 우측에 배치된 제 2그라운드 라인(760(b))에 비하여 상대적으로 좁은 폭을 가지고 있으며, 경성 인쇄회로기판(740)과 교차되는 지역에 복수개의 전도성 비아(705)들을 형성할 수 없다. 또 다른 예에 따른, 우측의 제 2그라운드 라인(760(b))은 제 1그라운드 라인(760(a))에 비하여 상대적으로 폭이 넓지만 큰 크기의 전도성 비아(705(c))들을 구비하기엔 좁다. 대신, 우측의 제 2그라운드 라인(760(b))에서 상대적으로 좁은 폭의 전도성 비아(705(d))들을 배치할 수 있다.

【0140】따라서 본 발명의 다양한 실시예에 따라, 제 2그라운드 라인(760(b))에 배치된 제 4전도성 비아(605(d))의 경우, 상기 일부 제 4전도성 비아(605(d))를 통하여 그라운드 라인(360)과 다른 층의 그라운드를 연결하더라도 라인로스 개선과 같은 성능은 향상될 수 있다.

【0142】도 8은 본 발명의 다양한 실시예에 따른 전자 장치의 사시도이다. 상기 전자 장치(10)는 스마트 폰이나 웨어러블 기기(wearable device)가 될 수 있다. 상기 다양한 실시예에 따른 전자장치(10)는 도 1 내지 도 7의 인쇄회로기판을 포함하는 전자 장치일 수 있다.

【0143】도 8을 참고하면, 전자 장치(10)의 전면(107)에는 디스플레이(101)가 형성될 수 있다. 디스플레이(101)의 상측으로는 상대방의 음성을 수신하기 위한 스피커 장치(102)가 형성될 수 있다. 디스플레이(101)의 하측으로는 상대방에게 전자 장치 사용자의 음성을 송신하기 위한 마이크로폰 장치(103)가 형성될 수 있다.

【0144】한 실시예에 따르면, 스피커 장치(102)가 설치되는 주변에는 전자 장치(10)의 다양한 기능을 수행하기 위한 적어도 하나의 부품(component)이 배치될 수 있다. 예를 들어, 배치되는 적어도 하나의 부품은 적어도 하나의 센서 모듈(104)을 포함할 수 있다. 이러한 센서 모듈(104)은, 예컨대, 조도 센서(예: 광센서), 근접 센서, 적외선 센서, 초음파 센서 중 적어도 하나를 포함할 수 있다. 한 실시예에 따르면, 부품은 카메라 장치(105)를 포함할 수도 있다. 한 실시예에

따르면, 부품은 전자 장치(10)의 상태 정보를 사용자에게 인지시켜주기 위한 LED 인디케이터(106)를 포함할 수도 있다.

【0145】 다양한 실시예에 따르면, 전자 장치(10)는 금속 베젤(110)(예: 금속 하우징의 적어도 일부 영역으로 기여될 수 있음)을 포함할 수 있다. 한 실시예에 따르면, 금속 베젤(110)은 전자 장치(10)의 테두리를 따라 배치될 수 있으며, 테두리와 연장되는 전자 장치(10)의 후면의 적어도 일부 영역까지 확장되어 배치될 수도 있다. 한 실시예에 따르면, 금속 베젤(110)은 전자 장치(10)의 테두리를 따라 전자 장치의 두께로 정의되며, 투프 형태로 형성될 수 있다. 그러나 이에 국한되지 않으며, 금속 베젤(110)은 전자 장치(10)의 두께 중 적어도 일부에 기여하는 방식으로 형성될 수도 있다. 한 실시예에 따르면, 금속 베젤(110)은 전자 장치(10)의 테두리 중 적어도 일부 영역에만 배치될 수도 있다. 한 실시예에 따르면, 금속 베젤(110)은 적어도 하나의 분절부(115, 116)를 포함할 수 있다. 한 실시예에 따르면, 각각의 분절부(115, 116)에 의해 분절된 단위 베젤부들은 적어도 하나의 주파수 대역에서 동작하는 안테나 방사체로 활용될 수 있다.

【0146】 다양한 실시예에 따르면, 금속 베젤(110)은 테두리를 따라 투프 형상을 가지며, 전자 장치(10)의 두께의 전부 또는 일부로 기여되는 방식으로 배치될 수 있다. 한 실시예에 따르면, 전자 장치(10)를 정면에서 보았을 경우, 금속 베젤(110)은 우측 베젤부(111), 좌측 베젤부(112), 상측 베젤부(113) 및 하측 베젤부(114)가 형성될 수 있다. 여기서, 상술한 하측 베젤부(114)는 한 쌍의 분절부(116)에 의해 형성된 단위 베젤부로써 기여될 수 있다.

【0147】 다양한 실시예에 따르면, 메인 안테나 장치는 전자 장치(10)의 하부 영역(메인부 안테나 영역)에 배치될 수 있다. 한 실시예에 따르면, 한 쌍의 분절부(116)에 의해 하측 베젤부(114)는 메인 안테나 방사체로 사용될 수 있다. 한 실시예에 따르면, 하측 베젤부(114)는 급전 위치에 따라 적어도 두 개의 작동 주파수 대역에서 동작하는 안테나 방사체로 기여될 수 있다. 예를 들어 하측 베젤부(114)는 Low Band(LB) 및 High/Mid Band(H/MB)대역을 지원하는 안테나의 일부일 수 있으며, 또는 High/Mid Band(H/MB) 대역을 지원하는 안테나의 일부일 수 있다.

【0148】 다양한 실시예에 따르면, 본 발명의 안테나 장치는 단순히 예시적인 구성일 뿐, 하측 베젤부(114)의 상술한 기능들이 또 다른 분절부(115)에 의해 분절된 상측 베젤부(113)에서 대체하여 수행되거나 함께 수행될 수도 있다.

【0149】 다양한 실시예에 따르면, 다이버시티(Diversity)부의 안테나 영역은 Diversity/MIMO(multiple input multiple output)용 안테나로 활용될 수 있다. 예를 들어, 상측 베젤부(113)는 LB, H/MB 대역을 지원하는 다이버시티 안테나의 일부일 수 있다.

【0150】 한 실시예에 따르면, 우측 베젤부(111) 또는 좌측 베젤부(112) 중 적어도 하나에 급전되어 안테나로서 동작할 수 있다. 예를 들어 우측 베젤부(111) 또는 좌측 베젤부(112)는 H/MB 또는 LB, H/MB 대역을 지원하는 안테나의 일부일 수 있다. 예를 들어, 메인부 안테나 영역에 포함된 우측 베젤부(111)또는 좌측 베젤부(112)를 포함하는 안테나는 메인 안테나로 동작할 수 있다. 다이버시티부 안테나 영역에 포함된 우측 베젤부(111)또는 좌측 베젤부(112)를 포함하는 안테나는 다

이버시디 안테나로 동작할 수 있다. 이하 안테나 구성에 대하여 구체적으로 설명한다.

【0152】 도 9는 본 발명의 다양한 실시예에 따른 전자 장치의 안테나의 배치 관계를 나타낸 간략도이다. 상기 전자 장치(10)는 스마트 폰이나 웨어러블 기기(wearable device)가 될 수 있다. 상기 다양한 실시예에 따른 전자장치(10)는 도 1 내지 도 7의 인쇄회로기판을 포함하는 전자 장치일 수 있다.

【0153】 본 발명의 다양한 실시예에 따른, 전자 장치(10)는 복수개의 안테나 방사체(230)를 포함할 수 있다. 예를 들어, 메인부(210)의 제 1안테나(231) 및 제 2안테나(232)와 다이버시티부(220)의 제 3안테나(233) 및 제 4안테나(234)로 구성될 수 있다.

【0154】 상기 제1안테나(231)는 LB 와 II/MB 대역을 지원하고 제2안테나(232)는 II/MB 대역을 지원할 수 있다. 다양한 실시예에 따르면, LB 대역이 비교적 파장이 길어 안테나 크기가 상대적으로 크며, 복수의 LB를 지원하는 안테나를 단말의 메인 부(210)에 추가로 포함 시키기 위해서는 실장 공간의 제약이 발생할 수 있어 하나의 안테나만 LB 대역을 지원하도록 구성될 수 있다.

【0155】 본 발명의 다양한 실시예에 따른, 다이버시티부(220)를 추가적으로 갖는 경우 제3안테나(233), 제4안테나(234)를 포함할 수 있다. 본 발명에서는 콜리레이션/아이솔레이션(Correlation/Isolation)특성을 높이기 위해 메인부(210)의 안

테나 방사체(230)는 단말의 하단부에 포함되며 다이버시티부(220)의 안테나 방사체(230)는 단말의 상단에 포함될 수 있다. 예를 들어, 단말의 하단부에 메인부(210)의 안테나 방사체(230)를 포함하고, 단말의 상단부에 다이버시티부(220)의 안테나 방사체(230)를 포함하는 경우, 복수의 LB대역 안테나를 포함하더라도 단말 내에서 가능한 안테나 격리 거리(Antenna Separation Distance)를 확보 할 수 있다.

【0156】 상기와 같은 구조로 H/MB대역 에서는 4개의 안테나로 수신이 가능하고 LB 대역에서는 2개의 안테나로 수신이 가능하다. 즉 H/MB 대역에서는 4th order diversity/MIMO 수행이 가능하고 LB 대역에서는 2nd Order diversity/MIMO 가 가능하다.

【0158】 도 10은 본 발명의 다양한 실시예에 따른 전자 장치의 안테나 장치의 구성도 및 인쇄회로기판의 구성도이다.

【0159】 도 10을 참조하면, 메인부(210)에 있는 메인 RF회로에서 접속부(370)와 제 2 인쇄회로기판(320)을 통해 제 1안테나(231)와 제2안테나(232)를 RF 신호선으로 연결하여 급전할 수 있다. 여기서 제 2 인쇄회로기판(320)은 전술한대로 제 1연성 인쇄회로기판(331) - 제 1경성 인쇄회로기판(341) - 제 2연성 인쇄회로기판(332) - 제 2경성 인쇄회로기판(342)으로 이루어질 수 있다. 안테나 급전부 또는 안테나 주변은 안테나 특성을 향상시키기 위해 그라운드가 제거 될 수 있다.

【0160】 본 발명의 다양한 실시예에 따라, 금속 베젤(110)은 전면에서 보았을 때, 우측 베젤부(111), 좌측 베젤부(112), 상측 베젤부(113), 하측 베젤부(114)를 포함할 수 있다. 한 실시예에 따르면, 상측 베젤부(113)는 일정 간격으로 형성된 한 쌍의 분절부에 의해 우측 베젤부(111) 및 좌측 베젤부(112)와 분리된 상태를 유지할 수 있다. 하측 베젤부(114)는 일정 간격으로 형성된 한 쌍의 분절부에 의해 우측 베젤부(111) 및 좌측 베젤부(112)와 분리된 상태를 유지할 수 있다. 한 실시예에 따르면, 한 쌍의 분절부는 유전체 재질로 형성될 수 있다. 한 실시예에 따르면, 한 쌍의 분절부는 금속 재질의 금속 베젤에 합성 수지 재질의 소재가 이중 사출되거나 인서트 몰딩되는 방식으로 형성될 수 있다. 그러나 이에 국한되지 않으며, 한 쌍의 분절부는 절연성을 갖는 다양한 재질의 소재가 적용될 수도 있다.

【0161】 본 발명의 다양한 실시예에 따르면, 하측 베젤부(114)는 소정의 제 1급전편이 하측 베젤부(114)와 일체로 형성될 수 있으며, 제 1급전편은 인쇄회로기판(PCB)의 제 1급전부에 의해 급전될 수 있다. 한 실시예에 따르면, 하측 베젤부(114)의 제 1급전편은 기판이 전자 장치에 설치되는 동작만으로 기판의 제 1급전부에 연결되거나, 별도의 전기적 연결 부재(예: C 클립 등)에 의해 전기적으로 연결될 수 있다.

【0162】 본 발명의 다양한 실시예에 따르면, 상기 인쇄회로기판에는 제 1급전 패드가 배치될 수 있으며, 제 1급전 패드는 하측 베젤부(114)의 제 1급전편과 전기적으로 연결될 수 있다. 한 실시예에 따르면, 제 1급전 패드에서 제 1급전부까지 제 1전기적 경로(예: 배선 라인)가 형성될 수 있다. 하측 베젤부(114)는 소정의

제 1급전편이 하측 베젤부(114)와 일체로 형성될 수 있으며, 제 1급전편은 인쇄회로기판(PCB)의 제 1급전부에 의해 급전될 수 있다. 한 실시예에 따르면, 하측 베젤부(114)의 제 1급전편은 기판이 전자 장치에 설치되는 동작만으로 기판의 제 1급전부에 연결되거나, 별도의 전기적 연결 부재(예: C 클립 등)에 의해 전기적으로 연결될 수 있다.

【0163】 본 발명의 다양한 실시예에 따르면, 상기 인쇄회로기판에는 제 1급전 패드가 배치될 수 있으며, 제 1급전 패드는 하측 베젤부(114)의 제 1급전편과 전기적으로 연결될 수 있다. 한 실시예에 따르면, 제 1급전 패드에서 제 1급전부까지 제 1전기적 경로(예: 배선 라인)가 형성될 수 있다.

【0164】 하측 베젤부는 LB, II/MB 또는 II/MB 대역을 지원하는 메인부(210)의 제 1안테나(231)의 일부일 수 있다. 좌측 베젤부(112)와 우측 베젤부(111)도 동일한 방법으로 급전될 수 있다. 좌측 베젤부(112)와 우측 베젤부(111)는 LB, H/MB 또는 H/MB 대역을 지원하는 메인부(210)의 제 2안테나(231)의 일부 일 수 있다.

【0165】 본 발명의 다양한 실시예에 따르면, 하측 베젤부(114)는 제 1급전편과 일정 간격으로 이격된 위치에 제 1접지편이 하측 베젤부(114)와 일체로 형성될 수 있으며, 제 1접지편은 인쇄회로기판(PCB)의 제 1접지부에 접지될 수 있다. 한 실시예에 따르면, 하측 베젤부(114)의 제 1접지편은 기판이 전자 장치에 설치되는 동작만으로 기판의 제 1접지부에 접지되거나, 별도의 전기적 연결 부재(예: C 클립 등)에 의해 전기적으로 연결될 수 있다.

【0166】 본 발명의 다양한 실시예에 따르면, 기판에는 제 1접지 패드가 배치

될 수 있으며, 제 1접지 패드는 하측 베젤부(114)의 제 1접지편과 전기적으로 연결될 수 있다. 한 실시예에 따르면, 제 1접지 패드에서 제 1접지부까지 제 2전기적 경로(예: 배선 라인)가 형성될 수 있다.

【0167】 본 발명의 다양한 실시예에 따르면, 하측 베젤부(114)는 소정의 제 2급전편이 하측 베젤부(114)와 일체로 형성될 수 있으며, 제 2급전편은 인쇄회로기판(PCB)의 제 2급전부에 의해 급전될 수 있다. 한 실시예에 따르면, 하측 베젤부(114)의 제 2급전편은 기판이 전자 장치에 설치되는 동작만으로 기판의 제 2급전부에 연결되거나, 별도의 전기적 연결 부재(예: C 클립 등)에 의해 전기적으로 연결될 수 있다.

【0168】 본 발명의 다양한 실시예에 따르면, 기판에는 제 2급전 패드가 배치될 수 있으며, 제 2급전 패드는 하측 베젤부(114)의 제 2급전편과 전기적으로 연결될 수 있다. 한 실시예에 따르면, 제 2급전 패드에서 제 2급전부까지 제 3전기적 경로(예: 배선 라인)가 형성될 수 있다. 하측 베젤부(114)는 LB,H/MB 또는 H/MB 대역을 지원하는 메인부(210)의 제 2 안테나(232)의 일부 일 수 있다.

【0169】 본 발명의 다양한 실시예에 따르면, 우측 베젤부(111)는 분절부와 일정 간격으로 이격된 위치에 제 2접지편이 우측 베젤부(111)와 일체로 형성될 수 있으며, 제 2접지편은 인쇄회로기판(PCB)의 제 2접지부에 접지될 수 있다. 한 실시예에 따르면, 우측 베젤부(111)의 제 2접지편은 기판이 전자 장치에 설치되는 동작만으로 기판의 제 2접지부에 접지되거나, 별도의 전기적 연결 부재(예: C 클립 등)에 의해 전기적으로 연결될 수 있다.

【0170】 본 발명의 다양한 실시예에 따르면, 기판에는 제 2접지 패드가 배치될 수 있으며, 제 2접지 패드는 우측 베젤부(111)의 제 2접지편과 전기적으로 연결될 수 있다. 한 실시예에 따르면, 제 2접지 패드에서 제 2접지부까지 제 4전기적 경로(예: 배선 라인)가 형성될 수 있다.

【0171】 예를 들어, 메인부(210)의 급전부와 급전 패드, 접지부와 접지 패드는 제 2 인쇄회로기판(320) 상에 배치될 수 있다. 메인부(210)와 다이버시티부(220)의 RF회로는 제 1 인쇄회로기판(Main PCB)(310) 상에 배치될 수 있다. 제 1 인쇄회로기판과 제 2 인쇄회로기판은 연성 인쇄회로기판으로 연결될 수 있다. 제 1 인쇄회로기판과 제 2 인쇄회로기판은 일체형일 수 있다.

【0172】 본 발명의 다양한 실시예에 따르면, 제 2 인쇄회로기판(320)에 포함된 제 2 경성회로기판(342)은 수직 선상에서 제 1 인쇄회로기판(Main PCB)(310) 보다 낮게 전자 장치(10)에 배치될 수 있다. 이를 통하여 제 2 경성회로기판(342)에 포함된 부품이 안테나와 더 이격될 수 있다. 그리고 USB Connector, Speaker 부와 같은 비교적 두꺼운 부품들도 제 2 경성회로기판(342) 상에 배치가 가능하다.

【0173】 본 발명의 다양한 실시예에 따르면, 메인부(210) RF 회로의 송수신 신호 또는 수신 신호는 제 2 인쇄회로기판(320)의 제 1,2급전부에 전달될 수 있다.

【0174】 다이버시티부(220)는 제 3안테나(233)와 제 4안테나(234)를 포함할 수 있다. 제 3안테나(233)는 상측 베젤부(113)를 일부 포함할 수 있으며 제 4안테나(234)는 좌측 베젤부(112) 또는 우측 베젤부(111)를 포함할 수 있다.

【0175】 제 3안테나(233)는 LB, II/MB 대역을 지원하고 제4 안테나(234)는 II/MB 대역을 지원할 수 있다. 다이버시티부(220)의 급전부와 급전 패드, 접지부와 접지 패드들은 제 1인쇄회로기판(Main PCB)(310) 상에 배치 될 수 있다. 다이버시티부(220)에서 급전부와 급전패드를 연결하는 전기적 경로들과 접지부와 접지패드를 연결하는 전기적 경로들은 제 1인쇄회로기판(Main PCB)(310) 상에 배치될 수 있다.

【0176】 다양한 실시예에 따르면, 안테나간 신호의 분리를 위해 메인부(210)에서 우측 베젤부(111)를 제 2안테나(232)로 사용하면 다이버시티부(220)는 반대방향인 좌측 베젤부(112)를 제 4안테나(234)로 사용할 수 있다. 또는, 메인부(210)에서 좌측 베젤부(112)를 제 2안테나(232)로 사용하면 다이버시티부(220)는 반대방향인 우측 베젤부(111)를 제 4안테나(234)로 사용할 수 있다.

【0178】 도 11은 본 발명의 다양한 실시예에 따라 제 2 인쇄회로기판을 이용하여 신호 전달을 나타내는 투시도이다. 도 11의 상기 제 2인쇄회로기판(320)은 상기 설명된 도 1 내지 도 7의 제 2 인쇄회로기판(320,420,520)과 동일한 것일 수 있다.

【0179】 도 11을 참고하면, 제 2 인쇄회로기판(320)은 상기 연성 인쇄회로기판(330)과 경성 인쇄회로기판(340)이 일체형으로 구성될 수 있다. 제 2 인쇄회로기판(320)은 제 1 연성 인쇄회로기판(331), 제 2연성 인쇄회로기판(332), 제 1경성

인쇄회로기판(341) 및 제 2경성 인쇄회로기판(342)을 포함할 수 있다.

【0180】 본 발명의 다양한 실시예에 따르면, 고주파 신호를 전달하는 전기 패스(350)는 RF 신호 이외에 다른 신호도 제 2 인쇄회로기판(320)을 이용해 전달할 수 있다. 다른 신호는, 예를 들어, 유에스비(USB) 신호, 이어잭(car jack) 신호, 마이크로폰(MIC) 신호, 스피커(speaker) 신호, 터치 키(touch key) 등을 포함할 수 있다.

【0181】 본 발명의 다양한 실시예에 따르면, 제 2경성 인쇄회로기판(342) 상에는 RF 신호와 연결되는 안테나 접속부(antenna contact)와 다른 신호들이 연결되는 유에스비 접속부(USB connector), 이어잭 접속부(earjack connector), 마이크로폰 접속부(MIC connector), 스피터 접속부(speaker contactor) 등이 포함된다. 예를 들어, RF 신호는 2개의 전기 패스(350)로 전송되며 각각 제 1전기 패스(351) 및 제 2전기 패스(352)일 수 있다. 제 1전기 패스(351) 및 제 2전기 패스(352)는 상호 특성에 영향을 주지 않기 위해, 제 1전기 패스(351) 및 제 2전기 패스(352) 사이에 그라운드 라인(360)을 배치할 수 있다. 또한, 상기 그라운드 라인(360)과 제 1경성 인쇄회로기판(341)이 교차하는 영역에서 전도성 비아(305)를 배치하여 다른 층의 그라운드와 연결할 수 있다. 제 2경성 인쇄회로기판(342) 후단에 연결되는 경성인쇄회로기판은 전자장치의 다양한 모듈일 수 있으며, 한 예로 터치 키(touch key)의 역할을 할 수 있다.

【0182】 본 발명의 다양한 실시예에 따르면, 상기 제 1전기 패스(351)와 연결되는 제 1연결부(342(a))는 제1 Antenna Contact, 제1 USB Connector, 제1

Earjack Connector, 제1 MIC Connector, 제1 Speaker Contact일 수 있고, 상기 제2 전기 패스(352)와 연결되는 제2 연결부(342(b))는 제2 Antenna Contact, 제2 USB Connector, 제2 Earjack Connector, 제2 MIC Connector, 제2 Speaker Contact일 수 있다.

【0184】 도 12 및 도 13은 제1연성 인쇄회로기판 및 제1경성 인쇄회로기판을 확대한 도면이다. 도 12 및 도 13의 상기 제2 인쇄회로기판(320)은 상기 다양한 실시예에 따라, 도 1 내지 도 7의 제2 인쇄회로기판(320, 420, 520, 620, 720)중 하나일 수 있다.

【0185】 도 12 및 13을 참조하면, 고주파 신호는 접속부(370)의 B to B TYPE 커넥터(A)에서 ANT Feeding(B, C)까지 연성 인쇄회로기판 상 마이크로 스트립 라인 또는 스트립 라인으로 연결될 수 있다. 또한, 커넥터 상에서 RF 신호가 전달되는 전기 패스(350) 좌우에는 그라운드 라인(306)이 존재하며, 이 그라운드 라인(306)은 전기 패스(350)와 같은 길이방향으로 전기 패스와 나란하게 ANT Feeding(B, C)까지 연결되도록 형성할 수 있다. 또한, 그라운드(306)상 전도성 비아(305)가 형성될 수 있다. 전기 패스(351) 우측에는 그라운드 라인과 일측의 신호선의 접촉을 피하기 위해 상기 전도성 비아(305)보다 작은 전도성 비아(307)가 형성될 수 있다.

【0187】 본 발명의 다양한 실시예에 따른 전자 장치에 있어서, 하우징, 상기 하우징 내의 인쇄회로기판으로서, 복수 개의 층을 포함하고 상기 복수 개의 층은 하나 이상의 전도층과 하나 이상의 절연층을 포함하는 인쇄회로기판, 상기 하우징의 적어도 한 부분으로 형성되거나 상기 하우징 내에 배치되는 제 1전기 부품, 상기 하우징 내 상기 인쇄회로기판 상부 또는 근처에 배치되는 제 2전기 부품으로서, 상기 제 1전기 부품과 분리되는 제 2전기 부품 및 상기 제 1부품으로부터 상기 제 2부품으로 연장하는 적어도 하나의 전기 패스로서, 적어도 한 부분은 상기 인쇄회로기판 상부 또는 내부로 지나가는 상기 전기 패스를 포함하며, 상기 인쇄회로기판은 전도성 비아들의 패턴을 포함하는 영역을 포함하고, 상기 각각의 비아는 상기 복수의 층들의 적어도 일부를 통하여 상기 하나 이상의 도전층들의 적어도 하나와 접촉하고, 상기 전기 패스는 상기 전도성 비아들과 접촉하지 않으면서 상기 영역을 통해 지나갈 수 있다.

【0188】 본 발명의 다양한 실시예에 따르면, 상기 제 1전기 부품은 안테나 방사체를 포함하고, 상기 제 2전기 부품은 통신회로를 포함할 수 있다.

【0189】 본 발명의 다양한 실시예에 따르면, 상기 안테나 방사체는 상기 하우징의 적어도 한 부분을 형성할 수 있다.

【0190】 본 발명의 다양한 실시예에 따르면, 상기 인쇄회로기판의 상기 영역 위에서 보여질 때, 상기 전도성 비아들의 패턴은 상기 전기 패스를 따라 하나 이상의 열로 정렬될 수 있다.

【0191】 본 발명의 다양한 실시예에 따르면, 상기 인쇄회로기판은 적어도 일

부가 연성 인쇄회로기판을 포함하며, 상기 연성 인쇄회로기판은 상기 전기 패스를 통해 신호를 전달 할 수 있다.

【0192】 본 발명의 다양한 실시예에 따르면, 제 1 인쇄회로기판 및 접속부를 통해 상기 제 1 인쇄회로기판과 연결되며, 적어도 하나의 연성 인쇄회로기판과 경성 인쇄회로기판을 포함하는 제 2 인쇄회로기판을 포함하고, 상기 전기 패스는 상기 접속부를 통해 고주파 신호를 전달하며, 상기 제 2 인쇄회로기판의 상부 또는 내부를 통과하여 지나갈 수 있다.

【0193】 본 발명의 다양한 실시예에 따르면, 상기 전기 패스 주변에 인접하게 배치된 그라운드 라인을 더 포함하며, 상기 그라운드 라인은 상기 경성 인쇄회로기판에서 상기 전도성 비아들을 통해 상기 하나 이상의 도전층과 연결될 수 있다.

【0194】 본 발명의 다양한 실시예에 따르면, 상기 제 2 인쇄회로기판은, 상기 제 1 인쇄회로기판과 통신회로를 통해 연결되는 제 1연성 인쇄회로기판, 상기 제 1연성 인쇄회로기판과 이격된 거리에 대면 배치되는 제 2연성 인쇄회로기판 및 상기 제 1연성 인쇄회로기판 및 제 2연성 인쇄회로기판 사이에 배치되며, 상기 전도성 비아들의 패턴을 포함하는 제 1경성 인쇄회로기판을 포함할 수 있다.

【0195】 본 발명의 다양한 실시예에 따르면, 상기 제 2연성 인쇄회로기판과 연결되고, 상기 하우징의 적어도 한 부분을 형성하는 상기 안테나 방사체의 일부가 배치되는 제 2경성 인쇄회로기판을 더 포함할 수 있다.

【0196】 본 발명의 다양한 실시예에 따르면, 상기 제 2 인쇄회로기판의 상기 그라운드라인에서, 상기 복수 개의 층 중 적어도 하나의 층은 라인의 전부가 그라운드로 구성된 그라운드 패턴을 형성할 수 있다.

【0197】 본 발명의 다양한 실시예에 따르면, 상기 제 1경성 인쇄회로기판의 전도성 비아들은, 상기 복수 개의 층을 전부 관통하는 제 1전도성 비아 및 상기 복수개의 층 중 일부층을 관통하는 제 2전도성 비아를 포함하며, 상기 제 1전도성 비아는 상기 전층 또는 일부층에 배치된 상기 그라운드 패턴과 상기 그라운드 라인을 연결할 수 있다.

【0198】 본 발명의 다양한 실시예에 따르면, 상기 경성 인쇄회로기판의 상기 그라운드 라인은 상기 전기 패스와 이격된 거리에 복수 개로 형성되며, 각각의 그라운드 라인 폭은 서로 다르게 구성될 수 있다.

【0199】 본 발명의 다양한 실시예에 따르면, 상기 그라운드 라인 폭을 따라 형성된 상기 전도성 비아들은 제 1크기의 홀을 갖는 제 3전도성 비아와, 상기 제 1크기와 서로 다른 제 2크기를 갖는 제 4전도성 비아를 포함할 수 있다.

【0200】 본 발명의 다양한 실시예에 따르면, 상기 제 2 인쇄회로기판에 배치된 상기 경성 인쇄회로기판 상부 또는 내측에는 전자 부품을 실장 가능한 공간을 형성할 수 있다.

【0201】 본 발명의 다양한 실시예에 따르면, 상기 접속부는 제 1 인쇄회로기판 또는 상기 제 2 인쇄회로기판 일측에 배치되며, 상기 제 2 인쇄회로기판 상부

또는 내부로 지나가는 상기 전기 패스가 상기 제 1 인쇄회로기판과 접속할 수 있다.

【0202】 본 발명의 다양한 실시예에 따르면, 상기 제 2 인쇄회로기판의 상기 제 1연성회로기판의 일단에 배치되는 제 1접속부 및 상기 제 2 인쇄회로기판의 상기 제 2연성회로기판의 일단에 배치되는 제 2접속부를 포함할 수 있다.

【0203】 본 발명의 다양한 실시예에 따르면, 상기 전자장치는 복수의 전기 패스들과, 복수의 안테나 방사체들을 포함하며, 상기 각각의 전기 패스는 상기 복수의 안테나 방사체들 중 상기 각각의 전기 패스에 대응하는 방사체에 연결되며, 상기 복수의 전기 패스들 사이에는 상기 그라운드 라인이 배치될 수 있다.

【0204】 본 발명의 다양한 실시예에 따르면, 상기 제 1 인쇄회로기판은 상기 제 1전기 부품의 적어도 일부를 포함하며 상기 안테나 방사체와 연결되고, 상기 제 2 인쇄회로기판은 상기 제 2전기 부품이 배치될 수 있다.

【0205】 본 발명의 다양한 실시예에 따르면, 상기 제 2 인쇄회로기판의 상기 연성 인쇄회로기판과 상기 경성 인쇄회로기판은 일체형으로 구성될 수 있다.

【0206】 본 발명의 다양한 실시예에 따르면, 상기 인쇄회로기판은, 상기 영역은 경성회로기판 영역이며, 상기 제1 전기부품과 상기 영역 사이에 배치된 제1 연성회로기판 영역과, 상기 제2 전기부품과 상기 영역 사이에 배치된 제2 연성회로기판 영역을 포함할 수 있다.

【0207】 본 발명의 다양한 실시예에 따르면, 전자장치에 있어서, 제 1 인쇄 회로기판, 접속부를 통해 상기 제 1 인쇄회로기판과 연결되며, 적어도 하나의 연성 인쇄회로기판과 적어도 하나의 경성 인쇄회로기판이 일체형으로 구비된 제 2 인쇄 회로기판, 상기 접속부를 통해 고주파 신호를 전달하며, 상기 제 2 인쇄회로기판의 상부 또는 내부를 통과하여 지나가는 전기 패스 및 상기 적어도 하나의 경성 인쇄 회로기판에 배치되어 상기 복수 개의 층간 그라운드를 연결하는 적어도 하나의 전 도성 비아를 포함할 수 있다.

【0209】 도 14는 본 발명의 다양한 실시예에서의, 네트워크 환경(11) 내의 전자 장치(10)를 나타낸다.

【0210】 도 14를 참조하면, 전자 장치(10)는 버스(180), 프로세서(120), 메모리(130), 입출력 인터페이스(150), 디스플레이(160), 및 통신 인터페이스(170)를 포함할 수 있다. 어떤 실시예에서는, 전자 장치(10)는, 구성요소들 중 적어도 하나를 생략하거나 다른 구성요소를 추가적으로 구비할 수 있다.

【0211】 버스(180)는, 예를 들면, 구성요소들(110-170)을 서로 연결하고, 구성요소들 간의 통신(예: 제어 메시지 및/또는 데이터)을 전달하는 회로를 포함할 수 있다.

【0212】 프로세서(120)는, 중앙처리장치(central processing unit(CPU)), 어플리케이션 프로세서(application processor(AP)), 또는 커뮤니케이션 프로세서

(communication processor(CP)) 중 하나 또는 그 이상을 포함할 수 있다. 프로세서(120)는, 예를 들면, 전자 장치(10)의 적어도 하나의 다른 구성요소들의 제어 및/또는 통신에 관한 연산이나 데이터 처리를 실행할 수 있다.

【0213】 메모리(130)는, 휘발성 및/또는 비휘발성 메모리를 포함할 수 있다. 메모리(130)는, 예를 들면, 전자 장치(10)의 적어도 하나의 다른 구성요소에 관계된 명령 또는 데이터를 저장할 수 있다. 한 실시예에 따르면, 메모리(130)는 소프트웨어 및/또는 프로그램(140)을 저장할 수 있다. 프로그램(140)은, 예를 들면, 커널(141), 미들웨어(143), 어플리케이션 프로그래밍 인터페이스(application programming interface(API))(145), 및/또는 어플리케이션 프로그램(또는 "어플리케이션")(147) 등을 포함할 수 있다. 커널(141), 미들웨어(143), 또는 API(145)의 적어도 일부는, 운영 시스템(operating system(OS))으로 지칭될 수 있다.

【0214】 커널(141)은, 예를 들면, 다른 프로그램들(예: 미들웨어(143), API(145), 또는 어플리케이션 프로그램(147))에 구현된 동작 또는 기능을 실행하는데 사용되는 시스템 리소스들(예: 버스(180), 프로세서(120), 또는 메모리(130) 등)을 제어 또는 관리할 수 있다. 또한, 커널(141)은 미들웨어(143), API(145), 또는 어플리케이션 프로그램(147)에서 전자 장치(10)의 개별 구성요소에 접근함으로써, 시스템 리소스들을 제어 또는 관리할 수 있는 인터페이스를 제공할 수 있다.

【0215】 미들웨어(143)는, 예를 들면, API(145) 또는 어플리케이션 프로그램(147)이 커널(141)과 통신하여 데이터를 주고받을 수 있도록 중개 역할을 수행할 수 있다.

【0216】 또한, 미들웨어(143)는 어플리케이션 프로그램(147)으로부터 수신된 하나 이상의 작업 요청들을 우선 순위에 따라 처리할 수 있다. 예를 들면, 미들웨어(143)는 어플리케이션 프로그램(147) 중 적어도 하나에 전자 장치(10)의 시스템 리소스(예: 버스(180), 프로세서(120), 또는 메모리(130) 등)를 사용할 수 있는 우선 순위를 부여할 수 있다. 예컨대, 미들웨어(143)는 상기 적어도 하나에 부여된 우선 순위에 따라 상기 하나 이상의 작업 요청들을 처리함으로써, 상기 하나 이상의 작업 요청들에 대한 스케줄링 또는 로드 밸런싱 등을 수행할 수 있다.

【0217】 API(145)는, 예를 들면, 어플리케이션(147)이 커널(141) 또는 미들웨어(143)에서 제공되는 기능을 제어하기 위한 인터페이스로, 예를 들면, 파일 제어, 창 제어, 영상 처리, 또는 문자 제어 등을 위한 적어도 하나의 인터페이스 또는 함수(예: 명령어)를 포함할 수 있다.

【0218】 입출력 인터페이스(150)는, 예를 들면, 사용자 또는 다른 외부 기기로부터 입력된 명령 또는 데이터를 전자 장치(10)의 다른 구성요소(들)에 전달할 수 있는 인터페이스의 역할을 할 수 있다. 또한, 입출력 인터페이스(150)는 전자 장치(10)의 다른 구성요소(들)로부터 수신된 명령 또는 데이터를 사용자 또는 다른 외부 기기로 출력할 수 있다.

【0219】 디스플레이(160)는, 예를 들면, 액정 디스플레이(liquid crystal display(LCD)), 발광 다이오드(light-emitting diode(LED)) 디스플레이, 유기 발광 다이오드(organic light-emitting diode(OLED)) 디스플레이, 또는 마이크로 전자기계 시스템(microelectromechanical systems(MEMS)) 디스플레이, 또는 전자종이

(electronic paper) 디스플레이를 포함할 수 있다. 디스플레이(160)는, 예를 들면, 사용자에게 각종 콘텐츠(예: 텍스트, 이미지, 비디오, 아이콘, 또는 심볼 등)을 표시할 수 있다. 디스플레이(160)는, 터치 스크린을 포함할 수 있으며, 예를 들면, 전자 펜 또는 사용자의 신체의 일부를 이용한 터치, 제스처, 근접, 또는 호버링 입력을 수신할 수 있다.

【0220】통신 인터페이스(170)는, 예를 들면, 전자 장치(10)와 외부 장치(예: 제 1 외부 전자 장치(12), 제 2 외부 전자 장치(14), 또는 서버(16)) 간의 통신을 설정할 수 있다. 예를 들면, 통신 인터페이스(170)는 무선 통신 또는 유선 통신을 통해서 네트워크(162)에 연결되어 외부 장치(예: 제 2 외부 전자 장치(14) 또는 서버(16))와 통신할 수 있다.

【0221】무선 통신은, 예를 들면, 셀룰러 통신 프로토콜로서, 예를 들면, LTE(long-term evolution), LTE-A(LTE Advance), CDMA(code division multiple access), WCDMA(wideband CDMA), UMTS(universal mobile telecommunications system), WiBro(Wireless Broadband), 또는 GSM(Global System for Mobile Communications) 등 중 적어도 하나를 사용할 수 있다. 또한, 무선 통신은, 예를 들면, 근거리 통신(164)을 포함할 수 있다. 근거리 통신(164)은, 예를 들면, WiFi(wireless fidelity), 블루투스(Bluetooth), NFC(near field communication), 또는 GNSS(global navigation satellite system) 등 중 적어도 하나를 포함할 수 있다. GNSS는 사용 지역 또는 대역폭 등에 따라, 예를 들면, GPS(Global Positioning System), Glonass(Global Navigation Satellite System), Beidou

Navigation Satellite System(이하 "Beidou") 또는 Galileo, the European global satellite-based navigation system 중 적어도 하나를 포함할 수 있다. 이하, 본 문서에서는, "GPS"는 "GNSS"와 혼용되어 사용(interchangeably used)될 수 있다. 유선 통신은, 예를 들면, USB(universal serial bus), HDMI(high definition multimedia interface), RS-232(recommended standard232), 또는 POTS(plain old telephone service) 등 중 적어도 하나를 포함할 수 있다. 네트워크(162)는 통신 네트워크(telecommunications network), 예를 들면, 컴퓨터 네트워크(computer network)(예: LAN 또는 WAN), 인터넷, 또는 전화 망(telephone network) 중 적어도 하나를 포함할 수 있다.

【0222】 제 1 및 제 2 외부 전자 장치(12, 14) 각각은 전자 장치(10)와 동일한 또는 다른 종류의 장치일 수 있다. 한 실시예에 따르면, 서버(16)는 하나 또는 그 이상의 서버들의 그룹을 포함할 수 있다. 다양한 실시예에 따르면, 전자 장치(10)에서 실행되는 동작들의 전부 또는 일부는 다른 하나 또는 복수의 전자 장치(예: 전자 장치(12, 14), 또는 서버(16))에서 실행될 수 있다. 한 실시예에 따르면, 전자 장치(10)가 어떤 기능이나 서비스를 자동으로 또는 요청에 의하여 수행해야 할 경우에, 전자 장치(10)는 기능 또는 서비스를 자체적으로 실행시키는 대신에 또는 추가적으로, 그와 연관된 적어도 일부 기능을 다른 장치(예: 전자 장치(12, 14), 또는 서버(16))에게 요청할 수 있다. 다른 전자 장치(예: 전자 장치(12, 14), 또는 서버(16))는 요청된 기능 또는 추가 기능을 실행하고, 그 결과를 전자 장치(10)로 전달할 수 있다. 전자 장치(10)는 수신된 결과를 그대로 또는 추가적으로

처리하여 요청된 기능이나 서비스를 제공할 수 있다. 이를 위하여, 예를 들면, 클라우드 컴퓨팅, 분산 컴퓨팅, 또는 클라이언트-서버 컴퓨팅 기술이 이용될 수 있다.

【0223】 이하에서 설명될 전자 장치(10)는 앞서 언급한 웨어러블 기기(wearable device), 노트북, 넷북, 스마트 폰(Smart Phone), 태블릿 PC, 갤럭시 탭, 아이패드 및 무선 충전 장치 중 어느 하나로 이루어질 수 있다. 본 실시예에서 상기 스마트 폰으로 이루어질 수 있다.

【0224】 본 발명의 다양한 실시예에 따른 상기 무선 충전 장치는 근거리에서 무선으로 전력을 송수신하여 전자 장치를 충전하는 장치를 말한다.

【0225】 더불어, 전자 장치의 디스플레이부는 베젤 영역을 최소화하여 디스플레이부를 좀더 키우면서 디자인을 고급스럽게 구현하거나, 플렉서블한 디스플레이부를 제공하거나, 볼록하거나 오목한 디스플레이부를 구현하기도 한다.

【0226】 즉, 상기 디스플레이부의 주변부가 밴딩되어, 화면 영역이 측면부까지 확대하여 사용하게 구비될 수 있다. 디스플레이부의 화면 영역이 밴딩되어 측면부까지 구비됨에 따라 화면 영역을 확대하여 사용하거나, 또는 측면부에 별도의 화면을 사용할 수 있으며, 디자인적으로 고급스러움은 구현할 수 있게 되었다. 다시 말해, 상기 디스플레이부는 제 1 화면 영역(view area)과, 상기 제 1 화면 영역의 양측면에 구비되는 제 2 화면 영역(view area)을 포함할 수 있다.

【0228】 도 15는 다양한 실시예에 따른 전자 장치(1901)의 블록도이다. 전자 장치(1901)는, 예를 들면, 도 14에 도시된 전자 장치(10)의 전체 또는 일부를 포함할 수 있다. 전자 장치(1901)는 하나 이상의 프로세서(예: AP(application processor))(1910), 통신 모듈(1920), 가입자 식별 모듈(1924), 메모리(1930), 센서 모듈(1940), 입력 장치(1950), 디스플레이(1960), 인터페이스(1970), 오디오 모듈(1980), 카메라 모듈(1991), 전력 관리 모듈(1995), 배터리(1996), 인디케이터(1997), 빛 보터(1998)를 포함할 수 있다.

【0229】 프로세서(1910)는, 예를 들면, 운영 체제 또는 응용 프로그램을 구동하여 프로세서(1910)에 연결된 다수의 하드웨어 또는 소프트웨어 구성요소들을 제어할 수 있고, 각종 데이터 처리 및 연산을 수행할 수 있다. 프로세서(1910)는, 예를 들면, SoC(system on chip)로 구현될 수 있다. 한 실시예에 따르면, 프로세서(1910)는 GPU(graphic processing unit) 및/또는 이미지 신호 프로세서(image signal processor)를 더 포함할 수 있다. 프로세서(1910)는 도 19에 도시된 구성요소들 중 적어도 일부(예: 셀룰러 모듈(1921))를 포함할 수도 있다. 프로세서(1910)는 다른 구성요소들(예: 비휘발성 메모리) 중 적어도 하나로부터 수신된 명령 또는 데이터를 휘발성 메모리에 로드(load)하여 처리하고, 다양한 데이터를 비휘발성 메모리에 저장(store)할 수 있다.

【0230】 통신 모듈(1920)은, 도 14의 통신 인터페이스(170)와 동일 또는 유사한 구성을 가질 수 있다. 통신 모듈(1920)은, 예를 들면, 셀룰러 모듈(1921), WiFi 모듈(1923), 블루투스 모듈(1925), GNSS 모듈(1927)(예: GPS 모듈, Glonass

모듈, Beidou 모듈, 또는 Galileo 모듈), NFC 모듈(1928) 및 RF(radio frequency) 모듈(1929)를 포함할 수 있다.

【0231】 셀룰러 모듈(1921)은, 예를 들면, 통신망을 통해서 음성 통화, 영상 통화, 문자 서비스, 또는 인터넷 서비스 등을 제공할 수 있다. 한 실시예에 따르면, 셀룰러 모듈(1921)은 가입자 식별 모듈(예: SIM 카드)(1924)을 이용하여 통신 네트워크 내에서 전자 장치(1901)의 구별 및 인증을 수행할 수 있다. 한 실시예에 따르면, 셀룰러 모듈(1921)은 프로세서(1910)가 제공할 수 있는 기능 중 적어도 일부 기능을 수행할 수 있다. 한 실시예에 따르면, 셀룰러 모듈(1921)은 커뮤니케이션 프로세서(CP: communication processor)를 포함할 수 있다.

【0232】 WiFi 모듈(1923), 블루투스 모듈(1925), GNSS 모듈(1927) 또는 NFC 모듈(1928) 각각은, 예를 들면, 해당하는 모듈을 통해서 송수신되는 데이터를 처리하기 위한 프로세서를 포함할 수 있다. 어떤 실시예에 따르면, 셀룰러 모듈(1921), WiFi 모듈(1923), 블루투스 모듈(1925), GNSS 모듈(1927) 또는 NFC 모듈(1928) 중 적어도 일부(예: 두 개 이상)는 하나의 integrated chip(IC) 또는 IC 패키지 내에 포함될 수 있다.

【0233】 RF 모듈(1929)은, 예를 들면, 통신 신호(예: RF 신호)를 송수신할 수 있다. RF 모듈(1929)은, 예를 들면, 트랜시버(transceiver), PAM(power amp module), 주파수 필터(frequency filter), LNA(low noise amplifier), 또는 안테나 등을 포함할 수 있다. 다른 실시예에 따르면, 셀룰러 모듈(1921), WiFi 모듈(1923), 블루투스 모듈(1925), GNSS 모듈(1927) 또는 NFC 모듈(1928) 중 적어도 하

나는 별개의 RF 모듈을 통하여 RF 신호를 송수신할 수 있다.

【0234】 가입자 식별 모듈(1924)은, 예를 들면, 가입자 식별 모듈을 포함하는 카드 및/또는 내장 SIM(embedded SIM)을 포함할 수 있으며, 고유한 식별 정보(예: ICCID(integrated circuit card identifier)) 또는 가입자 정보(예: IMSI(international mobile subscriber identity))를 포함할 수 있다.

【0235】 메모리(1930)(예: 메모리(130))는, 예를 들면, 내장 메모리(1932) 또는 외장 메모리(1934)를 포함할 수 있다. 내장 메모리(1932)는, 예를 들면, 휘발성 메모리(예: DRAM(dynamic RAM), SRAM(static RAM), 또는 SDRAM(synchronous dynamic RAM) 등), 비휘발성 메모리(non-volatile Memory)(예: OTPROM(one time programmable ROM), PROM(programmable ROM), EPROM(erasable and programmable ROM), EEPROM(electrically erasable and programmable ROM), mask ROM, flash ROM, 플래시 메모리(예: NAND flash 또는 NOR flash 등), 하드 드라이브, 또는 솔리드 스테이트 드라이브(solid state drive(SSD)) 중 적어도 하나를 포함할 수 있다.

【0236】 외장 메모리(1934)는 플래시 드라이브(flash drive), 예를 들면, CF(compact flash), SD(secure digital), Micro-SD(micro secure digital), Mini-SD(mini secure digital), xD(extreme digital), MMC(multi-media card) 또는 메모리 스틱(memory stick) 등을 더 포함할 수 있다. 외장 메모리(1934)는 다양한 인터페이스를 통하여 전자 장치(1901)와 기능적으로 및/또는 물리적으로 연결될 수 있다.

【0237】 센서 모듈(1940)은, 예를 들면, 물리량을 계측하거나 전자 장치(1901)의 작동 상태를 감지하여, 계측 또는 감지된 정보를 전기 신호로 변환할 수 있다. 센서 모듈(1940)은, 예를 들면, 제스처 센서(1940A), 자이로 센서(1940B), 기압 센서(1940C), 마그네틱 센서(1940D), 가속도 센서(1940E), 그립 센서(1940F), 근접 센서(1940G), 컬러(color) 센서(1940H)(예: RGB(red, green, blue) 센서), 생체 센서(1940I), 온/습도 센서(1940J), 조도 센서(1940K), 또는 UV(ultra violet) 센서(1940M) 중의 적어도 하나를 포함할 수 있다. 추가적으로 또는 대체적으로 (additionally or alternatively), 센서 모듈(1940)은, 예를 들면, 후각 센서(E-nose sensor), EMG 센서(electromyography sensor), EEG 센서(electroencephalogram sensor), ECG 센서(electrocardiogram sensor), IR(infrared) 센서, 홍채 센서 및/또는 지문 센서를 포함할 수 있다. 센서 모듈(1940)은 그 안에 속한 적어도 하나 이상의 센서들을 제어하기 위한 제어 회로를 더 포함할 수 있다. 어떤 실시예에서는, 전자 장치(1901)는 프로세서(1910)의 일부로서 또는 별도로, 센서 모듈(1940)을 제어하도록 구성된 프로세서를 더 포함하여, 프로세서(1910)가 슬립(sleep) 상태에 있는 동안, 센서 모듈(1940)을 제어할 수 있다.

【0238】 입력 장치(1950)는, 예를 들면, 터치 패널(touch panel)(1952),(디지털) 펜 센서(pen sensor)(1954), 키(key)(1956), 또는 초음파(ultrasonic) 입력 장치(1958)를 포함할 수 있다. 터치 패널(1952)은, 예를 들면, 정전식, 감압식, 적외선 방식, 또는 초음파 방식 중 적어도 하나의 방식을 사용할 수 있다. 또한, 터

치 패널(1952)은 제어 회로를 더 포함할 수도 있다. 터치 패널(1952)은 택타일 레이어(tactile layer)를 더 포함하여, 사용자에게 촉각 반응을 제공할 수 있다.

【0239】(디지털) 펜 센서(1954)는, 예를 들면, 터치 패널의 일부이거나, 별도의 인식용 쉬트(sheet)를 포함할 수 있다. 키(1956)는, 예를 들면, 물리적인 버튼, 광학식 키, 또는 키패드를 포함할 수 있다. 초음파 입력 장치(1958)는 마이크(예: 마이크(1988))를 통해, 입력 도구에서 발생된 초음파를 감지하여, 상기 감지된 초음파에 대응하는 데이터를 확인할 수 있다.

【0240】디스플레이(1960)(예: 디스플레이(160))는 패널(1962), 홀로그램 장치(1964), 또는 프로젝터(1966)를 포함할 수 있다. 패널(1962)은, 도 1의 디스플레이(160)와 동일 또는 유사한 구성을 포함할 수 있다. 패널(1962)은, 예를 들면, 유연하게(flexible), 투명하게(transparent), 또는 착용할 수 있게(wearable) 구현될 수 있다. 패널(1962)은 터치 패널(1952)과 하나의 모듈로 구성될 수도 있다. 홀로그램 장치(1964)는 빛의 간섭을 이용하여 입체 영상을 허공에 보여줄 수 있다. 프로젝터(1966)는 스크린에 빛을 투사하여 영상을 표시할 수 있다. 스크린은, 예를 들면, 전자 장치(1901)의 내부 또는 외부에 위치할 수 있다. 한 실시예에 따르면, 디스플레이(1960)는 패널(1962), 홀로그램 장치(1964), 또는 프로젝터(1966)를 제어하기 위한 제어 회로를 더 포함할 수 있다.

【0241】인터페이스(1970)는, 예를 들면, HDMI(high-definition multimedia interface)(1972), USB(universal serial bus)(1974), 광 인터페이스(optical interface)(1976), 또는 D-sub(D-subminiature)(1978)를 포함할 수 있다. 인터페이

스(1970)는, 예를 들면, 도 3에 도시된 통신 인터페이스(170)에 포함될 수 있다. 추가적으로 또는 대체적으로(additionally and alternatively), 인터페이스(1970)는, 예를 들면, MHL(mobile high-definition link) 인터페이스, SD(secure digital) 카드/MMC(multi-media card) 인터페이스, 또는 IrDA(infrared data association) 규격 인터페이스를 포함할 수 있다.

【0242】 오디오 모듈(1980)은, 예를 들면, 소리(sound)와 전기 신호를 쌍방향으로 변환시킬 수 있다. 오디오 모듈(1980)의 적어도 일부 구성요소는, 예를 들면, 도 1에 도시된 입출력 인터페이스(145)에 포함될 수 있다. 오디오 모듈(1980)은, 예를 들면, 스피커(1982), 리시버(1984), 이어폰(1986), 또는 마이크(1988) 등을 통해 입력 또는 출력되는 소리 정보를 처리할 수 있다.

【0243】 카메라 모듈(1991)은, 예를 들면, 정지 영상 및 동영상을 촬영할 수 있는 장치로서, 한 실시예에 따르면, 하나 이상의 이미지 센서(예: 전면 센서 또는 후면 센서), 렌즈, ISP(image signal processor), 또는 플래시(flash)(예: LED 또는 xenon lamp 등)를 포함할 수 있다.

【0244】 전력 관리 모듈(1995)은, 예를 들면, 전자 장치(1901)의 전력을 관리할 수 있다. 한 실시예에 따르면, 전력 관리 모듈(1995)은 PMIC(power management integrated circuit), 충전 IC(charger integrated circuit), 또는 배터리 또는 연료 게이지(battery or fuel gauge)를 포함할 수 있다. PMIC는, 유선 및/또는 무선 충전 방식을 가질 수 있다. 무선 충전 방식은, 예를 들면, 자기공명 방식, 자기유도 방식 또는 전자기파 방식 등을 포함하며, 무선 충전을 위한 부가적

인 회로, 예를 들면, 코일 루프, 공진 회로, 또는 정류기 등을 더 포함할 수 있다. 배터리 계이지는, 예를 들면, 배터리(1996)의 잔량, 충전 중 전압, 전류, 또는 온도를 측정할 수 있다. 배터리(1996)는, 예를 들면, 충전식 전지(rechargeable battery) 및/또는 태양 전지(solar battery)를 포함할 수 있다.

【0245】인디케이터(1997)는 전자 장치(1901) 또는 그 일부(예: 프로세서(1910))의 특정 상태, 예를 들면, 부팅 상태, 메시지 상태 또는 충전 상태 등을 표시할 수 있다. 모터(1998)는 전기적 신호를 기계적 진동으로 변환할 수 있고, 진동(vibration), 또는 햅틱(haptic) 효과 등을 발생시킬 수 있다. 도시되지는 않았으나, 전자 장치(1901)는 모바일 TV 지원을 위한 처리 장치(예: GPU)를 포함할 수 있다. 모바일 TV 지원을 위한 처리 장치는, 예를 들면, DMB(digital multimedia broadcasting), DVB(digital video broadcasting), 또는 미디어플로(mediaFlo™) 등의 규격에 따른 미디어 데이터를 처리할 수 있다.

【0246】본 문서에서 기술된 구성요소들 각각은 하나 또는 그 이상의 부품(component)으로 구성될 수 있으며, 해당 구성요소의 명칭은 전자 장치의 종류에 따라서 달라질 수 있다. 다양한 실시예에서, 전자 장치는 본 문서에서 기술된 구성요소 중 적어도 하나를 포함하여 구성될 수 있으며, 일부 구성요소가 생략되거나 또는 추가적인 다른 구성요소를 더 포함할 수 있다. 또한, 다양한 실시예에 따른 전자 장치의 구성요소들 중 일부가 결합되어 하나의 개체(entity)로 구성됨으로써, 결합되기 이전의 해당 구성요소들의 기능을 동일하게 수행할 수 있다.

【0248】 이상에서 설명한 본 발명의 다양한 실시예의 전자장치는 전술한 실시예 및 도면에 의해 한정되는 것은 아니고, 본 발명의 기술적 범위 내에서 여러 가지 치환, 변형 및 변경이 가능함은 본 발명이 속하는 기술분야에서 통상의 지식을 가진 자에게 있어 명백할 것이다.

【부호의 설명】

【0250】 하우징 : 100

인쇄회로기판 : 300

제 1전기 부품 : 230

제 2전기 부품 : 240

제 1 인쇄회로기판 : 310

제 2 인쇄회로기판 : 320

전기 패스 : 350, 450, 550, 650, 750

전도성 비아 : 305, 405, 505, 605, 705

그라운드 라인 : 360, 460, 560, 660, 760

경성 인쇄회로기판 : 330, 430, 530, 630, 730

연성 인쇄회로기판 : 340, 440, 540, 640, 740

접속부 : 370, 470, 570

【특허청구범위】

【청구항 1】

전자 장치에 있어서,

하우징;

상기 하우징 내의 인쇄회로기판으로서, 복수개의 층을 포함하고 상기 복수개의 층은 하나 이상의 전도층과 하나 이상의 절연층을 포함하는 인쇄회로기판;

상기 하우징의 적어도 한 부분으로 형성되거나 상기 하우징 내에 배치되는 제 1전기 부품;

상기 하우징 내 상기 인쇄회로기판 상부 또는 근처에 배치되는 제 2전기 부품으로서, 상기 제 1전기 부품과 분리되는 제 2전기 부품; 및

상기 제 1부품으로부터 상기 제 2부품으로 연장하는 적어도 하나의 전기 패스로서, 적어도 한 부분은 상기 인쇄회로기판 상부 또는 내부로 지나가는 전기 패스를 포함하며,

상기 인쇄회로기판은 전도성 비아들의 패턴을 포함하는 영역을 포함하고, 상기 각각의 비아는 상기 복수의 층들의 적어도 일부를 통하여 상기 하나 이상의 도전층들의 적어도 하나와 접촉하고,

상기 전기 패스는 상기 전도성 비아들과 접촉하지 않으면서 상기 영역을 통해 지나가는 것을 특징으로 하는 전자 장치.

【청구항 2】

제 1항에 있어서,

상기 제 1전기 부품은 안테나 방사체를 포함하고, 상기 제 2전기 부품은 통신회로를 포함하는 것을 특징으로 하는 전자 장치.

【청구항 3】

제 1항에 있어서,

상기 안테나 방사체는 상기 하우징의 적어도 한 부분을 형성하는 것을 특징으로 하는 전자 장치.

【청구항 4】

제 1항에 있어서,

상기 인쇄회로기판의 상기 영역 위에서 보여질 때, 상기 전도성 비아들의 패턴은 상기 전기 패스를 따라 하나 이상의 일로 정렬되는 것을 특징으로 하는 전자 장치.

【청구항 5】

제 1항에 있어서,

상기 인쇄회로기판은 적어도 일부가 연성 인쇄회로기판을 포함하며, 상기 연

성 인쇄회로기판은 상기 전기 패스를 통해 신호를 전달하는 것을 특징으로 하는 전자 장치.

【청구항 6】

제 1항에 있어서, 상기 인쇄회로기판은

제 1 인쇄회로기판; 및

접속부를 통해 상기 제 1 인쇄회로기판과 연결되며, 적어도 하나의 연성 인쇄회로기판과 경성 인쇄회로기판을 포함하는 제 2 인쇄회로기판을 포함하고,

상기 전기 패스는 상기 접속부를 통해 고주파 신호를 전달하며, 상기 제 2 인쇄회로기판의 상부 또는 내부를 통과하여 지나가는 것을 특징으로 하는 전자 장치.

【청구항 7】

제 6항에 있어서,

상기 전기 패스 주변에 인접하게 배치된 그라운드 라인을 더 포함하며,

상기 그라운드 라인은 상기 경성 인쇄회로기판에서 상기 전도성비아들을 통해 상기 하나 이상의 도전층과 연결되는 것을 특징으로 하는 전자 장치.

【청구항 8】

제 6항에 있어서, 상기 제 2 인쇄회로기판은,

상기 제 1 인쇄회로기판과 통신회로를 통해 연결되는 제 1연성 인쇄회로기판;

상기 제 1연성 인쇄회로기판과 이격된 거리에 대면 배치되는 제 2연성 인쇄회로기판; 및

상기 제 1연성 인쇄회로기판 및 제 2연성 인쇄회로기판 사이에 배치되며, 상기 전도성 비아들의 패턴을 포함하는 제 1경성 인쇄회로기판을 포함하는 것을 특징으로 하는 전자 장치.

【청구항 9】

제 8항에 있어서,

상기 제 2연성 인쇄회로기판과 연결되고, 상기 하우징의 적어도 한 부분을 형성하는 상기 안테나 방사체의 일부가 배치되는 제 2경성 인쇄회로기판을 더 포함하는 것을 특징으로 하는 전자 장치.

【청구항 10】

제 7항에 있어서, 상기 제 2 인쇄회로기판의 상기 그라운드라인에서,

상기 복수개의 층 중 적어도 하나의 층은 라인의 전부가 그라운드로 구성된

그라운드 패턴을 형성하는 것을 특징으로 하는 전자 장치.

【청구항 11】

제 10항에 있어서, 상기 제 1경성 인쇄회로기판의 전도성 비아들은,

상기 복수개의 층을 전부 관통하는 제 1전도성 비아 및 상기 복수개의 층 중 일부층을 관통하는 제 2전도성 비아를 포함하며,

상기 제 1전도성 비아는 상기 전층 또는 일부층에 배치된 상기 그라운드 패턴과 상기 그라운드 라인을 연결하는 것을 특징으로 하는 전자 장치.

【청구항 12】

제 10항에 있어서,

상기 경성 인쇄회로기판의 상기 그라운드 라인은 상기 전기 패스와 이격된 거리에 복수 개로 형성되며, 각각의 그라운드 라인 폭은 서로 다른 것을 특징으로 하는 전자 장치.

【청구항 13】

제 10항에 있어서,

상기 그라운드 라인 폭을 따라 형성된 상기 전도성 비아들은 제 1크기의 홀을 갖는 제 3전도성 비아와, 상기 제 1크기와 서로 다른 제 2크기를 갖는 제 4전도

성 비아를 포함하는 것을 특징으로 하는 전자 장치.

【청구항 14】

제 7항에 있어서,

상기 제 2 인쇄회로기판에 배치된 상기 경성 인쇄회로기판 상부 또는 내측에
는 전자 부품을 실장 가능한 공간을 형성하는 것을 특징으로 하는 전자 장치.

【청구항 15】

제 6항에 있어서,

상기 접속부는 제 1 인쇄회로기판 또는 상기 제 2 인쇄회로기판 일측에 배치
되며, 상기 제 2 인쇄회로기판 상부 또는 내부로 지나가는 상기 전기 패스가 상기
제 1 인쇄회로기판과 접속하는 것을 특징으로 하는 전자 장치.

【청구항 16】

제 15항에 있어서, 상기 접속부는,

상기 제 2 인쇄회로기판의 상기 제 1연성회로기판의 일단에 배치되는 제 1접
속부 및 상기 제 2 인쇄회로기판의 상기 제 2연성회로기판의 일단에 배치되는 제 2
접속부를 포함하는 것을 특징으로 하는 전자 장치.

【청구항 17】

제 6항에 있어서,

상기 전자장치는 복수의 전기 패스들과, 복수의 안테나 방사체들을 포함하며,

상기 각각의 전기 패스는 상기 복수의 안테나 방사체들 중 상기 각각의 전기 패스에 대응하는 방사체에 연결되며,

상기 복수의 전기 패스들 사이에는 상기 그라운드 라인이 배치되는 것을 특징으로 하는 전자 장치.

【청구항 18】

제 6항에 있어서,

상기 제 1 인쇄회로기판은 상기 제 1전기 부품의 적어도 일부를 포함하며 상기 안테나 방사체와 연결되고,

상기 제 2 인쇄회로기판은 상기 제 2전기 부품이 배치되는 것을 특징으로 하는 전자 장치.

【청구항 19】

제 6항에 있어서,

상기 제 2 인쇄회로기판의 상기 연성 인쇄회로기판과 상기 경성 인쇄회로기

판은 일체형으로 구성되는 것을 특징으로 하는 전자 장치.

【청구항 20】

제 1항에 있어서, 상기 인쇄회로기판은,

상기 영역은 경성회로기판 영역이며,

상기 제1 전기부품과 상기 영역 사이에 배치된 제1 연성회로기판 영역과,

상기 제2 전기부품과 상기 영역 사이에 배치된 제2 연성회로기판 영역을 포함하는 것을 특징으로 하는 전자 장치.

【청구항 21】

전자장치에 있어서,

제 1 인쇄회로기판;

접속부를 통해 상기 제 1 인쇄회로기판과 연결되며, 복수 개의 층으로 구성된 적어도 하나의 연성 인쇄회로기판과 적어도 하나의 경성 인쇄회로기판이 일체형으로 구비된 제 2 인쇄회로기판;

상기 접속부를 통해 고주파 신호를 전달하며, 상기 제 2 인쇄회로기판의 상부 또는 내부를 통과하여 지나가는 전기 패스; 및

상기 적어도 하나의 경성 인쇄회로기판에 배치되어 상기 복수 개의 층간 그라운드를 연결하는 적어도 하나의 전도성 비아를 포함하는 것을 특징으로 하는 전

제출 일자 : 2016-06-13

자 장치.

【요약서】

【요약】

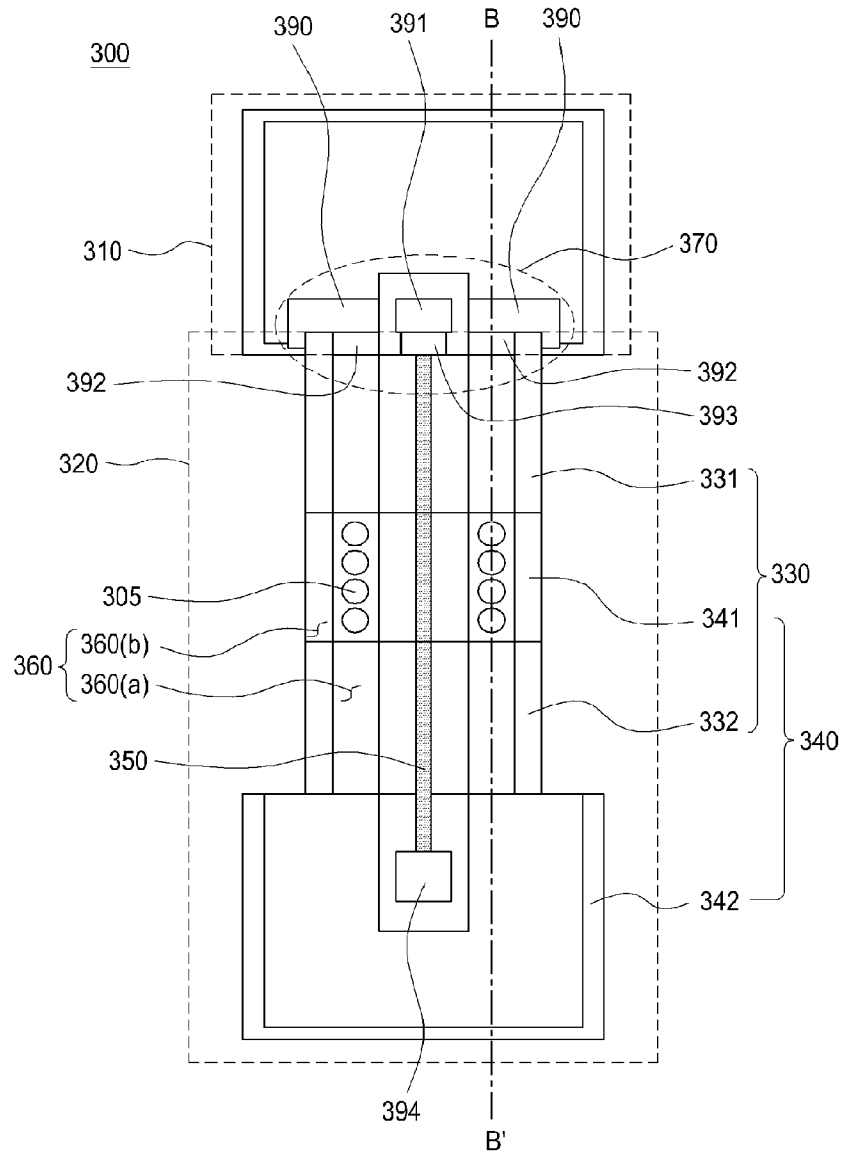
본 발명에 따른 전자 장치는 하우징; 상기 하우징 내의 인쇄회로기판으로서, 복수개의 층을 포함하고 상기 복수개의 층은 하나 이상의 전도층과 하나 이상의 절연층을 포함하는 인쇄회로기판; 상기 하우징의 적어도 한 부분으로 형성되거나 상기 하우징 내에 배치되는 제 1전기 부품; 상기 하우징 내 상기 인쇄회로기판 상부 또는 근처에 배치되는 제 2전기 부품으로서, 상기 제 1전기 부품과 분리되는 제 2전기 부품; 및 상기 제 1부품으로부터 상기 제 2부품으로 연장하는 적어도 하나의 전기 패스로서, 적어도 한 부분은 상기 인쇄회로기판 상부 또는 내부로 지나가는 전기 패스를 포함하며, 상기 인쇄회로기판은 전도성 비아들의 패턴을 포함하는 영역을 포함하고, 상기 각각의 비아는 상기 복수의 층들의 적어도 일부를 통하여 상기 하나 이상의 도전층들의 적어도 하나와 접촉하고, 상기 전기 패스는 상기 전도성 비아들과 접촉하지 않으면서 상기 영역을 통해 지나갈 수 있다. 이외에도 다양한 다른 실시예들이 가능하다.

【대표도】

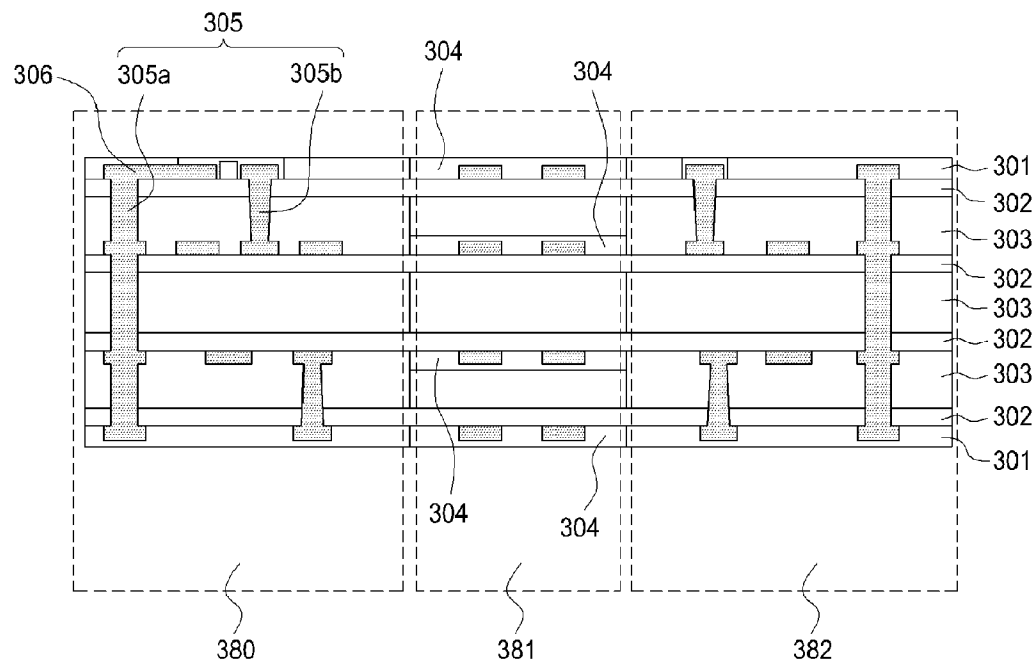
도 1

【도면】

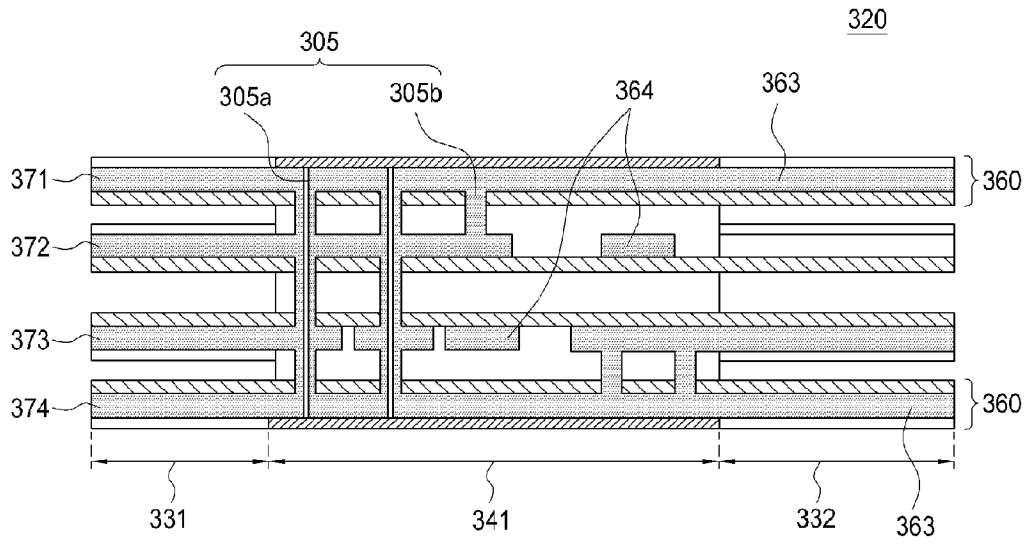
【도 1】



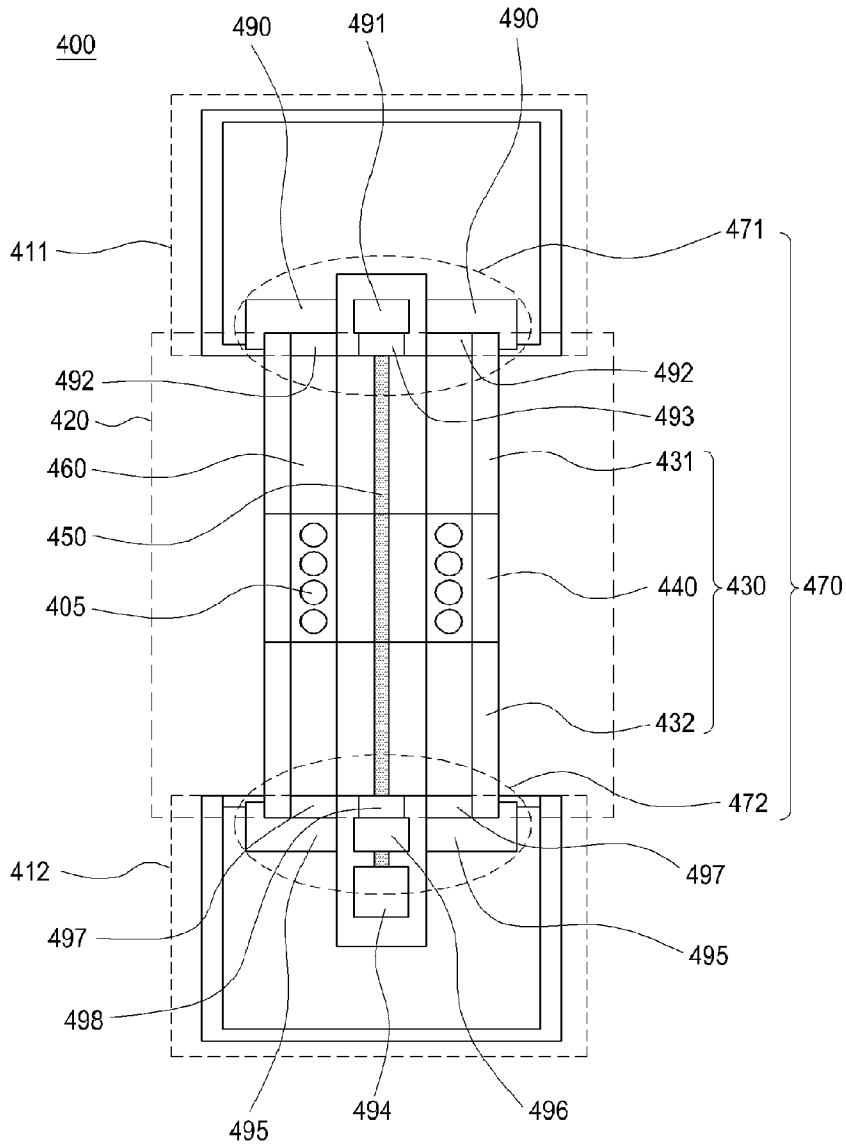
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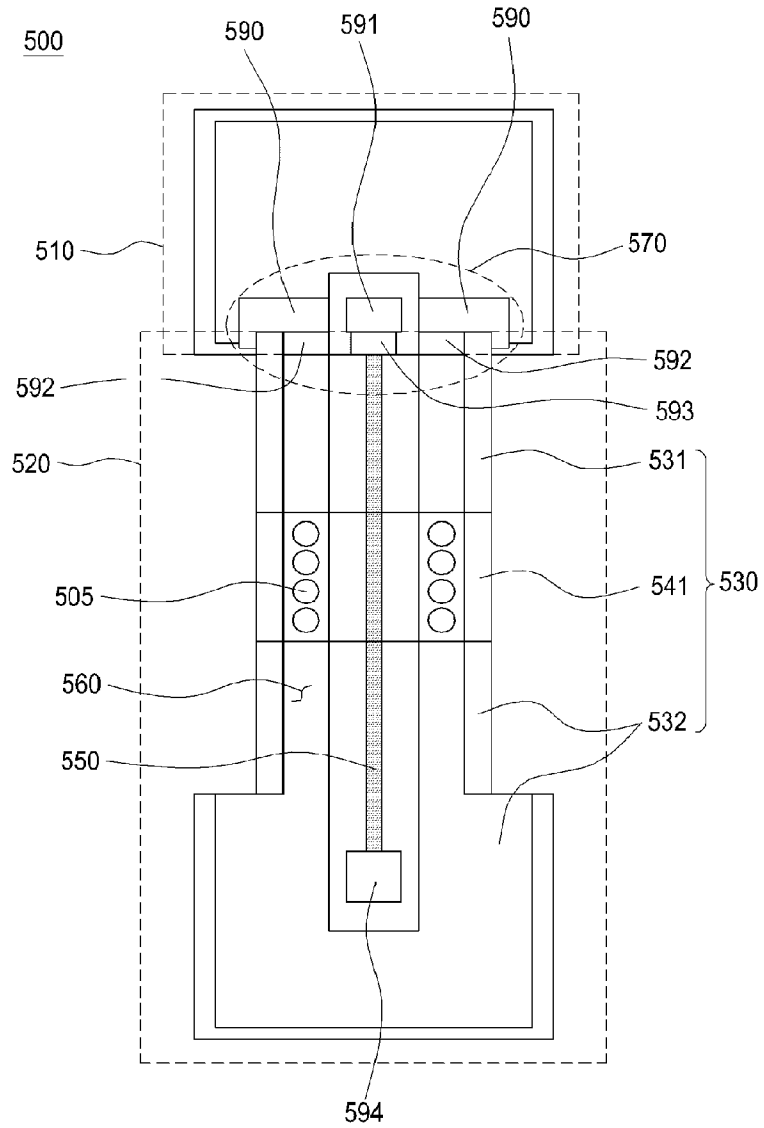
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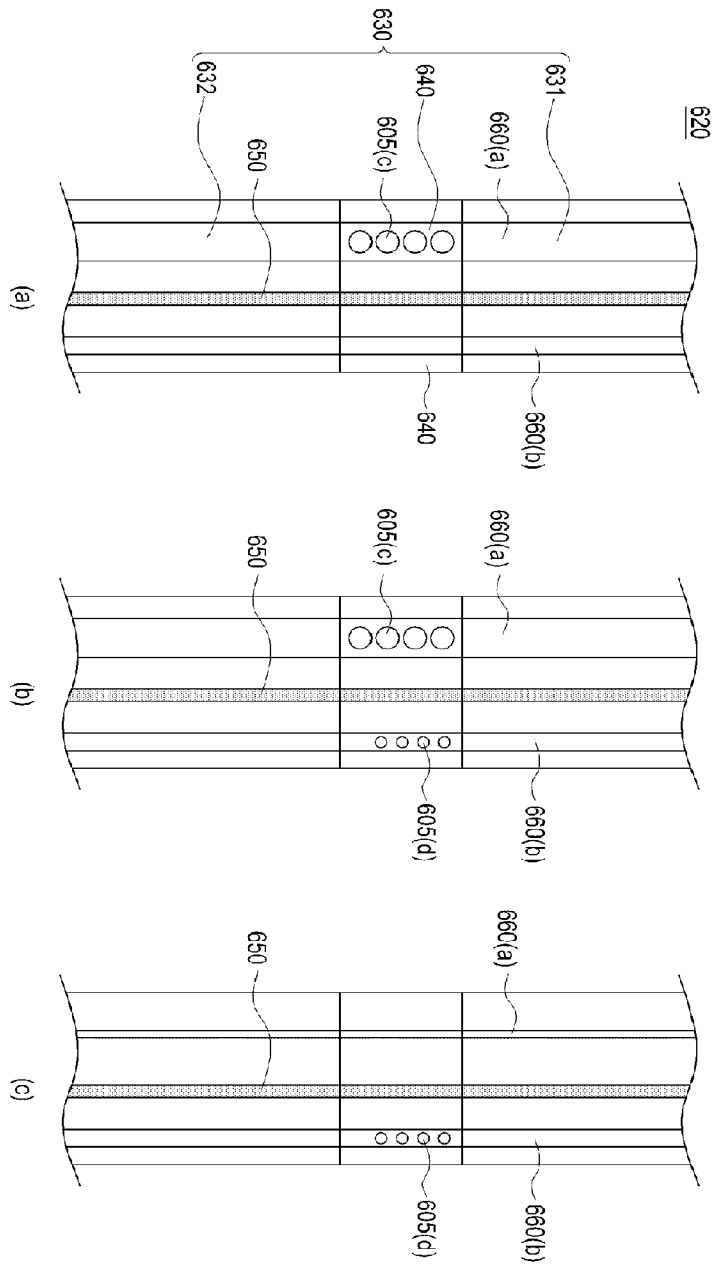
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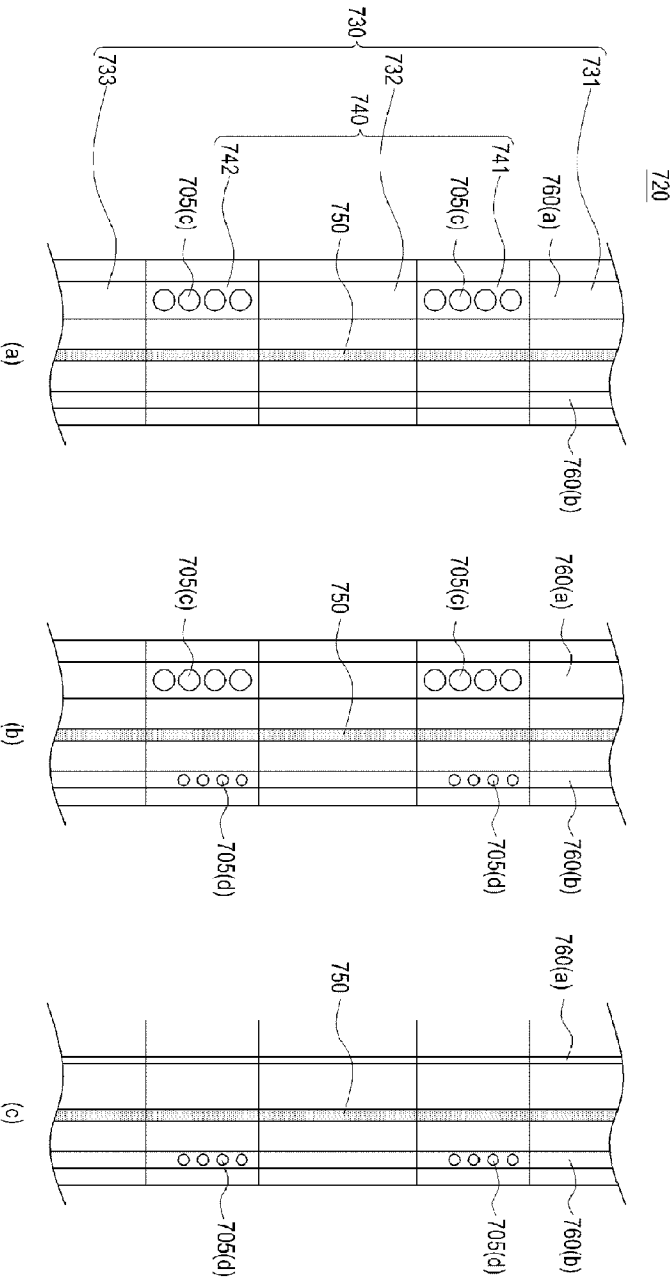
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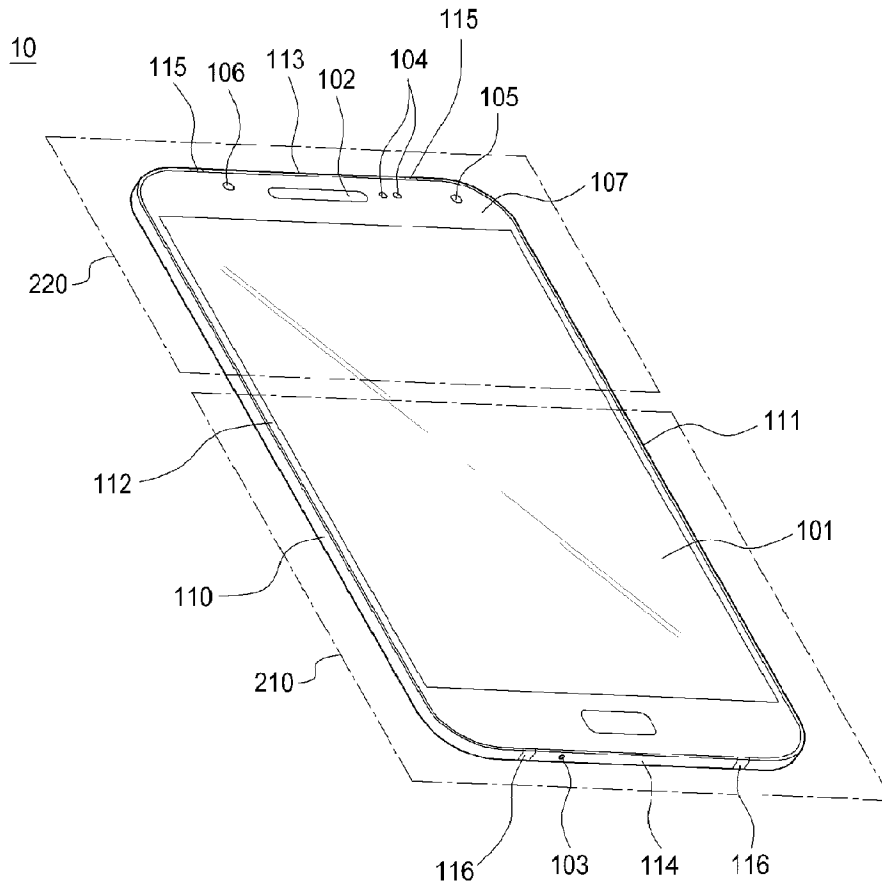
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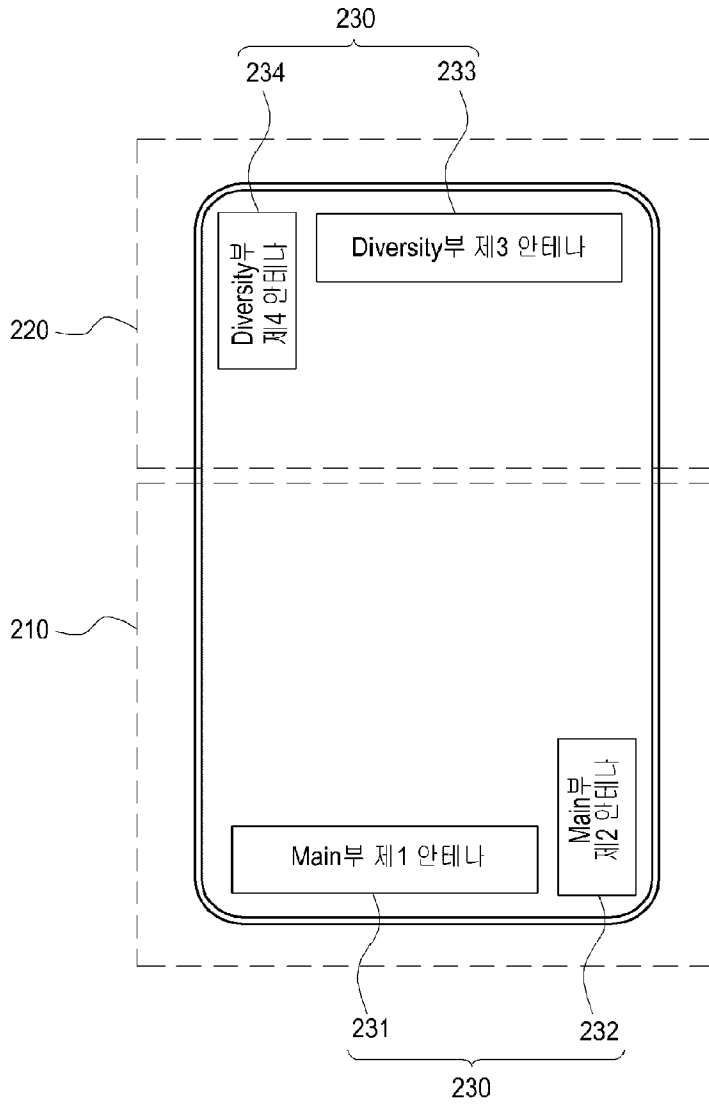
【도 7】



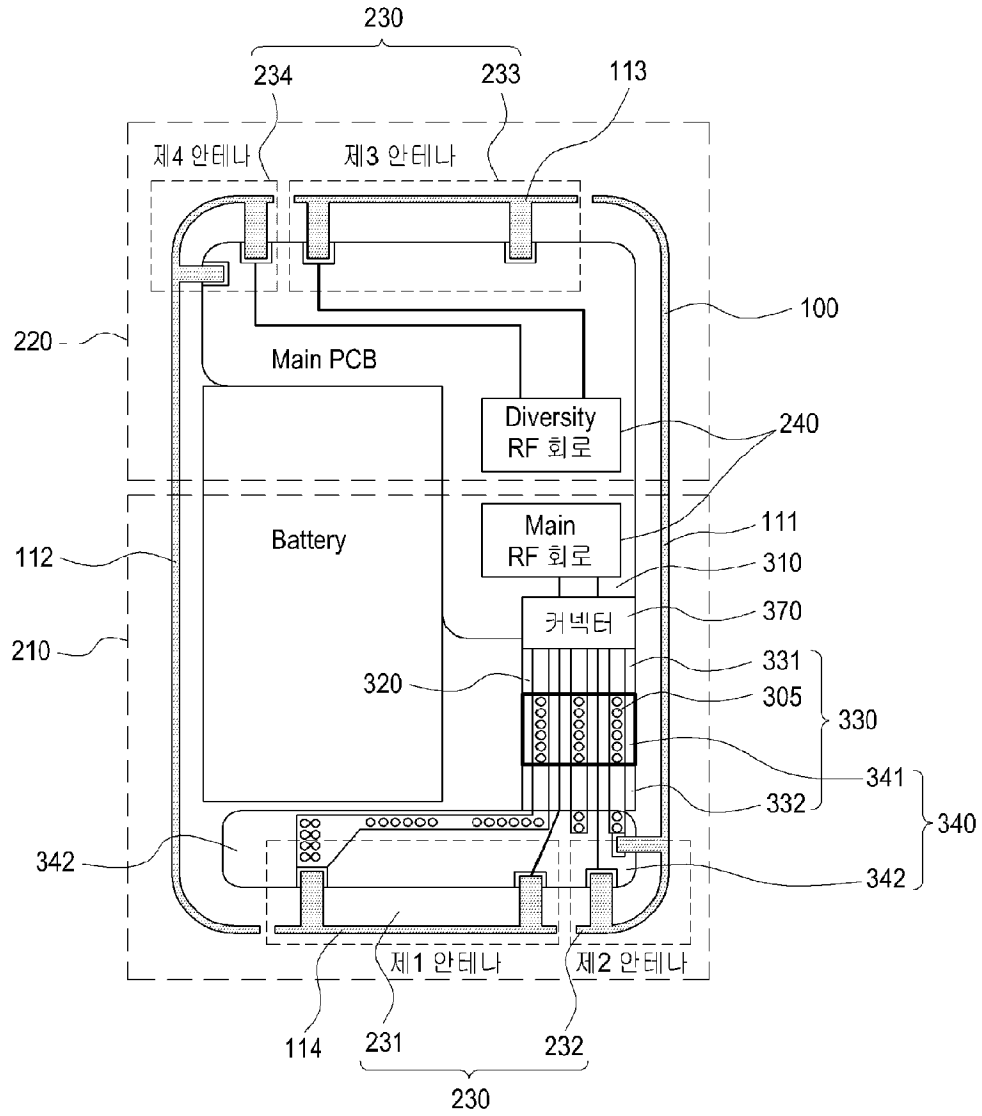
【도 8】



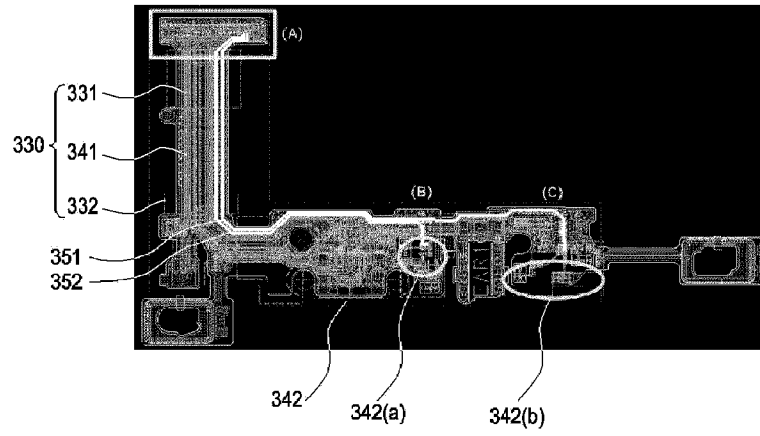
【도 9】



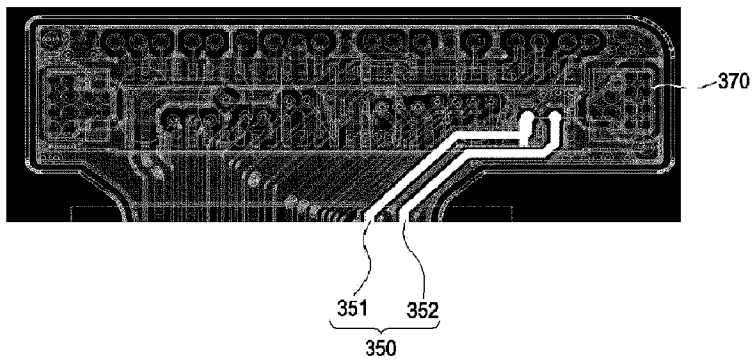
【도 10】



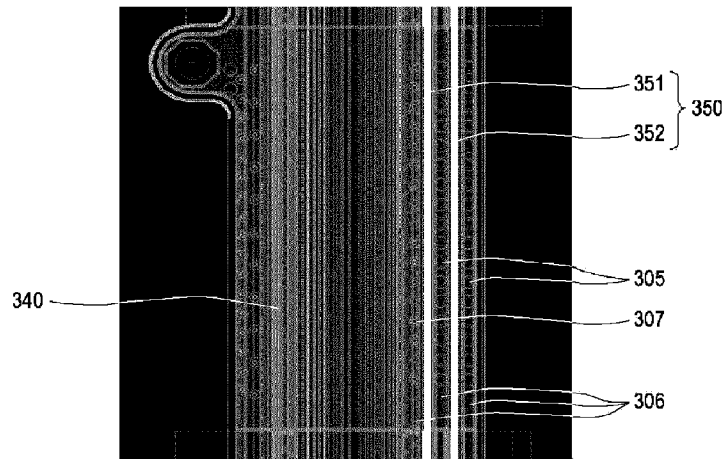
【도 11】



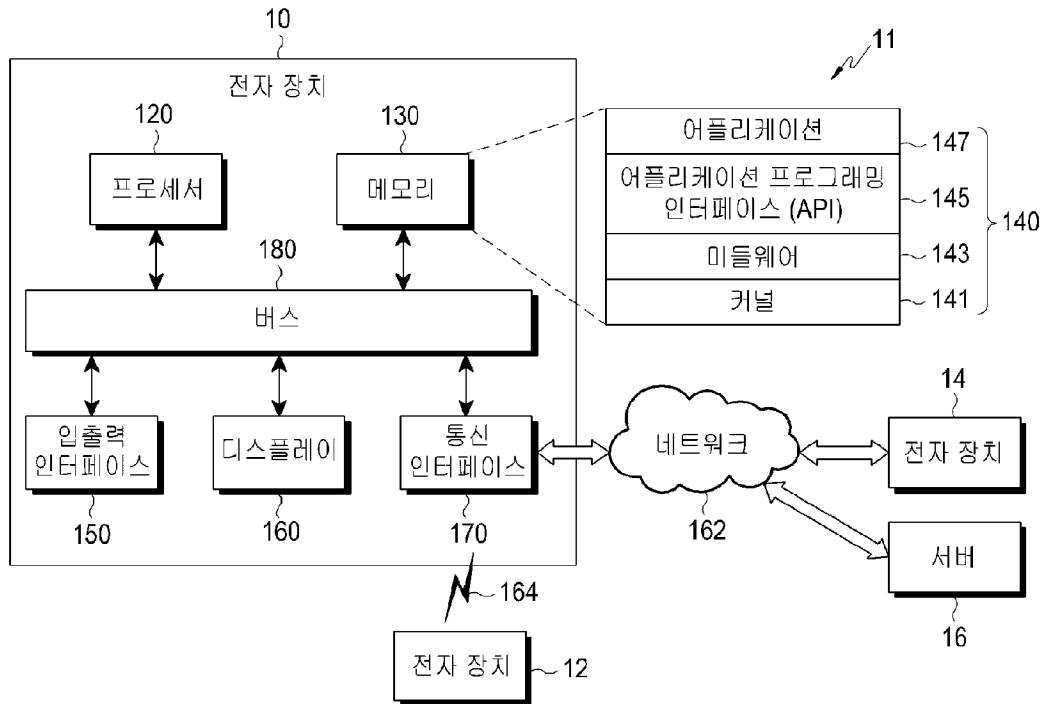
【도 12】



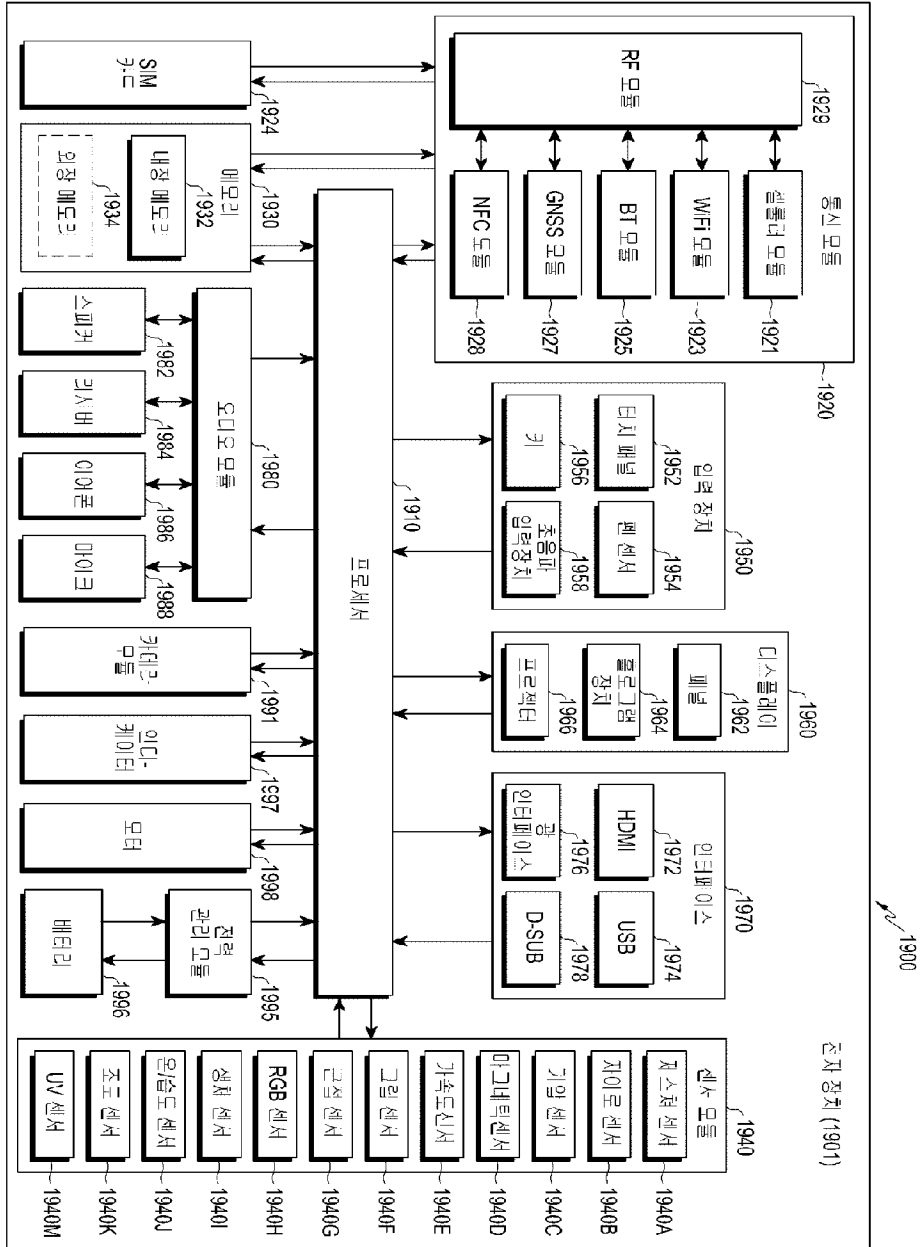
【도 13】



【도 14】



【도 15】



PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

66547 7590 12/11/2020
THE FARRELL LAW FIRM, P.C.
Paul J Farrell
290 Broadhollow Road
Suite 210E
Melville, NY 11747

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

Paul J. Farrell (Typed or printed name)
Paul J. Farrell (Signature)
March 11, 2021 (Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/939,708	07/27/2020	Han-Min CHO	678-5477 CON	8134

TITLE OF INVENTION: ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	03/11/2021

EXAMINER	ART UNIT	CLASS-SUBCLASS
AYCHILLHUM, ANDARGIE M	2848	361-790000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. Use of a Customer Number is required.

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 The Farrell Law Firm, P.C.

2

3

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

Samsung Electronics Co., Ltd

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Republic of Korea

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☒ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☒ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 50-4053

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment. NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature

Typed or printed name

Paul J. Farrell

Date March 11, 2021

Registration No. 33,494

Electronic Patent Application Fee Transmittal

Application Number:	16939708			
Filing Date:	27-Jul-2020			
Title of Invention:	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT			
First Named Inventor/Applicant Name:	Han-Min CHO			
Filer:	Paul J. Farrell/Deanna Marman			
Attorney Docket Number:	678-5477 CON			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
UTILITY APPL ISSUE FEE	1501	1	1200	1200

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1200

Electronic Acknowledgement Receipt

EFS ID:	42155605
Application Number:	16939708
International Application Number:	
Confirmation Number:	8134
Title of Invention:	ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT
First Named Inventor/Applicant Name:	Han-Min CHO
Customer Number:	66547
Filer:	Paul J. Farrell/Deanna Marman
Filer Authorized By:	Paul J. Farrell
Attorney Docket Number:	678-5477 CON
Receipt Date:	11-MAR-2021
Filing Date:	27-JUL-2020
Time Stamp:	14:50:02
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1200
RAM confirmation Number	E20213AE50222203
Deposit Account	504053
Authorized User	Deanna Marman

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

37 CFR 1.16 (National application filing, search, and examination fees)

37 CFR 1.17 (Patent application and reexamination processing fees)

37 CFR 1.19 (Document supply fees)
37 CFR 1.20 (Post Issuance fees)
37 CFR 1.21 (Miscellaneous fees and charges)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Miscellaneous Incoming Letter	Transmittal_of_Priority_Document.pdf	34845	no	1
			d72be5b8ee598ec55fea8c3bb1f79b6e32026dde		
Warnings:					
Information:					
2	Interim Copy of the Foreign Priority Document	KR_1020160073355.pdf	2554149	no	96
			968585da4c169bf90d19988ead75ee1fbd94a01e		
Warnings:					
Information:					
3	Issue Fee Payment (PTO-85B)	Issue_Fee_Transmittal.pdf	112211	no	1
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Warnings:					
Information:					
4	Fee Worksheet (SB06)	fee-info.pdf	30363	no	2
			a5bc46453daeff7729877578988302cafe7568e		
Warnings:					
Information:					
Total Files Size (in bytes):			2731568		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/939,708	07/27/2020	Han-Min CHO	678-5477 CON	8134

7590 03/22/2021
THE FARRELL LAW FIRM, P.C.
Paul J Farrell
290 Broadhollow Road
Suite 210E
Melville, NY 11747

EXAMINER	
AYCHILLHUM, ANDARGIE M	
ART UNIT	PAPER NUMBER
2848	

DATE MAILED: 03/22/2021

PRIORITY ACKNOWLEDGMENT

- ☒ 1. Receipt is acknowledged of priority papers submitted under 35 U.S.C. 119. The papers have been placed of record in the file.
- ☐ 2. Applicant's claim for priority, based on papers filed in parent Application Number _____ submitted under 35 U.S.C. 119, is acknowledged.
- ☐ 3. The priority papers, submitted _____, after payment of the issue fee are
- ☐ acknowledged
- While the priority claim or certified copy filed will be placed in the file record, neither will be reviewed and the patent when published will not include the priority claim.
- See 37 CFR 1.55(a)(2).
- ☐ not acknowledged since the processing fee in 37 CFR 1.17(i) has not been received.
- ☐ 4. For utility and plant applications filed on or after November 29, 2000, the priority claim is not entered because the claim was not presented within the time limit required by 37 CFR 1.55(a)(1). A petition to accept a delayed claim for priority under 35 U.S.C. 119(a) - (d) or (f), or 365(a) may be filed. See 37 CFR 1.55(c) and MPEP 201.14(a).

/TMC/

571-272-4200 or 1-888-786-0101
Application Assistance Unit
Office of Data Management

FOR

PTO/SB/08a (07-09)

Approved for use through 07/31/2016. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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Substitute for form 1449/PTO

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**

(Use as many sheets as necessary)

Sheet 1 of 2

Complete if Known

Application Number	not yet assigned - GAU: 2848
Filing Date	herewith
First Named Inventor	Han-Min CHO et al.
Art Unit	not yet assigned 16/939,708 -
Examiner Name	not yet assigned /ANDARGIE M
Attorney Docket Number	678-5477 CON

GAU: 284
MAYCHILLH**U. S. PATENT DOCUMENTS**

Examiner Initials*	Cite No. ¹	Document Number	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1	US- 2014/0225806	08-2014	LEE et al	
	2	US- 2009/0260859	10-2009	PAI et al.	
	3	US- 8,633,394	01-2014	Lai et al.	
	4	US- 8,143,526	03-2012	Hsu	
	5	US- 8,022,309	09-2011	Pai et al.	
	6	US- 7,781,680	08-2010	Pai et al.	
	7	US- 8,263,873	09-2012	Choi et al.	
	8	US- 2009/0294155	12-2009	Suzuki et al.	
	9	US- 2013/0063914	03-2013	Lin et al.	
	10	US- 2011/0094784	04-2011	Muro et al.	
	11	US- 2015/0223323	08-2015	O'Flynn	
	12	US- 2013/0257659	10-2013	Darnell et al.	
	13	US- 2012/0229346	09-2012	Chen et al.	
	14	US- 2006/0270472	11-2006	Chen et al.	
	15	US- 4,845,311	07-1989	Schreiber et al.	
	16	US- 2015/0234426	08-2015	Bailey	
	17	US- 2014/0147128	05-2014	Han	
		US-			
		US-			

FOREIGN PATENT DOCUMENTS

Examiner Initials*	Cite No. ¹	Foreign Patent Document	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages Or Relevant Figures Appear	T ⁶
		Country Code ³ -Number ⁴ -Kind Code ⁵ (if known)				
	18	KR 101261983	05-08-2013			
	19	KR 20150062059	06-05-2015			
	20	EP 2 180 769	04-28-2010			
	21	KR 20140136634	12-01-2014			
	22	KR 101305518	09-06-2013			
	23	KR 1020130081450	07-17-2013			

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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patent Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST.3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 2 hours to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /A.M.A/

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16939708	M327	03/22/2021	678-5477 CON

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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/939,708	04/13/2021	10978789	678-5477 CON	8134

66547 7590 03/24/2021

THE FARRELL LAW FIRM, P.C.

Paul J Farrell
290 Broadhollow Road
Suite 210E
Melville, NY 11747

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

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Mar 25, 2021 04:38:54 AM

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Application	Document	Mailroom Date	Attorney Docket No.
16939708	ISSUE.NTF	03/24/2021	678-5477 CON

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CERTIFICATE OF SERVICE

Investigation No. 337-TA-1478

The undersigned certifies that on September 10, 2026, true and correct copies of the foregoing document were served upon the following parties as indicated below:

The Honorable Lisa R. Barton Secretary to the Commission U.S. INTERNATIONAL TRADE COMMISSION 500 E Street, SW, Room 112 Washington, DC 20436	☐ Via Hand Delivery ☐ Via Overnight Delivery ☐ Via First Class Mail ☒ Via Electronic Filing (EDIS) ☐ Via Electronic Mail
The Honorable Doris Johnson Hines Administrative Law Judge U.S. INTERNATIONAL TRADE COMMISSION 500 E Street, S.W. Washington, D.C. 20436 Email: JohnsonHines1478@usitc.gov	☐ Via Hand Delivery ☐ Via Overnight Delivery ☐ Via First Class Mail ☒ Via Electronic Filing (EDIS) ☒ Via Electronic Mail
<i>Counsel for Respondents Ouraring Inc. and Oura Health Oy</i> Jasjit S. Vidwan James A. Fussell, III Saqib J. Siddiqui Bryan C. Nese MAYER BROWN LLP 199 K. Street NW Washington, DC 20006 Tel: 202-263-5282 Robert G. Pluta Nicholas H. Ciulla MAYER BROWN LLP 71 S. Wacker Drive Chicago, IL 60606 Tel: 312-782-0600 Email: Oura-ITC@mayerbrown.com	☐ Via Hand Delivery ☐ Via Overnight Delivery ☐ Via First Class Mail ☐ Via Electronic Filing (EDIS) ☒ Via Electronic Mail

/s/ Brian A. Treggs

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